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Piezoelectric Materials

Structure, Properties
and Applications

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Wesley G. Nelson
Editor

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PIEZOELECTRIC MATERIALS: STRUCTURE, PROPERTIES AND APPLICATIONS

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MATERIALS SCIENCE AND TECHNOLOGIES

PIEZOELECTRIC MATERIALS: STRUCTURE, PROPERTIES AND APPLICATIONS

WESLEY G. NELSON
EDITOR



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PREFACE

Piezoelectricity is the ability of certain crystalline materials to develop an electric charge proportional to a mechanical stress. Since its discovery, the piezoelectricity effect has found many useful applications, such as the production and detection of sound, generation of high voltages and frequency, microbalances, and ultra fine focusing of optical assemblies. It is also the basis of a number of scientific instrumental techniques and everyday uses such as acting as the ignition source for cigarette lighters and push-start propane barbecues. This book gives a review of the state of art in polycrystalline piezoelectric ceramic materials, which covers the processing, properties, characterization and applications of piezoelectric ceramic materials.

Chapter 1- Piezoelectricity is the ability of certain crystalline materials to develop an electric charge proportional to a mechanical stress, which is called the direct piezoelectric effect discovered by Curie brothers in 1880. Soon it was realized that materials showing this phenomenon must also show the converse piezoelectric effect: a geometric strain/deformation proportional to an applied voltage. Typical crystals (e.g., quartz, tourmaline and Rochelle salt) exhibit the piezoelectric effect. Since its discovery the piezoelectricity effect has found many useful applications, such as the production and detection of sound, generation of high voltages and frequency, microbalances, and ultra fine focusing of optical assemblies. It is also the basis of a number of scientific instrumental techniques with atomic resolution such as the scanning probe microscopy, and everyday uses such as acting as the ignition source for cigarette lighters and push-start propane barbecues. However, the traditional piezoelectric single crystals suffer from the disadvantages such as weak piezoelectric effect, low mechanical strength, sensitivity to moisture, and very narrow operated temperature range. Compared to the traditional single crystals, electrically poled polycrystalline ferroelectric ceramics, such as barium titanate (BaTiO_3) and lead zirconium titanate (PZT), offer the advantages of large and stable piezoelectric effects, high strength and ease of fabrication in general, especially into complex shapes and large area pieces. Nowadays, they become the dominant piezoelectric materials in the fields of piezoelectric applications such as actuators, sensors, and transducers in intelligent systems and smart structures, dominating the world market today. Therefore, the aim of this chapter is giving a review of the state of art in polycrystalline piezoelectric ceramic materials, which covers the processing, properties, characterization, and applications of piezoelectric ceramic materials; whereas single crystals, polymeric piezoelectric materials and organic/inorganic piezoelectric composites are outside the scope of this review. This chapter is organized in the following way. In the first part, the history and processing of piezoelectric ceramic materials are briefly introduced, and then the

general characteristics of piezoelectric ceramic materials are described with an emphasis on the piezoelectric parameters, compositions and properties, and piezoelectric constitutive relationships. Characterization methods for piezoelectric properties and ferroelectric domain structures of piezoelectric ceramic materials are addressed in the third part. Finally, various applications of piezoelectric ceramic materials in ultrasonic actuators, sensors, transducers, and active vibration controlling, are described, and the personal perspectives towards future trends of piezoelectric ceramic materials are also presented.

Chapter 2- Stress-biased piezoelectric composites represent a family of piezoelectric ceramic-based actuators and transducers that offer a variety of performance advantages compared to more traditional unimorph devices. In this chapter, the history, fabrication, performance attributes and potential applications for these devices are discussed. The devices are typically fabricated using piezoelectric and elastic layers at temperatures between 150 and 300°C. The different coefficients of thermal expansion of the layers typically result in doming of the composite during cooling and the development of an in-situ stress-profile across the piezoelectric layer. These characteristics increase the robustness of the composite devices and may enhance domain wall motion contributions to piezoelectric response, yielding high electromechanical deformation. To date, a variety of composite devices based on different layers, processing strategies, and design philosophies have been developed to take advantage of these characteristics. Potential applications for the devices are also discussed, as are the underlying aspects of the devices that contribute to their performance.

Chapter 3- Since the discovery of piezoelectric effect and relaxor behavior, the technology of using piezoelectric and electrostrictive ceramic materials as solid-state actuators for small and precise mechanical movement devices has undergone considerable investigation and development. Despite of their enormous applications, the direct extensional strain in most piezoelectric or electrostrictive ceramic materials is at best a few tenths of one percent. Thus it is essential to develop the method to amplify the strain of devices. Some well-known techniques for producing large displacement in these materials include two categories. The first category is related to the shape and structure modifications such as unimorph, bimorph bender moonie and rainbow actuator. However, each of these structures has its limitations in regards to size, weight, maxim displacement, load-bearing capability, high performance, and low voltage driving for using digital equipment such as digital camera, cellular phone etc.. The second category is resonance vibration-related mechanism such as inch-worm and ultrasonic motors. When the stress/strain capabilities of all of these technologies are considered, it is evident that there is still a real need for additional technique which will produce even higher strain or displacement and will use soft or hard piezoelectric material. At the same time, the device must sustain reasonable stress and resonance driving condition. This chapter describes such a technology which contains the newly developed structure, manufacturing process, properties and applications.

Chapter 4- There has been extensive research to study and improve the integration of piezoelectric elements in smart materials and structures since the early 1990s. The unique properties of piezoceramics to efficiently generate an electric potential from an induced mechanical strain while reversibly, produce stress from an applied voltage, appropriately placed these elements as favoured sensors and actuators. Research is also motivated by commercial interests in noise and vibration control, shape control, and structural damping, particularly for civil and aerospace applications. In the last decade, the field has extended to structural health monitoring for the detection and quantitative evaluation of in-service and

manufacturing defects. This short review arrives in a timely manner to outline the present progress of the technology by compiling the studies conducted and summarising the significant contributions made. Fabrication and manufacturing methods for piezoceramic attachment are consolidated while addressing the aspects of material selection, safe design, performance and durability. This is followed by the descriptions of testing techniques available to simulate in-service loading and to evaluate the manufactured smart product through various measured parameters. Important test results and analysis from past studies are presented to understand the failure modes and the operation limits. Literature on key modelling approaches to study the integrated piezoceramic transducers are briefly highlighted, showing the evolution of the simple analytical one-dimensional models to more complex and numerical two/three-dimensional models.

Chapter 5- The continuous technological advances have required materials with properties that conventional material cannot display. Material property combinations are being the focus to the development of composite materials, which are considered a multiphase material that exhibits properties of the constituent phases. One interesting material to be studied as sensing material is the composite made of ferroelectric ceramic and polymeric matrix as a two phases composite material. In that case the combinations properties intended are the high piezo and pyroelectric activities of the dense ceramic with the impact resistant, flexibility, formability and low densities of the polymer. Considering the way that the ceramic is dispersed in the polymeric phase, according to Newnham [1] there are 10 possible connectivities, which indicate the way that each phase is self-connected in the composite. The composite studied here is made of a modified lead titanate (Pz34) ceramic and polyether-ether-ketone (PEEK) high performance polymer matrix. The composite film was obtained hot pressing both phases at 360 °C during 2 h, and 12 MPa of pressure, being essentially a 0-3 composite that means the dispersed ceramic grains are not self-connected. Using the piezoelectric property the composite film was used to detect acoustic emission (AE), which is a transient elastic wave generated by sudden deformation taken place in materials under stress. AE can be applied for evaluating the “health” of structures in a non-destructive way. The composite film was surface mounted on an aluminum panel and ball bearing drop and pencil lead break were used as simulating AE sources. On the pyroelectric side, the composite film was used as sensing element in a pyroelectric chamber to measure X-ray intensity in the orthovoltage range (120 KV_p – 300 KV_p). Experimental results indicate that the combination properties of ceramic and polymer phases provide an alternative sensing material to be used as piezo and pyroelectric sensor.

Chapter 6- A simple piezoelectric floating mass transducer (PFMT) for implantable middle ear hearing devices (IMEHDs) is studied. The transducer that is implanted in the middle ear consists of a PMN-PT multi-layered piezoelectric actuator and a metal case. The proposed PFMT can be attached to the long process of the incus using a metal clamp, and vibrates the ossicular chain according to the force originating from the expansion and contraction of the piezoelectric actuator. This chapter presents the design and an evaluation of the proposed PFMT as a new type of vibrating transducer. The design factors considered include the dimensions, generated force, weight, power consumption, and time-frequency characteristics. An *in vitro* experiment using a human temporal bone confirms that the PFMT excitation at 10 V_{pp} can generate vibration displacements equivalent to a sound pressure of about 94 dB SPL below 1 kHz and about 140 dB SPL at 8 kHz. The power consumption of the PFMT to produce a sound stimulation equivalent to 94 dB SPL is 65 µW at 1 kHz.

Therefore, the proposed PFMT is expected to be useful as an IMEHD transducer with the advantages of simple surgical implantation, no interference from external electromagnetic fields, and improved productivity.

Chapter 7- Mechanical, electrical, and thermal fields are coupled in most of the physical problems. Thermal effects could be important when structures containing piezoelectric materials need to operate at high or low temperature environments. Such environments are quite possible in the applications of smart materials and structures. An area where piezoelectric materials may provide dramatic advantages is in the development of piezoelectric composite structures with the capability to sensor thermally induced distortions and to actively compensate for adverse thermo-mechanical conditions. Typical applications of such structures are envisioned in the thermal distortion management of propulsion components and/or space structures. The analysis of the thermal fracture process of piezoelectric materials could provide information to improve the design of piezoelectric devices operated in thermal environments. Thermoelastic behavior of piezoelectric solids without defects has been studied by many researchers [e.g. 1-9].

Thermo-piezoelectric materials with defects, such as holes and interface cracks, were studied [10, 11]. The problem of a two-dimensional piezoelectric material with an elliptic cavity under uniform heat flow was investigated [12]. Thermally induced fracture problem for a piezoelectric laminate having a crack subject to uniform electric and temperature fields was solved in [13]. Stress intensity factors for internal penny-shaped cracks in isotropic materials have been provided in Handbook [14], where the crack face temperature is maintained or the crack-face thermal flux is given. In the case of piezoelectric materials, crack tip singularities can include electric fields and electric displacements. The problem is important but is more complicate. Accordingly, there is a need to provide the thermal stress and electric displacement intensity factors for various thermal load conditions. The expressions should be in closed-form so that the material designers and scientists and directly use them.

This Chapter provides an exact solution for a thermal penny-shaped crack in a piezoelectric medium. An arbitrarily distributed temperature field is applied on the crack faces. Closed-form expressions for the stress and electric displacement intensity factors at the crack border are given. Both impermeable and permeable crack-face electric boundary conditions are considered. Further, crack tip field intensity factors for a given thermal flux on the crack faces are also obtained. The solutions are mainly provided for transversely isotropic materials, but the results for isotropic materials and dielectrics are also given. Also introduced is the design of a smart functionally graded thermopiezoelectric composite structure.

Chapter 8- It is generally well understood that thermal shock conditions, which arise during sudden heating or cooling of a solid, can result in very high stresses. If the thermal transient is severe enough, sudden fracture may occur. The degree of damage and strength degradation of ceramics subjected to fluctuating thermal environments is a major limiting factor in relation to service requirements and lifetime performance. Early studies to quantify the severity of thermal shock in a plate of finite thickness attempted to draw a correlation between the magnitude of maximum tensile stress on the plate surface and the likelihood of failure. This approach assumes that there are no pre-existing surface flaws. A more sophisticated fracture analysis incorporates an edge crack in the transient stress analysis and thus the degree of severity of thermal shock is characterized in terms of stress intensity factor which is a function of crack size and relevant heat transfer and thermo-elastic coefficients [1-3].

The strength of a material under a transient thermal stress state depends on a number of material properties including thermal expansion coefficient, thermal conductivity, thermal diffusivity, fracture toughness and tensile strength, elastic properties, etc. For piezoelectric ceramics, more parameters can affect transient thermal fracture. Generally, the application of piezothermoelasticity to the fracture of piezoelectric materials under thermal environment involves the solution of two distinct boundary-value problems. The first is a standard potential problem in the classical theory of thermal conduction, and the second is the determination of the mechanical and electrical fields caused by the uneven heating of the various points within the solid. This chapter is mainly concerned with the stress and electric fields in piezoelectric materials subjected to steady state or transient thermal loads. Section 7.2 gives a case study for piezoelectric material strip suddenly exposed to an environment of different temperature. The poling direction of the strip is parallel to its surfaces. In section 7.3, a piezoelectric material strip whose poling direction is perpendicular to its surfaces is investigated. Section 7.4 investigated an electrically conductive surface crack in a piezoelectric half space under a transient heating (or cooling) on its surface. Crack propagation behavior is discussed and the thermal shock strength of the piezoelectric medium is established. In section 7.5, an internal crack is studied for a piezoelectric cylinder under a transient heating on its outer surface. The transient thermal stress and electric displacement fields, and the associated thermal fracture problem are obtained. The maximum thermal stress and electric displacement, and the maximum thermal stress and electric displacement intensity factors are given. Also obtained are the transient crack opening displacement and the transient electric displacement inside the crack. The strength and fracture of the cylinder are evaluated. Finally, some conclusions are drawn in Section 7.6.

Chapter 1

PIEZOELECTRIC CERAMICS MATERIALS: PROCESSING, PROPERTIES, CHARACTERIZATION, AND APPLICATIONS

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ABSTRACT

Piezoelectricity is the ability of certain crystalline materials to develop an electric charge proportional to a mechanical stress, which is called the direct piezoelectric effect discovered by Curie brothers in 1880. Soon it was realized that materials showing this phenomenon must also show the converse piezoelectric effect: a geometric strain/deformation proportional to an applied voltage. Typical crystals (e.g., quartz, tourmaline and Rochelle salt) exhibit the piezoelectric effect. Since its discovery the piezoelectricity effect has found many useful applications, such as the production and detection of sound, generation of high voltages and frequency, microbalances, and ultra fine focusing of optical assemblies. It is also the basis of a number of scientific instrumental techniques with atomic resolution such as the scanning probe microscopy, and everyday uses such as acting as the ignition source for cigarette lighters and push-start propane barbecues. However, the traditional piezoelectric single crystals suffer from the disadvantages such as weak piezoelectric effect, low mechanical strength, sensitivity to moisture, and very narrow operated temperature range. Compared to the traditional single crystals, electrically poled polycrystalline ferroelectric ceramics, such as barium titanate (BaTiO_3) and lead zirconium titanate (PZT), offer the advantages of large and stable piezoelectric effects, high strength and ease of fabrication in general, especially into complex shapes and large area pieces. Nowadays, they become the dominant piezoelectric materials in the fields of piezoelectric applications such as actuators, sensors, and transducers in intelligent systems and smart structures, dominating the world market today. Therefore, the aim of this chapter is giving a review of the state of art in polycrystalline piezoelectric ceramic materials, which covers the processing, properties, characterization, and applications of piezoelectric ceramic materials; whereas single crystals, polymeric piezoelectric materials and organic/inorganic piezoelectric composites

are outside the scope of this review. This chapter is organized in the following way. In the first part, the history and processing of piezoelectric ceramic materials are briefly introduced, and then the general characteristics of piezoelectric ceramic materials are described with an emphasis on the piezoelectric parameters, compositions and properties, and piezoelectric constitutive relationships. Characterization methods for piezoelectric properties and ferroelectric domain structures of piezoelectric ceramic materials are addressed in the third part. Finally, various applications of piezoelectric ceramic materials in ultrasonic actuators, sensors, transducers, and active vibration controlling, are described, and the personal perspectives towards future trends of piezoelectric ceramic materials are also presented.

Keywords: Piezoelectric ceramic materials; processing; properties and characterization; applications.

1. INTRODUCTION

Piezoelectricity or pressure electricity is the phenomenon discovered in 1880 by the Curie brothers who were the first to demonstrate the generation of electricity (surface charges) on well prepared crystals of quartz as a result of mechanical pressure. Inversely, when a voltage is applied across a piezoelectric material, it can undergo a mechanical distortion in response. Typical crystals such as quartz, tourmaline, and Rochelle salt, exhibit piezoelectric effect, they have some applications in piezoelectric devices such as sonars. However, the traditional single crystal materials suffer from some disadvantages which limit their use to some extent. For example, they mostly exhibit only a weak piezoelectric effect, usually have low mechanical strength. Some ones are very sensitive to moisture and their operated temperature range is often limited. Compared to the traditional piezoelectric single crystals, piezoelectric ceramics, such as electrically poled barium titanate (BaTiO_3) and lead zirconium titanate (PZT) ceramics, exhibit large piezoelectric effects (high electromechanical coupling). They are mechanically strong, hard, chemically inert and immune to humidity. As a consequence, they become the present primary commercial piezoelectric materials, and hence are widely used as actuators and sensors in intelligent system and smart structures [1,2].

The beginning of the twentieth century gave the birth to most of the classic applications of piezoelectrics, such as quartz sonars, resonators and accelerometers. After the World War II and following the discovery of PZT, the advances made in piezoelectric materials allowed the development of numerous applications based on the tailored piezoelectric properties. High-performance of piezoelectric ceramic materials now open up new possibilities for “energy harvesting,” making use of ambient motion and vibration to generate electricity where batteries or other power sources are impractical or unavailable [3].

Concerning the amplitudes of piezoelectric ceramic materials and device applications existing already, today we may further expect continuous future development and fascinating new novel application areas. In this chapter, we will present an overview of the state of art in piezoelectric ceramic materials, which covers their processing, properties, characterization, and applications. We first briefly introduce the history and processing of piezoelectric ceramic materials, then describe the general characteristics of piezoelectric ceramic materials with a focus on the compositions and properties, piezoelectric parameters, and piezoelectric constitutive relationships, followed by the most common characterization methods for

piezoelectric properties and ferroelectric domain structures, and finally some potential applications of piezoelectric ceramic materials in actuators, sensors and transducers, are presented, and the personal perspectives towards future trends of piezoelectric ceramic materials are also given out.

2. HISTORY AND PROCESSING OF PIEZOELECTRIC CERAMIC MATERIALS

2.1. History of Piezoelectricity

Piezoelectricity is a property of a group of materials that was discovered in 1880 by Pierre and Jacques Curie during their study of the effects of pressure on the generation of electrical charge by crystals such as quartz, tourmaline, and Rochelle salt. In 1881, the term “piezoelectricity” was first suggested by W. Hankel, and the converse effect was mathematically deduced by Gabriel Lipmann from fundamental thermodynamic principles [4]. The schematic diagram for illustration of direct and converse piezoelectric effects is shown in Figure 1(a) and 1(b), respectively [5]. For the next few decades, piezoelectricity remained something of a laboratory curiosity. More work was done to explore and define the crystal structures that exhibited piezoelectricity. This culminated in 1910 by the publication of Woldemar Voigt's *Lehrbuch der Kristallphysik* (a textbook on crystal physics), which described the 20 natural crystal classes capable of piezoelectricity, and rigorously defined the piezoelectric constants using tensor analysis [6]. However, the complexity of the science of piezoelectricity made it difficult to mature to practical applications until a few years later. In 1917 Paul Langevin and his coworkers developed an ultrasonic submarine detector, which consisted of a transducer made of thin quartz crystals carefully glued between two steel plates, and a hydrophone to detect the returned echo. By emitting a high-frequency chirp from the transducer, and measuring the amount of time it takes to hear an echo from the sound waves bouncing off an object, one can calculate the distance to that object. This success opened up the opportunities for piezoelectric materials in underwater applications and a host of other applications such as ultrasonic transducers, microphones, and accelerometers. In 1935, Busch and Scherrer discovered piezoelectricity in potassium dihydrogen phosphate (KDP) and its isomorph. The KDP family was the first discovered major family of piezoelectrics and ferroelectrics. From 1940 to 1943, unusual dielectric properties such as an abnormally high dielectric constant were found in BaTiO_3 independently by Wainer and Salmon, Ogawa, and Wul and Golman. After its discovery, compositional modifications of BaTiO_3 led to improvement in temperature stability and high voltage output. Piezoelectric transducers based on BaTiO_3 ceramics became well established in a number of devices. BaTiO_3 ceramics has unusually high dielectric constant due to its ferroelectric (permanent internal dipole moment) nature, thus ushering in a new class of ferroelectrics with the ABO_3 perovskite structure. As shown in Figure 2, the unit cell of BaTiO_3 consists of a corner-linked network of oxygen octahedra with Ti^{4+} ions occupying B sites within the octahedral cage and the Ba^{2+} ions situated in the interstices (A site) created by the linked octahedral [7]. Below the Curie temperature, there is a structural distortion to a lower-symmetry phase accompanied by the shift off-center of the small cation (Ti^{4+}), as shown in Figure 2(c). Displacement occurs

along the c axis in a tetragonal structure, although it should be understood that it can also occur along the orthogonal a or b axes as well. The views of “polarization up” and “polarization down” (representing 180° polarization reversal) show two of the six possible permanent polarization positions. The spontaneous polarization derives largely from the electric dipole moment created by this shift.

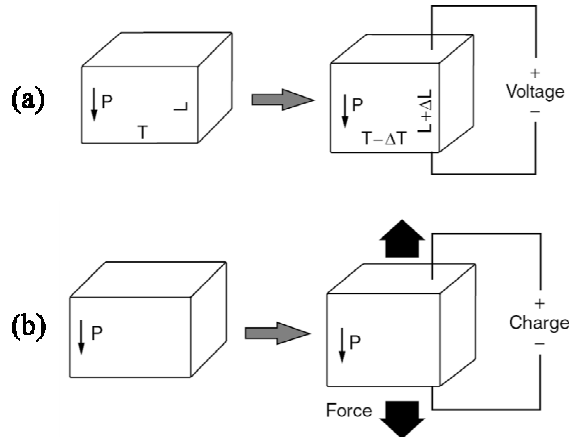


Figure 1. Schematic diagrams of the direct and converse piezoelectric effect: (a) an electric field applied to the material changes its shape; (b) a stress on the material yields an electric field across it. Reproduced with permission from [5], Kholkin, A.; Jadidian B.; Safari, A. In *Encyclopedia of Smart Materials*, Schwartz, M.; Eds.; ISBN 0-471-17780-6; John Wiley & Sons, Inc.: New York, NY, 2002, Vol.1, pp139 -148.

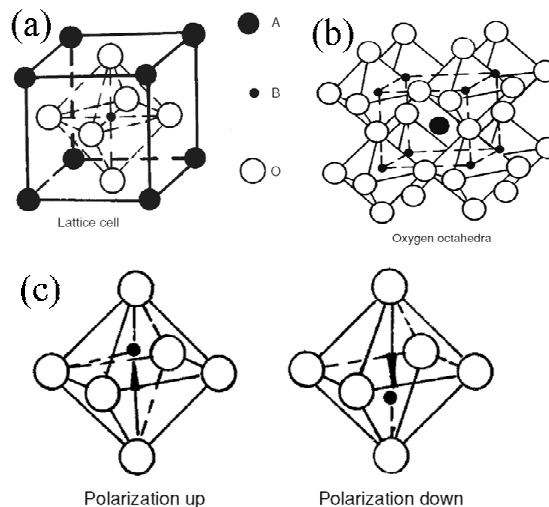


Figure 2. (a) Unit cell of ABO_3 perovskite, (b) oxygen octahedra, and (c) 180° polarization reversal for two of the six possible polarization states produced by displacement of the central cation in the tetragonal plane. Reproduced with permission from [7], Cady, W. G. *Piezoelectricity: An introduction to the theory and applications of electromechanical phenomena in crystals*; Dover Publications: New York, NY, 1964, pp1-20.

In the 1950s, Jaffe and his coworkers found that the PZT system could exhibit strong piezoelectric effects. The maximum piezoelectric response was found in the PZT compositions near the morphotropic phase boundary (MPB) between the rhombohedral and tetragonal phases. Since then, the PZT system containing various additives has become the dominant piezoelectric ceramic for a variety of applications. The phase diagram of PZT pseudo-binary is shown in Figure 3(a), where T_c line is the boundary between the cubic paraelectric phase and the ferroelectric phases [8]. A significant feature of the PZT solid solution is the MPB, which divides the region of ferroelectric phase into two parts: a tetragonal phase region (on the Ti-rich side) and a rhombohedral phase region (on the Zr-rich side). In the PZT system, at room temperature, the MPB occurs close to Zr/Ti=53/47. The MPB represents an abrupt structural change within a solid solution with variation in composition but nearly independent of temperature. Usually it occurs because of the instability of one phase (such as ferroelectric tetragonal phase) against the other (ferroelectric rhombohedral phase) at a critical composition where two phases are energetically very similar but elastically different. The MPB compositions have mixed symmetries and therefore are easily poled in the polycrystalline form. Moreover, the phase boundary reduces anisotropy energy, lowers the domain wall energy, and thus, increases the domain wall mobility. In turn, this provides a high extrinsic domain wall contribution to the electromechanical properties [9]. It is noticed that the dielectric constant, piezoelectric and electromechanical behavior attain maximum in the vicinity of the MPB composition, as shown in Figure 4 [8].

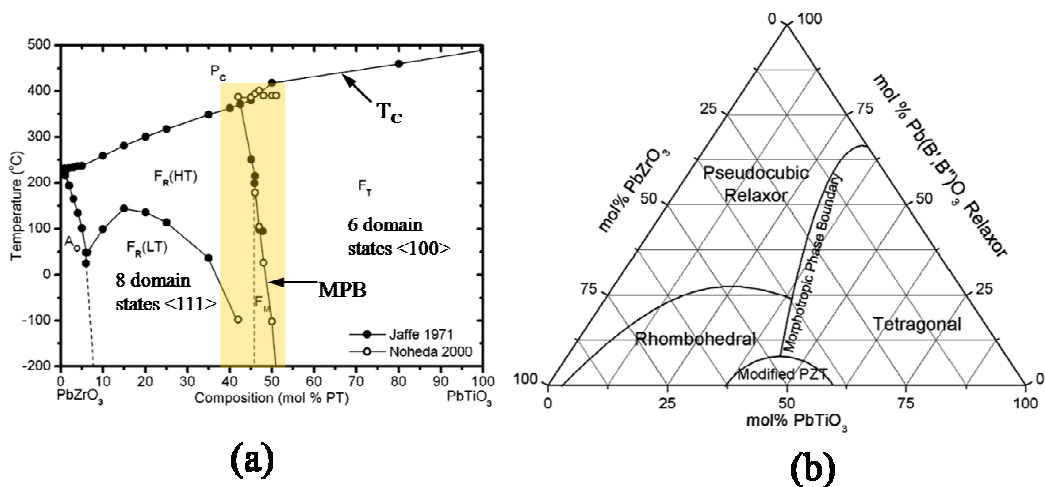


Figure 3. (a) Phase diagram of PbZrO_3 - PbTiO_3 system. P_c : paraelectric cubic, F_r : ferroelectric tetragonal, $F_r(\text{HT})$: ferroelectric rhombohedral (high temperature form), $F_r(\text{LT})$: ferroelectric rhombohedral (low temperature form), A_o : antiferroelectric orthorhombic, T_c : cubic temperature, and MPB: morphotropic phase boundary. Close to the MPB, a stable monoclinic phase is discovered. Reproduced with permission from [8], Jaffe, B.; Cook, W. R.; Jaffe, H. Piezoelectric ceramics; ISBN 0-12-379550-8; Academic Press: New York, NY, 1971, pp136-152; and [21], Noheda, B.; Gonzalo, J. A.; Cross, L. E.; Guo, R.; Park, S. E.; Cox, D. E.; Shirane, G. Tetragonal-to-monoclinic phase transition in a ferroelectric perovskite: The structure of $\text{PbZr}_{0.52}\text{Ti}_{0.48}\text{O}_3$. Phys Rev B. 2000, 61, 8687-8695. (b) Phase diagram of PbZrO_3 - PbTiO_3 - $\text{Pb}(\text{B}_1\text{B}_2)\text{O}_3$ ternary system. Reproduced with permission from [8], Jaffe, B.; Cook, W. R.; Jaffe, H. Piezoelectric ceramics; ISBN 0-12-379550-8; Academic Press: New York, NY, 1971, pp136-152.

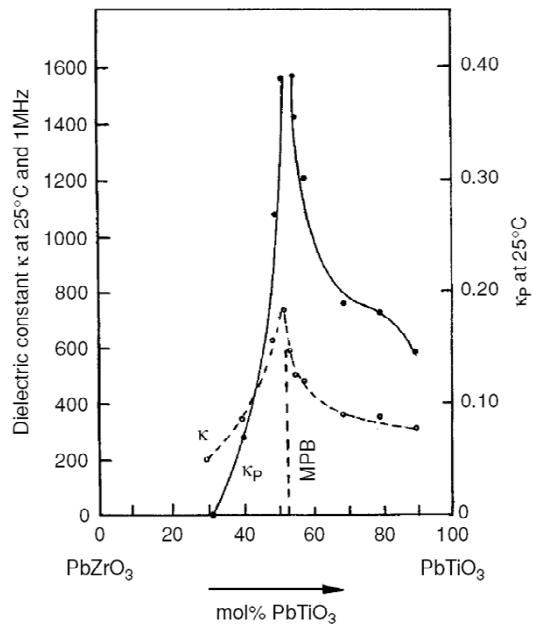


Figure 4. Dielectric constant (κ) and electromechanical coupling factor (κ_p) for the $\text{PbZrO}_3\text{-PbTiO}_3$ piezoelectric ceramic system. Reproduced with permission from [8], Jaffe, B.; Cook, W. R.; Jaffe, H. Piezoelectric ceramics; ISBN 0-12-379550-8; Academic Press: New York, NY, 1971, pp136-152.

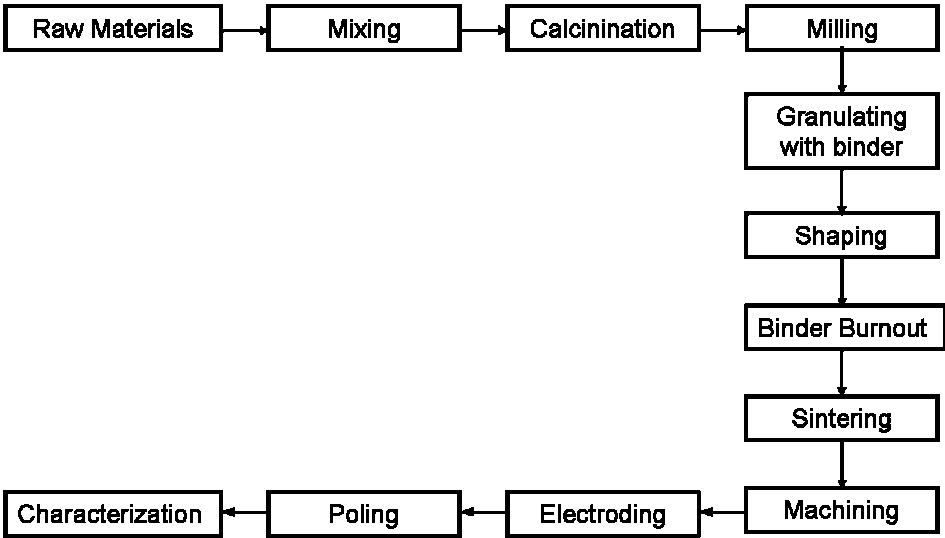


Figure 5. A flowchart of a typical oxide manufacturing process for piezoelectric ceramics.

Starting around 1965, several Japanese companies focused on developing the PZT-based ternary solid solutions, which are also considered as the basic compositions for piezoelectric ceramics. This involves the addition of a complex B-site lead perovskite, with general chemical formula $\text{Pb}(\text{B}_1\text{B}_2)\text{O}_3$ [10,11]. Here B_1 and B_2 have to stoichiometrically balance to a +4 valence, and then involve cations such as Zn^{+2} , Mg^{+2} , Co^{+2} , Sc^{+3} , etc., and larger valence

cations such as Nb^{+5} , Ta^{+5} , W^{+6} , etc. These complex lead perovskites are frequently relaxor ferroelectric materials, such as $\text{Pb}(\text{Ni}_{1/3}\text{Nb}_{2/3})\text{O}_3$, $\text{Pb}(\text{Mg}_{1/3}\text{Nb}_{2/3})\text{O}_3$, $\text{Pb}(\text{Zn}_{1/3}\text{Nb}_{2/3})\text{O}_3$. These compounds also have a MPB and enhanced piezoelectric properties at the boundary. When all combined in a ternary phase diagram, a MPB exists continuously, from one binary to the other, as illustrated in Figure 3(b) [8]. In the case of solid solutions with $\text{Pb}(\text{Ni}_{1/3}\text{Nb}_{2/3})\text{O}_3$, there are added advantages in the enhanced sintering kinetics, which lower the firing temperature and allow better control over volatile species, such as the PbO . The success of the Japanese effort attracted other nations, and today the needs and uses the PZT-based ternary piezoelectric ceramics extend from medical applications to the communications field, and to military applications and the automotive field. A review of the early history of piezoelectricity can be found in the work of Cady [7], and in 1971, Jaffe et al. published a book on piezoelectric ceramics [8] that is still one of the most referenced works on piezoelectricity. Extended reviews on the current and emerging piezoelectric materials, technology, and applications, can be found in reference [12].

The concept of piezoelectricity in solids can be understood from their internal structures. To explain it better, let us consider a single crystallite (a small single crystal less than $100\text{ }\mu\text{m}$ in average diameter) from a polycrystalline ceramic. This crystal has a define chemical composition, and consists of ions (atoms with positive or negative charges) that are constrained to occupy positions in a specific repeating relationships to each other, thus building up the structure of lattice of the crystal. The smallest repeating unit of the lattice is called the unit cell, and the specific symmetry possessed by the unit cell determines whether it exhibits piezoelectricity. Among the 32 point groups, 21 classes are non-centrosymmetric (a necessary condition for piezoelectricity to exist) and only 20 of these are piezoelectric. Although one class (cubic class 432) lacks a center of symmetry, it does not permit piezoelectricity because of other combined symmetry elements. Furthermore, for those materials that are piezoelectric but not ferroelectric (i.e., they do not possess spontaneous polarization), the stress itself is the only means by which the dipoles are generated. However, for ferroelectric materials with a spontaneous polarization whose direction can be switched by an applied field, they always exhibit piezoelectricity. The piezoelectric effect is linear and reversible, and the magnitude of the polarization is dependent on the magnitude of the stress and the sign of the charge produced is dependent on the types of stresses such as tensile or compressive [7]. The utilization of the piezoelectric effect in polycrystalline ferroelectric ceramics is realized by a poling process, in which an external electric field can orient the ferroelectric domains within the grains, thus produce a ceramic material acting very similar to a single crystal with both ferroelectric and piezoelectric properties. Before poling, polycrystalline ferroelectric ceramics with random grain orientation belong to the $\infty\infty\text{m}$ Curie group, and they do not possess any piezoelectric properties because of the random orientations of the ferroelectric domains in the polycrystalline ceramics. However, after poling, their Curie group becomes as ∞m , and they have a spontaneous polarization component in the direction of the poling field. To achieve the best piezoelectric properties without electrical breakdown, the poling electric field and temperature must be optimized. However, requirements vary drastically with the composition of piezoelectric ceramics.

Because of the high property coefficients and unique structural characteristics of MPB compositions, this feature is exploited in many commercial compositions. In the 1970s, some speculations concerning the reasons for this maximum in coupling at the MPB have been proposed based on thermodynamic theory [13], and in the 1980s, the ferroelectric group at

Penn State leadered by Cross has dedicated a continuing effort to formulating an adequate phenomenology of piezoelectricity in PZT system, and their work is documented in a sequence of papers [14-18]. Recently, a stable monoclinic phase is discovered in the ferroelectric PZT system close to the MPB [19,20], as shown in Fig. 3(a), which provides a new perspective to view the rhombohedral-to-tetragonal phase transformation in PZT. The joint experimental and theoretical works showed that the key point for explaining the giant piezoelectric properties inside the MPB (i.e., the huge effect of a weak electric field on the direction and magnitude of polarization) is the fact that in a monoclinic phase the polarization is no longer restricted along the a linear direction ([111] or [100] for instance) but can rotate freely in a plane. In fact, up to now several monoclinic phases have been discussed and evidenced. These phases are easily visualized and related to other ferroelectric phase via a picture proposed by Fu and Cohen [20]. An investigation of several compositions around the MPB has suggested a modification of the PZT phase diagram, as shown in Figure 3(a) [21]. High resolution X-ray powder diffraction measurements on the poled PZT ceramic samples close to the MPB boundary have also shown that for both rhombohedral and tetragonal compositions the piezoelectric elongation of the unit cell does not occur along their polar directions but along those directions associated with the monoclinic distortion, which provides the first direct evidence for the origin of the very high piezoelectricity in PZT [22].

2.2. Processing of Piezoelectric Ceramic Materials

The fabrication of most bulk piezoelectric ceramics starts with powder preparation. The powders are then pressed to the required shapes and sizes, and the green shapes are in turn processed to mechanically strong and dense ceramics. The more important processes that influence the product characteristics and properties are powder preparation, calcining and sintering. The next steps are machining, electroding and poling (application of a dc field to orient the ferroelectric domains and induce piezoelectricity). The electromechanical properties of piezoelectric ceramics are largely influenced by their processing conditions. A flowchart of a typical manufacturing process for piezoelectric ceramics with oxides as starting materials is shown in Figure 5. First, high purity raw oxide materials are accurately weighed according to their desired ratio and then are mechanically or chemically mixed. During the calcination step, the solid phases react to form the piezoelectric phase. After calcination, the solid mixture is milled to fine particles. Shaping is accomplished by a variety of ceramic processing techniques, including powder compaction, tape casting, slip casting, and extrusion. During the shaping operation, organic materials are typically added to the ceramic powder to improve its flow and binding characteristics. The organic is then removed at a low-temperature (500-600°C) burnout step. After organic removal, the ceramic structure is fired to an optimum density at an elevated temperature. Lead containing ceramics such as PbTiO_3 , PZT, $\text{Pb}(\text{Mg}_{1/3}\text{Nb}_{2/3})\text{O}_3$ are fired in sealed crucibles in an optimized PbO atmosphere to prevent lead loss above 800°C.

Due to the randomness of the grains in as-prepared polycrystalline piezoelectric ceramics, the poling process is necessary to induce piezoelectricity in polycrystalline ferroelectric ceramics, as schematically shown in Figure 6 [5]. It is the most critical step among the total fabrication process of piezoelectric ceramics, which can be taken place with the specimens

immersed in transformer oil at a temperature of $100 \sim 150\text{ }^{\circ}\text{C}$, while applying a static electric field of $2.5 \sim 4.5\text{ MV/m}$ along a desired direction for a period of $10 \sim 20$ minutes, to align the ferroelectric domains. In the poled condition, the ferroelectric ceramics, exhibit a spontaneous polarization with a component in the direction of the applied field. To obtain the best piezoelectric properties, the temperature and applied static voltage must be optimized in poling process. The poling temperature is limited by the leakage current, which can cause an increase in the internal temperature leading to thermal breakdown, while the electric field is limited by the breakdown strength of the ceramic. Higher fields can be used if they are applied as a succession of short pulses. In another poling method, called corona poling, high voltages of order of 10^4 V are applied to either a single needle, or an array of needles, with their tips located a few millimeters from the ceramic surface, while the opposite surface of the ceramic is earthed, to develop a high electric field in the ceramics. The corona poling method has many advantages over conventional poling, such as the capability of continuous poling for mass production and the use of samples with a larger surface area. Furthermore, it carries a diminished risk of electrical breakdown, because the poling charge cannot be quickly channeled to a 'weak spot', as it could be when using metallic electrodes. This method has been successfully used for poling piezoelectric ceramic-polymer composites such as PZT-epoxy [23]. Due to the symmetry limitations, the alignment of the ferroelectric domains along the direction of the poled field in poled ceramics is never complete. However, the end result is a ceramic whose net polarization along the poling axis has sufficiently high piezoelectric properties. For example, depending on the type of crystal structure involved, the degree of poling can be quite high, ranging from 83% for the tetragonal phase to 86% for the rhombohedral phase, and to 91% for the orthorhombic phase. The degree of poling is also increased in ascending order from polycrystalline ferroelectric ceramics, to poled ferroelectric ceramics, to single crystal ferroelectrics and to single-domain single crystals.

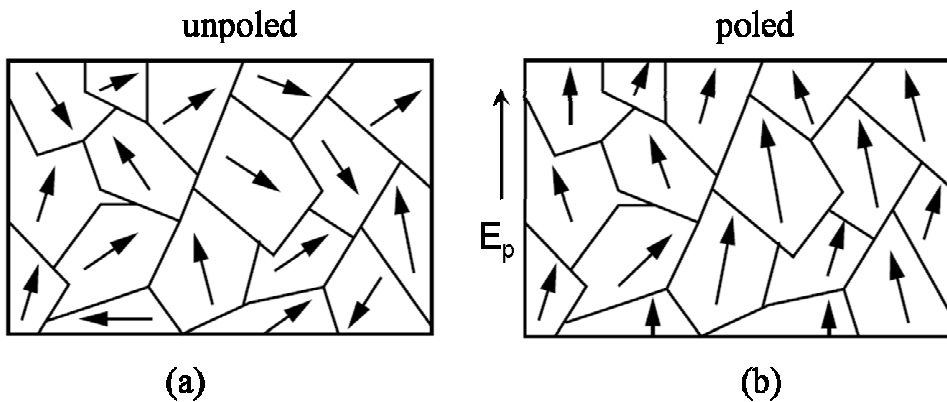


Figure 6. Schematic diagram of the poling process in piezoelectric ceramics. (a) In the absence of an electric field, the domains have random orientation of polarization; (b) the polarizations within the domains are aligned in the direction of the electric field. Reproduced with permission from [5], Kholkin, A.; Jadidian B.; Safari, A. In *Encyclopedia of Smart Materials*, Schwartz, M.; Eds.; ISBN 0-471-17780-6; John Wiley & Sons, Inc.: New York, NY, 2002, Vol.1, pp139 -148.

3. PROPERTIES OF PIEZOELECTRIC CERAMIC MATERIALS

3.1. Piezoelectric Parameters

The parameters that are of interest when considering the electromechanical effects of piezoelectric materials, are the piezoelectric coupling factor κ (e.g. κ_{33} , κ_{31} , κ_p and κ_t), mechanical quality factor (Q_m), frequency constant (N_i), and piezoelectric coefficients, such as the d and g coefficients which describe the interaction between mechanical and electrical behavior of piezoelectric ceramics. The effective electromechanical coupling coefficient κ_{eff} describes the ability of the ceramic transducer to convert one form of energy to another, which is defined by the equation:

$$\kappa_{\text{eff}}^2 = \frac{\text{mechanical energy converted to electrical energy}}{\text{input mechanical energy}} \quad (1)$$

or

$$\kappa_{\text{eff}}^2 = \frac{\text{electrical energy converted to mechanical energy}}{\text{input electrical energy}} \quad (2)$$

This parameter is a function in equations for electrical/mechanical energy conversion efficiency in actuators, in bandwidth and insertion loss in transducers, and signal processing devices, and in the location and spacing of critical frequencies of resonators. Since the energy conversion is always incomplete, κ^2 is (and thus also κ) is always lower than 1.0.

The effective coupling coefficient κ_{eff} is related to the values of f_m and f_n , and can be described as

$$\kappa_{\text{eff}}^2 \approx \frac{f_n^2 - f_m^2}{f_n^2} \quad (3)$$

Values for f_m and f_n can be readily measured by using a suitable bridge, which are the frequencies for the minimum and maximum impedance Z of the circuit as a whole, respectively. The approximations in equation (3) are good provided that the Q_m value for the resonator is high enough, for example greater than 100.

The planar coupling coefficient κ_p is related to the parallel and series resonant frequency by

$$\frac{\kappa_p^2}{1 - \kappa_p^2} = f(J_0, J_1, \nu \frac{f_p - f_s}{f_s}) \quad (4)$$

Where J_0 and J_1 are Bessel function and ν is Poisson's ratio. κ_{31} can be also calculated from

$$k_{31}^2 = \frac{1-\nu}{2} k_p^2 \quad (5)$$

The mechanical quality factor Q_m , representing the degree of mechanical loss of piezoelectric resonator at resonance, is defined as

$$Q_m = 2\pi \frac{\text{stored mechanical energy at resonance}}{\text{mechanical dissipated energy per resonant cycle}} \quad (6)$$

The Q_m can be obtained from the following equation:

$$Q_m = \frac{f_p^2}{2\pi f_s |Z_m| (C_0 + C_1) (f_p^2 - f_s^2)} \quad (7)$$

Where $|Z_m|$ is the minimum impedance at resonance, C_0 and C_1 are the capacitance shown in Figure 7(a), respectively.

The frequency constant N_l is defined by the following equation

$$N_l = l \times f_r = \frac{1}{2} \sqrt{\frac{Y}{\rho}} \quad (8)$$

Where l is the length of piezoelectric ceramic thin plate, f_r is the resonance frequency in the length direction, Y is the Young's modulus, and ρ is the density.

These values of the piezoelectric properties of a material can be derived from the resonance behavior of suitably shaped specimens subjected to a sinusoidally varying electric field. To a first approximation, the behavior of the piezoelectric specimen close to its fundamental resonance can be represented by an equivalent circuit, as shown in Figure 7(a) and Figure 7(b). The frequency response of the circuit is shown in Figure 7(c), in which various characteristic frequencies are identified [24]. The functions f_r and f_a are the resonant and anti-resonant frequencies when the reactance of the circuit is zero ($X_e=0$); f_s is the frequency at which the series arm has zero reactance ($X_1=0$); f_p is the frequency when the resistive component R_e is maximum; f_m and f_n are respectively the frequencies for the minimum and maximum impedance Z of the circuit as a whole. Piezoelectric vibrators with electrodes covering their two flat faces are used to measure the properties of piezoelectric ceramics. A more common geometry is a thin disc of diameter d electroded over both faces and poled in a direction perpendicular to the faces. In these disk-shaped specimens the resonance is focused on a radial mode excited through the piezoelectric effect across the thickness of the disc. The details about the determination of piezoelectric coefficients can be found in IRE standards on piezoelectric crystals: measurements of piezoelectric ceramics (Proc. IRE 49(7), 1161-1169(1961).

3.2. Compositions and Properties

In order to meet some stringent requirements for specific applications, piezoelectric ceramics under different doping conditions, and hence possessing different characteristics, have been developed for various applications. For example, some commercial piezoelectric ceramics with different compositions have been developed for the following four main uses of piezoelectrics:

- (a) the generation of charge at high voltages;
- (b) the detection of mechanical vibrations and for actuators;
- (c) the control of frequency;
- (d) the generation of acoustic and ultrasonic vibrations.

The 1st use requires a combination of a high g coefficient with resistance to damage to either electrical or mechanical properties by mechanical stress. The 2nd requires high piezoelectric g coefficients combined with low permittivity, and the 3rd requires properties stable with both time and temperature, with minimal losses and a high coupling coefficient. The 4th requires a material having low losses when the fields necessary to generate vibrations of useful amplitude are applied.

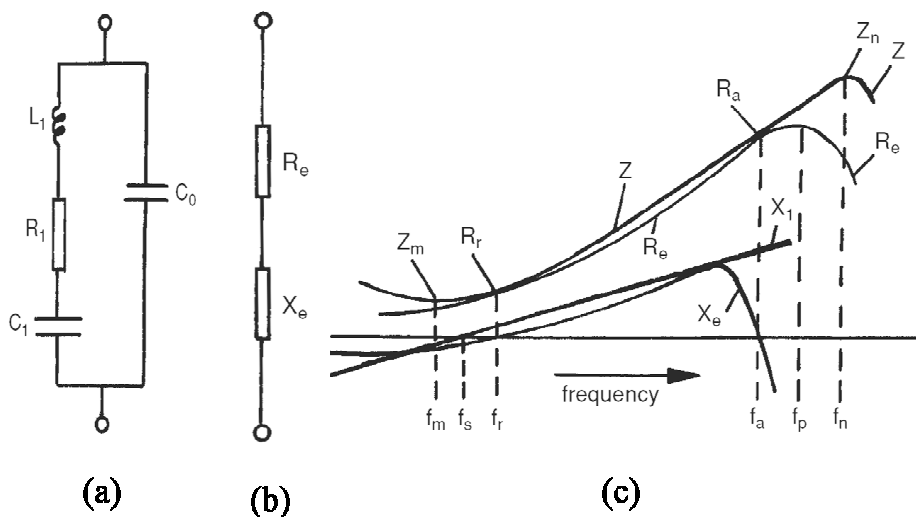


Figure 7. (a) Equivalent circuit for a piezoelectric specimen vibrating close to its fundamental resonance, (b) the equivalent series components of the impedance of (a), and (c) characteristic frequencies of the equivalent circuit, the differences between f_m , f_s and f_r , and between f_a , f_p and f_n are exaggerated. Reproduced with permission from [24], Zhu, X. H.; Meng, Z. Y. In *Encyclopedia of Smart Materials*, Schwartz, M.; Eds.; ISBN 0-471-17780-6; John Wiley & Sons, Inc.: New York, NY, 2002, Vol.1, pp1-16.

The techniques for modifying piezoelectric ceramics include element substitution and doping. In general, the term ‘element substitution’ implies that cations in perovskite lattice, e.g. Pb^{2+} , Zr^{4+} and Ti^{4+} , are replaced partially by other cations with the same chemical valence and similar ionic radii as those of the replaced ions. The new substituent cation usually

occupies the same position of the replaced cation in the perovskite lattice and a substitutional solid solution is thus formed; whereas the term ‘doping’ implies that some ions with chemical valences different from those of the original ions in the lattice, or some compounds with a chemical formula of $A^+B^{5+}O_3$ and $A^{3+}B^{3+}O_3$, are added to PZT ceramics.

From a global perspective, there are essentially four types compositional modifiers [25]. The first type are higher covalent substitutions (donor dopants) on A and/or B sites (such as La^{3+} replacing Pb^{2+} , or Nb^{5+} replacing Zr^{4+} or Ti^{4+}) to counteract the natural p-type conductivity of PZT and, thus, increase the electrical resistivity of the materials by at least three orders of magnitude. The donors are usually compensated by the formation of A-site vacancies. The donor-doped PZT piezoelectric ceramics are usually called ‘soft’ piezoelectric ceramics, meaning that they are easily depoled and driven nonlinear. The main features of which include square hysteresis loops, low coercive fields, high remanent polarization, high dielectric constants and dielectric loss, maximum coupling factors, high mechanical compliance, and reduced aging.

The second modified type are lower valent substitutions (acceptor dopants) on A and/or B sites (such as Fe^{3+} replacing Zr^{4+} or Ti^{4+}). These are compensated by the formation of oxygen vacancies. The acceptor-doped PZT piezoelectric ceramics are usually called ‘hard’ piezoelectric ceramics, because of their much enhanced linearity and high drive. The main features of which are poorly developed hysteresis loops, lower dielectric constants and dielectric loss, lower compliances and higher aging rates.

The third modified type are the isovalent substitutions on A and/or B sites (such as Ba^{2+} or Sr^{2+} replacing Pb^{2+} or Sn^{4+} replacing Zr^{4+} or Ti^{4+}). Such isovalent substitutions usually produce a broadening of the temperature-dependent properties, an increase in dielectric permittivity and a reduction of the Curie temperature, but cause no significant change in coupling coefficient, aging rate, volume resistivity, or low amplitude mechanical or dielectric loss.

The last types of compositional modifiers are more difficult to be classified and are called ‘thermally variable’. They can perhaps exist in more than a single valence state and in more than one type of ionic site. The feature of piezoelectric ceramics doped with variable valence additives is much improved temperature and time stability of the resonance frequency. These materials are typically used in electric wave filters or resonators where high temperature and time stability of resonance frequency is required.

The largest class of piezoelectric ceramics is made up of mixed oxides that contain corner-sharing octahedra of O^{2-} ions. The most technologically important materials in this class are perovskites that have the general formula ABO_3 , where $A = Na, K, Rb, Ca, Sr, Ba$, or Pb , and $B = Ti, Sn, Zr, Nb, Ta$, or W . Some piezoelectric ceramics that have this structure are $BaTiO_3$, $PbTiO_3$, and PZT (the most common piezoelectric ceramic in use today). By means of compositional modifications it is possible to adjust the properties to a remarkably wide range of requirements, and manufacturers supply a range of grades tailored to different fields of application. Details on this topic are found in the book by Jaffe [8] and reference [26].

3.3. Piezoelectric Constitutive Relationships

When writing the constitutive equations for a piezoelectric material, account must be taken of changes of strain and electrical displacement in three orthogonal directions caused by an interaction between the mechanical and electrical behavior of a piezoelectric material. Thus, the piezoelectric equations governing the direct piezoelectric effect and the converse piezoelectric effect can be written respectively [26],

$$D_i = \varepsilon_{ij}^T E_j + d_{ijk} T_{jk} \quad (9)$$

$$S_{ij} = \underline{d}_{ijk} E_k + s_{ijkl}^E T_{kl} \quad (10)$$

where D_i is the electric displacement (first-rank tensor); ε_{ij}^T the dielectric permittivity (a second-rank tensor); E_j the electric field (first-rank tensor); d_{ijk} and $\underline{d}_{ijk}$ the piezoelectric coefficient (third-rank tensor); T_{kl} the stress described by a second-rank tensor; S_{ij} the strain also described by a second-rank tensor; and s_{ijkl} the elastic coefficients (fourth-rank tensors). Superscripts T and E indicate that the dielectric permittivity ε_{ij} and the elastic constant s_{ijkl} are measured under conditions of constant stress and constant electric field, respectively. In general, a vector (formally regarded as a first-rank tensor) has three components, a second-rank tensor has nine components, a third-rank tensor has 27 components, and a fourth-rank tensor has 81 components. Not all of the tensor components are independent. Between equations (9) and (10) there are 45 independent tensor components, 21 for the elastic compliance s_{ijkl}^E , six for the dielectric permittivity ε_{ij}^T , and 18 for the piezoelectric coefficient, d_{ijk} .

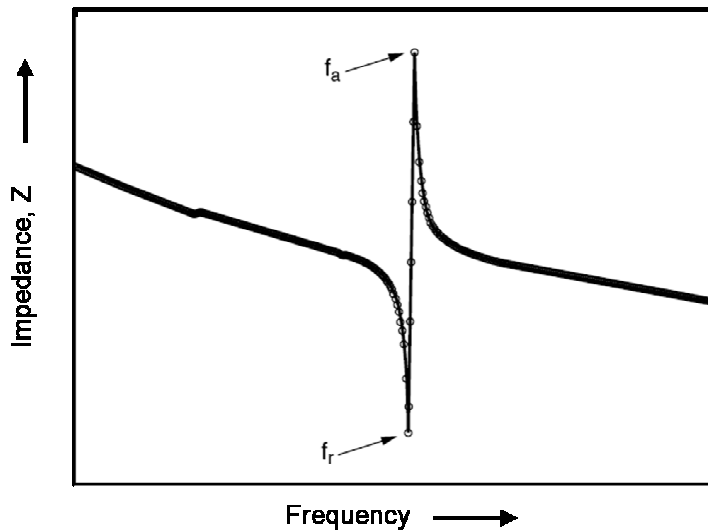


Figure 8. Impedance of a piezoelectric ceramic at resonance. Reproduced with permission from [28], Jordan, T. L.; Ounaies, Z. In *Encyclopedia of Smart Materials*; Schwartz, M., Eds.; ISBN 0-471-17780-6; John Wiley & Sons, Inc.: New York, NY, 2002, Vol.1, pp 162-173.

Fortunately, the crystal symmetry and the choice of reference axes can further reduce the independent numbers. Here the discussion is restricted to poled polycrystalline ceramics, which has only ∞ -fold symmetry in a plane normal to the poling direction. The symmetry of such a poled ceramic is therefore described as ∞mm , equivalent to 6 mm the hexagonal symmetry system. A standard convention for describing such a piezoelectric plate-like geometries (length l , width w and thickness t) is by setting subscript 1 corresponding to the length direction, subscript 2 corresponding to the width direction, and subscript 3 corresponding to the thickness direction (or the poling direction). The shear planes are indicated by the subscripts 4, 5, and 6 and are perpendicular to directions 1, 2, and 3, respectively. This simplifies the notations introduced before, where a 3-subscript tensor notation ($i, j, k = 1, 2, 3$) is replaced by a 2-subscript matrix notation ($i = 1, 2, 3$ and $j = 1, 2, 3, 4, 5, 6$), and a 2-subscript tensor notation ($i, j = 1, 2, 3$) is replaced by a 1-subscript matrix notation ($i = 1, 2, 3, 4, 5, 6$). A shear strain such as S_4 is a measure of the change of angle between the two initially orthogonal axes in the plane perpendicular to axis 1. The first subscript of the d constant gives the “electrical” direction (field or dielectric displacement), and the second gives the component of mechanical deformation or stress. For example, d_{31} is the coefficient relating the field along the polar axis to the strain perpendicular to it, while d_{33} is the corresponding coefficient for both strain and field along the polar axis. The planar isotropy of poled ceramics is expressed in their piezoelectric constants by the equalities $d_{32} = d_{31}$ (an electric field parallel to the poling axis 3 interacts in the same way with axial stress along either the 2 axis or the 1 axis) and $d_{24} = d_{15}$ (an electric field parallel to the 2 axis interacts in the same way with a shear in the 2, 3 plane as a field along the 1 axis with a shear in the 1, 3 plane). Similar relationships hold for the elastic constants because of isotropy in the plane perpendicular to the polar axis.

4. CHARACTERIZATION METHODS FOR PIEZOELECTRIC CERAMIC MATERIALS

4.1. Characterization of Piezoelectric Properties

Up to date, different methods have been developed to characterize the piezoelectric properties of piezoelectric ceramic materials. One method is the resonance technique, which involves measuring the characteristic resonance frequencies when a suitably shaped specimen is driven by a sinusoidal electric field. To a first approximation, the behavior of a poled ceramic sample close to its fundamental resonance frequency can be represented by an equivalent circuit, as shown in Figure 7(a). Another one is the direct technique, which is widely used to evaluate the sensor capabilities of piezoelectric materials at sufficiently low frequencies. Mechanical deformations can be applied in different directions to obtain different components of the piezoelectric tensors. In a simplest case, metal electrodes are placed on the major surfaces of a piezoelectric sample normal to its poling direction. Thus, the charge produced on the electrodes with respect to the mechanical load is proportional to the longitudinal piezoelectric coefficient d_{33} and the force F exerted on the ceramic sample: $Q = d_{33} F$. The charge can be measured by a charge amplifier using an etalon capacitor in the feedback loop. Details of the resonance technique and direct technique are described below.

4.1.1. Resonant method and equivalent circuit

The property values of the piezoelectric ceramic materials can be derived from the resonance behavior of suitably shaped specimens subjected to a sinusoidally varying electric field. Since any body of mass has its own characteristic frequencies at which it prefers to resonate, so when excited at this resonant frequency, f_r , a piezoelectric ceramics will resonate freely with a greater amplitude than at other frequencies. Following this resonant frequency is an anti-resonant frequency, f_a , where the impedance of the body is at a maximum and the oscillation amplitude is at a minimum. A typical resonance plot of impedance versus frequency for a piezoelectric ceramics near a resonance is shown in Figure 8 [28]. Notice the resonant frequency, f_r , at the point of minimum impedance and the anti-resonant frequency, f_a , at the point of maximum impedance. The measurement of the characteristic frequencies provides the means to evaluate the piezoelectric and elastic properties of a piezoelectric ceramics. Different vibrational modes of a ceramics, such as thickness or planar, give insight to the different constants associated with that modes. At resonance, a piezoelectric element can be modeled by an equivalent circuit, as shown in Figure 7(a). This circuit is commonly referred to as Van Dyke's Model and is recommended by the IEEE Standard on Piezoelectricity [29]. An alternate model proposed by Sheritt et al. [30], eliminates the resistance, R_1 , and instead represents the remaining components as complex to better characterize the losses associated with certain piezoelectric elements, especially polymers. All discussion presented below will assume Van Dyke's Model. Below f_r above f_a , the piezoelectric ceramics behaves capacitively; however between these two frequencies, the ceramics behaves inductively. This model is only valid near the resonance. Additionally, the resonance must be sufficiently isolated from other modes to eliminate the effects of any adjacent modes. To assure isolation of the resonance, sample geometry must be chosen carefully. Table 1 lists the suitable geometries for measuring the different piezoelectric and elastic coefficients [28].

Earlier literatures have suggested several circuits for measuring f_r and f_a of a piezoelectric ceramics [31-33]. These circuits usually consist of an oscillator for exciting the sample, a voltmeter or other device for measuring current through the circuit, and additional discrete components. To find f_r , the frequency of the oscillator is varied until the maximum current is detected through the circuit. Similarly, for f_a , the frequency of minimum current is determined. Note that there are actually six characteristic frequencies that may be identified for a particular resonance; they include f_m and f_n , the frequencies of maximum and minimum impedance; f_r and f_a , the resonant and anti-resonant frequencies; and f_p and f_s , the parallel resonant frequency and series resonant frequency. IEEE Standard 177 [32] identifies these six frequencies and establishes that for many cases, including piezoelectric ceramics, one can assume that $f_m = f_s = f_r$ and $f_a = f_p = f_n$. For lossy materials, such as some piezoelectric thin films, this assumption can introduce appreciable errors, so the six frequencies should be considered separately. The magnitude of the minimum impedance Z_m may be determined by substituting an adjustable resistor into the circuit for the ceramic at the previously identified frequency and adjusting the resistance until the voltmeter reading is the same as for the ceramic. Today, fully integrated impedance analyzers are commercially available to make this type of measurement, allow ones to choose an equivalent circuit model, and obtain the values of the discrete components of the equivalent circuit along with f_r and f_a . Commercial off-the shelf software is also available now which can be used in conjunction with an analyzer to

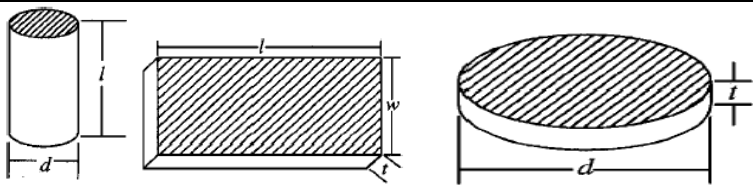
evaluate the impedance information and calculate the relative material properties of a piezoelectric device [34].

4.1.2. Direct methods for measuring piezoelectric parameters

Direct measurements of the piezoelectric constants are also possible and have been used to quantify the direct and converse effects in ceramic samples. Direct methods are also used to investigate the behavior of a ceramic in regard to hysteresis, nonlinearity, frequency response, aging, thermal behavior, and other characteristics that are not resolved by previous methods. These methods typically apply a known input to the ceramic, either an electric field or a force, and record the corresponding output, either a deformation or a charge under various conditions. These methods are in contrast to the bulk material characterization using the electrical resonance techniques described before. Many times, direct measurements are carried out on a ceramic that has been configured as a sensor or actuator. Typical processing may include electroding, laminating, applying preload, mounting, and other assembly procedures to adapt the material effectively for use as a sensor or actuator. These measurements aid the researcher in modeling the behavior of the piezoelectric device and allow efficient integration of the devices into real-world applications. Displacements of piezoelectric actuators are measured to determine the magnitude and sign of the relationship between the applied electric field and the strain developed, that is, the converse effect. For a PZT wafer, this corresponds to the d_{ij} coefficient; however, for bending type actuators, this relationship does not correlate directly with any of the measured properties for out-of-plane bending using the resonance techniques. It can be seen that when the ceramic is free to expand ($T_k = 0$), the strain is a function only of the product of the applied field E_i and the d_{ij} coefficient

$$S_j = d_{ij} E_i \quad (11)$$

Table 1. Sample geometries used for measuring material properties. Reproduced with permission from [28], Jordan, T. L.; Ounaies, Z. In *Encyclopedia of Smart Materials*; Schwartz, M. M.; Eds.; ISBN 0-471-17780-6; John Wiley & Sons, Inc.: New York, NY, 2002, Vol.1, pp 162-173

geometry			
dimension and property			
Dimensional requirements	long, slender, length poled rod; $l > 3d$	thin, flat plate, thickness poled; $l > 3.5 t, w$	thin flat disc, thickness poled $d > 10 t$
Properties measured	$k_{33}, \tan \delta, S_{33}^D;$ $S_{33}^E, d_{33}, g_{33}, Q_M;$	$K_3^T, k_{31}, S_{11}^D;$ $S_{11}^E, d_{31}, g_{31}, Q_M$	$K_3^T, k_p, \tan \delta, Q_M;$

Careful attention must be paid to the boundary conditions of the ceramic to ensure that this assumption is valid. In a plot of the strain as a function of applied field, the slope yields an average value of d_{ij} . Typically, these measurements are made by using a non-contacting displacement transducer to reduce the effects of loading on the actuator. Laser-based and other optical or capacitive displacement measurement techniques are most commonly used [35-37]. Displacements may range from submicron levels for single PZT wafers to the centimeter level for bending type actuators. For very small displacements, an optical-lever type measurement system or interferometric techniques [38] have been used to resolve the displacement of the ceramic. Direct application of either foil or optical strain gages has also been used for measuring the actuator strain. These measurements may be either static or dynamic, depending on the measurement system and the intended application of the ceramic. If dynamic measurements are made, excitatory frequencies should be at least an order of magnitude less than any resonant frequency of the device to ensure linear behavior and boundary conditions suitable for the intended measurement.

Another direct method used to measure piezoelectric constants is based on the direct piezoelectric effect [39]. Here, a known load is either applied to or lifted off a ceramic at rest. The resulting charge, which accumulates on the electrodes, is then measured as a voltage across a capacitor in parallel with the ceramic, or the current from the ceramic can be integrated directly. If E_i is 0 (short circuit), then equation (9) reduces to

$$D_i = d_{ij}T_j \quad (12)$$

Knowing the applied stress and measuring the electric displacement, the appropriate d_{ij} coefficient can be obtained. If a piezoelectric ceramic is immersed in a liquid and the pressure of the liquid is varied, then the piezoelectric coefficient d_h can be quantified by measuring the voltage on a large capacitor in parallel with the ceramic. This coefficient represents the response of the ceramic to hydrostatic pressure applied equally to all axes. Convention has dictated that electrodes are perpendicular to the 3 direction for the d_h coefficient. The d_h coefficient is related to the other d coefficients for a ceramic by the equation,

$$d_h = d_{33} + 2d_{31} \quad (13)$$

The frequency response of the device may be obtained by varying the frequency of the excitatory voltage to the ceramic while measuring the displacement. Typical resonant frequencies of bulk ceramic material are in the kilohertz to megahertz range depending on the mode of vibration, whereas resonant frequencies of bender types (unimorph or bimorph) may be less than 100 Hz. For maximum strain, a piezoelectric actuator can be excited at its natural frequency; however, this nonlinear behavior must be taken into account if the actuator is to be used across a range of frequencies. Careful attention must also be paid to the instrumentation system's dynamic response in both amplitude and phase distortions, when making dynamic measurements. Measurement systems have their own frequency response characteristics which must be separated from the response of the ceramic under test. It is also noticed that hysteresis phenomenon is always present in all piezoelectric materials, such a behavior is due to the lossy nature of the ceramic where the current trails the applied voltage by an angle α related to the loss tangent of the material. For actuators, this means that the absolute displacement depends on the excitatory voltage and frequency and also on whether the

voltage is increasing or decreasing. To characterize the amount of hysteresis in a ceramic, a sinusoidal voltage is applied to the device, and the displacement is recorded. By plotting the displacement versus driving voltage, as shown in Figure 9, the hysteretic behavior of the ceramic can be observed [28]. The amount of hysteresis (usually expressed in percent) is defined as the largest difference between the maximum and minimum displacement for any voltage divided by the total displacement. Of note in Figure 9, is the fact that, as the peak voltage is increased, the amount of hysteresis also increases for any given voltage.

The methods described above can be used either separately or together to investigate the dielectric, piezoelectric, and elastic properties of a ceramic. Resonant techniques, which are the preferred method of measurement in the IEEE standard, are easy to implement, and the associated frequencies can be measured accurately. There is even commercially available hardware and software to assist in these measurements and the evaluation of material properties. However, these methods do not explain any nonlinear behavior that is present in the ceramic. Dependence of material properties on the frequency and amplitude of the applied voltage are among these nonlinear effects. Direct measurements of the piezoelectric constants can quantify the material properties under different operating conditions and provide insight beyond the standard linear behavior predicted by resonance techniques. These methods though, are usually more rigorous in their requirements for material handling and instrumentation.

4.2. Characterization of Ferroelectric Domain Structure

Ferroelectric domain structures have several important effects on the behavior of piezoelectric ceramics. As a perovskite in ceramic form cools through its Curie point, it contracts isotropically since the orientations of its component crystals are random. However, the individual crystals will have a tendency to assume the anisotropic shapes required by the orientation of their crystal axes. This tendency will be counteracted by the isotropic contraction of the cavities they occupy. Therefore, a complex system of differently orientated domains are formed to minimize the elastic strain energy within the crystals [40]. It is believed that the formation of 90° domains is helpful to minimize both electrostatic and elastic strain energies, whereas the 180° domains only reduce the electrostatic energy. In the tetragonal structure, domains are usually designated as either *a* domain (with polarization direction perpendicular to the viewing direction) or *c* domain (with polarization direction parallel to the viewing direction). Considering the crystallographic directions of two neighboring domains, four alternative ways can be derived from two general domain types, i.e. 90° *a*-*a*, 90° *a*-*c*, 180° *a*-*a* and 180° *c*-*c* domains. The two alternatives in each of the two domain types are structurally identical, they only differ by the viewing direction chosen to illustrate the domain configurations. The diagnostic features for identifying the 90° *a*-*a* domains are: (a) the domain boundaries lying on the (011) plane appearing edge-on, and (b) diffraction spot splitting along [011] when viewed from [100]. As viewed from [010] direction, the characteristic features of the 90° *a*-*c* domains therefore become: (a) the domain boundaries lying on the (011) plane exhibiting asymmetric δ -fringes, and (b) spot splitting along [001] direction [41].

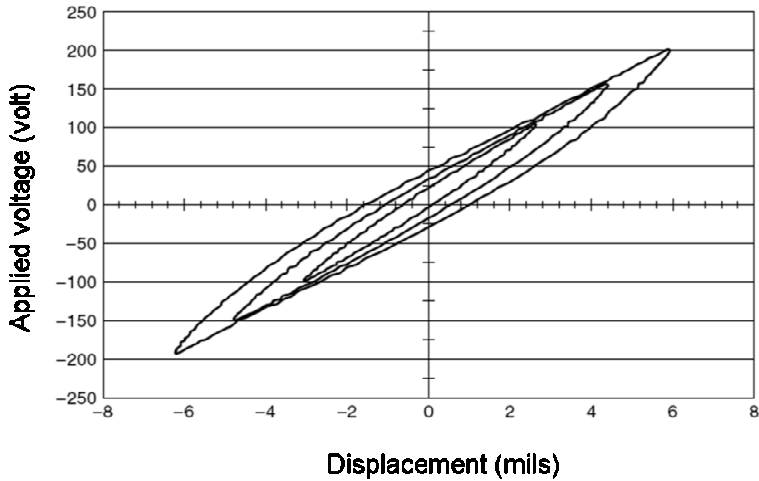


Figure 9. Strain hysteresis of a piezoelectric ceramic unimorph. Reproduced with permission from [28], Jordan, T. L.; Ounaies, Z. In *Encyclopedia of Smart Materials*; Schwartz, M.; Eds.; ISBN 0-471-17780-6; John Wiley & Sons, Inc.: New York, NY, 2002, Vol.1, pp 162-173.

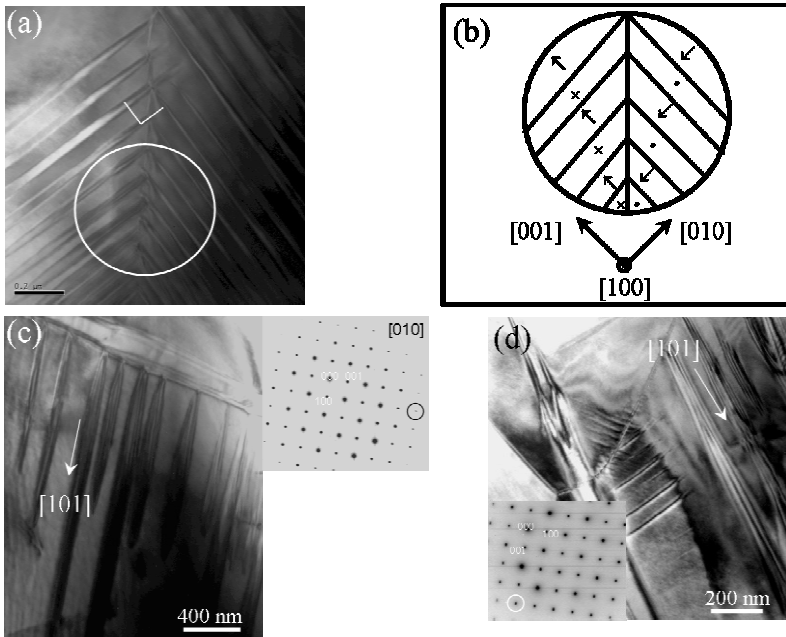


Figure 10. Ferroelectric domain structures observed in the $(\text{Pb}_{0.985}\text{Bi}_{0.01})(\text{Ni}_{1/4}\text{Zn}_{1/12}\text{Nb}_{2/3})_{0.6}(\text{Zr}_{0.1}\text{Ti}_{0.9})_{0.4}\text{O}_3$ piezoelectric ceramics with tetragonal structure. (a) Herringbone-like domain configuration, and (b) the possible assignment of the polarization direction to each domain, which is consistent with Figure (a). (c) and (d) Wedged-shaped 90° a - c domains exhibiting the δ -fringe contrast, viewed from $[010]$ direction. Reproduced with permission from [54], Zhu, X. H.; Zhu, J. M.; Zhou, S. H.; Liu, Z. G.; Ming, N. B.; Meng, Z. Y., Domain structures and their morphology evolution in the $\text{Pb}(\text{Ni}_{1/3}\text{Nb}_{2/3})\text{O}_3$ - PbTiO_3 - PbZrO_3 piezoelectric ceramics modified by bismuth and zinc substitutions, *J Am Ceram Soc.* 2008, 91, 227-234, Copyright © 2008, American Ceramic Society.

The application of a sufficiently strong field will orient the 180° domains in the field direction, as nearly as the orientation of the crystal axes allows. The field will also have an orientating effect on 90° domains in the tetragonal phase piezoelectric ceramics and on 71° and 109° domains in the rhombohedral phase, but the response will be limited by strain situation within and between the crystals. There will be overall change in the shape of the ceramic body with an expansion in the field direction and contraction at right angles to it. When the field is removed the strain in some regions will cause the polar orientation to revert to its previous direction, but a substantial part of the reorientation will be permanent. To better understand the piezoelectric properties of piezoelectric ceramics, their ferroelectric domains structures should be clearly classified.

Up to date, extensive transmission electron microscopy (TEM) studies of ferroelectric domains have mainly carried out in BaTiO_3 [41-43], PbTiO_3 [44,45] and modified PZT ceramics [46-48]. However, only a few papers deal with the structural aspects of ferroelectric domains in the $\text{Pb}(\text{Ni}_{1/3}\text{Nb}_{2/3})\text{O}_3$ - PbZrO_3 - PbTiO_3 (PNN-PZ-PT) ternary system [49-53], and the evolution of ferroelectric domain structure in this piezoelectric ceramics as a function of the relaxor PNN content and the $\text{Zr}/(\text{Ti} + \text{Ti})$ ratio, are not well understood. As is known that many macroscopic physical properties of piezoelectric ceramics such as piezoelectric coefficients, electromechanical coupling factors, dielectric constant, dielectric loss, depend strongly on the ferroelectric domain structures. The excellent dielectric and electromechanical properties were previously reported in the Bi- and Zn-modified PNN-PZ-PT piezoelectric ceramics, and a piezoelectric soft behavior (decrease of coercive electric field) and dielectric behavior transition from a normal to relaxor ferroelectrics were also observed when the relaxor PNN content or the $\text{Zr}/(\text{Zr} + \text{Ti})$ ratio was increased [49,50]. To better understand the superior electromechanical properties of such piezoelectric ceramics, Zhu et al. [50-52,54] have investigated the ferroelectric domain structures and their morphology evolution in the Bi- and Zn-modified PNN-PZ-PT piezoelectric ceramics as a function the relaxor $\text{Pb}(\text{Ni}_{1/3}\text{Nb}_{2/3})\text{O}_3$ content and the $\text{Zr}/(\text{Zr} + \text{Ti})$ ratio by TEM and selected area electron diffraction, and then followed by a complete analysis of the nature of the domain walls based on the predicated twinning planes for the formation of domains in these piezoelectric ceramic system. Figure 10(a) is a TEM image of ferroelectric domain structure in the $(\text{Pb}_{0.985}\text{Bi}_{0.01})(\text{Ni}_{1/4}\text{Zn}_{1/12}\text{Nb}_{2/3})_{0.6}(\text{Zr}_{0.1}\text{Ti}_{0.9})_{0.4}\text{O}_3$ piezoelectric ceramics with tetragonal structure, which demonstrates a herringbone-like domain configuration [54], in which the parallel bands with average width of 75 nm are observed on the both sides of the herringbone domain structure, and the observed domain boundaries intersect each other at either 90° or 45° . And the possible assignment for the polarization direction to each domain is given in Figure 10(b), which is fully consistent with the TEM image shown in Figure 10(a). The herringbone-like configuration of ferroelectric domains is not the only one type observed in this ceramics. On the contrary, wedge-shaped domains, like the ones shown in Figure 10(c) and 10(d) viewed from $[010]$ direction, are frequently observed, which are the common domain features in the ferroelectric phase of most ferroelectrics. The wedges point in a direction close to $[101]$, and the domain configuration is produced by sets of misoriented $\{110\}$ walls. The presence of δ -fringe contrast in the wedge-shaped domains, and the diffraction spot splitting along the $[100]$ direction in the corresponding SAED patterns (insets in Figure 10c and Figure 10d), indicate that the wedge-shaped domains are the 90° a - c domains. This type of domain appears as the 90° domain walls terminate at the piezoelectric

ceramic grain boundary, as seen in Figure 10(c) and Figure 10(d). Similar ones were observed in the polycrystalline BaTiO₃ ceramics. Its cause is not yet clear although redistributing the excess polarization charges at the grain boundary was proposed [43]. In the composition of (Pb_{0.985}Bi_{0.01})(Ni_{1/4}Zn_{1/12}Nb_{2/3})_{0.3}(Zr_{0.5}Ti_{0.5})_{0.7}O₃ close to the MPB, their domain structures exhibit quite different features, as shown in Figure 11. Within the grains A and C, fine 90° domain texture where the coherence of the dipoles is disrupted on a scale of 45 - 50 nm, was observed, whereas in the grain B, an intricate domain structure was observed, which exhibits wavy domain patterns. The lengths of the domains were about 0.35 μm and their widths were about 35 nm. Furthermore, the domain walls do not seem to prefer any one set of crystallographic planes unlike those observed in the grains A and B with 90° domains walls lying mainly on {110} planes. The waviness in the morphology of domain pattern observed in the grain B was due to continuous bending of the domain orientation between various equivalent directions on a length scale of 0.35 μm. In the composition of (Pb_{0.985}Bi_{0.01})(Ni_{1/4}Zn_{1/12}Nb_{2/3})_{0.4}(Zr_{0.5}Ti_{0.5})_{0.6}O₃ with rhombohedral structure, wavy domain patterns were observed, as shown in Figure 12 viewed from [011] direction [54]. The lengths of the domains were less than 0.6 μm and their widths were approximately 80 nm. The domain feature for the compositions at/near the MPB between the ferroelectric tetragonal and rhombohedral phases with high property coefficients, is a 90° domain texture mixed with an intricate domain structure. The different domain configurations observed in the ferroelectric tetragonal and rhombohedral phases are due to their different accommodations of local elastic strain fields. The piezoelectric soft behavior and a normal - relaxor ferroelectric behavior transition observed in this piezoelectric ceramic system, are closely related to the above domain morphology evolution as increasing the PNN content or the Zr/(Zr+Ti) ratio [54].

5. APPLICATIONS OF PIEZOELECTRIC CERAMIC MATERIALS

Piezoelectric materials can provide coupling between electrical and mechanical energy and thus have been extensively used in a variety of electromechanical devices. Both direct and inverse piezoelectric effects can be used for applications of piezoelectric ceramics. In general, the use of the direct piezoelectric effect can generate a charge or high voltage by applying a compressive stresses; whereas by using the converse piezoelectric effect, small displacements can be generated by applying an electric field to a piece of ceramics. Acoustic and ultrasonic vibrations can be generated by an alternating field tuned at the mechanical resonant frequency of a piezoelectric device and can be detected by amplifying the field generated by vibration incident on the material, which is usually used for ultrasonic transducers. The flexor transducer consists of two piezoelectric ceramic thin plates poled in opposite directions and can be used in gramophone pick-ups and ultrasonic accelerometers. The generation of surface waves enables filters and other devices to be made for use at frequencies exceeding 1GHz.

Applications of piezoelectric materials have now expanded into many fields since the discovery of the effect by the Curie brothers in 1880. Significant progress in applications was made possible after the discovery of PZT ceramic materials. Piezoelectric devices can be divided into four general categories: generators, sensors, actuators, and transducers depending of what type of physical effect used. For all of these basic functionalities, different designs

are available. A good overview on the different design types and possibilities for applications can be found in reference [55]. Table 2 summarizes the different effects and the designs which are candidates for application in transportation industry [56]. In the subsequent sections, some typical examples are chosen for detailed discussion, not only because of its technical importance but also its providing the opportunity to illustrate important ideas.

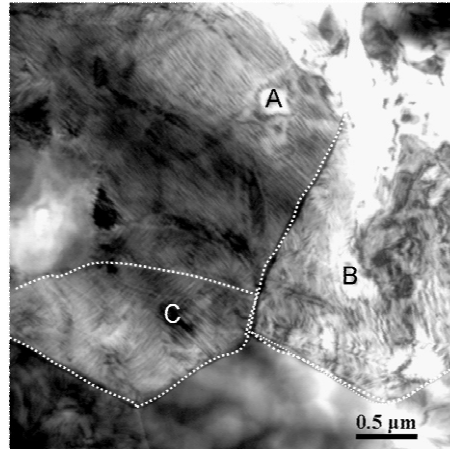


Figure 11. Ferroelectric domain structures observed in the composition of $(\text{Pb}_{0.985}\text{Bi}_{0.01})(\text{Ni}_{1/4}\text{Zn}_{1/12}\text{Nb}_{2/3})_{0.3}(\text{Zr}_{0.5}\text{Ti}_{0.5})_{0.7}\text{O}_3$ close to the MPB. Reproduced with permission from [54], Zhu, X. H.; Zhu, J. M.; Zhou, S. H.; Liu, Z. G.; Ming, N. B, Meng, Z. Y, Domain structures and their morphology evolution in the $\text{Pb}(\text{Ni}_{1/3}\text{Nb}_{2/3})\text{O}_3\text{-PbTiO}_3\text{-PbZrO}_3$ piezoelectric ceramics modified by bismuth and zinc substitutions, J Am Ceram Soc. 2008, 91, 227-234, Copyright © 2008, American Ceramic Society.

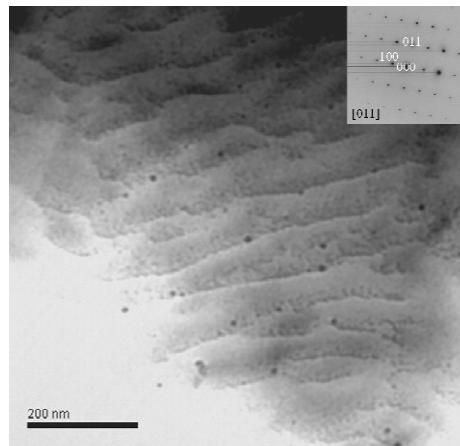


Figure 12. Ferroelectric domain structures observed in the composition of $(\text{Pb}_{0.985}\text{Bi}_{0.01})(\text{Ni}_{1/4}\text{Zn}_{1/12}\text{Nb}_{2/3})_{0.4}(\text{Zr}_{0.5}\text{Ti}_{0.5})_{0.6}\text{O}_3$ with rhombohedral structure. Reproduced with permission from [54], Zhu, X. H.; Zhu, J. M.; Zhou, S. H.; Liu, Z. G.; Ming, N. B, Meng, Z. Y, Domain structures and their morphology evolution in the $\text{Pb}(\text{Ni}_{1/3}\text{Nb}_{2/3})\text{O}_3\text{-PbTiO}_3\text{-PbZrO}_3$ piezoelectric ceramics modified by bismuth and zinc substitutions, J Am Ceram Soc. 2008, 91, 227-234, Copyright © 2008, American Ceramic Society.

Table 2. Applications of piezoelectric materials and designs. Reproduced with permission from [56], Nuffer, J.; Bein, T. Application of piezoelectric materials in transportation industry. Global Symposium on Innovative Solutions for the Advancement of the Transport Industry, 4-6 October, 2006, San Sebastian, Spain

Design/effect direction	Direct piezoeffect		Indirect piezoeffect	Both effects
	Sensor	Generator	Actuator	transducer
Bulk material (d ₃₃ - effect)	Accelerometers Knock sensors Pressure/force sensor	High voltage spark igniters		Ultrasonic sonar devices Distance meters Ultrasonic materials charac-terisation
Multilayer (d ₃₃ - effect)		Energy harvesting	Active Vibration Reduction Nano – Positioning High–force–actuation	
Patch (monolithic) (d ₃₁ - effect)	Dynamic strain sensor	Energy harvesting	Active Vibration Reduction	
Patch (fibre composites) (d ₃₁ and d ₃₃ - effect)	Dynamic strain sensors	Energy harvesting	Active Vibration Reduction	
Bi-/trimorph d ₃₁ + bimorp effect			Textile machines Fans	
Special designs		Cymbal transducers for Energy harvesting	Ultrasonic motors	Transformers

5.1. Piezoelectric Actuators

As very high voltages correspond to only tiny changes in the width of the piezoelectric ceramics, this width can be changed with better-than-micrometer precision, making piezoelectric ceramics the most important tool for positioning objects with extreme accuracy, thus their use in actuators [26]. Using the converse piezoelectric effect, a small displacement can be produced by applying an electric field to a piezoelectric material. Atomic force microscopes and scanning tunneling microscopes employ converse piezoelectricity to keep the sensing needle close to the probe. Vibrations can be generated by applying an alternating electric field. There is a demand in advanced precision engineering for a variety of types of actuators that can adjust position precisely (micropositioning devices), suppress noise vibrations (dampers), and drive objects dynamically (ultrasonic motors). These devices are used in areas, including optics, astronomy, fluid control, and precision machinery. Piezoelectric strains induced by an electric field are used for actuator applications. Figure 13 shows the design classification of ceramic actuators [57]. Simple devices composed of a disk or a multilayer structure use the strain induced in a ceramic by the applied electric field directly. Complex devices do not use the induced strain directly but use the amplified displacement through a special magnification mechanism such as a unimorph, bimorph or moonie. The most popularly used multilayer and bimorph structures have the following characteristics: The multilayer structure does not have a large displacement (10 μm) but has

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advantages in generation force (1 kN), response speed (10 μ s), lifetime (10^{11} cycles), and electromechanical coupling factor k_{33} (0.70). Unimorph and bimorph structures are defined by the number of piezoelectric ceramic plates: only one ceramic plate is bonded onto an elastic shim, or two ceramic plates are bonded together. The bimorph structure causes bending deformation because each piezoelectric plate bonded together produces extension or contraction in an electric field. In general, there are two types of piezoelectric bimorphs: the antiparallel polarization type and the parallel polarization type. A metallic sheet (called a shim) is occasionally sandwiched between the two piezoelectric plates to increase reliability; the structure can be maintained even if the ceramics fracture. Using the bimorph structure, a large magnification of displacement can be easily obtainable. The bimorph type has a large displacement (300 μ m) but has disadvantages in generation low force (1 N), slow response speed (1 ms), short lifetime (10^8 cycles), and small electromechanical coupling factor k_{eff} (0.10). They are commonly used for VCR head-tracking actuators due to their large displacements. An auto-tracking scan system that uses a piezoelectric actuator can make its head following the recording track driven in both still and quick modes. Furthermore, piezo-bimorph type camera shutters are also widely commercialized by Minolta.

A composite actuator structure called a “moonie” is developed to amplify the small displacements induced in piezoelectric ceramics. The moonie consists of a thin multilayer element and two metal plates that have narrow moon-shaped cavities bonded together, as shown in Figure 13(c) [57]. This device has characteristics intermediate between the conventional multilayer and bimorph actuators; it has an order of magnitude larger displacement (100 μ m) than the multilayer, much larger generative force (100 N), and quicker response (100 μ s) than the bimorph.

5.2. Ultrasonic Motor

An ultrasonic motor is an example of a piezoelectric actuator that uses resonant vibration. It is based on the concept of driving a rotor by mechanical vibration excited on a stator, *via* the piezoelectric effect. The ultrasonic motor consists of a high-frequency power supply, a vibrator, and a slider. The vibrator is composed of a piezoelectric driving component and an elastic vibratory part, and the slider is composed of an elastic moving part and a friction coat. The unique features of the ultrasonic motors are high output torque, quick response, large holding torque without energy dissipation and magnetic field-free operation. These special properties result from the use of piezoelectric ceramics, ultrasonic vibration and the friction force between the stator and rotor. Types of piezoelectric motors include the well-known travelling-wave motor used for auto-focus in reflex cameras, inchworm motors for linear motion, and rectangular four-quadrant motors with high power density (2.5 watt/cm³) and speed ranging from 10 nm/s to 800 mm/s. All these motors work on the same principle. Driven by dual orthogonal vibration modes with a phase shift of 90°, the contact point between two surfaces vibrates in an elliptical path, producing a frictional force between the surfaces. Usually, one surface is fixed causing the other to move. In most piezoelectric motors the piezoelectric ceramics is excited by a sine wave signal at the resonant frequency of the motor. Using the resonance effect, a much lower voltage can be used to produce a high vibration amplitude. Numerous types of ultrasonic motors have been proposed and they can

be classified by the way of elliptic particle motion created in the stator. Here, only two types of ultrasonic motors (standing-wave type and the propagating-wave type) are introduced.

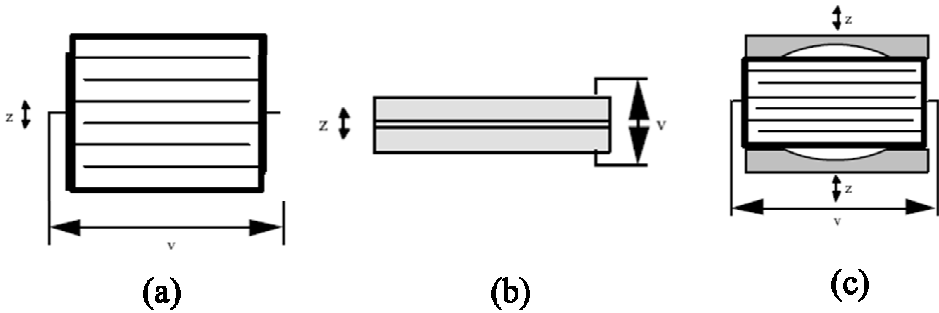


Figure 13. Structures of piezoelectric ceramic actuators. (a) Multilayer, (b) bimorph, and (c) moonie structures. Reproduced with permission from [57], Uchino, K.; Ito, Y. In *Encyclopedia of Smart Materials*; Schwartz, M.; Eds.; ISBN 0-471-17780-6; John Wiley & Sons, Inc.: New York, NY, 2002, Vol.1, pp148-162.

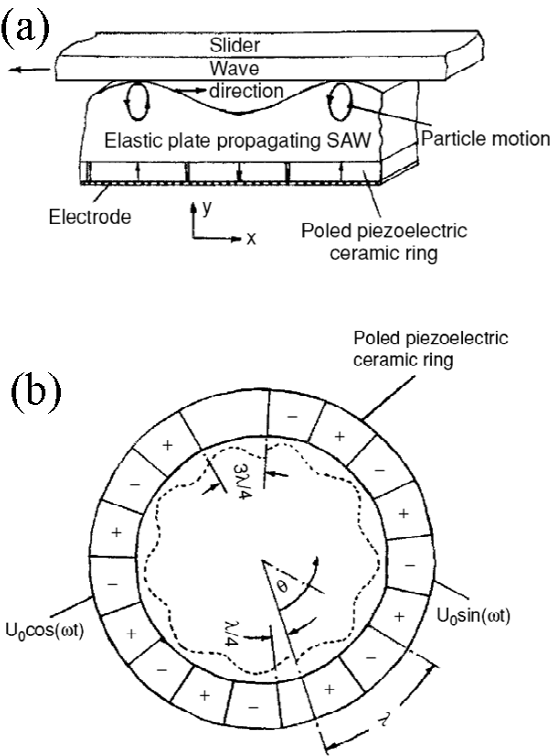


Figure 14. Operational principle of the propagating-wave type ultrasonic motor. (a) side view, (b) plan view showing the poled segments and how temporal and spatial phase differences are established. Reproduced with permission from [58], Moulson, A. J.; Herbert, J. M. *Electroceramics*; ISBN 0-412-294907; Chapman & Hall: London, UK, 1993, pp 304-305.

The standing-wave type is sometimes referred to as a vibratory-coupler type or a “woodpecker” type, where a vibratory piece is connected to a piezoelectric driver and the tip portion generates flat-elliptical movement. Attached to a rotor or slider, the vibratory piece provides intermittent rotational torque or thrust. In general, the standing-wave type has high efficiency, but lack of control in both clockwise and counterclockwise directions, is still a problem. In comparison, a wave in the traveling-wave type motor is excited by piezoelectric elements bonded to the stator. Such traveling wave is obtained by superimpose of two standing natural flexural waves of equal amplitude but differing in phase by 90° both spatially and temporally. The temporal difference is obtained by using one wave generated by the voltage $U_0 \sin(\omega t)$ and the other by $U_0 \cos(\omega t)$; the spatial phase difference results from the $3\lambda/4$ and $\lambda/4$ gaps between the two poled segments, as shown in Figure 14 [58]. The two standing waves can be written as

$$y_1 = y_0 \sin(n\theta) \sin(\omega t) \quad (14)$$

and

$$y_2 = y_0 \cos(n\theta) \cos(\omega t) \quad (15)$$

where y_1 and y_2 are the vertical displacements of the ring surfaces, θ is the angular displacement round the ring and n is the number of wavelengths accommodated around the ring. The resultant wave is given by

$$y = y_1 + y_2 = y_0 \{ \sin(n\theta) \sin(\omega t) + \cos(n\theta) \cos(\omega t) \} = y_0 \cos(\omega t - n\theta) \quad (16)$$

which indicates a surface wave travelling with velocity of ω/n . As shown in Fig. 14(b), the PZT ring is divided into 16 positively and negatively regions and two asymmetric electrode gap regions are used to generate a 9th mode propagating wave at 44 kHz. By propagating the traveling elastic wave induced by the thin piezoelectric ring, a ring-type slider in contact with the “rippled” surface of the elastic body bonded onto the piezoelectric elements is driven in both rotational directions by exchanging the sine and cosine voltage inputs. Another advantage is its thin design, which makes it suitable for installation in cameras as an automatic focusing device [59]. Canon utilized the “surfing” motor for a camera automatic focusing mechanism, installing the ring motor compactly in the lens frame. It is worth noticing that the stator elastic ring has many teeth, which can magnify the transverse elliptical displacement and improve the speed. This type motor has some advantages over the conventional electromagnetic motor, such as silent driving manner (due to ultrasonic frequency driving), no gear mechanism for speed reduction, and energy saving. A recent review on the various principles of miniature piezoelectric ceramics motors and suitable fabrication process can be available [60].

5.3. Piezoelectric Ceramic-Based Sensors

A piezoelectric ceramic sensor is a device that uses the piezoelectric effect to measure pressure, acceleration, strain or force by converting them to an electrical signal. The principle

of operation of a piezoelectric ceramic sensor is that a physical dimension, transformed into a force, acts on two opposing faces of the sensing element. Depending on the design of a sensor, different "modes" to load the piezoelectric element can be used: longitudinal, transversal and shear [61]. Piezoelectric sensors have proven to be versatile tools for the measurement of various processes. They are used for quality assurance, process control and process development in many different industries.

From the Curies' initial discovery in 1880, it took until the 1950s before the piezoelectric effect was used for industrial sensing applications. Since then, the utilization of this measuring principle has experienced a constant growth and can be regarded as a mature technology with an outstanding inherent reliability. It has been successfully used in various critical applications as for example in medical, aerospace and nuclear instrumentation. The rise of piezoelectric technology is directly related to a set of inherent advantages. The high modulus of elasticity of many piezoelectric materials is comparable to that of many metals and goes up to 105 N/m². Even though piezoelectric sensors are electromechanical systems that react on compression, the sensing elements show almost zero deflection. This is the reason why piezoelectric sensors are so rugged, have an extremely high natural frequency and an excellent linearity over a wide amplitude range.

One disadvantage of piezoelectric sensors is that they cannot be used for true static measurements. A static force will result in a fixed amount of charges on the piezoelectric material. While working with conventional readout electronics, imperfect insulating materials, and reduction in internal sensor resistance will result in a constant loss of electrons, and yield a decreasing signal. Elevated temperatures cause an additional drop in internal resistance; thus at higher temperatures only piezoelectric materials that maintain a high internal resistance can be used. Anyhow, it would be a misconception that piezoelectric sensors can only be used for very fast processes or at ambient conditions. In fact, there are numerous applications that show quasi-static measurements while there are other applications that go to temperatures far beyond 500°C.

Based on the piezoelectric technology various physical quantities can be measured; the most common are pressure and acceleration. Detection of pressure variations in the form of sound is the most common sensor application, e.g. piezoelectric microphones (sound waves bend the piezoelectric material, creating a changing voltage) and piezoelectric pickups for electric guitars. In which, a thin membrane and a massive base are used, ensuring that an applied pressure specifically loads the elements in one direction. For accelerometers, a seismic mass is attached to the piezoelectric ceramic elements. When the accelerometer experiences a motion, the invariant seismic mass loads the elements according to Newton's second law of motion $F = ma$. The main difference in the working principle between these two cases is the way forces are applied to the sensing elements. In a pressure sensor, a thin membrane is used to transfer the force to the elements, while in accelerometers the forces are applied by an attached seismic mass.

Sensors often tend to be sensitive to more than one physical quantity. Pressure sensors show false signal when they are exposed to vibrations. Sophisticated pressure sensors therefore use acceleration compensation elements in addition to the pressure sensing elements. By carefully matching those elements, the acceleration signal (released from the compensation element) is subtracted from the combined signal of pressure and acceleration to derive the true pressure information. Vibration sensors can also be used to harvest otherwise

wasted energy from mechanical vibrations. This is accomplished by using piezoelectric materials to convert mechanical strain into usable electrical energy [62].

When operated in a hydrostatic environment, bulk piezoelectric ceramics are typically poor choices as underwater acoustic sensors in the audio to low ultrasonic frequency range (i.e. below 5100 kHz). As receivers, the sensitivity is low because of the crystal symmetry of the poled ceramic. The piezoelectric charge coefficient in the hydrostatic mode, d_h , is equal to the sum $d_{33} + d_{31} + d_{32}$ of the ceramic. However, $d_{31} = d_{32}$, and for the various PZT compositions, $d_{33} \approx -2d_{31}$; therefore, d_h is almost equal to zero [63]. Since the piezoelectric voltage coefficient, g_h , is equal to $d_h/\epsilon_r\epsilon_0$, the dielectric constant, ϵ_r , for most ferroelectric ceramics is large over 1000, so the g_h coefficient is subsequently small. As a consequence, the voltage generated by an incoming pressure wave is very low. In order to improve the electroacoustic performance of a poled ceramic, it must be configured in such a way so that the effect of the hydrostatic pressure is minimized. This, it usually takes the form of air backing one side of the ceramic element, encapsulating part of the ceramic in a soft polymer to absorb a portion of the hydrostatic stress, or incorporating air spaces into the transducer itself. A comprehensive review on the current trends and historical development of piezoelectric sensors and sensor materials technology is given in reference [64]. The reader is referred to these texts for a more complete overview on the primary piezoelectric sensor configurations.

5.4. Ultrasonic Transducer

Transducer performance requirements are more and more demanding due to a wide range of imaging modalities, such as flow and harmonic modes, which are now integrated in echographic systems. Transducer performance depends on its internal structure and on the constitutive material properties, the most important of which is the active piezoelectric element. For the piezoelectric element, in particular for medical applications, two of the most important material parameters are the effective electromechanical coupling coefficient k_{eff} of the main vibration mode used and the acoustic impedance Z . The classical single-element transducer particular for medical imaging applications, is based on a piezoelectric plate or disc poled along the thickness direction, whose thickness defines the resonance frequency of the device. The plate, typically a piezoelectric ceramic, has an acoustic impedance (i.e. around 33 MRA) much higher than of biological tissues (close to that of water, i.e. 1.5 MRA). This large difference leads to an acoustical mismatch and a poor axial resolution. Consequently, other layers are added to the active layer [65]. Figure 15 shows a typical geometry of the basic ultrasonic transducer [57]. In the front (i.e. between the piezoceramic and the propagation medium), a matching layer is used. The thickness of a matching layer is generally around a quarter-wavelength at the resonance frequency, and its acoustical impedance is intermediate between those of the piezoceramic and tissues. The use of a matching layer thus improves the sensitivity of the transducer. Moreover, since the acoustical energy can better flow towards the tissues, the duration of acoustical resonance in the active layer is decreased. Consequently, the matching layer also improves axial resolution. One or more matching layers are used to increase sound transmissions into tissues. On the rear face of the active element, a thick layer is usually added. It is referred to as the backing, which is

attached to the transducer rear to damp the acoustic return wave and to reduce the pulse duration. The closer its acoustical impedance is to that of the active layer, the more energy is lost. The consequence is a lower sensitivity but a higher axial resolution. Thus, a trade-off has to be performed for each application. The attenuation coefficient and the thickness of the backing layer must be sufficient so that no energy can be radiated back to the active layer, which would produce parasitic echoes. For medical ultrasonic imaging, in general, broadband piezoelectric transducers should be used. The broad bandwidth response corresponds to a short pulse length that results in better axial resolution. Three factors are important in designing broad bandwidth transducers. The first is acoustic impedance matching, which is effectively coupling the acoustic energy to the body. The second is high electromechanical coupling coefficient of the transducer. The third is electrical impedance matching, which is effectively coupling electrical energy from the driving electronics to the transducer across the frequency range of interest.

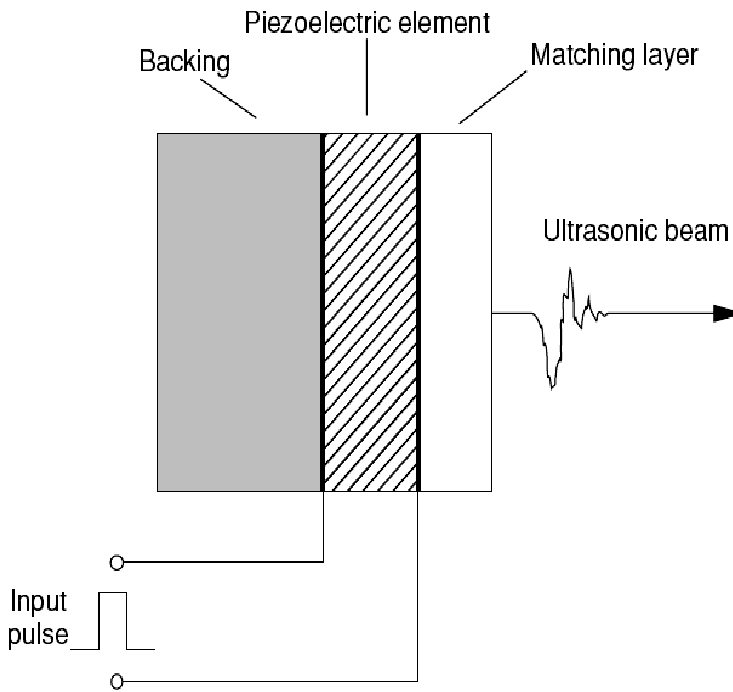


Figure 15. Schematic diagram of the fundamental transducer for acoustic imaging. Reproduced with permission from [57], Uchino, K.; Ito, Y. In *Encyclopedia of Smart Materials*; Schwartz, M.; Eds.; ISBN 0-471-17780-6; John Wiley & Sons, Inc.: New York, NY, 2002, Vol.1, pp148-162.

Ultrasonic transducers often operate in a pulse-echo mode [66]. The transducer converts electrical input into an acoustic wave output. The transmitted waves propagate into a body, and echoes are generated that travel back to be received by the same transducer.

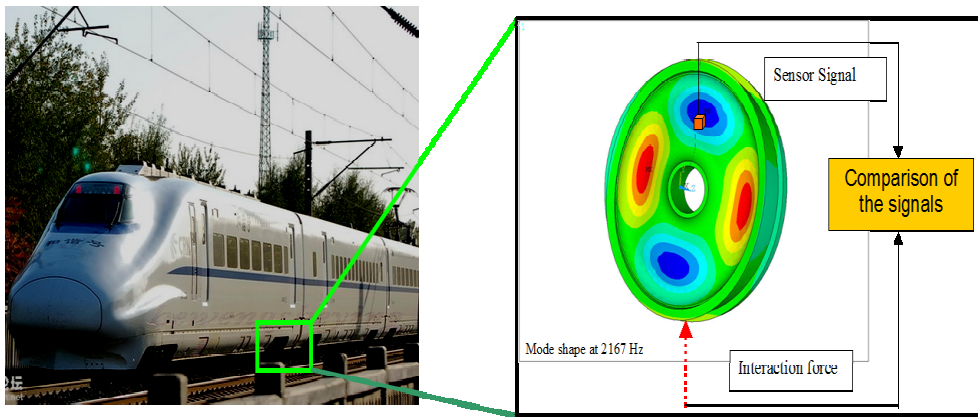


Figure 16. The method used for the assessment of the roughness of the wheel. A piezoelectric sensor detects the vibrations of the wheel, leading to an assessment of its wear status (source: TU Darmstadt). Reproduced with permission from [56], Nuffer, J.; Bein, T. Application of piezoelectric materials in transportation industry. Global Symposium on Innovative Solutions for the Advancement of the Transport Industry, 4-6 October, 2006, San Sebastian, Spain.

These echoes vary in intensity according to the type of tissue or body structure, and thereby create images. An ultrasonic image represents the mechanical properties of the tissue, such as density and elasticity. We can recognize anatomical structures in an ultrasonic image because the organ boundaries and fluid-to-tissue interfaces are easily discerned. Ultrasonic imaging can also be done in real time. This means that we can follow rapidly moving structures such as the heart without motional distortion. In addition, ultrasound is one of the safest diagnostic imaging techniques. It does not use ionizing radiation like X rays and thus is routinely used for fetal and obstetrical imaging. Useful areas for ultrasonic imaging include cardiac structures, the vascular system, the fetus, and abdominal organs such as the liver and kidney. In brief, it is possible to see inside the human body by using a beam of ultrasound without breaking the skin. There are various types of transducers used in ultrasonic imaging. Mechanical sector transducers consist of single, relatively large resonators that provide images by mechanical scanning such as wobbling. Multiple array transducers permit the imaging systems to access discrete elements individually and enable electronic focusing in the scanning plane at various adjustable penetration depths by using phase delays. The two basic types of array transducers are linear and phased (or sector). Linear array transducers are used for radiological and obstetrical examinations, and phased array transducers are useful for cardiological applications where positioning between the ribs is necessary.

Nowadays the universally familiar ultrasound non-invasive imaging system becomes the preminent method for in-utero foetal imaging. This technology, unknown 30 years ago, is now made possible by the development of piezoelectric-based array transducers and their miniaturization, since increasing resolution requires increasing number of sensors inside a same detector size. Today 2D detector arrays are commercially available, and continuing advances will soon provide real 3D imaging systems. Other novel medical application using piezoelectric ceramics include the development of focused surgery transducers and non-invasive medical therapies.

5.5. Active Vibration Damping

Piezoelectric actuators and sensors are also useful for active vibration control. Researchers at TU Darmstadt in Germany have been investigating ways to reduce vibrations by attaching piezoelements. When the material is bent by a vibration in one direction, the system responds to the bend and sends electric power to the piezoelement to bend in the other direction. In a demonstration at the Material Vision Fair in Frankfurt in November 2005, several panels were hit with a rubber mallet, and the panel with the piezoelement immediately stopped swinging. The company Siemens AG in Germany has recently developed damping units based on multilayer piezoactuators for trains using the tilting technology. That is a wear detection system for train wheels. The idea is to detect the changes in the vibration behavior of the entire wheel caused by the surface changes on the rolling contact area, as schematically shown in Figure 16 [56]. Piezoelectric sensors are placed on distinct areas of the wheel, changing the displacements on the surface into electrical signals. The challenging is to adequately define the correlation between modal behavior of the wheel and the measured signals as well as the sensitivity of the overall concept to the comparably small changes in modal parameters. Active damping of train bogies is necessary to improve the performance of such high speed trains. The optimization of speed versus trajectory control coupled with comfort and security questions represent the key issues of this technology. Although the use of PZT-based multilayered piezoelectric damping was not straightforward for such high load applications, significant progress was recently achieved. Other “lighter” applications of active and adaptive embedded damping units have been developed for use in vibration reduction/control inside tennis rackets and skis.

The application of adaptive composite systems in structural systems is vast and diverse. Several active vibration suppression concepts have been investigated by a program shared between Daimler-Benz Aerospace Military Aircraft (DASA), Daimler-Benz Forschung (DBF), and Deutsche Forschungsanstalt für Luftfahrt (DLR) [67]; two concepts using aerodynamic control surfaces and two concepts using piezoelectric components. In the DASA concept, a thin surface of piezoactuators is set out to flatten the dynamic portion of the combined static and dynamic maximum bending moment loading case directly in the shell structure. The second piezo concept by DLR involves preloaded PZT block actuators at structural fixtures. Both piezoelectric strategies were aimed at straight open-loop performance related to concept weight penalty and input electric power.

6. FUTURE OUTLOOK OF PIEZOELECTRIC CERAMIC MATERIALS

Presently PZT-based piezoelectric ceramics are leading materials for piezoelectric transducers due to their excellent piezoelectric properties. However, more recently, there is growing concern regarding the toxicity in lead-containing devices driven by the result of restriction of hazardous substances directive regulations. To address this concern, future research will be focused on finding lead-free compounds that have piezoelectric properties similar to those of PZT. In 2004, a group of Japanese researchers led by Yasuyoshi Saito discovered a sodium potassium niobate (KNN) composition with properties close to those of PZT, including a high T_c [68]. Bismuth ferrite (BiFeO_3) is also a promising candidate for

replacement lead-based ceramics. To meet the extremely diverse applications, hybrid piezoelectric composites are urgently required to be prepared by incorporating piezoelectric ceramics with other advanced materials. The piezoelectric composites show some unique properties or functions such as excellent dynamical response, high sensitivity to weak hydrostatic waves, damage resistance and control, which can be utilized to tailor or tune the overall performance of a smart structural system. Numerous efforts and exploratory approaches have been made to develop piezoelectric composites, such as piezoelectric ceramics-metal, piezoelectric ceramics-polymer, piezoelectric ceramics-shape memory alloy composites [69-71]. The piezoelectric composites have been evolved and become one of the potential advanced composites for smart systems. Numerical modeling and computer simulations of the piezoelectric composites, in conjunction with some experimental characterization efforts, will further lead to the optimization of technical factors, such as structural design, geometry, processing parameters, and material selection, and in turn the improvement of the overall performance of the piezoelectric composites. A major issue in piezoelectric composites is the evolution of stresses, permanent strains and cracks owing to the mismatch in thermal expansion and other physical properties of the dissimilar components in the composites. To solve the above problems, development of functionally graded composites with gradients in compositions, microstructure and properties through one or more layers, will be very demanding objective. Such an approach offers a number of advantages over the traditional methods of tailoring the compliance of composite materials or structural elements, and opens up new horizons for novel applications [7]. In addition, the research will continue toward the development of more resilient piezoelectric ceramic materials used for operation under severe external conditions (temperature, pressure, harsh chemical environments). This will further improve their potential application in space and deep ocean exploration, as well as in noise cancellation in airplanes and helicopters.

7. CONCLUSION

In this chapter, we present an overview of the state of art in piezoelectric ceramic materials, which includes their processing, properties, characterization, and the potential applications. More than one century after the discovery of piezoelectricity, piezoelectric ceramics have become commercially viable. Among the fabrication process of piezoelectric ceramics, poling process is the most critical step, which is necessary to induce the piezoelectricity in the polycrystalline ferroelectric ceramics. To meet with stringent requirements for specific applications, compositional modifications of piezoelectric ceramics with different doping conditions are possible to adjust the properties to a remarkably wide range of requirements. Characterization of the piezoelectric properties of piezoelectric ceramics is crucial for establishing the relationship between the manufacturing process and ceramic performance, which enables ones to adjust the manufacturing process of the piezoelectric ceramics to produce tailored materials. Insights gained through characterization have led to many new devices and uses. Significant progress has been made in the applications of piezoelectric ceramics since the discovery of PZT ceramics. Various potential applications of piezoelectric ceramic materials in ultrasonic actuators, sensors, transducers,

and active vibration controlling, are presented in this chapter, and the personal perspectives towards future trends of piezoelectric ceramic materials are also presented.

ACKNOWLEDGMENTS

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Chapter 2

STRESS ENGINEERED PIEZOELECTRIC COMPOSITES

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ABSTRACT

Stress-biased piezoelectric composites represent a family of piezoelectric ceramic-based actuators and transducers that offer a variety of performance advantages compared to more traditional unimorph devices. In this chapter, the history, fabrication, performance attributes and potential applications for these devices are discussed. The devices are typically fabricated using piezoelectric and elastic layers at temperatures between 150 and 300°C. The different coefficients of thermal expansion of the layers typically result in doming of the composite during cooling and the development of an in-situ stress-profile across the piezoelectric layer. These characteristics increase the robustness of the composite devices and may enhance domain wall motion contributions to piezoelectric response, yielding high electromechanical deformation. To date, a variety of composite devices based on different layers, processing strategies, and design philosophies have been developed to take advantage of these characteristics. Potential applications for the devices are also discussed, as are the underlying aspects of the devices that contribute to their performance.

I. OVERVIEW AND HISTORY

Stress-biased composite devices have been under investigation and development since the early 1990's, and compared to more traditional electromechanical actuators, the devices demonstrate unique performance characteristics. Studies of these devices are ongoing in numerous industrial and academic laboratories around the world, with researchers focusing on fundamental investigations of the devices, characterization of performance attributes, and development of the devices for various applications. These investigations have resulted in

numerous reports documenting the manufacture and performance of the devices, and have suggested possible device applications. Significant patent literature regarding the devices also exists. This chapter will review the literature regarding these devices and will explore various aspects of the manufacture and performance of devices.

The basic performance parameters that characterize an actuator system are displacement, force, frequency of operation, lifetime, weight, size and power consumption. Most piezoelectric actuators have excellent operational frequency bandwidth and can generate large forces but normally demonstrate a small strain response. Typical plate piezoelectric actuators or direct extensional devices have strains on the order of 0.1%, which severely limits their applicability. Traditionally, this problem has been addressed by utilizing the strain characteristics of piezoelectrics along with an internal or external mechanical amplifying mechanism to produce larger displacements, e.g., stacks, unimorph/bimorph benders and flextensional actuators as shown in Figure 1. In stacked actuators, thin plates of piezoelectric disks are physically stacked and an electric field (in the same direction as the poling direction, denoted by the small arrow) is applied to the individual elements. This architecture results in a linear increase in displacement with the number of individual elements. The force increases linearly with the area of the actuator.

Bender actuators (Figure 1b) have a thin poled piezoelectric plate bonded to one side (unimorph) or both sides (bimorph) of a metal sheet. When the piezoelectric layer in a unimorph is driven to expand or contract, the elastic nature of the metallic substrate resists this dimensional change, leading to a bending deformation [1]. In the case of bimorph benders, an opposing field is applied to each of the two piezoelectric layers resulting in opposite strains. This opposing strain in the piezoelectric layers causes a bending moment in the device, resulting in flexing of the bender. The tip deflection of the unimorph/bimorph benders is dependent on the applied field, thickness and length of both the piezoelectric and metal layers.

Flextensional devices, like cymbals (Figure 1c), use an external metallic membrane to convert and amplify small radial displacements of the piezoelectric disk (indicated by horizontal arrows) into large axial displacement (indicated by vertical arrows) of the metal membranes. The displacement and force generation characteristics of these devices depend on the endcap dimensions, including cavity depth (d_c), cavity diameter at the apex and base, thickness of the metal sheet (t_c), Young's modulus of the endcaps, piezoelectric material, type and dimensions of the bond. The strain (displacement) and stress (blocking force) ranges of these different technologies are compared in Figure 2 [2]. Although the above devices display improved displacement characteristics compared to simple piezoelectric plates, technologies that produce even greater strains (displacements) are highly desired because of the number of possible applications that require much higher strain levels than currently achieved.

A new class of piezoelectric transducers, called stress-biased actuators, was developed in the 1990's to improve upon the performance characteristics of unimorphs. The approach developed was to pre-stress the active ceramic in the flextensional device, thus increasing the extrinsic domain wall contribution to piezoelectric response. Consequently, the strain of the flextensional device was increased. The concept of taking advantage of enhanced extrinsic domain wall motion was first proposed by Zhang and Cross (1989) in bimorph actuators made with single crystal barium titanate [3]. The bimorph was fabricated by bonding a barium titanate (BT) single crystal plate to a metal shim above the Curie temperature of 140°C using a conductive epoxy. During cooling, the length of the single crystal is controlled

by the length of the metal plate due to the epoxy bonding. The stress generated in the single crystal during cooling is partially relieved by adjusting the number of a-(100) and c-(001) domains due to the ferroelastic nature of the single crystal (see Figure 3). When such a bimorph is subjected to an electric field applied perpendicular to the thickness of the device, the a-domains in the device (domains whose polarization vector is parallel to the length of the device) switch to c-domains (domains whose polarization vector is oriented perpendicular to the device surface and parallel to the field). This process is referred to as 90° domain switching or 90° domain wall motion, and it results in a large lateral strain in the BT crystal, thus bending the actuator. The domain state of this device before and after field application is shown in Figure 3 [3]. These domain bimorphs exhibited a much higher displacement to device length ratio compared to conventional bimorph actuators because of the increased 90° domain switching. Although this technological approach increased device displacement, device operation was still limited to frequencies lower than a few kilohertz.

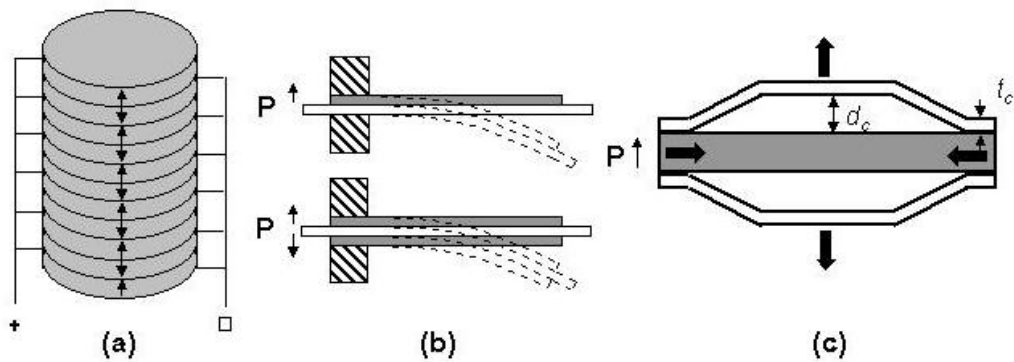


Figure 1. Different displacement amplification mechanisms: (a) stacks, (b) benders (unimorph (top) and bimorph (bottom)) and (c) flextensional devices (cymbals). For all examples, the piezoelectric layer is shown as dark and metal layers are shown as light. In (b), the blocks denote the clamped end of the device.

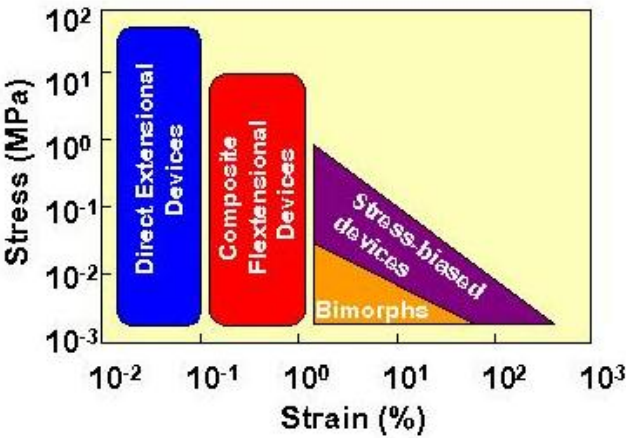


Figure 2. Strain (Displacement) and stress (blocking force generated) ranges of different piezoelectric actuator technologies available in the commercial market. Adapted from [2].

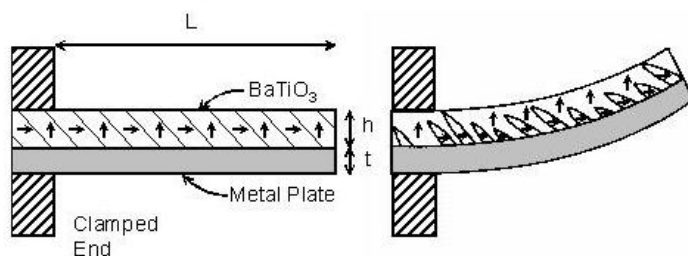


Figure 3. Typical domain configuration of a BT crystal domain bimorph under no applied field (left) and an applied field (right). Reproduced from [3].

Since the origin of bimorph domain actuators, the idea of taking advantage of ferroelastic/ferroelectric effects to increase domain wall motion contribution to device response has been under extensive investigation and has led to the development of innovative and technologically significant devices. The stress-biased devices that have been developed to date (in chronological order) are RAINBOW (1994), THUNDER[®] (1996), CERAMBOW (1996), CRESCENT (1997), LIPCA (2000), PRESTO (2006) and Stress-Biased Cymbals (2007).

Pioneering work in this field was carried out by Haertling, who invented the RAINBOW (Reduced And Internally Biased Oxide Wafer) device in 1994 [2, 4]. In these devices, the concept of increasing the 90° domain wall motion in single crystal BT was further extended to polycrystalline lead zirconate titanate materials [5]. The use of affordable, poled polycrystalline ceramic materials compared to single crystals suggested technologically viable devices might be feasible. The associated patent for the process was licensed and large scale manufacturing was investigated [6, 7].

Using a unique fabrication process, Haertling managed to develop an in-situ stress profile in polycrystalline lead zirconate titanate and lanthanum-modified lead zirconate titanate (PZT and PLZT, respectively) wafers, which in turn modified the domain configuration. Rainbows are manufactured by high-temperature chemical reduction (975°C) of one major surface of a PZT wafer to form a stressed composite structure. After manufacture, the final composite consists of an unreduced, piezoelectrically active layer and a reduced, metal rich (electrically conductive) layer. The improved performance characteristics of these devices have been studied extensively and are attributed to increased domain wall motion resulting from the stress profile in the piezoelectric layer created during the manufacturing process [4, 8-12]. The main advantage of these devices is flexural strain levels approaching 500% (compared to the device thickness). Shortcomings of the devices are fragility and a limited ability to withstand large point loads or mechanical shock. A summary of some of the literature regarding these devices is reported in Table I.

The other stress-biased device that has received significant attention is the Thunder[®] (THin UNimorph Driver and SENsoR) [23-29] actuator developed and patented by NASA. These devices remain commercially available through several vendors. Thunder[®] actuators represent an improved version of the Rainbow devices and are formed by a relatively simple process. These are relatively small, rugged devices capable of generating high flexural strains [27] and are produced by stacking the component layers (PZT, polyimide, metallic substrate and metallic electrode) in a laminate structure and heating them to temperatures between 200

- 325°C in an autoclave, followed cooling to room temperature. This manufacturing process results in a domed composite device that possesses an in-situ stress profile and domed shape similar to that of Rainbow devices. One reason behind the greater commercial relevance of Thunder[®] devices is the improved ruggedness that can be attributed to the elastic nature of the metallic backing substrate. A second factor is enhanced manufacturing reproducibility, e.g., improved control of layer thickness and bond characteristics compared to Rainbow devices. Thunder[®] devices have also been extensively studied, with most publications reporting on displacement characteristics under different mechanical and electrical conditions.

Table I. Summary of key publications for Rainbow devices

Author(s)	Summary of Publication(s)
Furman et al. [8, 9]	Measured the static displacement response and resonance frequencies of different vibrational modes of Rainbows under various boundary conditions and compared values with those predicted from finite element analysis using the commercial software package, ABAQUS.
Elissalde and Cross [10, 11]	Reported on the quality factors and elastic constants of the ceramic and chemically reduced layers by measuring the resonance frequency of different modes. Also investigated the structure-property relationship of the Rainbow ceramic-cermet composite.
Li et al. [12]	Modeled and profiled the stress distribution that exists along the thickness and length of the Rainbow devices using finite element analysis and compared to the experimentally derived stress levels from the XRD patterns.
Wise et al. [13]	Studied the feasibility of using Rainbow actuators to replace the motors used in the scanning mechanism of Fourier Transform Spectrometer.
Dausch [14-16]	Compared the ferroelectric fatigue behavior of Rainbow actuators and conventional bulk ceramics using XRD. Studied the fatigue characteristics of devices made of hard and soft PZT's under both low and high fields. Investigated compositional effects on displacement degradation as a function of time under different loads.
Wang and Cross [17-19]	Used composite beam theory and measured the resonance frequency to calculate the Young's modulus of the reduced layer. Reported the kinetics and thermodynamics of the reduction process involved in the fabrication of Rainbow devices at different temperatures and reduction times. Also studied the tip deflection and blocking force and compared measured values to predictions from theoretical models.
Schwartz et al. [20, 21]	Reported on the effects of stress on the initial domain configuration (a- and c-domain populations) and switching response by varying the thickness of the reduced layer (dependent on the reduction time). Estimated the d31 coefficients as a function of applied field and reduced layer thickness using the tip deflection at various fields and unimorph theory.
Hyer and Jilani [22]	Used thin plate theory to predict the manufactured shape (saddle or domed) of circular rainbows devices with different thickness ratio (reduced/unreduced) and radius to thickness ratio of the PZT wafer before reduction

Table II summarizes the literature on Thunder[®] actuators and sensors.

Table II. Summary of key publications for Thunder[®] devices

Author(s)	Summary of Publication(s)
Mossi et al. [23-25]	Studied the performance characteristics of Thunder devices fabricated with aluminum substrates of different thickness. Measured actuator displacement under different mechanical loads and electrical fields.
Hyer and Jilani [26]	Predicted the deformation characteristics of Thunder [®] devices during cooling in the manufacturing process using theoretical models.
Mulling et al. [27]	Reported on the displacement characteristics of Thunder [™] devices under mechanical loads and the importance of boundary conditions.
Ounaies et al. [28]	Measured the displacement of Thunder [®] devices as a function of frequency, operating temperature, various metal to piezoelectric thickness ratios at low and high applied fields.
Schwartz and Narayanan [29]	Reported an increase in the displacement response of Thunder [®] devices modified using mechanical pre-loads.
Navapan et al. [30]	Used Moiré interferometry to measure the local strain along the thickness of the Thunder [®] device to predict the d-coefficient variation along the thickness of the piezoelectric layer.
Nothwang et al. [31]	Studied the displacement, fatigue and lifetime of Thunder [®] devices with different widths and substrate thickness under various mechanical loads.
Taleghani and Campbell [32] Kennedy et al. [33] Aimmanee and Hyer [34]	Used finite element models to predict the manufactured shape and displacements of various Thunder [®] devices

Other novel stress-biased devices developed in the 1990’s were based on the Thunder[®] fabrication process, exploring variations in the bond and processing temperature. In Cerambow devices, Sn/Pb (60/40) solder was used to bond a PZT wafer to a brass disk. As before, the thermal expansion mismatch of the component layers was exploited to induce the desired pre-stress [35]. “Crescent” devices are fabricated using a polymer to bond a piezoelectric wafer to a metal substrate between 200 - 400°C and rapidly cooled to room temperature [36]. More recent developments in the area of stress-biased actuators involve the utilization of innovative materials and techniques to fabricate these devices, e.g., LIPCA, PRESTO and stress-biased cymbals, however, these devices have yet to see the depth of study of the Rainbow and Thunder[®] devices. LIPCA (LIghtweight Piezo-composite Curved Actuator) devices are manufactured to reduce the weight of typical Thunder[®] actuators by replacing the metallic substrate with lightweight fiber reinforced plastic layers. The key reports of these devices are reported in Table III. PRESTO (PRE-STressed electrOactive component) devices, which were only developed as recently as 2006, use a sintering process for bonding the passive and piezoelectric layers [43]. As with Rainbow and Thunder[®] devices, thermal expansion differences between the passive and piezoelectric layers result in pre-stressing the PZT. Stress-biased cymbals (SBC), developed by the authors (2007), use endcaps made of a flat-trained shape memory alloy (SMA) bonded to the piezoelectric disk [44]. The pre-stress in the piezoelement is induced by heating the composite above the

austenite finish temperature (A_f) of the SMA. Presto and SBC's are recently developed devices, and therefore, have not been extensively investigated.

Table III. Summary of key publications for LIPCA devices

Author(s)	Summary of Publication(s)
Yoon et al. [37, 38]	Reported on the design principle and manufacture of these devices and reported device weight could be reduced by up to 40% compared to similar Thunder® devices by replacing the metal substrate with a fiber composite layer. Investigated the displacement degradation of LIPCA devices with time under free and mechanically loaded conditions.
Goo and Koon [39]	Reported the displacement and fundamental resonance frequency of LIPCA devices as a function of the thickness of the PZT layer
Haris et al. [40]	Used laminated beam model to predict the displacement characteristics of LIPCA devices
Lee et al. [41]	Used piezo-shell element and material nonlinearity to predict the displacement under different electric fields
Kim et al. [42]	Evaluated the displacement characteristics of various LIPCA's made with different stacking sequences of their component layers and modeled the stress profile along the thickness of these devices.

The authors have previously proposed a new device called shape memory unimorph (SMU) that has the potential to realize a range of other possible stress-biased devices [45]. SMU's are manufactured by replacing the metal substrate in Thunder® devices by shape memory alloys pre-trained to the necessary domed structure at 550°C. The substrate is then flattened at room temperature while bonding the PZT using an epoxy. The prestress is induced by heating the composite above A_f to attain the pre-trained dome structured composite. These devices may demonstrate improved performance compared to similar stress-biased actuators because the pre-stress magnitude is independent of the thermal expansion coefficient difference and thickness of the component layers, as for Thunder® and Rainbow, or similar devices. The advantages of this approach are control of the pre-stress magnitude (dome height), shape restoration of the deformed device during mechanical loading and simple process [45]. The different fabrication processes involved to make the various stress-biased are detailed below.

II. TYPES OF STRESS-BIASED DEVICES AND FABRICATION

In general, stress-biased unimorphs (and composite flextensional devices) consist of a metallic or fiber composite substrate, a piezoelectric layer and a bond to hold the two layers together. In all cases, the substrate has a higher thermal expansion coefficient than the electroactive layer. Most of the stress-biased actuators, except SBC, are fabricated at elevated temperatures ranging between 150 and 975°C and utilize the thermal expansion mismatch of the component layers as the driving force to pre-stress the piezoelectric layer when the bond solidifies during cooling cycle of the fabrication process. This results in the characteristic curvature in these devices. Typical stress-biased actuators of different shapes and sizes are

shown in Figure 4. The fabrication processes that are used to manufacture these devices are detailed below.



Figure 4. Stress-biased Thunder[®] devices that are manufactured in different shapes and sizes [24].

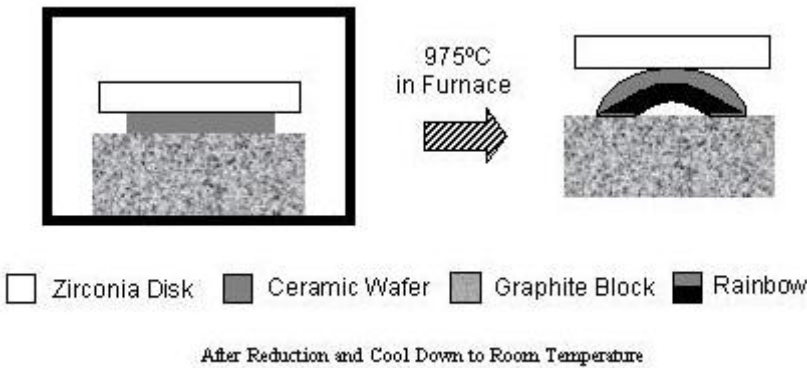


Figure 5. Representation of the manufacturing process of a Rainbow ceramic. The zirconia setter is placed on top of the piezoelectric ceramic wafer, which is placed in air at 975°C on a graphite block to form a Rainbow. The reduced portion of the Rainbow is shown in solid black [6, 46].

(a) Rainbow

Rainbow [2, 4, 6, 7] ceramics were invented by Haertling at Clemson University but are no longer commercially available. They were manufactured by Aura Ceramics Inc during the late 1990’s. Rainbows are monolithic structures that are produced by chemically reducing one side of a lead-containing piezoelectric ceramic (PZT, PLZT, etc.) at an elevated temperature. In this process, the piezoelectric ceramic is placed on a graphite block and inserted into a

furnace preheated to 975°C. A zirconia plate is placed on top of the piezoelectric ceramic to shield that side of the piezoceramic from chemical reduction [4, 6, 7].

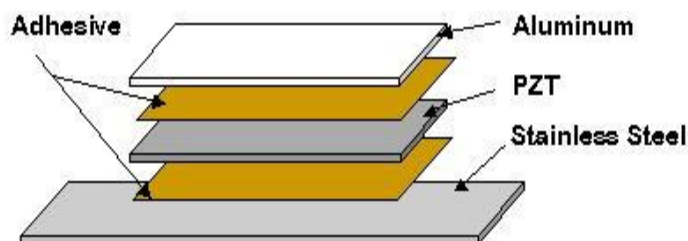


Figure 6. Schematic of the component layers of a Thunder[®] actuator. Typical layer thicknesses for a commercial Face International Corporation TH-8R device: aluminum – 2mils; LaRC-SI – 1mil; PZT – 8mils; and stainless steel support – 6mils.

The kinetics of this reduction process, which involves the formation of metallic lead and produces a “cermet” layer, has been reviewed by Wang and Cross [18]. Along the interface between the piezoelectric and graphite block, oxygen is removed from the ceramic, creating a composite structure with a non-piezoelectric layer (the chemically reduced material) and a piezoelectric ceramic layer. When this structure is cooled to room temperature, the difference in thermal expansion coefficients between the ceramic and reduced layers creates a stressed dome-shaped structure [2, 4, 6, 7], as illustrated in Figure 5 [6, 46]. The thickness of the reduced layer, and therefore, the stress magnitude, is directly dependent on the time and temperature of reduction. It is reported that the ratio of the reduced to unreduced layer should be ~ 0.3 for optimum performance.

(b) Thunder[®]

Thunder[®] [23, 27, 28] actuators are based on a NASA patent and are commercially available through a variety of manufacturers [27]. Thunder[®] actuators are produced by bonding a thin piezoelectric ceramic to a metal sheet using a high temperature curable polyimide adhesive, referred to as LaRC-SI, as shown in Figure 6. In this process, the polyimide material is either spray coated onto both surfaces of the piezoelectric ceramic and the metal sheet, [23, 47] or polyimide films fabricated through a different manufacturing process are layered between the ceramic and metal as a sandwich [47]. The coated materials are then layered together and autoclaved at 250 - 325°C to cure the polyimide. As in the Rainbow process, internal stresses are induced in the Thunder[®] actuator due to the thermal expansion mismatch between the ceramic and metal that occurs upon cooling, producing a domed stress-biased actuator [47, 48]. The bond between the layers is essential because it makes possible the induced pre-stress in the piezoelectric layer and improves the robustness of the device. Thunders have a domed shape, the degree of which depends on the type and thickness of its constituent materials (metal substrate, PZT wafer), the length of the device and the type of adhesive used [47]. Different metals, such as stainless steel, aluminum, brass, copper and various hard and soft PZT's have been utilized in the fabrication of Thunder[®] devices.

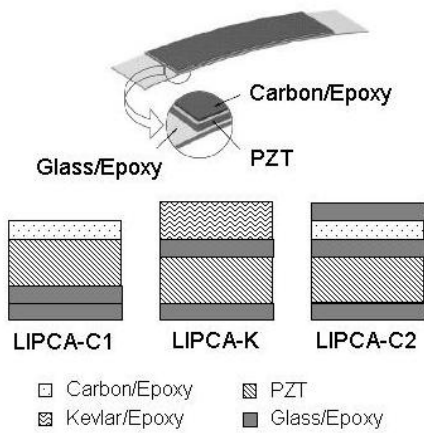


Figure 7. Stacking sequence of different LIPCA devices. Adapted from [51].

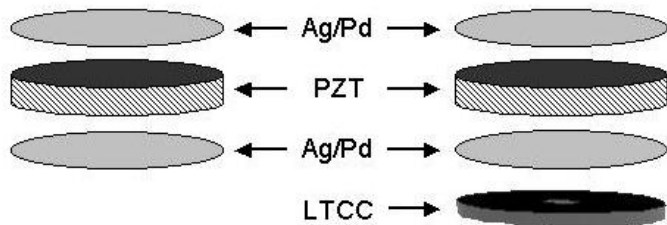


Figure 8. Different types of PRESTO devices made using (a) conductive Ag/Pd paste (left) and (b) LTCC layer (right).

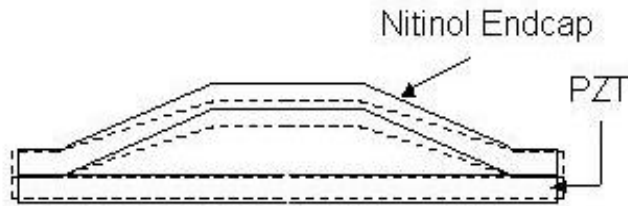


Figure 9. Cross-sectional view of Stress-biased Cymbals (SBC). Typical representation of a shape memory endcap bonded to a PZT disk before (solid lines) and after (broken lines) the pre-stressing process [44].

(c) Cerambow

Cerambow devices, developed by Haertling et al. [35], like most stress-biased unimorphs utilize a metal substrate to create a pre-stress in the piezoelectric ceramic but at a lower processing temperature. They are fabricated by soldering a thin brass sheet to an electroded piezoceramic like PLZT. Soldering is carried out by preheating brass disks on a hotplate to ~ 250°C and uniformly applying Sn/Pb (60/40) solder. The electroded PLZT disk is then placed

on top of the solder and a preheated steel block is used to hold the PLZT disk in place. The whole setup is then cooled to room temperature and the thermal expansion difference between the metal and PLZT results in bending of the CERAMBOW device. The edges are sanded to remove any excess solder and avoid electrical breakdown during testing or application.

(d) Crescent

Crescent actuators, developed by Chandran et al. [36], are fabricated in a similar fashion to Thunder[®] devices, but use a thermally cured polymeric (epoxy) adhesive as the bonding agent compared to LaRC-SI used for Thunder[®] devices. The PZT disk is attached to the stainless steel sheet using an epoxy between 200 - 400°C and cured for 0.5 to 6 hrs. The device is air cooled to room temperature during which time it attains its domed shape. The devices cooled from 250°C are reported to display superior displacement characteristics compared to the ones that was fabricated at 350°C [36].

(e) LIPCA

LIPCA is a variation of the Thunder[®] device developed by Yoon and co-workers [49]. These devices use fiber reinforced composite materials as the passive layers to reduce device weight without compromising force generation and displacement capabilities. A variety of LIPCA devices (LIPCA-C1, LIPCA-C2, LIPCA-K, LIPCA-CK, LIPCA-CKK) have been manufactured with different component layers and variations in stacking sequence [42]. The LIPCA-C1 [42] device consists of three layers: a bottom glass-epoxy layer (0.18mm), middle PZT layer (0.25mm) and top carbon-epoxy (0.1mm) layer. The LIPCA-K device [42] possesses a similar stacking sequence as LIPCA-C1, except that the carbon-epoxy layer is replaced with a Kevlar-epoxy layer as shown in Fig.7 [51].

LIPCA devices can be easily manufactured by stacking the component layers in the required order in a flat mold. The PZT wafers used are electroded and copper strip wires are attached using silver paste prior to stacking in the mold. The stacked layers are then vacuum bagged and cured at 177°C for 2 hrs in an autoclave or oven [37, 38]. In general, these devices develop a radius of curvature during cooling due to the thermal expansion mismatch between the glass-epoxy ($\alpha = 14.2 \times 10^{-6}/^{\circ}\text{C}$) and the PZT ($\alpha = 2 \times 10^{-6}/^{\circ}\text{C}$) layer which can be observed after removing the device from the mold [37, 50]. The difference in the stacking order of LIPCA-C2 puts the PZT fully into compression. The neutral axis (between compressive and tensile stress) is shifted just above the PZT layer, as compared to that of LIPCA-C1, where the neutral axis resides in the PZT layer.

(f) Presto

Presto devices were developed by Juuti and co-workers [43] in an attempt to use sintering phenomena for bonding while also utilizing the shrinkage and coefficient of thermal expansion differences between the component layers to induce pre-stress in the piezoelectric layer. Two different techniques were used to manufacture these devices as shown in Figure 8. In the first case (Figure 8a), a conductive Ag/Pd (silver-palladium) layer, 24.5mm in diameter and ~ 20 - 60μm thick, is screen printed onto one side of the PZT disk (25mm in diameter and 250 or 375μm thick). After drying, the single-sided, electroded disk is sintered at 850°C in a belt furnace to yield an Ag-Pd layer between 12 and 33μm in thickness. The top electrode

was either a layer of silver paint or the same Ag/Pd paste with a final thickness of 12 μ m after firing.

In the second case (Figure 8b), pre-stressing was achieved by laminating a commercially available DuPont 951 Low Temperature Co-fired Ceramic (LTCC) tape (green thickness of 114 μ m and fired thickness of 96 μ m) with the PZT disk and sintering. Prior to lamination, both the top and bottom electrodes (20 μ m) were first screen printed onto the PZT with Ag/Pd paste and fired at 850°C to attain a final thickness of 12 μ m [43]. The green LTCC tape was laser cut to fit the PZT disk with a hole at the center for the electrical connection to the bottom electrode. This LTCC layer was then stacked onto one side of the electroded PZT disk and laminated in an isostatic laminator at 20MPa and 70°C for 10 minutes. This laminated structure was then fired at 875°C in a belt furnace. In both cases, the devices attain a dome shape during cooling due to the shrinkage difference and thermal expansion mismatch of the Ag/Pd or LTCC ceramic and PZT layers, which is characteristic of most stress-biased unimorphs. The actuators were then poled at room temperature with a field of 2.5MV/m for 30 minutes [43].

(g) *Stress-biased Cymbals*

A novel method of pre-stressing the electroactive element using cymbal endcaps made of shape memory alloy has been reported by the authors. In this method, a flat trained 0.25mm thick Nitinol sheet with an austenite finish temperature (A_f) of 45°C was used to make the cymbal shaped endcap by stamping and cutting the Nitinol sheet in a specially designed steel die. The flanges of the metal endcap and the surface of the electrodes on the PZT disk (25mm dia, 0.2mm thick) were roughened with SiC abrasive paper to improve the bond strength and cleaned with acetone to remove the residual metal and SiC particles. Cyanoacrylate was used to bond the metal endcaps to the electroded PZT disk. After applying the epoxy, the endcap was then assembled with the PZT and cured under a load of approximately 50N applied to the flanges during curing at room temperature. After bonding the endcap to the PZT disk, the device was heated above the A_f to recover the flat trained shape of the SMA; i.e., the cavity depth of the end cap decreases as shown in Figure 9 [44]. This shape recovery process is opposed by the bond between the endcap flanges and PZT, resulting in a radial tensile stress in the PZT disk. The magnitude of the radial stress can be controlled by the heating time (final cavity depth), initial cavity depth and thickness of the SMA [44].

III. PERFORMANCE ANALYSIS

Stress-biased actuators offer enhanced displacement performance compared to unimorph and bimorph actuators, as well as traditional direct extensional actuators. These materials can demonstrate flexural strains on the order of 1 – 500%, with flexural strain being defined as the tip deflection divided by the thickness of the ceramic layer, and moderate force generation capabilities that approach 1MPa. These strain values are an order of magnitude higher than typical flextensional piezoelectrics, although the ultra-high displacements are strongly dependent on physical boundary conditions [27].

Although the specifics of the fabrication procedures differ for each of the stress-biased devices, typically, except for SBC devices, they attain a domed structure after processing. The

driving force for the doming of the devices is the thermal expansion mismatch between the two layers, i.e., the metal/fiber composite layer (or the reduced layer for Rainbow devices) has a greater thermal expansion coefficient than the piezoelectric layer. Therefore, during cooling, the devices dome, yielding an actuator that has a convex shape when viewed from above. As the devices dome, lateral stresses within the piezoelectric layer are developed. The stress profile and state (tensile, compressive or both, with a neutral plane contained within the piezoelectric layer) depend upon device architecture, and fabrication conditions. The specific nature of the stress profile and magnitude has been the focus of intense investigation due to the significant role of stress on domain wall contributions to piezoelectric response. The magnitude and characteristics of the stress imparted during the manufacturing process depends on the thermal expansion mismatch of the component layers, mechanical properties of the metallic substrate (Young's modulus and Poisson's ratio), temperature of the manufacturing process, and bond characteristics.

Generally, it is believed that the enhanced electromechanical response of these devices is dependent on the in-situ stress profile that is imparted during the manufacturing process and the presence of tensile stress in the upper region of the PZT that facilitates enhanced domain wall motion. As documented by Navapan [30, 52] the superimposition of a stress profile on the piezoelectric layer results in a varying piezoelectric coefficient along the thickness of the PZT layer. This concept of taking advantage of pre-stress effects on electromechanical response is the primary aspect of stress-biased actuators that has fueled their interest and development. It should also be noted, however, that the presence of elastic layers on either surface of the PZT layer for devices such as Thunder[®] actuators is believed may play a significant role in device reliability. As discussed below, investigations regarding stress state, magnitude, and gradient continue as related to the specific role of stress on device performance and reliability.

(a) General considerations

There have been numerous studies that have attempted to investigate the factors that contribute to the improved performance of stress-biased actuators. Most reports suggest that the enhancement in displacement response is due to the increase in 90° domain switching [4, 12] in the upper region of the piezoelectric layer when a field is applied. The lateral tensile stresses that are present create an altered domain population compared to a simple poled piezoelectric plate. When an electric field is applied to the device, the enhanced domain switching that occurs contributes to a higher effective piezoelectric coefficient and greater electromechanical response. When the field is removed the pre-existing stress state results in a return to the original altered domain population. The in-situ stress state and applied electric field thus result in a combined ferroelectric/ferroelastic behavior that dictates electromechanical response. Contributing factors to these response mechanisms are discussed in detail below.

Other researchers believe that an equally important aspect of these devices, rather than their enhanced electromechanical response, is their robustness, which is imparted by the metal or fiber backing substrate. Rainbow devices exhibit large displacement characteristics but are fragile and prone to fracture under a point load due to the brittle nature of the cermet layer. On the other hand, Thunder[®] devices can sustain large point loads without fracture due to the elastic nature of the backing metallic (e.g., stainless steel) substrate and the polymeric

bond. More recently, FEA simulations suggest that the entire piezoceramic in these devices is in a state of compression after the manufacturing process and even during operation under quasistatic and dynamic fields [30, 32-34, 52]. The magnitude of the compressive stress is dependent on the thermal expansion coefficient mismatch, thickness of the component metallic substrate and electrode layers and the field applied during operation. Therefore, it is believed that the improved performance and lifetime of these devices are a direct result of the fact that the entire piezoceramic is in a state of compression, rather than a tensile state, which is more likely to contribute to failure of the piezoceramic.

Most studies of these devices have been carried out to understand the role of stress on domain populations and switching. To further understand the primary mechanism that is responsible for the enhanced performance of these devices, it is necessary to review the different contributions (extrinsic and intrinsic) to piezoelectric response and the effect of different stress fields (tensile and compressive) on intrinsic and extrinsic contributions to piezoelectric response.

(b) The role of stress on intrinsic and extrinsic contributions to piezoelectric response, dielectric, and mechanical response

The piezoelectric response of ferroelectric ceramics is a result of contributions from two mechanisms. The first is the lattice contribution, which originates from the piezoelectric effect within each individual ferroelectric domain. Here, changes in unit cell dimensions occur as a result of the force that the applied electric field exerts on the ionic species in the lattice, as demonstrated in Figure 10. The second contribution, often called the extrinsic mechanism, is largely due to the displacement of non-180° domain walls [53, 54]. High strains are achieved through non-180° domain switching because of the large anisotropy of the crystal lattice (1 - 6%, depending on composition) and the accompanying deformation that occurs upon reorientation of the polar axis of the domain.

In many ferroelectric ceramics, such as lead zirconate titanate and barium titanate, the piezoelectric response is dominated by domain wall contributions [53-56]. A review of data presented for barium titanate in the classic text by Jaffe, Cook, and Jaffe gives an indication of the relative importance of the two mechanisms at low electric field (e.g., $\sim 0.1\text{V/mm}$) [55]. Single crystal, single domain BaTiO_3 , whose response is solely intrinsic in nature, demonstrates a piezoelectric d_{31} coefficient of -34.5pm/V , while poled polycrystalline BaTiO_3 demonstrates a d_{31} coefficient of -79 pm/V [21]. Under high field conditions ($> 0.1\text{kV/mm}$), the relative importance of the extrinsic response mechanism is even greater. For example, in BaTiO_3 , the contribution from the extrinsic effect to the total strain response can be 100 times greater than the intrinsic effect [46, 58]. It has also been noted that the dielectric constant for a purely intrinsic response, including 180° domain switching, was 300, while the contribution for a purely extrinsic 90° domain switching material was 4500 for single crystal BaTiO_3 [46, 58]. Reorientation of the polarization vector contributes to these significant differences in dielectric constant. Studies of PZT show that the extrinsic effect contributes 60% to 70% of the piezoelectric response observed experimentally [59] for standard poled polycrystalline ceramics operated under low applied electric fields.

The extrinsic effect is complicated, because it involves interactions on several length scales, i.e., interactions between ions, domains, and even different phases (as in PZT) [59]. Therefore, it is difficult to develop a detailed microscopic picture to account for the extrinsic effect. Arlt et al. have proposed a formal phenomenological theory dealing with linear

electromechanical properties related to 90° domain wall motion [57, 59], and numerous investigators have explored domain effects for the past 50 years.

Because of the high strains associated with non- 180° domain switching, different strategies have been sought to take further advantage of these effects in order to develop actuators with greater electromechanical response. As mentioned previously, the earliest effort in this area appears to have been undertaken by Zhang and Cross and involved the development of what the authors described as a “domain bimorph actuator” [3]. The strategies that followed involving the superimposition of stress fields within the piezoelectric layer to alter the domain configuration and non- 180° domain switching characteristics under applied field led to the development of the family of stress-biased actuators that are the focus of this chapter.

Since the piezoelectric layer in stress-biased devices of common geometries is believed to be under significant compressive and tensile stresses, the effects of stress on the domain populations and domain switching characteristics of the materials must be considered, in addition to the effects of applied electric field. In the discussion below, a material composition in a tetragonal phase field is assumed. Analogous explanations are possible for non-tetragonal ferroelectric compositions. As noted above, domain population and domain switching can be induced both by electric field and mechanical stress; i.e., the piezoelectric materials used in the fabrication of these devices are both ferroelectric and ferroelastic.¹ A schematic illustration of the ferroelastic switching of a single domain in a tetragonal PZT under compression and tension is shown Figure 11. It may be noticed that only 90° switching can be induced by the application of a purely mechanical stress since it serves to relieve the applied stress (recall that the unit cell is slightly anisotropic and elongated in the poling direction). In a polycrystalline ferroelectric body, such 90° switching contributes to depolarization of the material; domains initially oriented parallel to the poling direction (c-domains) are reoriented perpendicular to the poling direction (a-domains) due to the applied stress. It should also be noted that depolarization can be induced by compressive stresses acting parallel to, or tensile stresses acting perpendicular to, the poling direction. In fundamental studies of stress effects on domain population and mechanical (stress-strain) response, the first stress condition has been more extensively considered [62-64].

While a simple mechanical stress may prove detrimental,² it can be used to advantage when coupled with electrical loading. In such conditions, the mechanical stress would tend to depole the material by increasing the a-domain population by ferroelastic switching, while the electric field would polarize the material by increasing the a-c domain ferroelectric switching. Such behavior can result in a large strain response due to the anisotropic nature of the unit cell. Extensive reports have been published regarding stress effects on piezoelectric properties and the response of ferroelectric materials; selected reports pertinent to this subject are reviewed below.

The effect of a transverse tensile stress on field induced 90° -domain reorientation (for an applied field parallel to the poling direction) in tetragonal PZT near the morphotropic phase boundary was investigated *in situ* using X-ray diffraction (XRD) by Li et al. [60]. These

¹ It should be noted that electrostrictive compositions have also been utilized successfully in the fabrication of stress-biased actuators, but here we focus on the piezoelectric-based devices

² The mechanical stress conditions shown in Figure 11 may serve to depole the ceramic, making it piezoelectrically inactive. Also, an in-situ tensile stress may result in a ceramic with lower mechanical strength.

authors determined that a transverse tensile stress altered the $I_{(002)}/I_{(200)}$ ratio, under zero and various applied electric fields. These results were then used to estimate the piezoelectric d_{31} coefficient. It was reported that at $E = 0.9\text{ kV/mm}$, the estimated $d_{31,\text{domain}}$ (i.e., the extrinsic contribution to electromechanical response) changed from -200 pm/V at zero stress, to -350 pm/V at 75 MPa of lateral tensile stress [60]. This $\sim 75\%$ enhancement in d_{31} due to increased 90° domain switching confirms that control of electrical and mechanical boundary conditions may be employed to achieve higher electromechanical response. Stated otherwise, through control of boundary conditions, ferroelectric and ferroelastic effects may be used together to enhance the electromechanical response of a piezoelectric ceramic.

Similar enhancements in domain wall motion in tetragonal soft PZT were observed by Zhou [61] for uniaxial compressive stress ($< 50\text{ MPa}$) applied parallel to the direction of polarization. Enhancements in d_{31} and d_{33} of 62% and 57%, respectively, were observed at low fields ($\pm 0.1\text{ kV/mm}$) [35], as reproduced in Figure 12, while enhancements of 50% and 48% were observed at high fields ($\pm 2\text{ kV/mm}$) [61]. Zhou [61] also observed a further increase in the piezoelectric charge coefficients with the application of bias field. It was also reported that the differential dielectric constant increased from $\sim 7,400$ to $\sim 16,000$ for an increase in compressive axial pre-stress of 35 MPa , suggesting greater domain wall motion due to the pre-stress. It can also be noted that the coercive field (inset of Figure 12) decreases with increasing compressive pre-stress suggesting the softening of the material (indirect evidence to increased domain switching). Analogous results were reported by Yimmirun et al. [62].

Sherrit et al. [63] investigated the variations of the d_{33} coefficient of soft PZT ceramics as a function the amplitude and frequency of the applied stress. The d_{33} coefficient was measured under quasistatic conditions using a step stress profile applied along the polar axis. The d_{33} coefficient increased from 200 to 430 pC/N when the stress was increased from 0 to 70 MPa . Lynch [64] reported an increase in d_{33} from 1188 to 4200 pC/N for an increase in the uniaxial compressive stress (parallel to the poling direction) from 0 to 30 MPa . Fett [65] and coworkers observed an increase in the dielectric constant measured at 1 kHz from $\sim 1,750$ to $2,250$ under an applied compressive stress of 100 MPa parallel to the poling direction. Finally, Viehland and coworkers [66] reported an increase in the ΔP (change in polarization), strain, d_{33} , coupling coefficient, acoustic power and hysteretic loss with increasing compressive stress, again suggesting the importance of stress on domain wall motion in hard PZT materials.

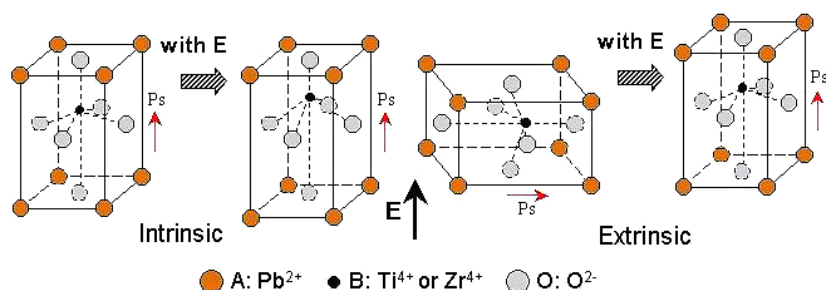


Figure 10. Unit cell schematic showing the exaggerated representations of intrinsic (shown for a unit cell) and extrinsic contribution (shown for a domain) to the response of ferroelectric materials under an electric field.

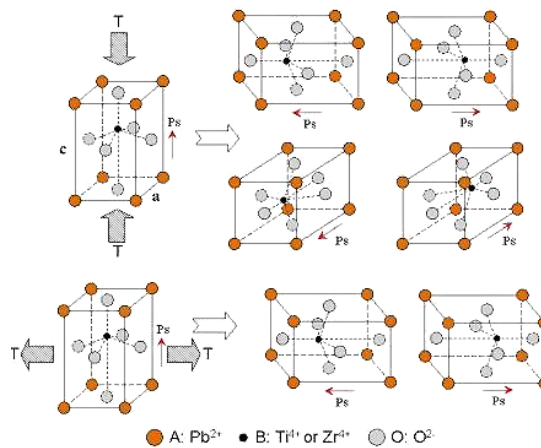


Figure 11. Schematic illustration of ferroelastic domain switching induced by stress. The two polarized regions on the left represent the original polarization orientations prior to the application of the stress that is shown. The six polarized regions on the right represent possible dipole orientations following stress application. 'T' denotes the stress direction.

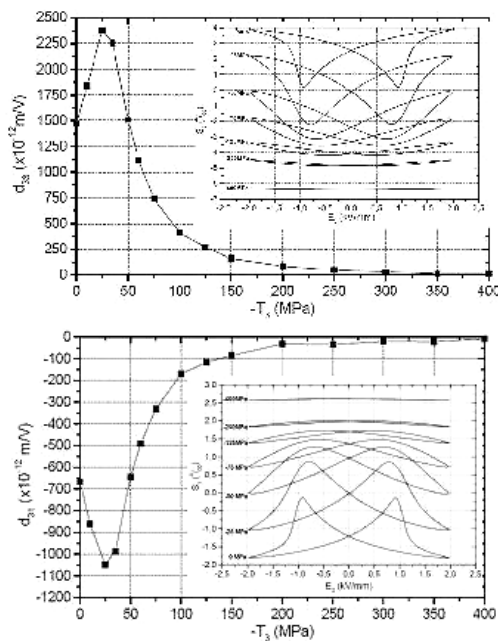


Figure 12. Effect of uniaxial compressive stress along the direction of polarization of soft PZT on the d_{33} and d_{31} coefficients measured at low fields ($\pm 0.1 \text{ kV/mm}$) as reported by Zhou [61]. Inset shows the actual butterfly strain loops ($\pm 2 \text{ kV/mm}$) measured under different levels of compressive stress. Reproduced from [61].

Although domain wall motion increases the strain response in a material, it is also responsible for the non-linear and hysteretic behavior of the device, which makes it more difficult to model, and complicates the utilization of these devices in many applications. This non-linear behavior can also be observed in Figure 12 for d_{33} , d_{31} , coercive field and remanent

strain (s_1 , s_3) with increasing magnitude of compressive stress. Figure 13 shows the non-linear stress-strain and stress-polarization behavior of a typical PZT material under a compressive stress along the direction of polarization (analogous to a tensile stress perpendicular to the polarization direction) [61]. The numbers 1-3 indicate the different domain configurations in the material at various points along the D-T or S-T curves. At position (1), all the domains are aligned parallel to the direction of the poling field, therefore creating a remanent strain and polarization in the material. At low stress levels ($< 75\text{MPa}$ in this case), the material behaves linearly, with polarization and strain changing proportionally to the applied stress. Upon crossing a particular stress value, known as the coercive stress, domains gradually start to realign permanently (irreversible) in random directions, or close to the direction perpendicular to the applied stress, thus resulting in the observed non-linear behavior in both the strain and polarization response. After sufficient stress is applied, position (2), the domain switching process saturates, resulting in a depoled material. Upon removal of the stress, only a small fraction of the domains switch back to their original orientations, indicated by the near vertical path traced from position 2 to 3 in the polarization curve [61]. Because of this behavior, material properties like Young's modulus and piezoelectric charge coefficients are non-linear functions of electric field and stress.

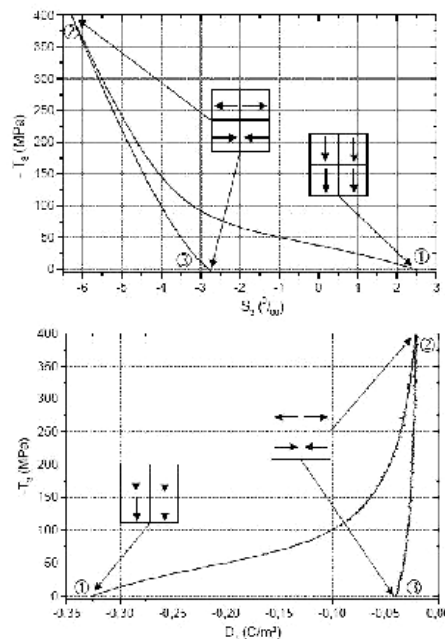


Figure 13. Effect of uniaxial compressive stress along the direction of polarization of soft PZT on the strain and polarization as reported by Zhou [61]. Corresponding domain states at different regions are indicated in the schematic. Reproduced from [61].

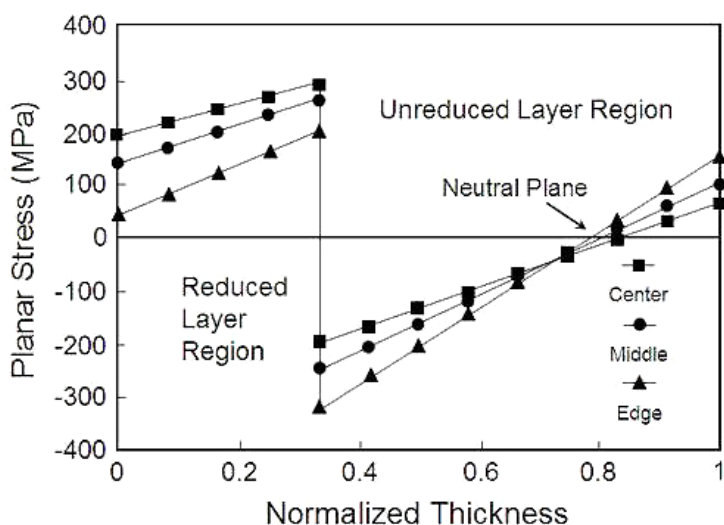


Figure 14. Change of planar stress versus thickness at three different locations across a circular Rainbow having a thickness ratio (reduced/unreduced) of $\frac{1}{2}$. Adapted from [12].

The studies noted above describe the electrical, piezoelectric, and mechanical behavior of ferroelectric/ferroelastic materials under comparatively simple (and controlled) boundary conditions. We next discuss how these effects relate to the observed performance of stress-biased actuators. Similar behavior occurs, though the more complicated boundary conditions and changes in stress state and domain population under dynamic operational conditions make a thorough analysis of the factors that contribute to performance more difficult.

(c) Domain wall contributions to performance

Since Thunder[®] and Rainbow actuators are similar in their concept and operation, the prevailing stress states in the PZT layer are believed to be similar. The variation of the planar stress through the thickness of a Rainbow as reported by Li et al. is given in Figure 14 [12]. For both Rainbow and Thunder[®] actuators, the tensile stress within the surface region of the piezoelectric layer is reported to approach 400MPa [48]. This value was estimated using finite element analysis without accounting for the ferroelastic response of the PZT, and is predicted based on a linear model of material response that does not account for the domain reorientation processes noted in Figure 11, which contribute to a significant softening of material response. It is estimated that the maximum tensile stress would approach values ranging between 30 - 50MPa if ferroelasticity is considered in the calculations [48]. Irrespective of material softening behavior, the general nature of the curve is believed to be accurate although the maximum stress values are likely lower than those indicated.

Figure 14 shows the entire reduced layer is in tension, which is signified by the positive values on the vertical scale. There is a neutral plane in the unreduced (piezoelectric) layer where stress is zero, however, the unreduced layer is in tension from the neutral plane to the surface. The portion between the neutral plane and the reduced/unreduced layer interface is in a compressive state and the magnitude of the stress varies linearly through the thickness of the device, with an abrupt transition occurring at the interface [12]. Investigations of Thunder[®] actuators suggest that they have a similar stress profile [52], the only differences

being that the reduced layer is replaced by stainless steel and lower stress levels are expected in Thunder[®] devices because of the lower processing temperatures. Lastly, it is important to note that the stress magnitude and profile in these devices will be strongly influenced by device geometry (i.e., rectangular versus circular, layer thickness ratio, etc.).

The high stress levels within the surface region of the ceramic enhance the effective piezoelectric response by forcing the domains to align preferentially in the a-direction (parallel to the surface). Numerous authors have confirmed these changes in domain population. Typical x-ray diffraction (XRD) patterns reported by Dausch for bulk poled PZT-5A and Rainbows fabricated with the same material are illustrated in Figure 15. These patterns illustrate the (001) and (100) reflections that are representative of the c and a-domain populations on the surface of the device, respectively. It can be observed that the c/a peak intensity ratio for the poled bulk material is ~ 3.4 , while it is ~ 1.3 for poled Rainbow device, confirming the increased a-domain population in the surface region of the device where lateral tensile stresses exist. This altered domain configuration, together with the limited amount of domain pinning (by stress) that has been observed in these devices, results in greater 90° domain switching, which is the primary mechanism that contributes to the enhanced electromechanical performance of these devices [20]. To our knowledge, the reasons for the limited domain pinning by stress in these devices have not been explored.

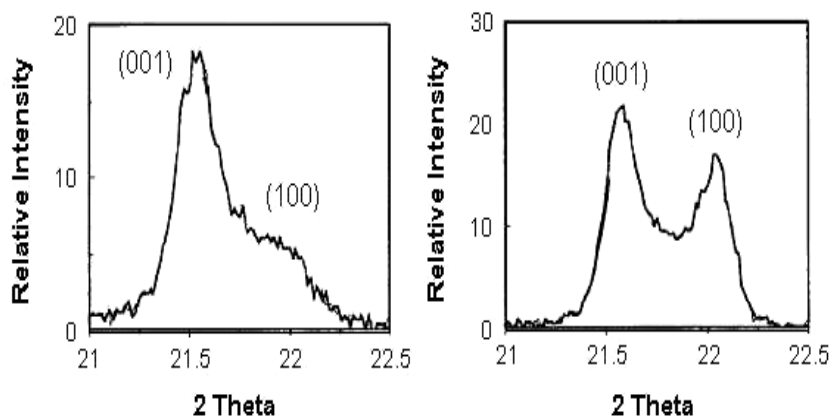


Figure 15. XRD of poled bulk ceramic (left) and Rainbow devices (right) made of tetragonal PZT-5A material showing the (001) and (100) peaks which represents the c- and a-domain populations on the surface. Adapted from [14].

At the interface with the reduced layer, more domains are aligned in the c-direction (perpendicular to the interface, i.e., parallel to the poling direction) [46]. This region of the device is thus expected to respond to applied fields via more of an intrinsic (single domain, single crystal) nature than even a poled polycrystalline ceramic. Because of this behavior, the observed performance of stress-biased devices is more likely due to effects that occur within the surface region of the actuator. These effects on domain response and effective piezoelectric coefficient of Thunder[®] actuators under different pre-stress magnitudes were further confirmed with Moiré interferometry studies carried out by Navapan [52], which are discussed below.

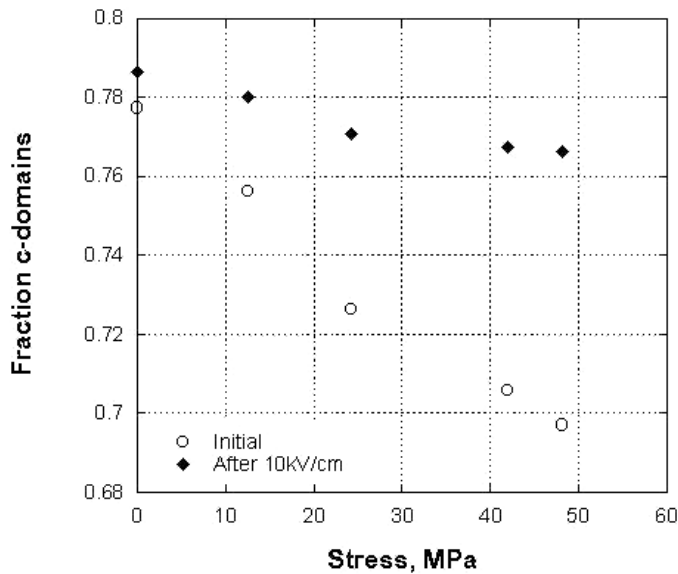


Figure 16. Domain configuration of poled Rainbow devices, under different stress levels, before (initial) and during application of a 1 kV/mm electric field. Fraction c-domains obtained by in-situ XRD measurements of the (200) and (002) peak areas [20].

The x-ray diffraction (XRD) results illustrated in Figure 15 show that the a-domain population in the surface region of Rainbow actuators is higher than that in the corresponding poled polycrystalline ceramic. Since Rainbow and Thunder[®] actuators have a similar stress profile, it is concluded that the a-domain population in the surface region of Thunder[®] actuators is also high. Preliminary x-ray diffraction studies of Thunder[®] devices support this conclusion [52]. In addition to domain population, enhancements in electromechanical response are also dependent on enhanced 90° domain switching. Schwartz and coworkers [20] have presented an analysis of the enhanced domain switching in the surface region of these devices, which is reproduced in Figure 16. In this figure, it may be seen that with increased stress level, as developed through changes in processing conditions, domain switching increases from approximately 1% ($T = 0$ MPa) to 7% ($T = 50$ MPa). This is calculated based on the difference in the fraction of c-domains under field vs. in the free field condition. It may be observed that the fraction of c-domains is reduced under increasing lateral tensile stress, as anticipated (i.e., the material is more highly depoled as stress increases), but that the lateral stress does not result in a corresponding decrease in domain switching. Rather, an increase in 90° domain switching is observed, which contributes to higher electromechanical response.

(d) Other contributions to performance and performance attributes

While the focus of many of the investigations on stress-biased actuators has been on the role of the stress state, it is important to note that other device aspects, such as mass loading (from the metal or reduced cermet layer of the composite) [20, 48], and engineering mechanics aspects (due to the similarity to unimorph structures) [1, 19, 21] have also been reported as contributing factors for the observed greater displacement response. Many reports

have been published on the performance characteristics of Rainbow and Thunder[®] devices compared to other stress-biased devices like LIPCA, Presto and SBC's. Some of the properties of these devices reported are discussed herein.

Rainbow devices exhibit very high displacements ($\geq 500\%$) and moderate load bearing capabilities (up to 10kg [2]) based on wafer thickness, as depicted in Figure 17. The strain (axial displacement/wafer thickness) decreases while the blocking force increases with an increase in the wafer thickness. Also shown in the figure are regions where Rainbow devices attain their different shapes, namely a saddle or dome shape, which is mainly dependent on the wafer thickness. Thinner wafers are more prone to be saddled while thicker wafers are typically domed due to the inherent mechanics of thin and thick plates [22].

The displacement response of Rainbow devices is highly dependent on the material composition (hard, soft, or electrostrictive) because of their significantly different response to a superimposed stress. Displacement response for Rainbow devices made of different piezoelectric (5.5/56/44 and 1/53/47) and electrostrictive (8.6/65/35)³ PLZT compositions are illustrated in Figure 17. It should be noted that very high displacements can be achieved for a moderate field of 10kV/cm (400V) with maximum displacements reaching 200 μ m. Displacements, as high as 1mm have been achieved for 0.2mm thick devices [2] and as high as 3mm for 100mm diameter wafers [4].

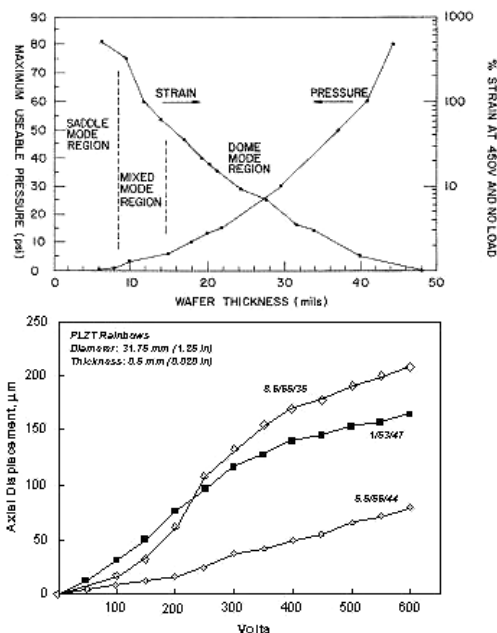


Figure 17. (Top) Effect of wafer thickness on the displacement and pressure of a 31.75mm diameter Rainbow made of PLZT 5.5/56/44. Reproduced from [6]. (Bottom) Displacement response as a function of applied voltage of rainbows made from different PLZT compositions [4].

³ Standard notation is used here for lanthanum doped lead zirconate titanate compositions. The first number indicates the mol % La substituted for Pb and the ratio, e.g., 56/44 indicates the relative amounts of $\text{PbZrO}_3/\text{PbTiO}_3$.

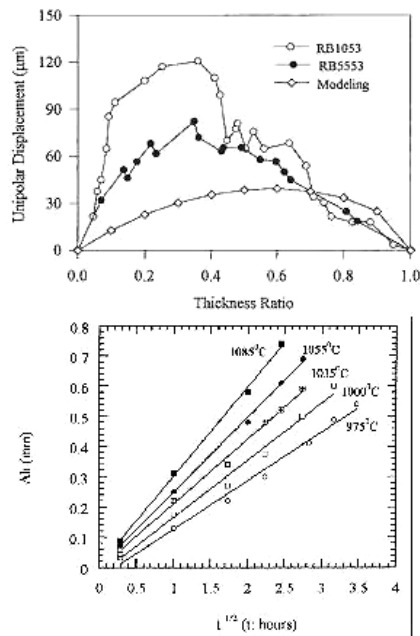


Figure 18. (Top) Effect of thickness ratio (reduced/unreduced) on the displacement of different Rainbows. Reproduced from [12]. (Bottom) Reduced layer thickness as a function of reduction time and temperature. Reproduced from [18].

The other critical parameter that determines the displacement response of the Rainbow (and Thunder-type) devices is the thickness ratio (reduced layer to total device thickness) [12], since it determines the magnitude of the stresses and the nature of the stress profile in the piezoelectric layer. Figure 18 illustrates the axial displacement of Rainbows with different thickness ratio with 0 and 1 representing an unreduced piezoelectric and completely reduced non-piezoelectric cermet layer, respectively. The thickness of the reduced layer can be varied accurately by controlling the reduction temperature and time [18] as shown in Figure 18. Maximum displacement response was typically observed for a thickness ratio of $\sim 1/3$ to $\sim 1/2$.

Interestingly, unimorph theory, which describes the tip deflection response of piezoelectric cantilevers under applied field, predicts similar general response characteristics for the effect of thickness ratio for non-stress-biased devices. This confirms that both geometry and stress effects must be considered to understand the performance of the stress-biased devices. Wang and Cross [19] have analyzed the performance of Rainbow actuators using unimorph theory. The approach used is based on the constituent equations for unimorphs and bimorphs [67], as later modified by Wang et al. [1]. The key equations are:

$$\delta = aF + bV \quad (1)$$

$$a = \frac{4s_{11}^p L^3}{wt_p^3} \frac{AB + 1}{1 + 4AB + 6AB^2 + 4AB^3 + A^2B^4} \quad (2)$$

$$b = \frac{3d_{31}L^2}{t_p^2} \frac{AB(B+1)}{1+4AB+6AB^2+4AB^3+A^2B^4} \quad (3)$$

$$A = \frac{E_m}{E_p} \quad (4)$$

$$B = \frac{t_r}{t_p} \quad (5)$$

The tip deflection of the cantilever is indicated by δ , the term a relates the tip deflection to an applied force, F , at the cantilever tip, and the term b describes the tip deflection per applied volt, V . w is the width of the cantilever, L is the length of the cantilever, t_p is the thickness of the piezoelectric layer, and t_r is the thickness of the reduced layer. The elastic compliance of the piezoelectric layer is given by S_{11}^p and the piezoelectric coefficient is d_{31} . The term A shows the effects of the ratio of the Young's moduli of the metallic (E_m) to the piezoelectric layer (E_p) on tip deflection, and the term B describes the effects of the ratio of the reduced layer to the piezoelectric thickness (t_r/t_p) on the tip deflection. Tip deflection for a given applied voltage may thus be described in terms of the ratio of the Young's moduli of the two layers, the piezoelectric coefficient, and the relative thickness of the two layers of the device.

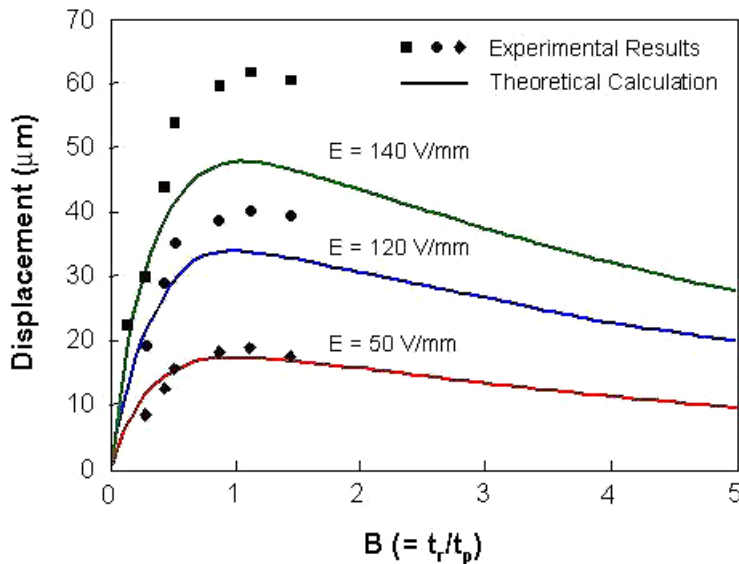


Figure 19. Tip deflection plotted as a function of thickness ratio for cantilever Rainbow actuators under different electric fields ($f = 5\text{Hz}$, $L = 38\text{mm}$, $t = 1.02\text{mm}$, and $d_{31} = 320\text{pm/V}$). Adapted from [21].

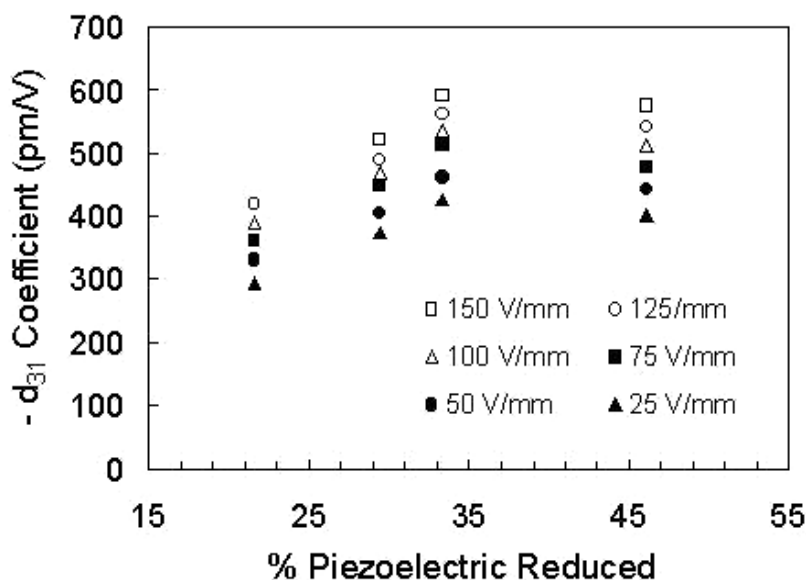


Figure 20. Calculated effective d_{31} coefficient of Rainbow actuators as a function of relative piezoelectric reduced under varying electric fields. Adapted from [21].

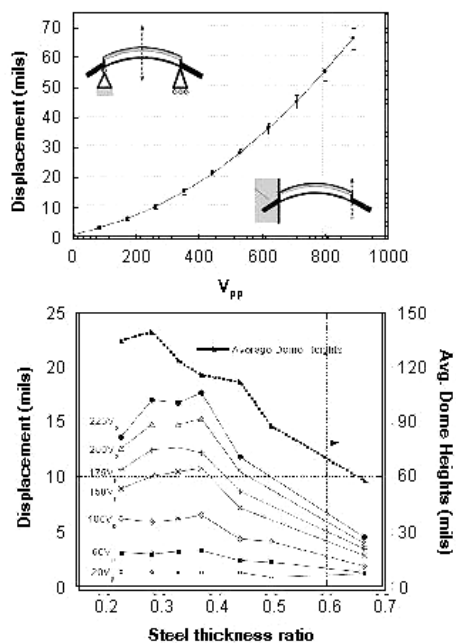


Figure 21. (Top) Simply supported displacement response under no load of a 6R (7.62 x 5.08 x 0.079cm with 0.254mm SS and 0.381mm thick PZT layers) Thunder® actuator under different peak-to-peak voltages at 1Hz. Adapted from [25]. (Bottom) Change in maximum displacement response of Thunder® devices with various thickness ratios (PZT: 8mils; SS: 3-20mils) at 1Hz. Adapted from [28].

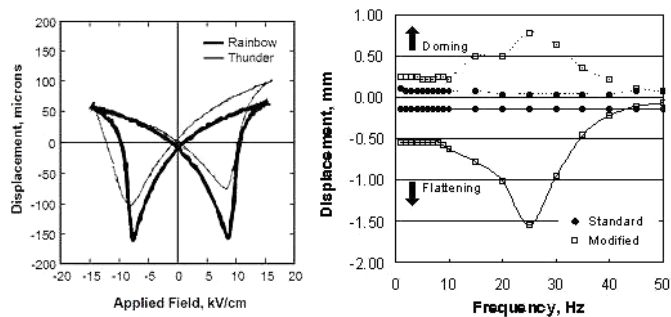


Figure 22. (Left) Strain loop of Rainbow and Thunder[®] device made of PZT at $\pm 15\text{kV/cm}$. Adapted from [47]. (Right) Bi-directional displacement response of standard and spring modified Thunder[®] TH-8R actuator (PZT: $\sim 0.2\text{mm}$, SS: $\sim 0.15\text{mm}$) under $200V_{pp}$ at various frequencies. Adapted from [29]. Zero on the displacement response axis represents the position of the dome height under no applied voltage.

The results of Wang and Cross in analyzing Rainbow actuator response with unimorph theory is reproduced in Figure 19. At low field, reasonable agreement between predicted and observed response is noted. In particular, effects of geometry are well captured by unimorph theory. However, unimorph theory becomes less predictive at higher fields, where enhanced extrinsic contributions are expected, yet the general shape of the curve as a function of layer thickness ratio is still effectively modeled. This seems to suggest that unimorph theory is effective in capturing geometry effects, but as expected, does not account for stress effects on electromechanical response, as would be expected.

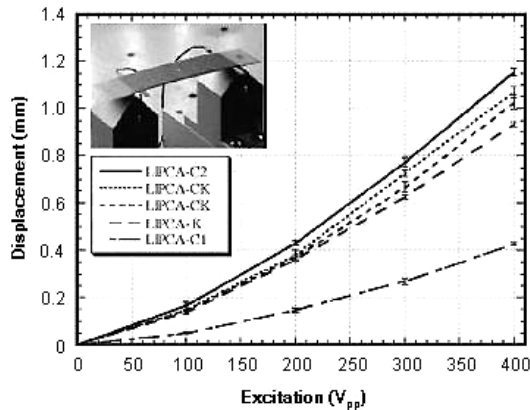


Figure 23. Displacement response of various LIPCA devices under different excitation voltages in simply supported condition at 1Hz. Adapted from [42].

In a subsequent study, Schwartz and coworkers [21] used unimorph theory to predict the piezoelectric coefficient as a function of geometry/stress state. The approach utilized by these authors was to assume unimorph theory could account for the tip deflection of the Rainbow actuators and to rearrange equation (3) to solve for the d_{31} coefficient of the devices. The results of this study of stress effects on d_{31} are shown in Figure 20. As demonstrated by Li et al., maximum tensile stress in the surface region of the piezoelectric is maximized for a

reduced layer/piezoelectric layer thickness ratio of $\sim 1/3$. As seen in this figure, the highest effective d_{31} coefficients are observed at approximately this thickness, which tends to confirm the suggested mechanism of stress enhanced domain switching. Expected effects of increasing applied electric field on piezoelectric coefficient are also noted in this figure.

The displacement response of stress-biased actuators is also strongly dependent on the clamping conditions. Typical displacement response of a simply supported Thunder[®] actuator is shown in Figure 21. The device behaves nearly linearly below $\sim 300V_{pp}$ and demonstrates significant non-linearity for voltages above this value. The maximum displacement (at $900V_{pp}$) achieved in this clamping configuration (simply supported) is $\sim 1.65\text{mm}$. Similar behavior was observed for a cantilever configuration with a maximum displacement of $\sim 3.2\text{mm}$ at $900V_{pp}$. As observed for Rainbow devices, the displacement response of Thunder[®] actuators also depends on the thickness ratio (metal substrate to total device thickness), as depicted in Figure 21. The displacement response attains a maximum value for a steel substrate/piezoelectric thickness ratio between $0.280 - 0.375$. Above this thickness ratio, the displacement decreases because of the reduced piezoelectric volume, i.e., a device consisting mostly inactive stainless steel.

Figure 22 shows typical butterfly strain loops for Rainbow and Thunder[®] devices. The important feature that should be noted in this figure is the asymmetric nature of the curves; i.e., the strain loop is tilted with respect to the horizontal axis. Typically, when a negative field (field direction opposite to poling direction) is applied to these devices the dome height is further increased (curvature increases), while a positive field (applied field aligned with the poling direction) causes the devices to flatten. When a positive field is applied to a Thunder[®] or Rainbow device, the piezoceramic layer expands in the thickness direction while contracting in the lateral (parallel to the surface) direction resulting in flattening. It can be seen that the Thunder[®] device is flattened to a greater extent compared to the corresponding increase in dome height under negative field. This suggests enhanced domain wall movement for the positive field condition compared to the negative field condition. It should also be noted that the mechanics aspects as related to the stiffness of the devices may also play a role in the asymmetric response.

Similar behavior in the displacement response (doming and flattening) has been reported by the authors for standard and spring modified Thunder[®] devices, as illustrated in Figure 22 [29, 68]. The standard Thunder[®] actuator exhibited a greater displacement response in the flattening direction by approximately a factor of 2 to 4 times, depending on the frequency. Also, the displacement response of spring modified devices was twice that of standard Thunder[®] devices. Displacement behavior similar to Thunder[®] devices (Figure 21) was observed for LIPCA devices (Figure 23). The displacement response of LIPCA devices ranged from $\sim 0.42\text{mm}$ to $\sim 1.16\text{mm}$ for an applied voltage of $\pm 200\text{V}$ at 1Hz .

The performance characteristics of stress-biased cymbal devices are also dependent on the in-situ stress state. Figure 24 shows the hysteresis and butterfly loops of stress-biased cymbals under increasing pre-stress levels (plotted here as decrease in cavity depth. See Ref. 44 for information regarding the relationship between stress level and decrease in cavity depth). It can be observed that the remanent polarization, coercive field and area under the loop decreases with increasing pre-stress level. A decrease of $\sim 37\%$ (0.964×10^6 to $0.602 \times 10^6 \text{ J/m}^3$) in the dissipation energy was observed for a decrease in the cavity depth of $100\mu\text{m}$. It can also be observed that the slope of the hysteresis loops (differential dielectric constant)

at $\pm 0.1 \text{ kV/mm}$ increased by $\sim 70\%$ with increasing pre-stress levels, indicating greater domain wall contributions to response. Enhancement in the strain response was also observed with increasing pre-stress levels. The slope of the butterfly strain loops (d_{33}) at $\pm 0.1 \text{ kV/mm}$ increased by $\sim 57\%$. Both the remanent and saturation strain increased with stress because the strain was measured at the apex of the endcap. The maximum displacement at the apex of the endcap increases by approximately 30% at $\pm 2 \text{ kV/mm}$ due to pre-stress [44]. Thus, similar effects of in-situ lateral tensile stresses are noted in devices that are geometrically different from Rainbow and Thunder[®] devices.

(e) Investigation and modeling of stress profile as a function of device geometry

Initial reports on modeling the stress profile of the Rainbow devices, shown in Figure 14, suggested that the surface of these devices was under a tensile stress and that this resulted in enhanced domain switching in these devices. Although this is true for Rainbow devices (because the top electrode is applied after device fabrication), this may not be the case for other devices (Thunder[®], LIPCA, or Presto). For these devices, the presence of a top metal electrode during the manufacturing process also contributes to the stress state in the piezoelectric layer. Navapan [52] used classical laminate theory (CLT) to predict thermally and electrically generated strains and calculated the corresponding stresses along the thickness of Rainbow devices, and Thunder[®] devices made with substrates of different thicknesses.

The results of these simulations are shown in Figure 25. Note that compared to the stress profile shown in Figure 14, the orientation of these has the bottom layer on the right, rather than the left. The thermal stress of a 3 layer Thunder[®] device (stainless steel substrate (SS)/polyimide bond layer (LaRC-SI)⁴/lead zirconate titanate (PZT); no top aluminum electrode during fabrication) was predicted to have a similar stress profile to Rainbow devices (Figure 14). The top surface of the PZT is under tension and the bottom part is under compression. In contrast, for a typical, commercially available Thunder[®] device, which is a 5 layer device with a stainless steel substrate, aluminum top electrode and two bond layers (SS/LaRC-SI/PZT/LaRC-SI/Al), the entire piezoceramic is predicted to be under compression. The stress magnitude at the top and bottom surface of the PZT in a 3 layer device is predicted to be $\sim 102 \text{ MPa}$ and $\sim -154 \text{ MPa}$, whereas it is predicted to be $\sim -41 \text{ MPa}$ and $\sim -187 \text{ MPa}$ for the 5 layer device.

The stress states (tensile or compressive) in these devices do not change with the application of either positive or negative electric field, though the predicted stress magnitudes vary considerably. When a positive electric field is applied to the 3 layer device, the device flattens, thus reducing the tensile stress magnitude on the surface; i.e., the neutral plane (plane of zero stress) in the PZT is shifted toward the surface. Under applied negative field, the neutral plane shifted toward the substrate. For the 5-layer device, the compressive stress magnitude is increased or decreased upon the application of a positive or negative field, respectively, as illustrated in Figure 25.

⁴ LaRC-SI (NASA Langley Research Center- Soluble polyImide) was developed at the NASA Langley Research Center. It is an amorphous thermoplastic and can be reformed at elevated temperature and pressures. It can be applied or deposited using various methods like spray, spin, dip, paint and doctor blade techniques.

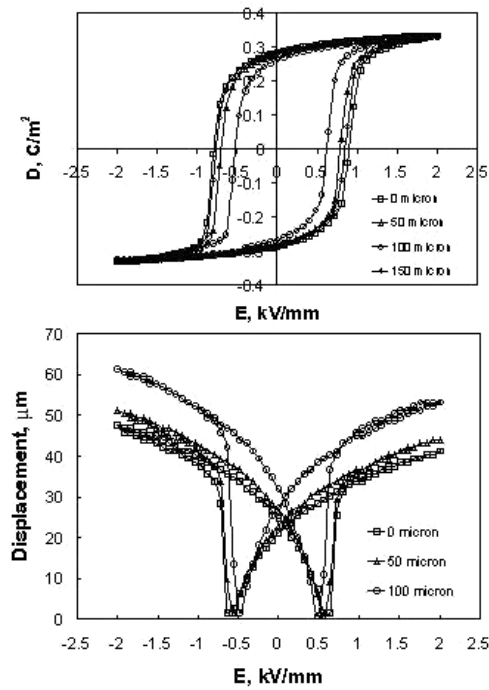


Figure 24. Hysteresis polarization (Top) and Butterfly strain (Bottom) loops of stress-biased cymbals (PZT: 25mm diameter and 0.2mm thick; Endcap: cavity depth-1.5mm, thickness-0.25mm) at different pre-stress levels (decrease in cavity depth). Adapted from [44].

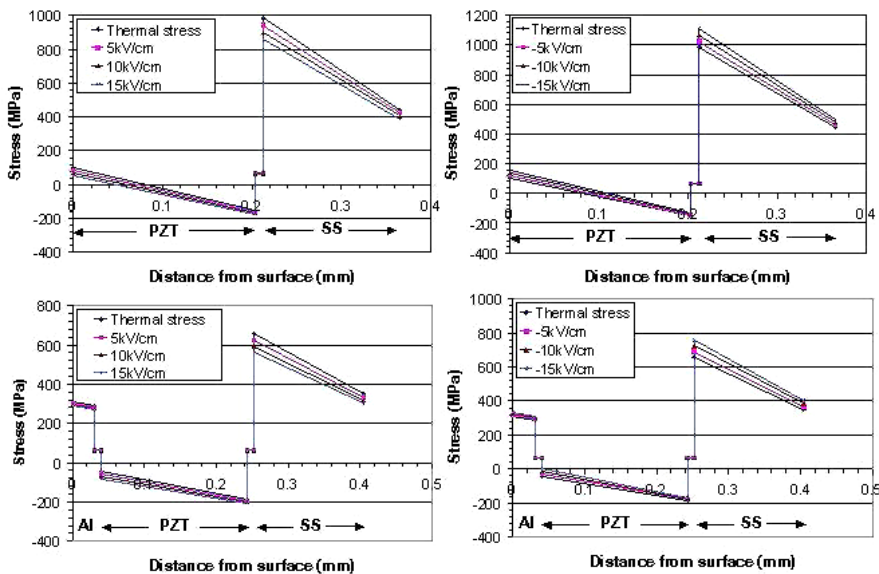


Figure 25. Stress profile of a 3-layer (top) and 5-layer (bottom) Thunder[®] device made with a 6-mil (~ 0.15 mm) stainless steel substrate and 8-mil (~ 0.2 mm) PZT wafer under positive (left) and negative (right) fields. Thermal stress represents the stress profile of the device after manufacturing (i.e., under 0 kV/cm field) [52].

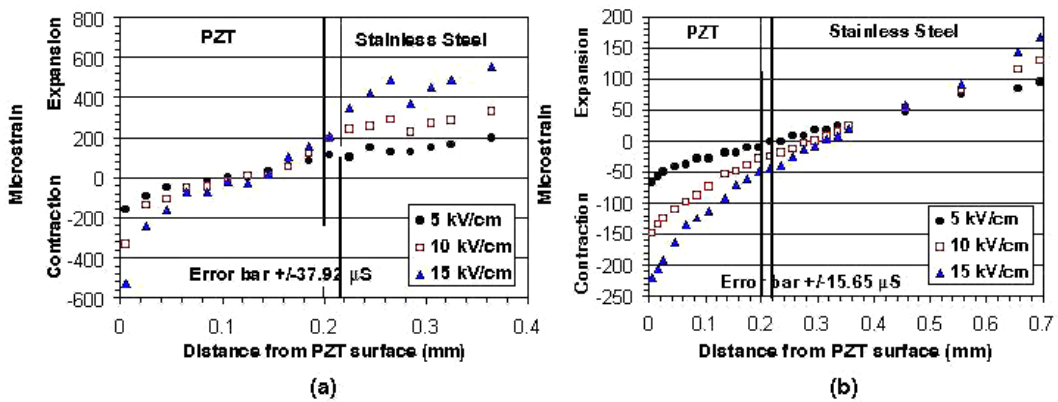


Figure 26. Local deformation response of (a) 6 mil and (b) 20 mil thick substrate Thunder® actuator under positive fields [30, 52]. PZT layer thickness in each device was 8 mils (~ 0.2 mm).

Similar residual thermal stress profiles were noted for LIPCA devices by Kim et al [42]. The basic design strategy of LIPCA devices is to place the piezoelectric layer completely into compression. It was predicted that the entire PZT in LIPCA-C2, -K, -CK and -CKK devices is under compression, while in LIPCA-C1 devices, the lower (approximately) 2/3 of the PZT is under compression and the top (approximately) one third is in tension. These authors noted that the measured displacements were 119% - 170% greater for the devices in compression (-C2, -K, -CK, -CKK) than devices that were partially under compression and tension (-C1). For an applied field of $\pm 200V$, the devices under compression showed displacements of $\sim 0.93 - 1.15mm$ compared to $\sim 0.43mm$ for LIPCA-C1. The implications of these results are discussed further below.

(f) Limitations in modeling associated with linear mechanical and piezoelectric behavior

Most theoretical analyses of these devices have been based on linear finite element modeling. These models do not account for field-dependent variations in extrinsic contributions to electromechanical response, altered domain populations in different regions of the device, or domain wall motion contributions to softening (decrease in Young's modulus). As a result, the stress levels in these models tend to overestimate stress magnitude, distribution of the electric field in the piezoelectric is not captured, and the magnitude of the electromechanical response is often not properly estimated. A significant opportunity remains for further work in this area.

Navapan-Traiphon and coworkers [30, 52] have attempted to estimate the variation in the piezoelectric coefficient as a function of position within the PZT ceramic in Thunder® devices using Moiré interferometry, and classical lamination theory in conjunction with a least squares fitting method. Moiré interferometry was used to determine the local lateral strain response across the thickness of the device. The influence of edge effects on local deformation was also accounted for in the analysis [30, 52]. Thunder® actuators with various piezoelectric/substrate thickness ratios were used to obtain the differences in stress profiles across the piezoelectric layer. The acquired data provided the fundamental information needed to determine the piezoelectric d_{31} coefficient as a function of position below the

surface by means of mathematical fitting (least squares method) in which the deformation response was modeled by classical laminate theory [52]. The influence of edge effects on local deformation was also accounted for in the analysis [52].

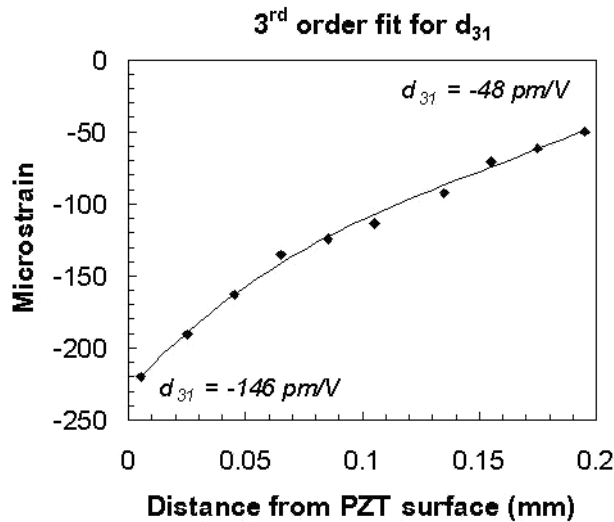


Figure 27. Typical 3rd order polynomial fit of measured strain data by linear least squares method [52].

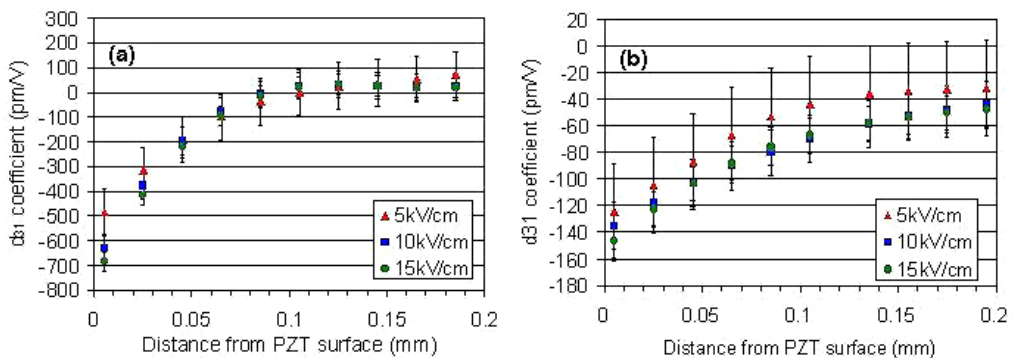


Figure 28. Distribution of piezoelectric d_{31} coefficient across the thickness of the PZT layer in (a) 6 mil and (b) 20 mil thick substrate Thunder[®] actuator expressed by the 3rd order polynomial fit [30, 52].

Results of these analyses are presented in Figure 26 through 28. Local strains across the thickness of Thunder[®] devices were evaluated at different applied positive fields and are shown in Figure 26(a) for an 8mil PZT/6mil substrate device and Figure 26(b) for an 8mil PZT/20mil substrate device. Recall that the “optimum” response device has a (substrate/piezoelectric layer) ratio of approximately 0.3 to 0.5. The device with the 20mil substrate thus represents a device with greater load bearing capacity, but also one in which the electromechanical response would be anticipated to be lower. In this figure, the negative (compressive) and positive (tensile) strains indicate lateral contraction and expansion, respectively, plotted as a function of depth below the surface of the device. Under applied field aligned with the poling direction (positive field), close to the surface of the PZT,

compressive strain was measured due to dimensional contraction in the plane of the surface by reorientation of a- to c-domains and through intrinsic effects. With increasing applied fields, higher strains were observed due to more domain switching and a higher intrinsic response contribution. Compressive strain close to the PZT surface of the 6mil-substrate device increased from approximately $160\mu\text{S}$ at 5kV/cm to $330\mu\text{S}$ at 10kV/cm and $530\mu\text{S}$ at 15kV/cm . For the 20mil-substrate device, compressive strain increased from approximately $70\mu\text{S}$ at 5kV/cm to $145\mu\text{S}$ at 10kV/cm and $285\mu\text{S}$ at 15kV/cm . The magnitude of compressive strain decreased non-linearly from the surface to the lower part of the PZT layer, suggesting greater domain contributions to response at the surface region compared to at the lower region. This result is in agreement with previous work [60] that showed enhancement of extrinsic effects by tensile stress, while extrinsic contributions were reduced by compressive stress. The degree of non-linearity of the curves increased with electric field, as would be expected for greater extrinsic contributions to response [59, 69]. The strain became tensile in the stainless steel layer due to the extensional response associated with flattening of the devices under positive electric fields. The distribution of strain in the substrate layer was relatively linear compared to the PZT layer, since only mechanical, and not ferroelastic effects were present. A greater degree of non-linearity in the response, particularly near the free surface, was observed, and the extent of non-linearity increased with applied electric field, most likely due to increased extrinsic contributions to response.

The strain data was then analyzed using classical lamination theory in conjunction with a least squares fit of the theory to extract local d_{31} coefficients. A linear least squares method was employed to model the experimentally determined local deformations by assuming the variation of the d_{31} coefficient as an n^{th} order polynomial in the thickness direction, z , of the PZT layer. Typical fitting results for the 3^{rd} order fit, which produced the best fit to the data, are shown in Figure 27. As suggested by the earlier work of Li et al. [12], the piezoelectric d_{31} coefficient is predicted to be a function of position below the PZT free surface. Also as predicted, higher d_{31} values are observed at the surface of the PZT compared to the interface with the substrate.

A summary of these investigations for the two devices is shown in Figure 28. Many of the expected trends for the effects of device architecture and stress level are seen in this figure. For the thicker substrate device (Figure 28b), lower stress magnitudes are expected. This would be expected to produce lower piezoelectric d_{31} coefficients and more linear piezoelectric response across the thickness of the PZT layer. Both of these trends are, in fact, observed.

In contrast, for the thinner substrate device (Figure 28a), higher stress levels in the PZT are predicted. This would be expected to contribute to enhanced extrinsic contributions to response and higher non-linearity in the d_{31} coefficient. Again, both of these trends are observed. The d_{31} response for this device is more non-linear than for the thick substrate device, and in the PZT surface region, significantly higher d_{31} values are predicted.

Lastly, it is interesting to note that the use of the least squares method to fit the strain data also allows for predictions of electric field effects on effective d-coefficient. A review of Figure 28 confirms that as higher electric fields are applied to the devices, higher effective d_{31} coefficients are obtained, presumably because of enhanced extrinsic (domain wall motion) contributions to response. The local deformation measurements and analysis of Navapan-Traiphol and co-workers are, thus, in-line with expectation, and perhaps more importantly, appear to confirm the early work of Li et al. [12], who had predicted non-linear piezoelectric

response as a function of position based on macroscopic device deformation (change in dome height) characteristics.

(g) Device reliability and lifetime

Although stress-biased devices show improved displacement characteristics, under some operational conditions, reliability and fatigue of the devices are of concern since the active layer is under significant stress. Researchers have studied the reliability of Rainbow, Thunder[®] and recently, LIPCA devices by cycling them electrically while monitoring the displacement degradation of the device as a function of time. These fatigue tests were carried out under different electrical (high and low field; with or without bias) and mechanical (with and without load) boundary conditions. It has been reported that these three devices can operate up to 10^7 cycles without significant degradation ($< 10\%$) in the displacement characteristics [15, 31, 38, 46, 70]. Typically, it has been observed that the fatigue or displacement degradation characteristics are highly dependent on the type of material (hard or soft), geometry, pre-stress and operational conditions, such as temperature, mechanical and electrical loads.

Dausch [14] compared the ferroelectric polarization fatigue of bulk ceramics and Rainbow actuators made from different soft ferroelectric (PZT-5H and PZT-5A) and electrostrictive (PLZT 9/65/35) compositions by measuring the hysteresis loops. It was observed that the remanent polarization decreased by 25% for bulk ceramics while it reduced by 33% and 53% for PZT-5H and PZT-5A, respectively, after 10^8 cycles. For the electrostrictive composition the remanent polarization degraded by about 8% for both the bulk and Rainbow devices. The reduced fatigue of the electrostrictive compositions may be attributed to absence of ferroelectric domains in the material.

Dausch also reported the change in the XRD peak intensities of the bulk ceramics and Rainbow devices before and after electrical cycling. The ratio of the peak intensities ($I_{\text{Fatigued}}/I_{\text{Virgin}}$) of Rainbow devices for different crystallographic planes (100), (001) were calculated from the XRD patterns. For the tetragonal PZT-5A composition, the peak intensities for the (001) and (100) planes decreased with electrical cycling. This suggests a stress relaxation caused by fatigue because the c/a ratio ($I_{(001)}/I_{(100)}$) increased from ~ 1.3 to ~ 1.7 , indicating a decrease in the a -domain population and an increase in the c -domain population on the surface.

Dausch et al. [15] also reported the displacement degradation of Rainbows based on soft (PZT-5A, PZT-5H), hard (PZT-4) and electrostrictive materials at low and high fields. At low fields ($< E_c$), the free displacement (no mechanical load) reduced by less than 10% for all Rainbow devices, even after 10^7 cycles. The displacement (under a 300g static mechanical load) reduced by 20-50% after 10^7 cycles with PZT-5A and PLZT 9/65/35 exhibiting the least fatigue. At high fields ($> E_c$), PLZT 9/65/35 showed the least fatigue, exhibiting only a 7% displacement degradation over 10^8 cycles, whereas Rainbows fabricated using PZT-5H and PZT-5A showed a significant degradation in displacement of 43% and 66%, respectively.

Nothwang et al. [31, 46, 70] studied the displacement fatigue of Thunder[®] actuators as a function of various factors, such as applied field, drive frequency, bias voltage, external stress (mechanical load), internal stress (geometry) and temperature using an equivalent circuit model (ECM) as a basis. The observed results were related to three primary fatigue mechanisms, namely, polarization fatigue, domain pinning and micro-cracking. Polarization

fatigue was examined by cycling the material under ambient conditions and measuring the capacitance. Polarization fatigue contributed to $< 5\%$ to the observed degradation of the device performance. Devices were tested under mechanical load and zero load conditions to investigate the domain pinning and micro-cracking mechanisms, respectively. Of the factors investigated, mechanical load led to significant fatigue in the displacement response, principally through domain pinning. It was reported that micro-cracking and domain pinning dominated at low loads ($< 20\text{g}$) and high loads ($> 50\text{g}$), respectively.

Yoon et al. [38] reported the fatigue characteristics of different LIPCA devices. The free displacement decreased by approximately 5% for both LIPCA-C1 and LIPCA-C2 devices after 10^7 cycles, whereas displacement under a static load of 2N decreased by $\sim 20\%$ for both the devices. The free displacement degradation was attributed to ferroelectric polarization fatigue of the PZT after repeated cycling while the degradation under load was attributed to failure of the glass/epoxy laminate.

IV. APPLICATIONS

Stress-biased actuators, like Rainbow and Thunder[®], have been investigated for a variety of applications, such as acoustic sound generators, loud speakers, active vibration dampers, pumps, positioners, switches, etc. [46, 53, 71-73]. Because of their performance characteristics, these devices find application where high strain is required and device space is restricted, or where power requirements must be minimized [46].

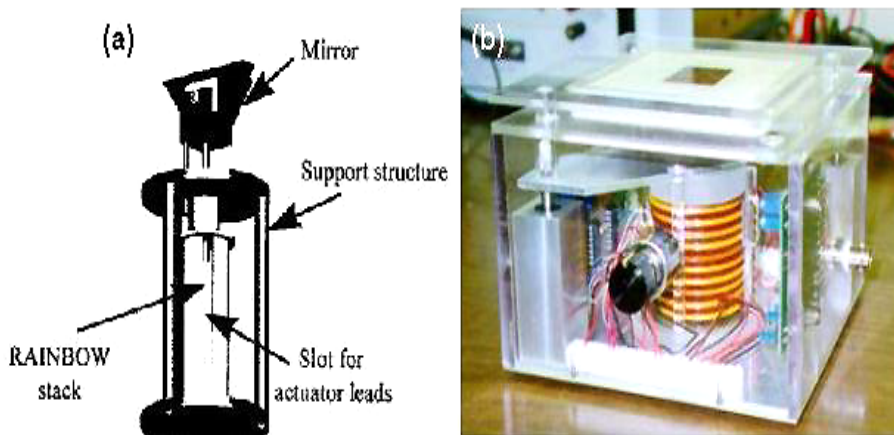


Figure 29. Typical applications of Rainbow devices (a) scan mechanism of a Fourier Transform Spectrometer. Reproduced from [13]; (b) microstrip patch antenna [74].

Wise reported that a stack of Rainbow devices could be used for optical translation in space-based spectrometer systems (shown in Figure 29a), where the actuators were required to move optical components weighing approximately 250grams through a one-centimeter travel [13]. The power consumed by this drive mechanism was less than 2 Watts, which is significantly less than state of the art motor systems [13].

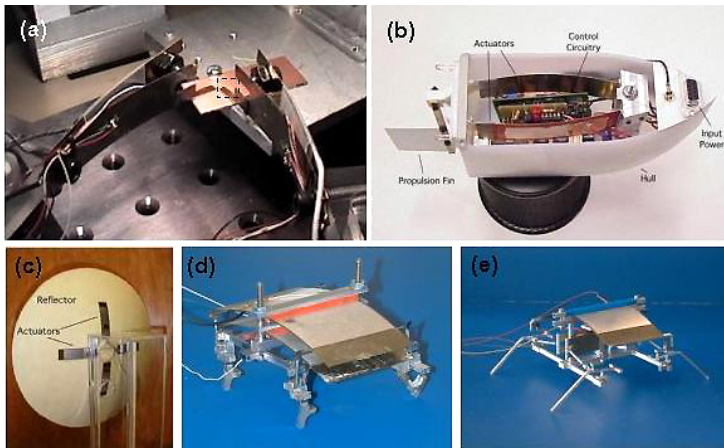


Figure 30. Various applications of Thunder[®] devices: (a) Tweezer arm for milli robots-inset showing the magnified view of the tweezer tip [75], (b) miniature swimming vehicle [76], (c) Aperture antenna [78], and (d,e) Crawling robotic insects [79, 80].

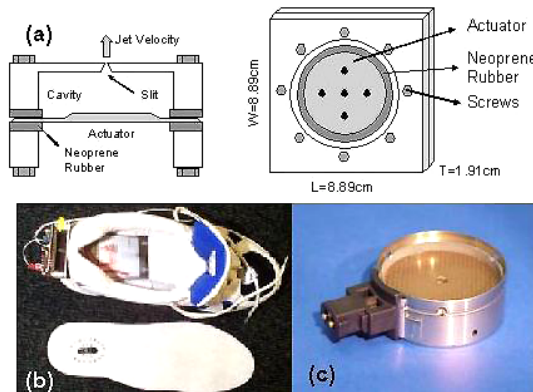


Figure 31. Additional applications of Thunder[®] devices: (a) synthetic jets [81, 82], (b) energy harvesting in shoes © 1998 IEEE [83], and (c) fuel injection system [85-87].

Kiely [74] successfully employed a stack of 20 Rainbow devices in clamshell configuration in microstrip patch antennas, as illustrated in Figure 29b. The narrow bandwidth of these antennas is normally increased by the addition of a parasitic element. The bandwidth can be further increased by varying the distance between the antenna and the parasitic element. Rainbows actuators were used in this application to change the distance between the parasitic patch and the antenna, thus varying the bandwidth and the radiation pattern of the system. It was reported that the bandwidth increased from 2.7% to 5.4% for a change in spacing of 4mm (1mm to 5mm) between the antenna and the parasitic patch [74].

Stress-biased actuators have also been used in a variety of other applications. Thunder[®] TH-8R actuators (~ 4cm long x ~ 1cm wide) are being used in the arms of a tweezers in milli-robots, which are used for micro manipulation [75], as illustrated in Figure 30a. Each tweezer arm consists of a Thunder[®] actuator and a base stainless sheet to which a tweezer tip is attached. The tweezer tips are placed perpendicular to each other and controlled by applying

voltage to the actuator. Various tasks like holding, folding, part rolling, pick and place, pivot grasp, re-grasp and alignment were successfully carried out with the tweezers.

Researchers are also working to introduce a new autonomous miniature-swimming vehicle that utilizes a propulsion system based on two Thunder[®] actuators as shown in Figure 30b [76]. The bending motions of the two curved piezoelectric beams (achieved by applying a sinusoidal voltage signal to both actuators) actuate a pendulum-type kinematic mechanism, which moves a fin and produces the swimming motion of the vehicle [76]. The steering of the vehicle is achieved by applying a DC field (which changes the center of actuation) to one of the actuators while the usual sinusoidal signal is applied to the other. Narayanan et al. [77] proposed the swimming motion (or efficiency) of the miniature vehicle could be increased by using modified Thunder[®] actuators that demonstrate greater actuation capabilities.

Still other applications are illustrated in Figure 30. Four Thunder[®] actuators attached to the back of an aperture antenna (shown in Figure 30c) modified the radiation pattern by actively deforming the shape of the reflector [78]. Thunders are also used in mesoscale crawling robotic insects [79, 80] developed by The Center for Intelligent Mechatronics at Vanderbilt University, as shown in Figure 30d,e.

For the high displacement applications noted above, Thunder[®] actuators are typically used to decrease the actuator weight and power consumption. Locomotion is generated by exciting the devices at different resonance frequencies [79, 80]. Researchers studied the velocity profiles of synthetic jets made with different piezoelectric actuators, including circular Thunder[®] actuators (63.5mm in diameter), as demonstrated in Figure 31a. It was observed that the drive signal, orifice shape and cavity volume had a significant effect on the jet velocity compared to drive voltage and frequency for the Thunder[®] devices [81, 82].

Thunder[®] devices were also evaluated in parasitic power harvesting in shoes (Figure 31b) and were also compared to other systems like PVDF and magnetic generators. It was observed that the average power (peak power reported in brackets) generated by the piezoelectric unimorphs were 1.8mW (~ 80mW) compared to 1.1mW (~ 20mW) for PVDF and 0.23mW (1.2mW) for magnetic generators [83].

The use of Thunders to stretch tune an optical fiber grating producing a shift in the transmitted wavelength of about 5.3nm has also been evaluated [84]. Thunder[®] devices have also been investigated in fuel injection applications [85-87]. Other potential uses of Thunder[®] devices include needle positioners in the textile industry, valves, compressors, vibrating elements, agitators, mixers, noise canceling devices, air flow controllers, airfoil shaping, nondestructive evaluation, pumps, beam and pipe stiffeners, auto focus mirrors, vibration control [45], harvesting energy [88], etc. Thunders also have a variety of ultrasonic applications like welding, cutting and cleaning [77].

Since the remaining devices (LIPCA, Presto, SBC and SMU) were developed more recently, application-oriented investigations have not yet been as detailed. Currently, LIPCA's are being explored for use in biomimetic wing sections of small unmanned aerial vehicles (UAVs) [89]. Although the applicability of these devices has only recently been explored, device performance characteristics show great promise for many applications and also replacing the current technology. For example, the SBC's may replace the cymbal devices where a higher displacement is required at a lower excitation voltage like underwater sound projectors. For such replacements, further investigation remains to demonstrate device utility and worthiness.

V. FUTURE DIRECTIONS

Since the invention of these devices more than a decade ago, there has been significant progress in a wide variety of areas leading to their development and implementation into a range of applications. The devices offer unique performance characteristics, most notably, displacement response, that will continue to be explored, and with time, lead to more widespread device utilization. The role of stress has been extensively explored, and to first order, seems to be reasonably well understood. A variety of device types has been designed to take advantage of controlled stress state and proof of concept has been illustrated. Fatigue behavior has been characterized under a range of conditions and the impact of mounting/boundary conditions of device performance has been evaluated. Finally, various groups have developed drive circuitry that enables compact power management for these devices. Based on the broad range of studies carried out, it seems reasonable to inquire about remaining opportunities. These, in the opinion of the authors, are plentiful, with both fundamental and developmental studies remaining to be completed.

First, and perhaps foremost, the devices are still not perceived as “off-the-shelf” components. Until that time, they will remain more the focus of inquiry than utilization. Efforts at device development have been focused, without major manufacturers being involved. While the devices have proven themselves, even in space-based applications, further development will be required for more widespread use. A number of small companies continue to pursue the development of these devices. They appear to be commercially viable, but have not yet achieved widespread commercial implementation. Work in this area will require additional studies on device performance under application specific conditions, an improved understanding of the power needs/consumption of the devices, and continued development of drive circuitry.

Second, there remain opportunities for fundamental investigation. While stress effects have been characterized, some aspects of device performance remain poorly understood. For example, the devices typically respond asymmetrically to applied field. It is not known whether this is due to a geometry effect, e.g., different requirements for doming vs. flattening, or whether changes in stress profile during dynamic operation also contribute to this response. Also, given the significance of the surface aluminum layer used in Thunder[®] devices on the predicted stress profile across the piezoelectric layer (without, apparently, concomitant changes in device electromechanical response), the specific role of stress on device response is, arguably, still poorly understood. It might be questioned whether tensile stresses are required, or whether stress gradient is more important in defining device performance than stress state. Similarly, the role of changes in stress magnitude and gradient during dynamic operation of the devices may be equally important, and as of present, have received little attention. The role of such effects on material response characteristics, such as 90° domain switching, need to be better understood, and the work of Jones and colleagues in this area may be helpful [90, 91]. These investigators are pursuing a detailed understanding of the interaction between ferroelectric and ferroelastic effects under dynamic stress and field conditions. Understanding this behavior under controlled conditions may assist in understanding such effects under the complex boundary conditions that exist during the operation of stress-biased devices. Such understanding might lead to devices with further enhanced displacement response characteristics, as well as greater device reliability. For

example, if stress gradient, and not stress magnitude/sign is the important factor, it might be possible to engineer stress-biased devices with both optimum gradient for performance, and optimum stress state for reliability through choice of device architecture parameters and materials. Device engineering along these lines will also require improved understanding of such effects as the preferred poling direction which appears to occur in these devices as well as material response characteristics such as flexoelectricity [92-95]. Systematic studies to investigate such effects should be designed and carried out.

Third, improved models for device characteristics (stress state and magnitude, electric field distribution, and position-dependent piezoelectric coefficients) are required. Non-linear material models need to be developed and an improved understanding of factors, such as the effectiveness of the bonding layer in Thunder[®] devices in transferring the stress associated with thermal expansion mismatch from the substrate to the PZT layer is required. Similar factors in other stress-biased devices require analogous improvements in understanding. However, data on non-linear material response under controlled conditions has become more readily available. This data might readily be utilized for the development of more accurate non-linear material models. These could be coupled to experimental investigations of the role of stress gradient, stress state, and dynamic changes in stress during operation to develop optimized stress-biased devices. Although commercialization of these devices has not been exploited, it should not be forgotten that they still demonstrate unique performance characteristics compared to other classes of electromechanical actuators.

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Chapter 3

PIEZOELECTRIC MATERIALS: STRUCTURE, PROPERTIES AND APPLICATIONS

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ABSTRACT

Since the discovery of piezoelectric effect and relaxor behavior, the technology of using piezoelectric and electrostrictive ceramic materials as solid-state actuators for small and precise mechanical movement devices has undergone considerable investigation and development. Despite of their enormous applications, the direct extensional strain in most piezoelectric or electrostrictive ceramic materials is at best a few tenths of one percent. Thus it is essential to develop the method to amplify the strain of devices. Some well-known techniques for producing large displacement in these materials include two categories. The first category is related to the shape and structure modifications such as unimorph, bimorph bender moonie and rainbow actuator. However, each of these structures has its limitations in regards to size, weight, maxim displacement, load-bearing capability, high performance, and low voltage driving for using digital equipment such as digital camera, cellular phone etc.. The second category is resonance vibration-related mechanism such as inch-worm and ultrasonic motors. When the stress/strain capabilities of all of these technologies are considered, it is evident that there is still a real need for additional technique which will produce even higher strain or displacement and will use soft or hard piezoelectric material. At the same time, the device must sustain reasonable stress and resonance driving condition. This chapter describes such a technology which contains the newly developed structure, manufacturing process, properties and applications.

I. TREND OF CERAMIC ACTUATOR

1. Trend of Ceramic Actuator

The basic design parameters that characterize the potential of any linear actuator are displacement, force, frequency and electric input power. Most actuators usually operate well in some of these categories but are typically poor in others. The conventional actuating methods of controlling the precise positioning or reducing the displacement deviation are classified into three types such as oil pressure cylinder, servo or step motor and voice coil type.. For example, hydraulic actuators have excellent force and displacement capabilities only at low frequencies, but which have a large size and slow response time and low accuracy. Similar to problem occurred in hydraulic actuators, the other methods mentioned above have some problems in using for various fields as follows

2. An Overview of Solid-State Actuator and High Strain Piezoelectric Actuator Structure

Nowadays, the needs for the devices which can control very small displacement and movement precisely are increasing in various fields such as optics and mechanics and so on. For example, application for an imaging process such as optic zooming, hand shaking protection in digital camera requires sub-micron order movement of lens.

On the other hand, there are other categories of actuator; that is solid state actuator which structure is simple compared with conventional actuator. The typical solid state actuators are classified into three categories by the type of input energies as follows.

1. Thermal expansion(Shape Memory Alloy: SMA): thermal input
2. Magnetostriction: magnetic field input
3. Piezostriction/electrostriction: electric field

SMA actuators are similar in that they can generate a large displacement and force, but their actuating frequency is extremely low. On the other hand, electromagnetic actuators have a good frequency range. Linear induction actuators using this have a excellent force and stroke output. However they are generally heavy and require significant electric current and have low efficiency. Among these actuators, the piezoelectric/ electrostrictive ceramic materials are the most promising materials for the solid state actuator because of their simple structure, quick response and high reliability. In spite of their merits, these ceramic actuators have still some problems to be solved for wide commercial use such as improvement of small strain, high driving voltage and poor reproducibility of position due to strain-electric field hysteresis. Especially, in order to solve the problem of the small strain, many device designs were proposed as presented in Figure 1. Simple devices directly use the longitudinally or transversely induced strain. The disk and multilayer types use of the longitudinal strain and the cylinder types utilize the transverse strain. Complex devices do not use the induced strain directly, but rather a magnified displacement, produced through a spatial magnification mechanism. Among these designs, the multilayer and bimorph types are the most commonly

used structures. For most piezoelectric actuators, the focus of research has been on an attempt to amplify the deflection of the material to produce it into a useful form. As a result, numerous novel, ingenious concepts have been created. Piezoelectric actuation methods are branched into three categories based on the amplification scheme: Externally amplified, internally amplified. Externally amplified actuators rely on an external mechanical part for increasing the displacement [2]. These types include hinge-lever, Moonie, Cymbal etc. Internally amplified actuators differ from externally amplified actuators in that they generate amplified strokes through the internal structure without the use of any mechanical parts. These types include piezo-stack, multilayer, bender, Rainbow. Frequency amplified actuators depend on an alternating voltage signal to generate resonance motion such as inchworm. The multilayer type produces only relatively modest displacements ($10\mu\text{m}$), it offers a respectable generative force, quick response speed. This multilayer actuator has been developed by several companies. However, this manufacturing process limits the general use because of expensive production line, manufacturing cost and fatigue crack between ceramic and electrode. The bimorph type provides larger displacement than multilayer type, but can only offer a relatively low generative force, much response time, shorter life and higher driving voltage. A more recent device, the moonie, was developed by Sugawara et al. in 1992 [1]. This metal-ceramic composite actuator is a type of flextensional transducer which is able to convert and amplify (~ 10 times) the radial displacement of disk into a linear axial motion. Displacements as high as $20\mu\text{m}$ have been achieved at a sustaining stress of 10kgf . The moonie has characteristics intermediate between the conventional multilayer and bimorph actuator. This device consists of a thin multilayer ceramic element and two metal plates with a narrow moon-shaped cavity bonded together as shown in Figure 2. Considering the stress/strain capabilities of above technologies, it is evident that there is still a necessity for another technology which can produce higher strain. At the same time, the device must sustain reasonable stress or load. According to this need, a new type of monolithic ceramic bender which is capable of achieving high axial displacement and sustaining moderate pressure has been developed. Known as a “rainbow” (reduced and internally biased oxide wafer) ceramic [3], this structure possesses a wide range of stress/strain characteristics than other types. In its most basic sense, rainbow technology consists of a new processing method for treating conventional, high-lead-containing piezoelectric or electrostrictive ceramic wafers, such as PZT, PLZT and PNN. This technology involves the high-temperature chemical reduction of one surface of a wafer, thereby producing a stress-biased, dome-like structure, as shown schematically in Figure 3. The stress achieved in the ceramic by means of the single-sided reduction process is a critical feature of the structure. The stress produces a state of tension toward the bottom or concave side of the wafer and compression toward the top side. Therefore, the wafer assumes either a dome or saddle shape, depending on the magnitude of the stress and the diameter-to-thickness ratio of the wafer. When the stress and the diameter-to-thickness ratio are high, the rainbow wafer takes on the higher-profile, saddle shape. The change in shape of the wafer after reduction is believed to be due to the reduction in volume of the reduced layer compared to the unreduced material. The change is also due to the differential thermal contraction between the reduced and unreduced layers on cooling to room temperature.

The reduced side of the piezoelectric serves as a mechanical support for the device, the source of internal stress, and one of the device electrodes. It is necessary to deposit only one additional electrode on top of the unreduced piezoelectric for operational purposes. However,

it is beneficial to also electrode the reduced layer to ensure good contact for electrical leads. After attaching leads to the two electrodes, the device is completed and ready for activation with an appropriate voltage. However, the largest displacements are usually achieved when driven with alternating current at frequencies less than 40Hz. In operation, the dome height of the rainbow varies as a function of the magnitude and polarity of the voltage. The axial motion thus produced is largely a consequence of the lateral contraction. Therefore, the material with higher d_{31} coefficient generates the higher displacement.

Table 1. The demerits of conventional actuator

	Demerits
Hydraulic actuator	Large size, Slow response time
Servo or step motor	Lack of precision
Voice coil	Small generating force

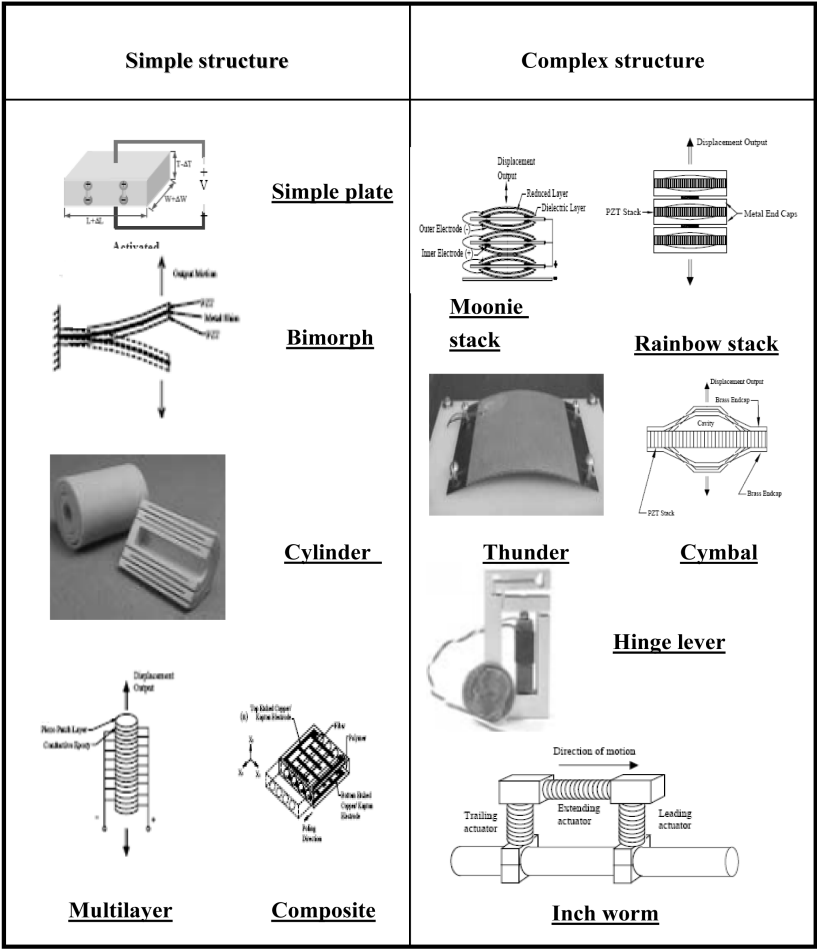


Figure 1. A classification of ceramic actuator based on structure [2].

II. NEW TYPE ACTUATOR FABRICATED BY POWDER INJECTION MOLDING

Conventional bimorph bending actuators are composed of two piezoelectric plates, or two piezoelectric plates and an elastic shim, bonded together. The bonding layer in the latter, however, causes both an increase in hysteresis and a degradation of the displacement characteristics, as well as delamination problem [4]. Furthermore, the fabrication process for such devices, which involves cutting, polishing, electroding and bonding step, is rather laborious and costly. Multilayer structure also can not avoid above problem. The multilayer structure is essential comprised of alternating ferroelectric and conducting layers, which are co-fired to produce a dense, cohesive unit. A ferroelectric layer sandwiched between a pair of electrodes constitutes a single ferroelectric element. Hundreds of these thick films are connected in parallel and electric field is supplied by the external electrodes. It is possible for this structure to generate the internal crack between the layers under bipolar drive caused by alternating strain. Therefore, to produce reliable device is the key issues.

Thus, there are needs for new type actuator without complicate process, layer structure, bonding process etc. Such a new type monolithic device has been produced from powder injection molding process which is suitable to mass production without any restrictions such as size, material etc. Piezoelectric and electrostrictive ceramic materials have been widely used as solid-state actuators [5-7]. Despite of their enormous applications, the direct extensional strain in most piezoelectric or electrostrictive ceramic materials is at best a few tenths of one percent. Thus, it is essential to develop the method to amplifying the strain of devices.

1. New Type Piezoelectric Transformer: Manufacturing Process, Structure and Properties of Dome Shaped Piezo-Tansformer

1.1 Introduction

As mentioned above, up to now, this Rainbow actuator has been generally used as solid state devices in non-resonance frequency range. However, this deoxidation process limits the use of a hard PZT by the formation of Pb vacancy, which deteriorates the dielectric/piezoelectric properties of a hard PZT. In addition, this process has inherited restriction on thickness (< 0.7 mm) and a large radius of curvature due to the interfacial strain between oxide and metal layer [2]. In 1994, Haertly reported a new type of monolithic ceramic bender, i.e., Rainbow actuator, which showed very high axial strain ($>1\%$) and sustaining moderate pressure (~ 0.6 MPa) [8-10]. These excellent properties were achieved by the formation of dome-like shape by the deoxidation process of one side plane of disc. Up to now, this Rainbow actuator has been generally used as solid state devices in non-resonance frequency range. However, this deoxidation process limits the use of a hard PZT by the formation of Pb vacancy, which deteriorates the dielectric/piezoelectric properties of a hard PZT. In addition, this process has inherited restriction on thickness (< 0.7 mm) and a large radius of curvature due to the interfacial strain between oxide and metal layer [11]. Thus, this actuator has problems on application to high power transducer such as piezoelectric transformer and small size ultrasonic motor. Furthermore, it is inevitable to generate toxic Pb

gases, which will be strictly controlled by WEEE (Waste Electrical and Electronic Equipment) and RoHS (Restriction of Hazardous Substances). Therefore, it becomes a critical issue to develop a new method for avoiding the above problems.

In order to use the piezoelectric ceramics in actuator, high density should be attained in the final product. Powder densification can be achieved through one of three methods: i) sintering with low density green body, ii) press to high density followed by sintering, or iii) simultaneously press and sintering using a full density technique such as hot isostatic press and hot press.

Polymeric binders are used in shaping the ceramic powders, holding their original shape in the sintering furnace until finishing densification. Several forming methods are in use such as powder injection molding, extrusion, tape casting etc. They are similar with regard to powder characteristics, binder formulations, dispersants, mixture homogeneity, rheology and the effects of processing variables on forming. Powder injection molding (PIM) has a five significant characteristics-low production cost, shape complexity, low tolerance, applicability to several materials and good properties of final product. PIM competes with other shaping technologies, namely plastic injection molding, die and investment casting, machining, cold isostatic compaction and slip casting. It overcomes the property limitations inherent to plastics, the shape limitation of traditional powder compaction, the costs of machining, the productivity limits of isostatic pressing and slip casting, and the defect and tolerance limitation of casting. Piezoelectric ceramic can be shaped using low pressures with the aid of an organic binder. Compaction relies on an external source for deforming the powders into a high density component that approaches the final geometry. The ways of delivering the pressure to the powder, the mechanical constraints and the rate of pressurization are the main process parameters which determine the resulting density. As previously discussed, I will propose new type actuator with large strain and explain its characteristics and application results in detail.

1.2 Manufacturing process

$0.03\text{Pb}(\text{Sb}_{0.5}\text{Nb}_{0.5})\text{O}_3\text{-}0.03\text{Pb}(\text{Mn}_{1/3}\text{Nb}_{2/3})\text{O}_3\text{-}0.465\text{PbTiO}_3\text{-}0.475\text{PbZrO}_3$ composition, which is generally used for high power ultrasonic devices, was used to fabricate piezoelectric devices. A preliminary study indicated that this composition was characterized by excellent dielectric/piezoelectric properties ($d_{33}\sim 290\text{pC/N}$, $k_{33}^T\sim 1300$, $k_p=0.62$, $Q_m\sim 2000$ at room temperature). In this experiment, two type piezoelectric samples were fabricated. One is a disc shape samples fabricated by conventional die press forming process and the other is a dome-type samples fabricated by Powder Injection Molding [PIM]. The dimension of the disc was a diameter of 28 mm and a thickness of 2.1mm. The dimension of the dome shape sample is the same as that of the disc except with a curvature radius of 18 mm. The detailed fabrication process was shown in Figure 2.

Since the conventional die press forming process is a well-known process, I will shorten that process and describe the PIM process at some length. Figure 3 shows a schematic of the individual steps involved in PIM forming.

The binder usually based on a common thermoplastic such as wax of polyethylene. The binder system consists of three components. An example binder that is molten at 150°C consists of 60% paraffin wax, 35% polypropylene, and 5% stearic acid.

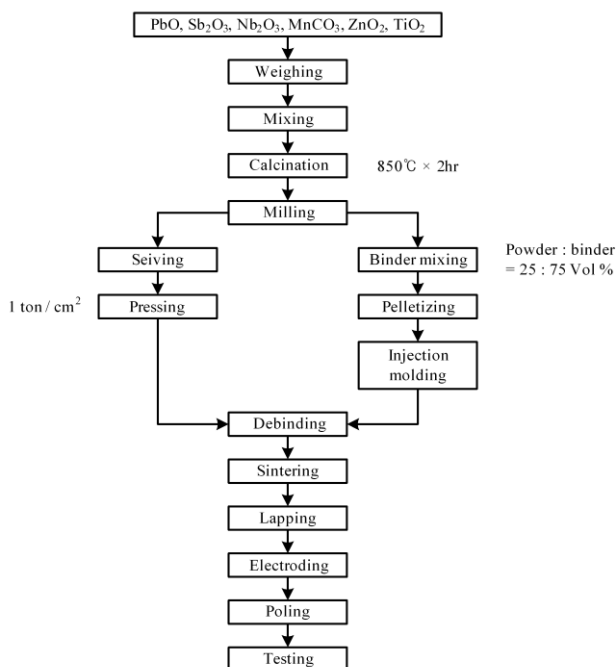


Figure 2. Flow sheet for sample preparation.

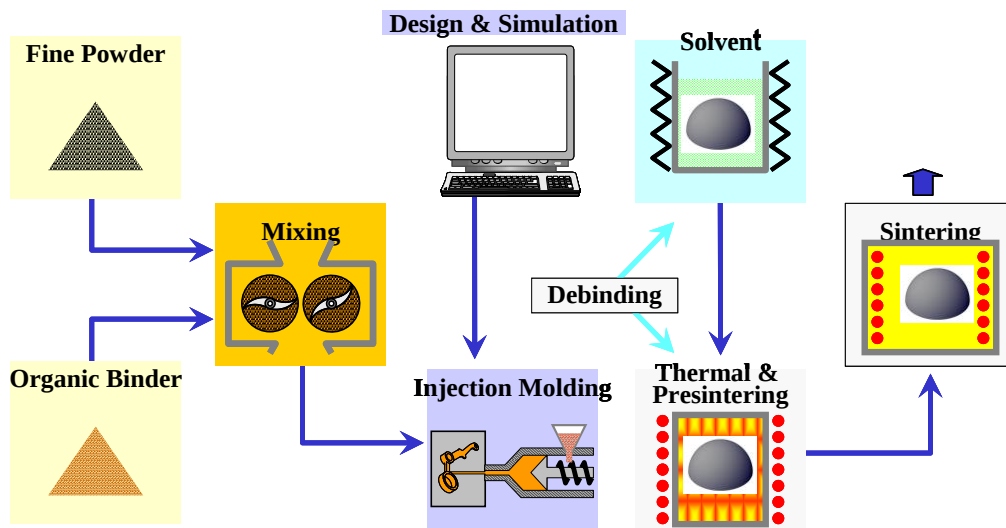


Figure 3. A schematic of the individual steps involved in PIM forming.

The binder content is near 40 vol.% for mixture. The particles of piezoelectric powder have sizes between 0.1 and 5mm with near spherical shape. The powder-binder mixture is pelletized and injection molded into dome shape. The polymer imparts viscous flow characteristics to the mixture to aid forming, die filling, and uniform packing. After injection molding, the binder is removed from component by a solvent and thermal debinding. The

next step is sintering, which can be incorporated directly into a thermal debinding cycle. But in case of PZT materials, debinding and sintering process should be separated because of Pb vaporization during the high temperature sintering. To limit PbO loss from the debinded green body, a PbO-rich atmosphere was maintained by placing an equimolar mixture of PbO and ZrO₂ inside a covered alumina crucible. The sintered specimens are then polished. In order to perform poling process and measure the electrical properties, silver paste was coated to form electrodes on both sides of the sample, and then subsequently fired at 650°C for 20min. During the poling, the randomly oriented domains are forced by the external electrical field to change their position, aligning in the direction of the applied field.

1.3 Characteristic the effects of geometrical factors on the step-up ratio in piezoelectric transformer

In recent years, the screen dimension of TFT-LCD TV has been continuously increased, which strongly requires miniaturized high power and efficiency piezoelectric transformer for LCD backlight inverter. But, the existing actuators cannot meet the requirements of high power piezoelectric transformer for the miniaturized LCD backlight inverter [12-14]. Therefore, this chapter aimed to explain a 3-dimensional piezoelectric actuator without deoxygenation process. The strain characteristics of the 3-dimensional piezoelectric actuator were analyzed by finite differential method (FEM) and measured by non-contacting displacement sensor. In order to estimate the applicability of this actuator to the high power piezoelectric transformer, the voltage step-up ratio and lighting experiment were also performed by printing ring type electrodes

1) Modeling

Electromechanical properties and strain vector for both disc and dome shape piezoelectric devices have been estimated by FEM modeling. The ATILA commercial package (ver. 6.0.0) for FEM was used to simulate electromechanical properties and strain vectors for the dome type piezoelectric devices.

In the case of piezoelectric ceramics, three piezoelectric, two dielectrics and five elastic coefficients permitted by symmetry are known to have non zero values. Up to now, the complete set of these values has been published in only a few ceramics. Fortunately, for PZT 4, all of the above values have been characterized and PSN-PMN-PZT should have similar values of PZT4 because PZT4 is a hard PZT [15]. The piezoelectric, dielectrics and elastic constants of PZT 4, used in the FEM modeling, is shown in table 2.

Figure 4 show the maximum tip displacement results for both dome and disk type specimens as a function of the applied electric field.

As shown in Figure 4, the amount of the displacement of the dome shape specimen is increased by geometrical shape effect. This indicates that the existence of de-oxidized layer like Rainbow structure [3] is not a critical factor to increase the displacement of the piezoelectric devices. At resonance frequency, the behavior of the strain vector in the dome shape device is presented in Figure 5. This figure shows that elongation and shear strain is found in outside planes except in top and edge planes. Thus, it is expected that this multi-directional strains will enhance the output electric power of a piezoelectric transformer.

Table 2. PZT4 data used for this modeling

S_{11}^E	S_{12}^E	S_{13}^E	S_{33}^E	S_{44}^E
$12.3 \cdot 10^{-12} \text{ m}^2/\text{N}$	$-4.05 \cdot 10^{-12} \text{ m}^2/\text{N}$	$-5.31 \cdot 10^{-12} \text{ m}^2/\text{N}$	$15.5 \cdot 10^{-12} \text{ m}^2/\text{N}$	$39.0 \cdot 10^{-12} \text{ m}^2/\text{N}$
D_{15}	d_{31}	d_{33}	$\epsilon_{11}^T/\epsilon^0$	$\epsilon_{11}^T/\epsilon^0$
$496 \cdot 10^{-12} \text{ C/N}$	$-123 \cdot 10^{-12} \text{ C/N}$	$289 \cdot 10^{-12} \text{ C/N}$	1475	1300

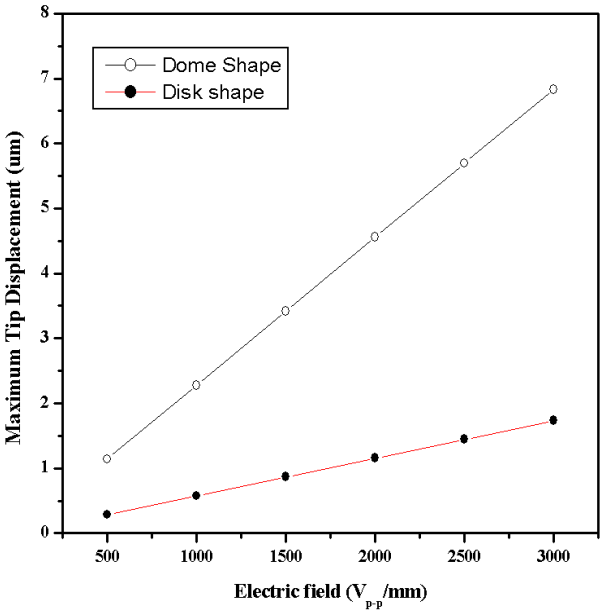


Figure 4. The maximum tip displacement results for both dome and disk type specimens using ATILA.

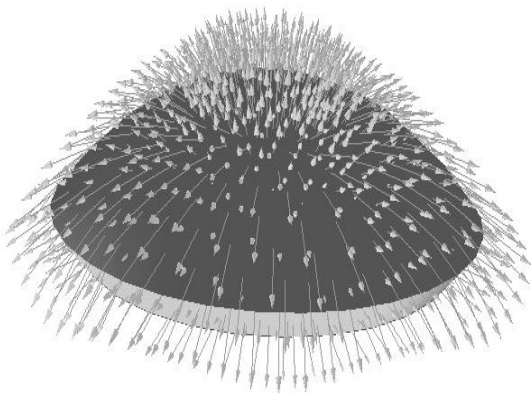


Figure 5. FEM analysis of displacement vectors after voltage applied at 1st resonance frequency.

2) Properties of Dome Shaped Piezoelectric Transformer

In order to evaluate the differences of the electromechanical displacement properties and step-up ratio between the disk shaped sample and dome shaped sample fabricated by conventional compacting process and PIM process, respectively. In addition to this, the microstructure of dome shaped sample was compared with that of disk sample under the same sintering condition to be convinced of densification.

(1) Microstructure

PIM is considered as one of the most effective methods for fabricating a complex structure. In order to confirm the applicability of PIM to piezoelectric materials, the microstructures of piezoelectric materials fabricated by PIM have been compared with those of disc type materials fabricated by dry press processing. When specimens were sintered at 1300°C for 2 hrs, the average grain size ($\sim 2\mu\text{m}$) of the PIM specimen is slightly larger than that ($\sim 1.5\mu\text{m}$) of the conventional die press processed specimen. However, both specimens do not show any noticeable pores, as shown in Figure 6. In general, the full densified structure of die press processed specimens is known to be difficult to obtain when 75 Vol% binders were added. On the contrary, the specimen processed by PIM shows a full densified structure. This strongly indicates that PIM process is the most promising method for a 3-dimensional complex structure composed of piezoelectric materials.

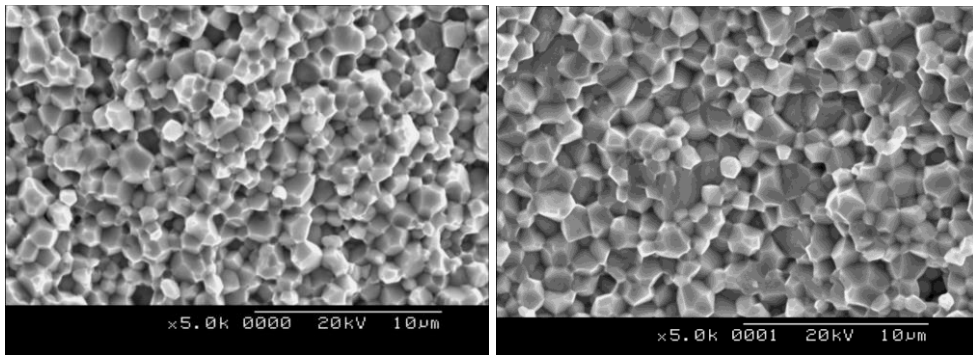


Figure 6. SEM micrographs of PSN-PMN-PZT sintered at 1573K for 2hr ; (a) disk type sample produced by die-pressing method , (b) dome shape sample produced by PIM method.

(2) Displacement Characteristics

For proving the validity of piezoelectric strain coefficient (d_{33} [pm/V]) by a strain measurement, d_{33} meter (HC materials, ZJ-3B) was used to measure piezoelectric charge coefficient (d_{33} [pC/N]) on disc type samples, and the magnitude of d_{33} was 280 [pC/N] (i.e. generally, d_{33} [pm/V] = d_{33} [pC/N]).

Figure 7 shows the schematic diagram of a test apparatus for the measurement of strain behaviors. Micro Laser Interferometer (Cannon, DS-80) was used as a non-contact displacement sensor and applied voltage on specimens generated by function generator (nF WF 1943A) and high voltage amplifier (TREK, 610E) was 0.5, 1, 1.5, 2, 2.5, 3kVp-p/mm at 3 Hz (sine wave form), and displacement/applied voltage signals were then measured by an oscilloscope (YOKOKAWA, DL1620).

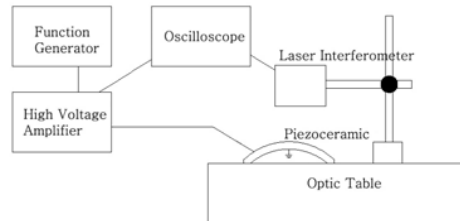


Figure 7. Experimental set-up for measuring displacement of dome shape piezoelectric device.

On the basis of the previous FEM modeling, the feasibility of dome type piezoelectric actuator has been carried out. The electromechanical displacement properties of both disc and dome type devices was measured at 3Hz and presented in Figure 8 (a), (b), (c).

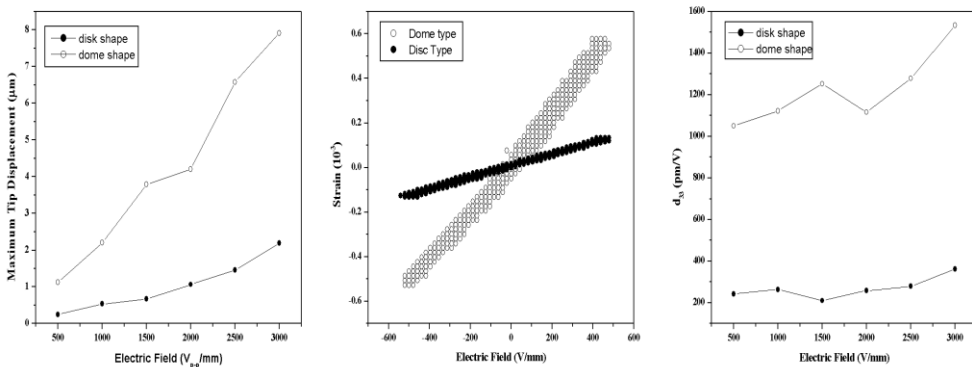


Figure 8. Comparison of electromechanical strain properties for both dome and disk shape piezoelectric samples as a function of electric field ; (a) maximum tip displacement (b) strain (c) piezoelectric strain coefficient (d_{33} [pm/V]).

As shown in Figure 8 (a), the average maximum tip displacement of the dome type device is 4.2 times higher than that of the disc type device. In order to obtain the piezoelectric strain coefficient (d_{33} [pm/V]) of two devices, strain-electric field values measured after applying electric field of ± 500 [V/mm] to the devices, as shown in Figure 8 (b), and then the d_{33} is calculated from the slope of this curve. As a result, the d_{33} of the disk and dome type device are 265 [pV/m] and 1125 [pV/m], respectively. Figure 8 (c) shows all results of d_{33} values measured as a function of electric field, these results indicate that the average d_{33} values of the disk and dome shape device are 269 [pV/m] and 1225 [pV/m], respectively. This shows that the average d_{33} value of the dome shape device is 4.5 times higher than that of the disc shape device, which is in accord with the FEM results.

(3) Step- Up Ratio and High Power Characteristics

Ring type electrode pattern was used to measure the step-up ratio of piezoelectric transformers composed of the dome shape actuator. $\pm 5 V_{p-p}$ square wave, generated by high speed bipolar amplifier (nF, HSA4052) connected with a function generator (nF, WF 1943A), was applied on the electrodes.

The output voltage was calculated by

$$V_{L(p-p)} = \frac{V_{s(p-p)}}{10k\Omega} \times R_L \quad (1)$$

where, $V_{L(p-p)}$ represents the peak to peak output voltage connected with load resistance(R_L) and $V_{s(p-p)}$ the peak to peak voltage with a standard resistance ($R_S=10 k\Omega$). Utilizing this output volatages, step-up ratio was determined by the following equation.

$$\text{Step-up ratio} = \frac{V_{L(p-p)}}{V_{in(p-p)}} \quad (2)$$

Where, $V_{in(p-p)}$ denotes the peak to peak input voltage applied on the dome shape piezoelectric transformer. For measuring the step-up ratio, an electrode pattern, shown in Figure 9, was fabricated.

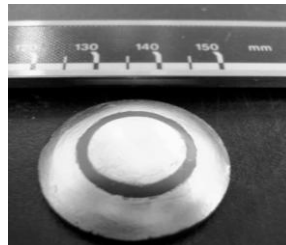


Figure 9. Real picture of dome shaped piezoelectric transformer.

The step-up ratios as a function of load resistance are presented in Figure 10.

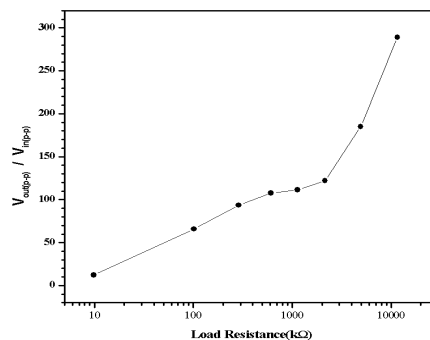


Figure 10. Step-up ratio of dome shaped piezoelectric transformer.

During the measurement, $\pm 5 V_{p-p}$ square wave was applied at resonance frequency ($\sim 98\text{kHz}$) and the $V_{L(p-p)}$ and step-up ratios were calculated following the method described in eq(1) and (2). As a result, the maximum step-up ratio is 290, which is 14 times higher than that of Rosen type device [16].

3) Application; High Power Piezoelectric Transformer

The high power characteristics of a dome shape piezoelectric transformer were estimated by the lighting test of 10 parallelepipeds connected 6 watt EEFL with dimension of $\phi 8 \times 630$ mm, which driving voltage and lamp resistance are $750V_{rms}$ and $130 k\Omega$, respectively. During the lighting test, $124 V_{rms}$ square wave, generated by high speed bipolar amplifier connected with a function generator, was applied, which is shown in Figure 11.



Figure 11. Light experiment of 60watt EEFL.

The input/output current & voltage waveforms are shown in Figure 12. The voltage is measured by 500:1 high voltage differential probe (Tektronix, P5200) and the setting value of oscilloscope is 1000:1. Thus, halves of the measured values represent the actual input/output voltages of the dome type transformer. Moreover, the input/output currents are measured by 10:1 current probe (Tektronix, P6021)

The efficiency is then calculated by the following equation.

$$Efficiency(\%) = \frac{V_{rms(output)} \times I_{rms(output)}}{V_{rms(input)} \times I_{rms(input)}} \quad (3)$$

From the measured waveform data, the calculated efficiency is found to be 96%. Therefore, this result supports that the high power characteristics of dome shape transformer is related to a geometrical effects and proves that this transformer is adequate to apply to high power devices.

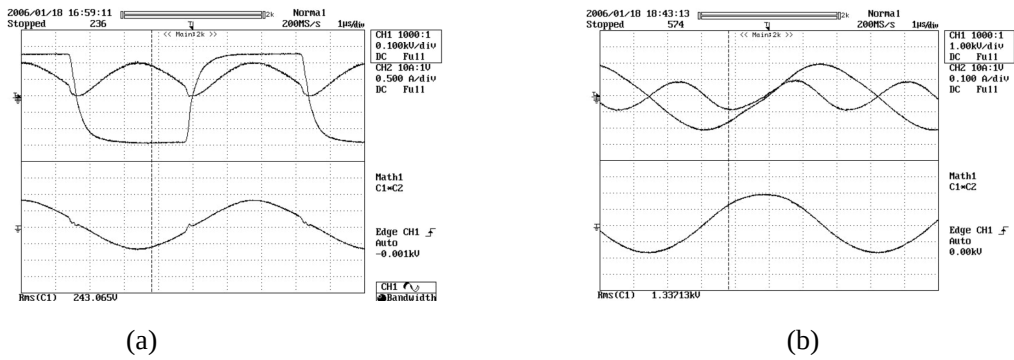


Figure 12. Detailed waveforms of input (a)/ output (b).

4) Conclusion

In order to investigate the applicability of PIM to hard piezoelectric materials, the microstructure of piezoelectric materials fabricated by PIM was compared with that of disc type material fabricated by dry press processing. The average grain size ($\sim 2\mu\text{m}$) of the PIM specimen was slightly larger than that ($\sim 1.5\mu\text{m}$) of the conventional die press processed specimen but the dense and poreless morphologies were observed in both two different samples. On the basis of the FEM modeling, the feasibility of dome type piezoelectric actuator was carried out. This showed that the average piezoelectric strain coefficient (d_{33} [pm/V]) of the dome shape device was 4.5 times higher than that of the disc shape device, which was in accord with the FEM results. The possibilities for high power piezoelectric transformer were also examined using dome shape actuator. The voltage step-up ratio and lighting experiment were performed by using ring type electrodes patterns. The observed maximum step-up ratio was 290, which was 14 times higher than that of Rosen type device and the output power exhibited about 60 watt with the 96% efficiency. Therefore, this result indicated that the high power characteristics of dome shape transformer were related to a geometrical effect and proved that 3-dimension dome shape transformer was adequate to apply to high power devices.

2. Compact Size Ultrasonic Linear Motor Using a Dome Shaped Piezoelectric Actuator

2.1 Introduction

An ultrasonic motor is a motor which uses ultrasonic vibration - a type of elastic vibration - to obtain a driving force, which then drives the motor using friction. Compared with a normal electromagnetic motor, which has a high efficiency but a low torque at a low speed, the ultrasonic motor can produce a high torque at a low speed with high efficiency. An ultrasonic motor is an attractive device in application for an imaging process such as optic zooming, hand shaking protection in digital camera. There are several ways to transform the piezoelectric vibration into the linear sliding movement such as non-self-moving and/or self-moving types [17]. However, the complex structure and circuit of conventional ultrasonic

linear motors limit the adaptation to the optic zoom of the mobile communication hand sets and compact digital camera.

Figure 13 is an exemplary construction of a **Dome Shaped Piezoelectric Linear Motor (DSPLM)**.

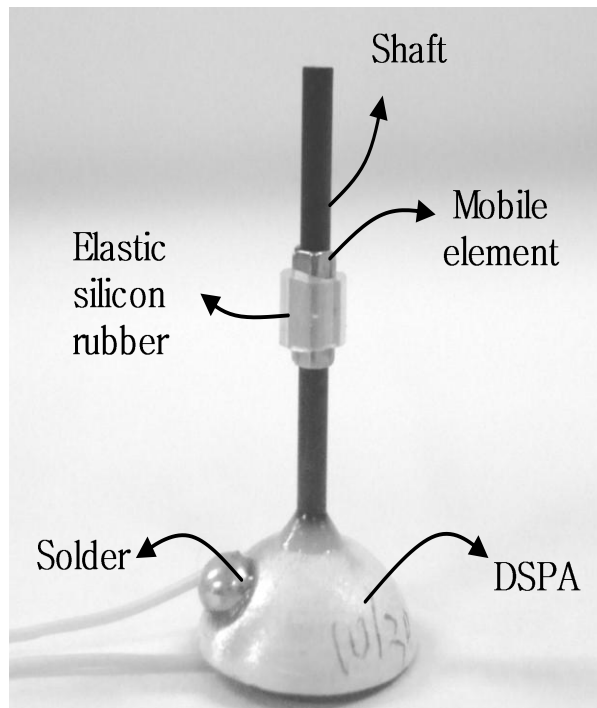


Figure 13. The photograph of DSPLM.

The motor consists of the three parts, which are actuating component, drive shaft and mobile element. The actuating component contains **dome-shaped piezoelectric actuator [DSPA]** fabricated by PIM. The displacement of actuating component is to transfer into the drive shaft. The shaft is located at the center of the dome-shaped piezoelectric actuator. The mobile element is couple with outer the shaft bounded by an elastomer. The frictional force can be controlled by the elastic force of silicon rubber.

2.2 Structure and properties of DSPLM

1) Modeling

As mentioned above chapter, the FEM code ATILA has been used in the simulation of the DSPA. ATILA is a FEM code developed for active structures such as piezoelectric materials and acoustic transducers. A three-dimensional model was used in which only DSPA combined with shaft was drawn and meshed as shown in Figure 14 (a), (b).

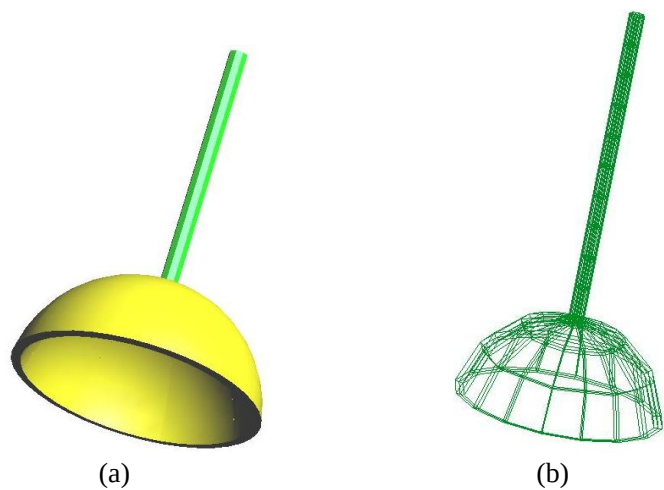


Figure 14. 3-dimensional structure combined with shaft (a) and Finite Element meshes used for DSPA analysis (b).

The new type linear motor called DSPLM has been defined with the aid of ATILA. It uses one dome shaped piezo ceramics of 9.86mm in diameter and 4.6mm in curvature of radius with thickness of 0.40mm. The shaft made of carbon fiber reinforced plastic is used to provide a linear motion and of which length was varied from 15mm to 30mm. The piezoelectric, dielectrics and elastic constants of PZT 5A, used in the FEM modeling, is shown in table 3 [15].

Table 3. PZT5A data used for this modeling

s_{11}^E	s_{12}^E	s_{13}^E	s_{33}^E	s_{44}^E
$16.4 \cdot 10^{-12} \text{ m}^2/\text{N}$	$-5.74 \cdot 10^{-12} \text{ m}^2/\text{N}$	$-7.22 \cdot 10^{-12} \text{ m}^2/\text{N}$	$18.8 \cdot 10^{-12} \text{ m}^2/\text{N}$	$47.5 \cdot 10^{-12} \text{ m}^2/\text{N}$
d_{15}	d_{31}	d_{33}	$\epsilon_{11}^T/\epsilon^0$	$\epsilon_{11}^T/\epsilon^0$
$584 \cdot 10^{-12} \text{ C/N}$	$-171 \cdot 10^{-12} \text{ C/N}$	$374 \cdot 10^{-12} \text{ C/N}$	1730	1700

Figure 15 shows the electrical impedance versus frequency, computed for a DSPLM having a different shaft length. It appears on these three graphs that 2 coupled modes exist; the first one is around 87kHz, the 2nd one is around 206 kHz under the free state of DSPLM. As shown in Figure 15, 1st resonance frequencies decrease slightly with increasing of shaft length. In order to increase Z-axis vibration, the bottom side of DSPLM was set clamped state by controlling the boundary condition of ATLLA program, and a resonance frequency is computed again at the same shaft length. As shown in Figure 16, the 1st resonance peak significantly appeared resulting in the slight increase of the 1st resonance frequency compared to non-clamped state and the 2nd resonance peak nearly disappeared. These results can predict that clamping state obviously increase the Z-axis vibration mode of the DSPLM by constraining the radial mode vibration.

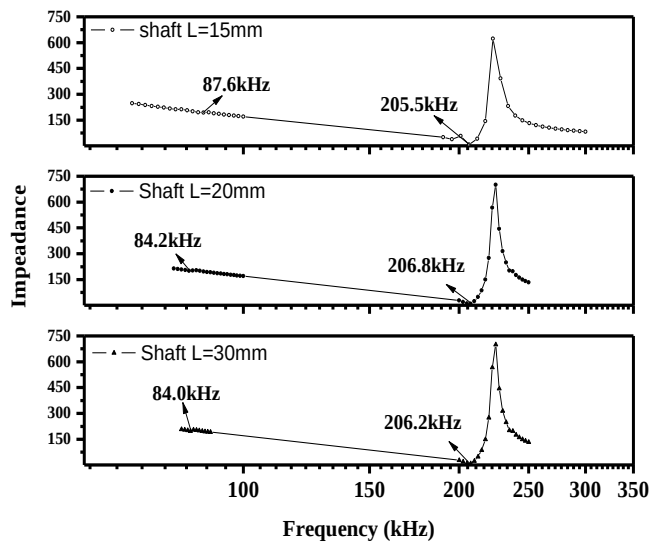


Figure 15. Simulated impedance versus frequency curves for the DSPLM with the non-clamped bottom side.

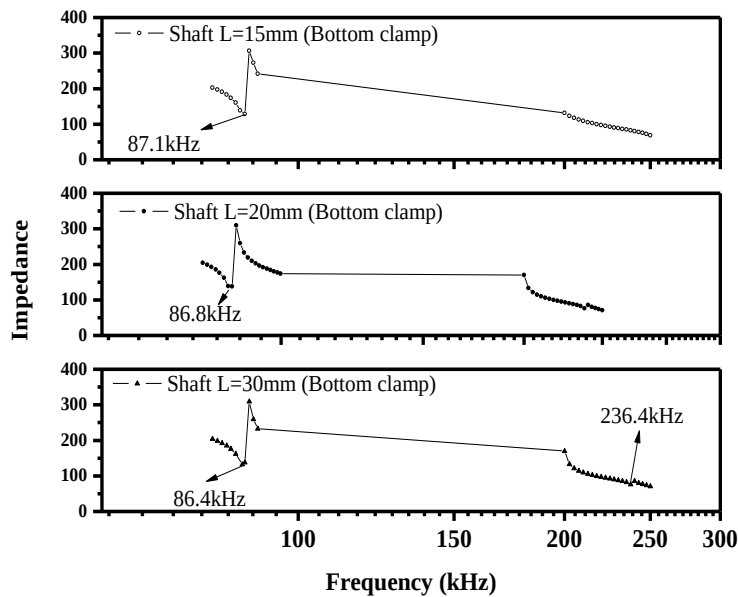


Figure 16. Simulated impedance versus frequency curves for the DSPLM with the clamped bottom side.

The vibration displacement simulation of the DSPLM was performed by ATILA FEM software to determine the maximum displacement for the shaft length in 30 mm at the 1st and 2nd resonance frequencies. The 1st and 2nd resonance frequencies correspond to λ and 3λ -mode, respectively.

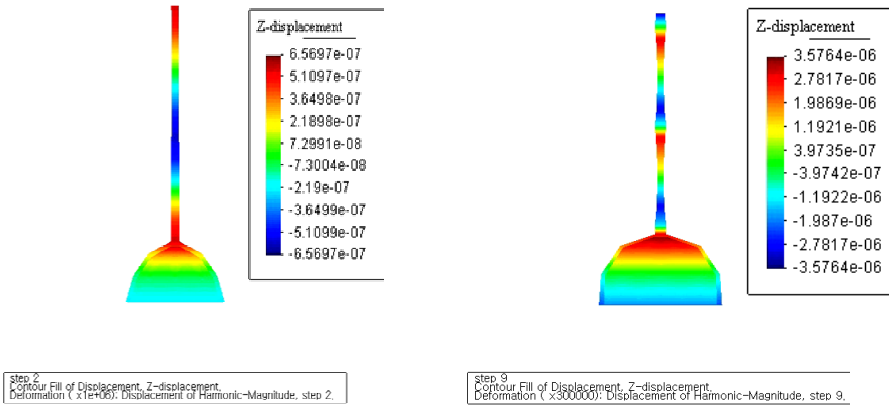


Figure 17. ATILA 3-D simulation results for maximum axial displacement at 1st and 2nd resonance frequency with the non-clamped bottom side.

Figure 17 (a), (b) shows the axial vibration displacements of the non-clamped sample computed at the 1st and 2nd resonance frequencies by applying a electric field of 70V_{p-p} to DSPLM. These results indicate that the maximum tip displacements in the Z-axis and radial mode resonance frequencies are ± 0.657 and $\pm 3.576 \mu\text{m}$, respectively. Note that the maximum tip displacement increases at the 2nd resonance frequency because of the lower impedance of radial vibration. On the other hand, in case of the bottom clamped DSPA, maximum tip displacement increases up to $\pm 6.72 \mu\text{m}$ at the 1st resonance frequency and maximum tip displacement at the 2nd resonance frequency decreases to $\pm 2.499 \mu\text{m}$ because of restricting the radial vibration mode due to bottom constraining force.

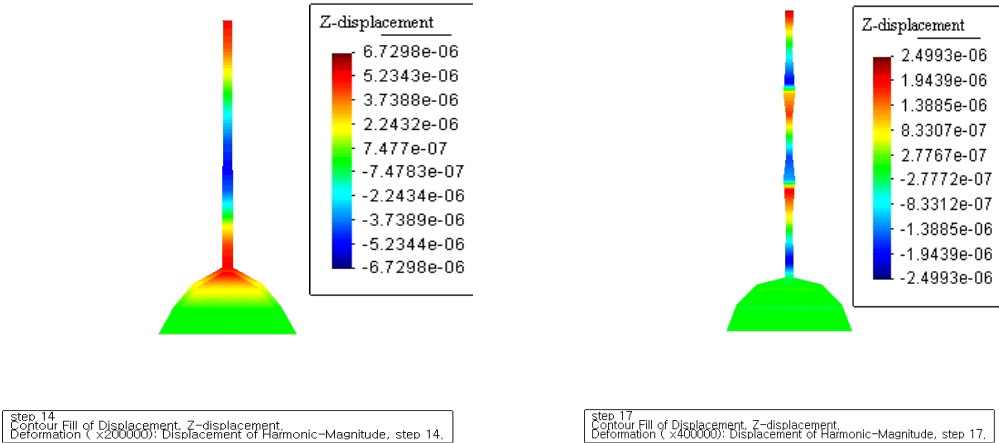


Figure 18. ATILA 3-D simulation results for maximum axial displacement at 1st and 2nd resonance frequency with the clamped bottom side.

From these computed results, it can be predicted that clamping state obviously increases the Z-axis displacement of the DSPA at the 1st resonance frequency.

2) Experimental Analysis of DSPLM

To verify that the PIM process is applicable to soft PZT materials, a $0.04\text{Pb}(\text{Sb}_{0.5}\text{Nb}_{0.5})\text{O}_3\text{-}0.46\text{PbTiO}_3\text{-}0.5\text{PbZrO}_3$ composition, which is generally used for a bimorph actuator, was used to fabricate DSPA. A preliminary study indicated that this composition is characterized by excellent dielectric/piezoelectric properties ($d_{33}\sim 550\text{pC/N}$, $k_{33}^T\sim 1450$, $k_p=0.7$ at room temperature) and high Curie temperature ($T_c=320$). In this experiment, DSPAs with diameters of 9.86mm, with the curvature radius of 4.53mm and a thickness of 0.40mm were fabricated using PIM. To evaluate the electromechanical properties, the sintered specimens were polished and electroded with silver paste (Metech Inc. #3288) fired at 650°C for 30 min, and then poled in a stirred silicone oil bath at 150°C by applying a dc electrical field of 2.5kV/mm for 40 min. The shaft was then bonded by epoxy in the center of the poled sample. In order to compare with simulation results, impedance curve over a frequency range from 10 to 300 kHz was measured using HP 4194A and the tip displacement behaviors at 1st resonance frequency was measured by applying an alternative square wave of $60\text{V}_{\text{p-p}}$ using the test equipments. The shaft of DSPA fixed at the top has length of 30 mm. Micro Laser Interferometer (Canon, DS-80) was used as a non-contact displacement sensor and driving voltage was applied by self-made circuit. Note that 1st and 2nd modes appear between 50 and 100 kHz under the clamped state as shown in Figure 19. Moreover, the 1st resonance frequency of the bottom-clamped DSPA decreases to 50 kHz.

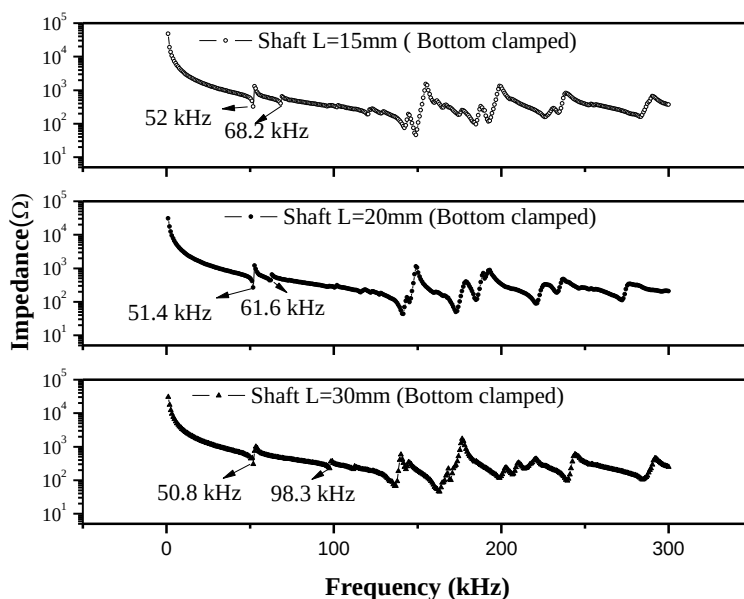


Figure 19. Experimental data of impedance versus frequency curves for the DSPLM with the clamped bottom side.

The decrease of 1st resonance frequency is caused by increasing the total mass of DSPA due to bottom-clamped ceramic element. In general, the effect of mass on the resonance frequency can be described by following equation.

$$f_r = \frac{1}{2\pi} \sqrt{\frac{S}{m}}$$

Where, f_r denotes the resonance frequency, S is the stiffness, m is the total mass. According to this equation, the resonance frequency decreases at the inverse square root of the total mass. The ceramic element combined with the bottom-side of DSPLM increased the total mass of DSPLM from 1.02 to 2.0. Consequently, the resonance frequency decreased to 50 kHz. In case of simulation, we only specified the boundary condition clamped at bottom-side of DSPA without the clamping ceramic. However, the clamping structure of experimentally manufactured sample was combined with bottom clamping ceramic element using the epoxy, which can make different resonance frequencies from simulation results. In spite of some discordance between resonance frequencies, it can be concluded that various vibration modes are caused by bottom-clamped ceramic element combined with DSPA and bonding material.

For comparing the results of simulation with those of experimental data, the dynamic displacements of DSPA at 1st resonance frequency were measured with Micro Laser Interferometer (Cannon, DS-80A) having a resolution of 0.08nm. Because of sampling speed limitation of displacement measuring instrument, the measurement of the displacement is limited to 1st resonance frequency (about 50 kHz). As shown in Figure 20, the tip displacement property of DSPA combined with shaft and bottom clamping ceramic at 1st resonance frequency was $\pm 7.48 \mu\text{m}$ under the electric field of $60V_{p-p}$

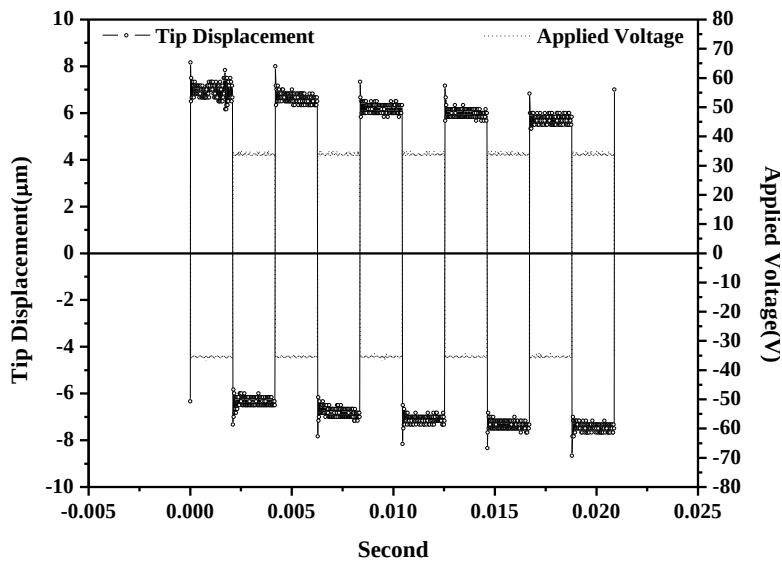


Figure 20. The Z-axial tip vibration displacements and applied voltage for DSPLM.

In spite of small size, the DSPA manufactured by PIM process has a large tip displacement, which is in accordance with the FEM results. This result indicates again that PIM process is very useful and simple method to make a large stain actuator.

3) Operation Principle of a Dome-Shaped Piezoelectric Linear Motor

The linear motion of the newly developed linear motor is operated by the principle of inertia displacement, which can be explained by a saw tooth electrical potential as shown in Figure 21 (a), (b). Figure 21 (a) shows the motions of a mobile element induced by the DSPA. When a mobile element locates at position l_a in Figure 21 (a), the electric potential is not applied, represented as “a” position in Figure 21 (b), which is called the initial state. During a gentle rise in the drive signal applied to the DSPA (a-b position in Figure 21 (b)), the DSPA will slowly expand. Accordingly, the mobile element coupled with the drive shaft by friction will move to the upper direction, represented as position l_b in Figure 21 (a).

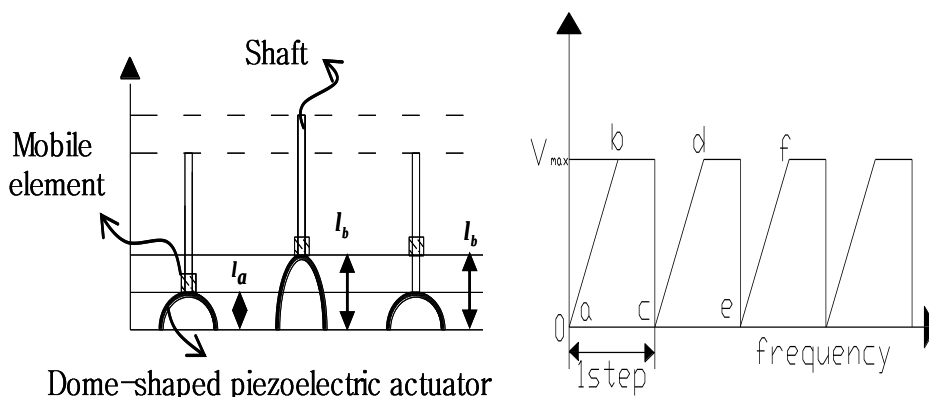


Figure 21. The principle of inertia displacement (a) and (b) saw tooth electrical potential.

As the electric potential suddenly falls to the “c” position in Figure 21 (b), which is the minimum electric potential, the DSPA quickly contracts towards the down direction. However, the mobile element remains in its second position “ l_b ” since its inertial force keeping it there exceeds the frictional coupling force. Therefore, the mobile element moves in position with the distance of $l_b - l_a$. Consequently, the mobile element will move repeatedly when the drive signal is applied for a desired number of times. If the slope of the saw tooth in the electrical potential is reversed, the direction of the mobile element would be reversed. It can be said that the DSPLM proposed here realizes the principle of inertial displacement. Moreover, it provides much higher reliability than multi-layer or bimorph-type actuators. [18, 19].

For testing the continuous linear motion of DSPLM under the 10 g load, the prototype sample was manufactured as shown on Figure 11. The actuating part is composed of DSPA combined with clamping element. The actuating part is driven by self-made PWM circuit. As shown in Figure. 22, the mobile element also consists of three parts, which are the rigid metal for the inertia weight (10g), such as alloy metal, a pair of L-shaped moving pieces made of phosphor bronze and plate spring fixed by bolt, which puts the pressure on the L-shaped plate spring in order to adjust the frictional force. Thus, mobile elements are compressed on the surface of the shaft by controlling the bolt passing through the one end of the plate spring.

The dynamic linear motion of DSPLM was measured using a self-made driving circuit, which generate $60V_{p-p}$ at a 1st resonance frequency of 50 kHz and controlled by customer I.C. Pulse width modulation method used is able to control the up and down moving direction.

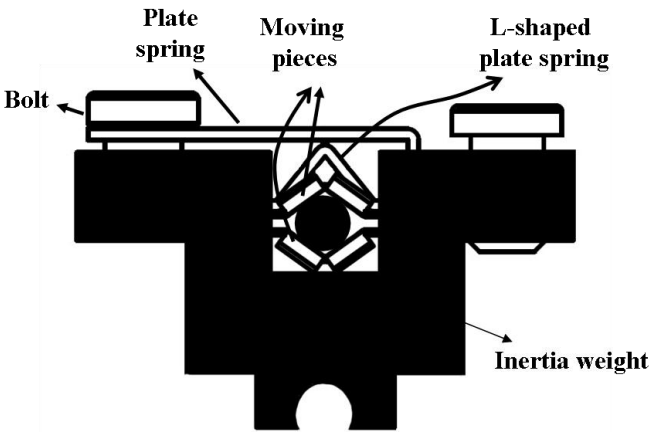


Figure 22. Detailed schematic view of mobile element.

As shown in Figure 23, the total displacement of DSPLM was 15.6 μm . From these result, dividing 5 into the total displacement, we can get the displacement per pulse of 3.12 μm under 10g load. Therefore, the controllable accuracy of DSPLM is 3.12 μm . Furthermore, we can observe the inertia and frictional behavior from the results of the step-like displacement. The rapid increasing of displacement indicates that frictional force is larger than inertia force and the mobile element move together with shaft during a wide pulse width applied to DSPLM. On the other hand, during a narrow pulse width applied to DSPLM, the increase of displacement becomes almost zero.

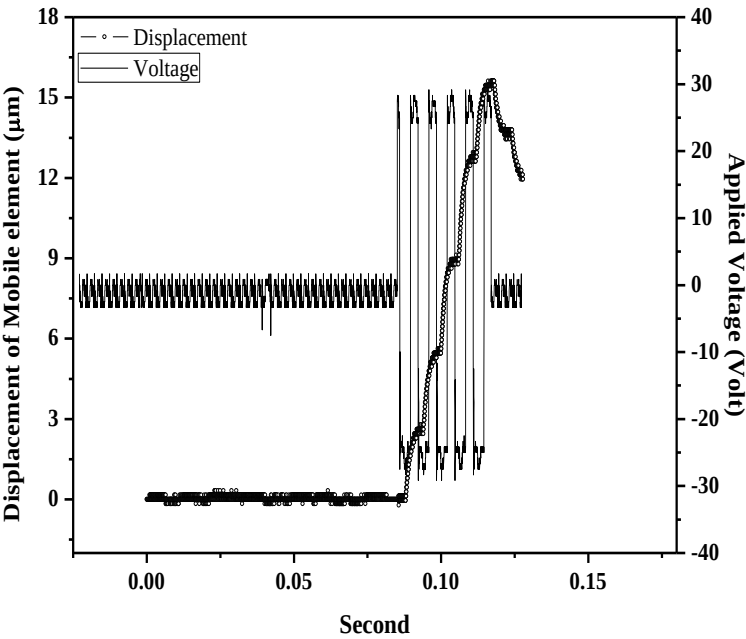


Figure 23. The total displacement of DSPLM according to apply a voltage at 1st resonance frequency.

This denotes that inertia force is larger than frictional force and the mobile element does not fall down with shaft. Comparing the results of Figure 19 with that of Figure 23, the displacement of DSPLM applied 10g decrease from 7.48 μm to 3.12 μm .

The continuous moving displacement properties of upper direction were measured at an applied 5 pulse with a Micro Laser Interferometer (Cannon, DS-80A) having a resolution of 0.08 nm. During the measurement, the DSPLM was attached to a test fixture and the laser beam was focused on the inertia weight combined with shaft. All test devices were located on the optic table to isolate them from external vibration. The applied voltage and displacement signal were measured by voltage probe and recorded in the oscilloscope.

The applied voltage and displacement signal were measured by voltage probe and recorded in the oscilloscope.

2.3 Application: a newly designed chopper for pyroelectric infrared sensor by using a dome-shaped piezoelectric linear motor (DSPLM)

In recent years, pyroelectric infrared-ray sensors have been utilized in a broad range of fields for a variety of purposes such as food temperature measurements during cooking in electronic ovens, detection of human body locations in certain areas cooled with air conditioners, light switch controls, and security systems.

The PIR sensors take advantage of pyroelectric effects obtainable with pyroelectric materials such as LiTaO_3 single crystal or PZT polycrystal. [20,21].

Pyroelectric materials have spontaneous polarization and always produce electric charges on surfaces. However, pyroelectric materials are kept electrically neutral in their stationary objects in an atmosphere where the electric charges are coupled with charges in the atmosphere. [22] When infrared rays are incident on pyroelectric materials, their temperatures are changed, and then destroying or changing the neutral state of the electric charges produced on the surfaces, thereby measuring quantities of incident infrared rays is possible. Generally, the incident infrared rays of PIR sensors should be periodically interrupted by means of a chopper which consists of a motor and revolving plate, in order to continuously measure the constant temperature of the portion to be detected. [23] However, since the conventional chopper is relatively large with high electric consumption and low reliability, a compact sized solid-state chopper is required.

This Chapter aims to propose and develop a newly designed chopper for a PIR sensor by using a dome-shaped piezoelectric linear motor (DSPLM) with a compact structure, higher reliability, and lower electric power consumption.

The operation characteristics of the PIR sensor, since stationary infrared rays were intermitted by the newly developed solid-state chopping system, were explained in detail.

1) Structure of Chopper Combined with DSPLM

Figure 24 shows an exemplary construction of a DSPLM and chopper. The DSPLM consists of four parts, the actuating component, drive shaft, mobile element and chopping plate. The actuating component contains a dome-shaped monolithic piezoelectric ceramic fabricated using PIM. The displacement of the actuating component is transferred to the drive shaft. The shaft is located at the center of the dome-shaped piezoelectric actuator and set to the fixture.

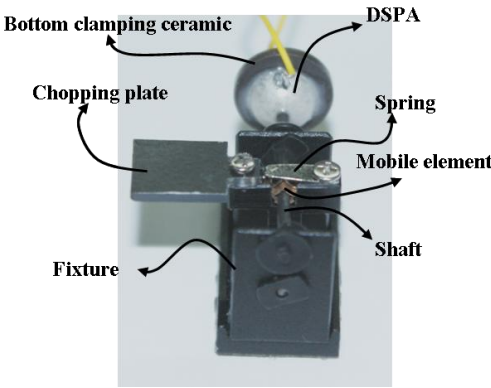


Figure 24. A Proto type structure of IR chopper combined with DSPLM.

The mobile element is coupled with the outer surface of the shaft bound by a spring made from elastic plate spring. The frictional force can be controlled by the elastic force of silicon rubber. The chopping plate is coupled with the outer surface of the spring. As shown in Figure 24, one end of chopping plate is manufactured into a rectangular shape in order to cover up the PIR sensor.

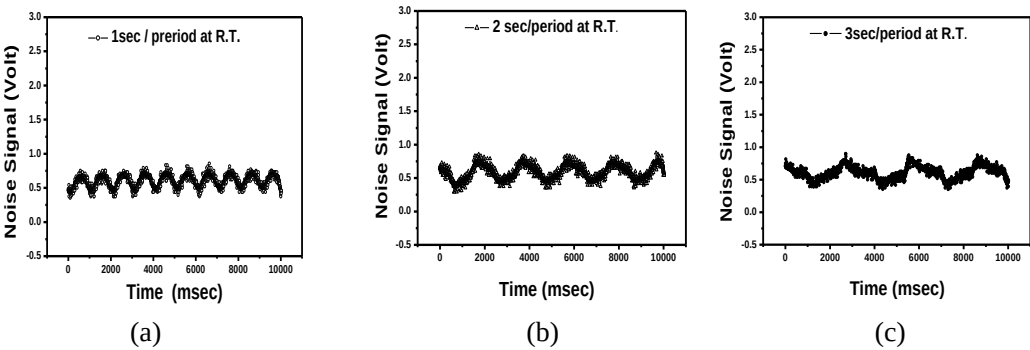


Figure 30. The chopping noise when the DSPLM chopper is operated at room temperature after removing the IR source: (a) 1sec/period, (b) 2sec/period, (c) 3sec/period.

2) The Effect of Chopping Periods On the Output Signal of a PIR Sensor

The system shown in Figure 25 was made to observe the characteristics of infrared detection of a chopper using a DSPLM. Infrared rays were emitted through a 30 mm diameter hole installed in front of a black body (ISOTECH, Model 9116) that kept the temperature of 310K similar with the temperature of the human body. The emitted IR was interrupted periodically by the DSPLM chopping plate connected with a driving circuit, and then the intermitted IR was incident on the PIR sensor (NICERA, dual-type sensor, Model SDA0254), which was 1 mm away from the chopping plate.

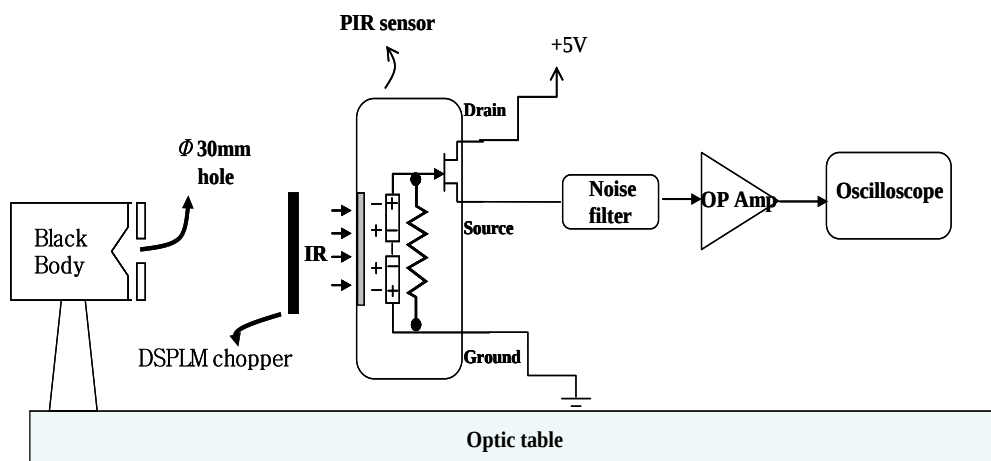


Figure 25. Block diagram of the overall experimental system and a basic structure of the PIR sensor.

In order to measure a signal from the PIR sensor, 5V from a DC power source was supplied to the drain and ground terminals, and the signal produced from a source terminal was transmitted into OP AMP (LM324N) through a noise filter and amplified 1000 times. The gain of amplifier was designed for having a maximum value at 1Hz. The amplified signals were measured and recorded by an oscilloscope. Figure 25 also shows a basic structure of the PIR sensor. Since the PIR sensor used in the experiment is composed of two pyroelectric elements in series (opposed configuration), this sensor has low sensitivity to ambient temperature variation, vibration and optical noise.

To estimate the intrinsic characteristics of the PIR sensor, when the stationary IRs emitted from black body was incident on the PIR sensor which was sustained at room temperature, the output signal was observed and evaluated.

Figure 26 shows the DSPLM chopping system for intermitting stationary infrared rays incident upon the PIR sensor.

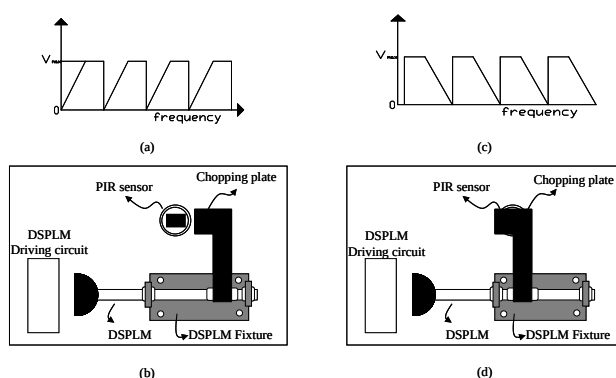


Figure 26. DSPLM chopping system for intermitting IR: (a), (c) driving waveform, (b), (d) chopping plate direction.

The PIR sensor and DSPLM chopper were mounted on the same board. When a waveform like that shown in Figure 26 (a) was applied to the DSPLM, the chopping plate

moved toward the right and the IR was incident on the PIR sensor as shown in Figure 24 (b). On the other hand, when a waveform such as in Figure 26 (c) was applied, the chopping plate moved toward the left and covered up the PIR sensor to block the IR. To measure the output characteristics as a function of chopping period, the operation period of the DSPLM chopper was set as shown in Figure 27. T_1 and T_3 are the times during which the DSPLM chopper operates and the PIR sensor is completely opened and closed, respectively, and they are fixed at 0.2 sec.

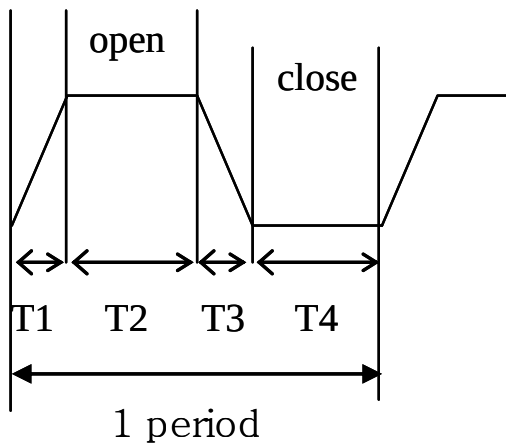


Figure 27. The operation period of the DSPLM chopper.

On the other hand, T_2 , T_4 are the holding times during which the PIR sensor is maintained in a completely opened and closed state, respectively, and are set to be the same time interval. The total time from T_1 to T_4 is defined as one period. The changes of output signals at various chopping periods (1 to 3 sec) were observed by applying stationary infrared rays. Under the same chopping periods, the chopping noise signals of the PIR sensor were measured by operating the DSPLM chopper at room temperature after removing the infrared rays.

3) Operational Characteristics of Modulated PIR Sensor

When the stationary IR emitted from black body was incident on the PIR sensor which was sustained at room temperature, the intrinsic output signal is as shown in Figure 28. Based on 0.6 V, the output signals with the upper and lower peaks are detected. This phenomenon is a basic characteristic curve of the dual-element type PIR sensor. As shown in Figure 25, since the PIR sensor was composed of two pyroelectric elements in series-opposed configuration, each element produces signals with a difference in phase when the IR is incident.

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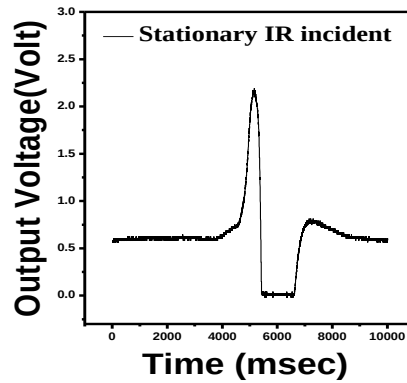


Figure 28. The intrinsic output signals.

Meanwhile, as the infrared rays are kept at a stationary state, the output peaks disappear. However, when the constant incident infrared rays are intermitted periodically by the DSPLM chopper, output signal responses for the PIR sensor at various intermitting periods are as presented in Figure 29 (a), (b), (c) .

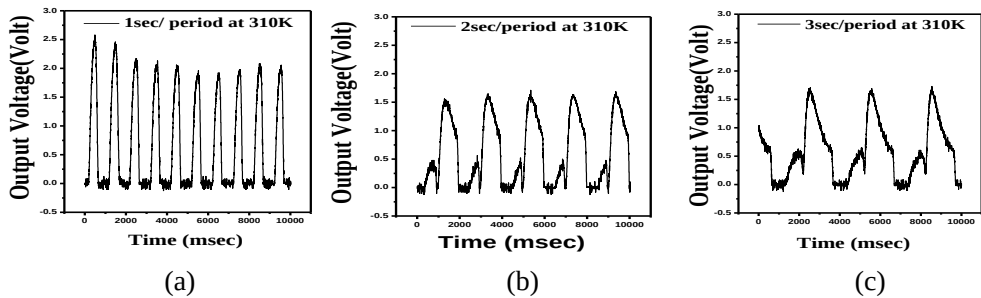


Figure 29. Output signal responses for the PIR sensor as a function of various intermitting periods at stationary incident IR (310K).

These results indicate that the stationary infrared rays incident to the PIR sensor are effectively interrupted. Consequently, pulse signals with constant peak value are obtainable for the stationary infrared rays. Furthermore, Figure 29 shows the changes of the output voltage curves when the intermitting period is increased from 1 to 3 seconds at intervals of 1 sec. Comparing with the average maximum output voltage of a second period, those of 2 and 3 second periods decreased as much as 0.5 V. A typical experiment will measure the output of a sensor as a frequency is varied. In this experiment, the response of PIR sensor is given by the responsivity equation: [23]

$$V = \eta P_0 p \omega A \frac{R_t}{\sqrt{1 + \omega^2 \tau_t^2}} \frac{R_e}{\sqrt{1 + \omega^2 \tau_e^2}} \quad (1)$$

Where η is the emissivity of surface of sensor, P_o is the peak value of the modulation of the infrared ray, p is the pyroelectric coefficient, ω is the angular frequency of radiation modulation, A is the sensor area, R_t and R_e are the thermal and electric resistances and τ_t and τ_e are the thermal and electrical resistances. According to the model proposed by Benjamin and colleagues [25], at very low frequencies, Eq. (1) reduces to a simple form where frequencies the voltage is proportional to ω (and at high frequencies the voltage is proportional to $1/\omega$). Since the intermitting period increases from 1 to 3sec, modulation frequency decrease from 1 to 0.33Hz. Therefore, average output voltage was decreased as increasing intermitting period.

As the intermitting period increases, two separated signals are obviously represented. When a chopping period is adequately longer than the thermal time constant of the pyroelectric elements in the PIR sensor, different signals generated from the two pyroelectric elements are separately represented as shown in Figure 27 (b), (c). However, as chopping period is shorter than the thermal time constant, the separated peaks overlap each other as shown in Figure 27 (a).

Figure 30 shows the chopping noise when the DSPLM chopper is operated at room temperature after removing the IR source. From the results, the average noise output voltage was 0.3 V, resulting in a 1/6 lower value compared with the output voltage when the infrared rays were incident. These results indicate that the signal-to-noise ratio has available values within 5.4 to 7.2. Moreover, when the DSPLM was in operation, the power consumption was less than 1 watt.

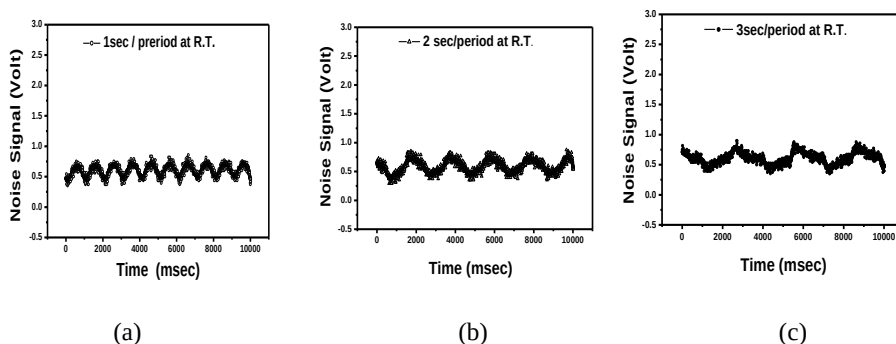


Figure 30. The chopping noise when the DSPLM chopper is operated at room temperature after removing the IR source: (a) 1sec/period, (b) 2sec/period, (c) 3sec/period.

2.4. CONCLUSION

As discussed previously, the DSPLM chopping system clearly demonstrates that stationary infrared rays are effectively intermitted. Therefore, a module of the continuously detectable PIR sensor that was small in size with reduced power consumption and high reliability was successfully realized using the DSPLM chopper.

In conclusion, to develop a compact and simple structured chopper for a PIR sensor, a smart ultrasonic linear motor was introduced for a new solid state chopper. When infrared

rays close to the temperature of the human body were stationary incident on the PIR sensor from the blackbody, pulse signals with constant peak value of 1.7 to 2.2V were observed, and the average noise output voltage was 0.3V, resulting in a 1/6 lower value compared with the output voltage when the infrared rays were incident, regardless of the intermitting period. Thus, the signal-to-noise ratio obtained has a value of 5.4 to 7.2. Moreover, it was shown that the power consumption was less than 1 watt when the DSPLM was in operation. Therefore, a module of the continuously detectable PIR sensor that was small in size with reduced power consumption and high reliability was successfully realized using the DSPLM chopper.

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Chapter 4

REVIEW OF INTEGRATED PIEZOCERAMIC TRANSDUCERS IN SMART MATERIALS AND STRUCTURES

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ABSTRACT

There has been extensive research to study and improve the integration of piezoelectric elements in smart materials and structures since the early 1990s. The unique properties of piezoceramics to efficiently generate an electric potential from an induced mechanical strain while reversibly, produce stress from an applied voltage, appropriately placed these elements as favoured sensors and actuators. Research is also motivated by commercial interests in noise and vibration control, shape control, and structural damping, particularly for civil and aerospace applications. In the last decade, the field has extended to structural health monitoring for the detection and quantitative evaluation of in-service and manufacturing defects. This short review arrives in a timely manner to outline the present progress of the technology by compiling the studies conducted and summarising the significant contributions made. Fabrication and manufacturing methods for piezoceramic attachment are consolidated while addressing the aspects of material selection, safe design, performance and durability. This is followed by the descriptions of testing techniques available to simulate in-service loading and to evaluate the manufactured smart product through various measured parameters. Important test results and analysis from past studies are presented to understand the failure modes and the operation limits. Literature on key modelling approaches to study the integrated piezoceramic transducers are briefly highlighted, showing the evolution of the simple analytical one-dimensional models to more complex and numerical two/three-dimensional models.

INTRODUCTION

The history of piezoelectricity dates back to the experiments demonstrated by Pierre Curie and Jacques Curie in 1880 where certain natural crystals including tourmaline, quartz, topaz and Rochelle salt were found to produce surface charges when subjected to mechanical stress. It was only a year later, Gabriel Lippmann derived mathematical evidence that the converse piezoelectric effect is possible, which was later confirmed in experiments by the Curie brothers. However, piezoelectricity did not take off until the First World War in 1917 when Paul Langevin invented the steel-quartz-steel sandwich transducer to detect underwater submarines using high-frequency waves. Research in underwater navigation and sonar intensified over the years with the discovery of stronger piezoelectric resonators made of ceramics like barium titanate in 1945 and later lead zirconium titanate (PZT) in 1954. A detailed account of the history of piezoelectric materials can be found in the paper by Ballato [[1]].

Ceramic piezoelectric materials are most widely used today and preferred over single crystals due to higher sensitivities as well as the flexibility to fabricate into various geometrical shapes and sizes, Figure 1 [[2]]. There have been constant improvements to the properties and the quality of piezoceramics.



Figure 1. Picture of various piezoceramic types, sizes and shapes.

By slightly changing the chemical composition, different grades of ceramic materials can be produced to serve a diverse range of applications. The stiffness of the piezoceramic can facilitate the actuating and sensing of signals without the need for amplification [[3]].

Coupled with low power requirements [[4]] these elements make very attractive self-contained and efficient transducers. Advancements in manufacturing especially in the quality of adhesives and bonding techniques consequently allow the integration of piezoceramics in smart materials and structures. There are various text references available today that document the design and application of these smart materials and structures [[5]-[7]]. Smart materials technology incorporating piezoceramics was pioneered in the aerospace industry when there was a need to control the noise and vibrations of large structures like satellites and solar array panels as well as the isolation of high precision equipments in optical and microgravity experiments [[8]-[12]]. Shape control is another established topic in this area where compensation of temperature and moisture desorption effects is important for sensitive space components, such as those applied in spacecraft antennas, optical benches and support structures for reflectors and telescopes [[13],[14]]. In adaptive air vehicles, smart materials are also gaining interests in active control through surface deformation and twisting of wings, stabilisers, tails and rotor blades to manipulate lift, reduce drag and improve aerodynamic performance [[15]-[18]].

Huston [[19]] conceived the possibility of smart civil structures that have the ability to sense and respond to external environmental hazards but only with efficient, inexpensive and reliable integrated transducers. This can be achieved with piezoceramics and following the lead of the aerospace industry, the technology was subsequently researched in civil engineering particularly for vibration suppression in resisting against seismic and wind loads [[20]-[23]]. The paper by Song et al. [[24]] provided a concise review of developments in vibration control for common construction members including cantilevered, truss, frame and cable-stayed structures with emphasis on piezoceramic applications. The value of smart materials and piezoceramics were further added with the introduction of these topics in a graduate level civil engineering course by Zhang [[25]].

In the last decade, piezoelectric materials have gained significant interests in structural health monitoring (SHM). The SHM systems developed with attached piezoceramic transducers are not only aimed to detect but also to characterise in-service damages and manufacturing flaws before they grow and propagate into critical failure modes. Ultrasonic tomography is one emerging field of study that can achieve quantitative evaluation of damages through imaging with an array of piezoceramic sensors [[26]-[29]]. The defects targeted in SHM are diverse and exist in a wide range of materials, e.g. fatigue cracks in concrete, corrosion in metals, delamination in monolithic laminates, and skin-core disbond in honeycomb sandwich structures. There are also research activities that exploit this technology to monitor the integrity of composite strengtheners and repair patches [[30]-[33]]. The prospect of piezoceramic application in SHM thus spans into a number of major industries including aerospace [[34]-[36]], civil [[37]-[39]], mining [[27],[40]], and naval [[41]].

This short review aims to collate the significant contributions from individual literature in utilising these established transducers so that this information can be readily available for implementation and further development. In the next section, important fabrication and manufacturing aspects of integrating piezoceramics to the host structure are presented. This is followed by an outline of testing techniques used in evaluating the performance and durability of the integrated transducers in smart materials and structures together with important key results. References to modelling approaches formulated to simulate and study the attached piezoceramic are also briefly pointed out in this review.

Integration Design Aspects

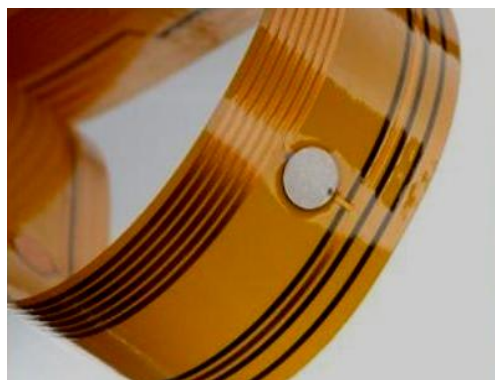
The simplest design is to bond the piezoceramic to the surface of the host with adhesive [[4]] but there are developments to embed the transducer within the structure [[42],[43]]. Embedded transducers are shielded from external damages as well as fulfil the role of an intelligent active system as part of a load bearing component. Piezoceramic integration is particularly suitable in composite laminates since the layered construction can accommodate the embedment within the structure. As composites gradually emerge to replace metals for weight reduction and corrosion resistance especially in the aerospace industry [[44]], this creates many research opportunities and interests in developing different embedment techniques for composite manufacturing. The common technique of embedment is to prior to curing either include the piezoceramic element directly into the lay up of the composite or cut-out a volume within a few layers of lamina plies to accommodate the transducer [[42],[45]]. Both these methods introduce a region of resin rich pockets around the transducer as well as material discontinuities, which can be a source of stress concentration and overall reduction in the strength of the structure [[46]-[48]]. Application of piezoceramic fibres encased in a thin active layer of epoxy for embedment [[49]-[52]], and interlacing the plies for load redistribution around the transducer element [[53]-[55]] are some ways being developed to address these issues.

The considerations involved in the piezoceramic integration design process are largely dependent on the application and requirements of the host structure. One important aspect that determines the selection of suitable transducer materials is the operating temperature of the structure. High temperature applications especially in aerospace structures travelling above subsonic velocities demand for piezoceramics with high Curie temperatures to ensure that depolarisation does not occur and piezoelectric properties are not lost when in-service [[56],[57]]. A heat conduction model can be used to estimate the temperature and thermal stress distribution of the piezoceramic element [[58]]. Composite manufacturing also involves high curing temperatures above 120°C that requires a good match in the coefficient of thermal expansion (CTE) among materials involved in the transducer embedment. This ensures that residual shear stresses induced around the resin pocket matrix during the curing and cooling processes are kept to a minimum to prevent premature failure of the transducer [[9]]. To service these high temperature requirements, PZT is typically preferred for its high Curie temperature above 200°C with a CTE range that match those of the material components used in advanced composites.

Geometry and size of the piezoceramic transducer are also selection aspects determined by the host structure. Surface conformation of the transducer in direct contact with the boundaries of the structure is crucial to provide a strong bond in the attachment as well as to optimise strain transfer. There exists a wide selection of bending piezoceramics fabricated from different manufacturing approaches that can accommodate curved structural geometries including unimorphs and bimorphs [[59]-[63]]. These variants in transducer construction are originally motivated by the need to produce larger displacements compared to single piezoceramic elements for shape control systems [[64],[65]]. The magnitude of the strain delivered by the transducer to the structure is also constrained by its thickness when the voltage range applied is limited [46]. There are studies that address the selection of the optimum transducer thickness for actuation purposes [[66],[67]]. A larger area of coverage can be achieved with an array of connected piezoceramic elements through a SMART

Layer®, Figure [[68],[69]]. This innovation consists of PZT elements sandwiched between two polyimide layers, which can adopt the complex contour of the structure and can either be surface mounted or embedded.

The influence of the adhesive layer is an important consideration in surface bonded piezoceramic elements and patches. The types of adhesives commonly used are epoxy, acrylic, polyurethane and cyanoacrylate formulations suitable for bonding of structural joints [[70]]. Much studies presented in literature, e.g. [[71]-[78]], are focused on developing analytical and numerical solutions to model the strain transfer between the transducer and the structure via the adhesive layer for structures in extension, bending or torsion. The adhesive thickness is a compromise between optimisation of the strain transfer for actuation through a thin layer and the reduction of the stress induced on the transducer by the host structure under in-service loading through a thick layer [[57],[79]]. Based on a parametric study accounting for impedance measurements sensitive to bonding degradation, Bhalla and Soh [[80]] recommended using high shear modulus adhesives to reduce the bond thickness, and applying small area-sized transducers to minimise the loss of admittance. Models for embedded piezoceramics were also presented within the same scope of references for performing similar analysis on the resin layer that separates the transducer from the laminate structure. Fundamental modelling approaches are briefly reviewed in a later section of this chapter.



(a)



(b)

Figure 2. (a) Flexibility of the SMART Layer® to conform to different structure contours. (b) Example of an array of piezoceramic elements in a SMART Layer®. (Courtesy of www.acellent.com).

A significant motivation in studying the influence of the adhesive layer is the ability to evaluate and understand the interface stresses and consequently design strategies to minimise the likelihood of debonding, a common failure mode for surface mounted transducers [[81]-[83]]. Analytical studies suggested that shear and peeling stresses at the interface are highly concentrated at the ends of surface bonded actuators where the structure in traction can result in adhesive failure or cracking in the brittle actuator [[84],[85]]. Based on a numerical analysis applying a high-order modelling approach, Rabinovitch and Vinson [[86]] revealed that compliant adhesives can significantly reduce the edge shear stresses in the bond layer. Experiments performed by Blackshire et al. [[87]] with an interferometric-based displacement field imaging technique [[88]], found transducer cracking events, bond degradation and performance variation for piezoceramic transducers bonded on aluminium plates subjected to accelerated dynamic vibrations, outdoor exposure, adverse temperature cycling and electrochemical exposure, simulated conditions typically found in operational aircraft environments. A follow-up analytical study conducted by Blackshire and Cooney [[89]] deduced that the load is asymptotically transferred to the transducer via the adhesive layer from its edges, moving toward the centre of the transducer at a rate determined by the shear modulus, adhesive thickness, substrate stiffness and transducer thickness. Tsoi and Rajic [[90]] also reported that PZT transducers bonded to an aluminium beam exhibited no change in the performance below 1000 microstrains cyclic loading through the observation of the electrical impedance [[91]]. Above this strain level, some variation in the impedance can be observed, which indicated deterioration in the bond layer.

Insulation is also an important requirement to shield the transducer from electrical interference especially in the case of an embedded piezoceramic that has a large surface area in contact with the host structure. Kapton tape and other polyimide films are common products used to insulate the integrated piezoelectric transducer [[92]-[94]] but these materials have low adherence properties. This problem can be solved by applying a B-stage adhesive coupling agent or by plasma activation of the combination of the adhesive and the insulating material to increase the peel resistance [[57]]. Thick adhesive layers can also function to encapsulate and protect the transducer from external strains [[95]]. A new approach to protect the piezoceramic element is through a thermoplastic buffer layer,

Figure 3, developed by the Composites Group at the University of Queensland, Australia [[96]]. The thermoplastic layer can function as an excellent insulator for its higher electrical resistance properties compared to polyimide films while producing a strong bond with the thermosetting resin of a composite either surface mounted or embedded through a Thermoset Composite Welding (TCW) technique [[97],[98]].

Piezoceramic transducers to function as actuators and sensors require connection to amplifiers and signal converters where direct links can be achieved via wires, leads and electrodes. The integration of the connectors is dependent on the design of the transducer construction, e.g. the piezoceramic fibre patch contains interdigitated electrodes [[49]] while the SMART Layer® technology employs a printed circuit on one of the polyimide films [[68]]. In surface mounted transducers, these connectors can be typically soldered or bonded to the piezoceramic. For embedded transducers, the paper by Ghasemi-Nejhad et al. [[42]] provided detailed descriptions of three manufacturing methods for facilitating the electrical connection, namely cut-out holes, molded-in holes and embedding. Among these techniques, embedding is the most popular approach described in literature due to the ease of manufacturing while the connection between the leads and the transducer is protected and

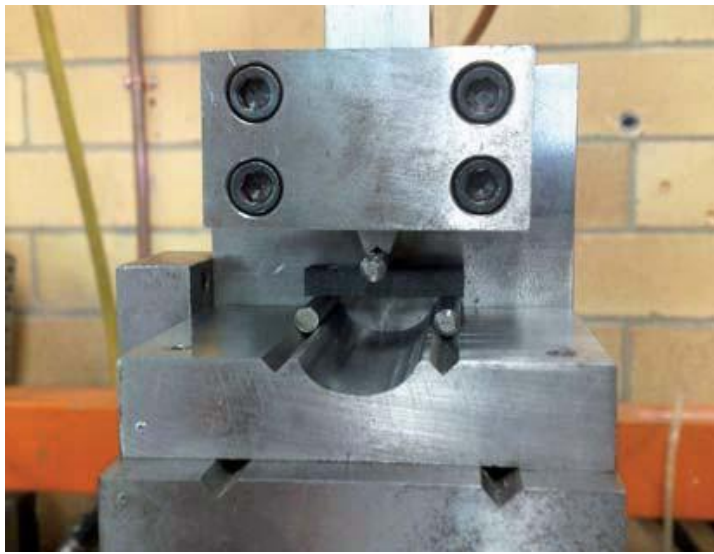
securely fastened within the matrix of the composite to minimise electrical failure [[99],[100]]. Wireless technology is also recently being explored in SHM with research investigating methods of designing the architecture of a wireless sensing unit that drives connected piezotransducers and transmits signals to a central data collection infrastructure [[101]-[103]]. An overview on this topic was provided by Lynch and Loh [[104]] where a section of the article has been dedicated to conceptual studies exploiting the wireless technology with piezoceramics.

Testing and Key Results

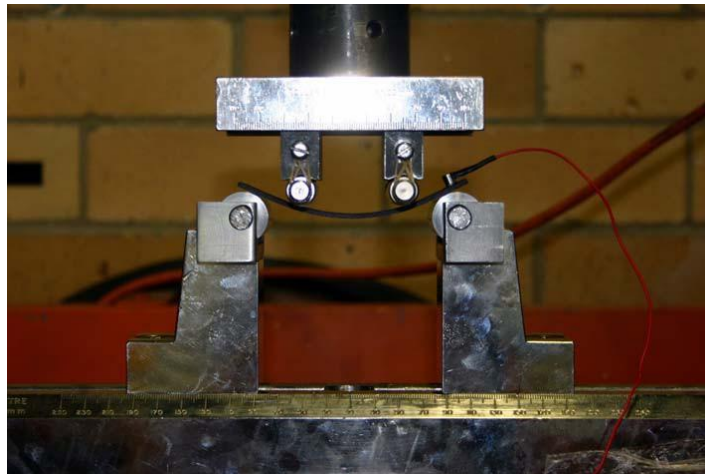
As there is a wide range of design selection aspects for integrating piezoceramics in materials, it is essential to test the quality of the attachment, integrity and performance of the smart structure under expected loading conditions. In surface mounted transducers, the adhesive layer is the major concern in testing techniques. Electrical impedance has been shown as a reliable indicator of the bond quality [[91]]. Blackshire and Cooney [[89]] have also demonstrated that using a laser interferometry system, imaging of the adhesive layer for disbonds can be achieved. Key results for surface mounted piezoceramic transducers have been presented in the previous section. If the transducer is embedded within a composite, its load bearing capabilities have to be checked while ensuring that the inclusion does not significantly affect the strength and exceed the allowable design limits of the original structure. Testing also functions not only to evaluate the material properties of the manufactured product but to determine the fatigue life and prolonged sustainability of the system by considering accelerated dynamic loading at high strain levels and temperature cycling [[87]]. Identified vulnerabilities in the smart structure can be valuable in improving future designs while positive results provide confidence for practical implementation and opportunities of commercialisation.



Figure 3. The PZT element with copper leads encapsulated in a thermoplastic buffer layer and attached on a laminate composite.



(a)



(b)

Figure 4. (a) Three-point bending fixture. (b) Four-point bending fixture with an acoustic emission sensor attached to the specimen. These test machines are located in the Material Testing Laboratory at the University of Queensland.

Tension and bending are the basic loading conditions for smart materials and structures. Tension tests can be conducted to measure the strength of the smart structure while bending tests are more suited for assessing the shear stress at the adhesive layer and laminar interfaces in contact with the integrated piezoceramic. Loading can be applied either monotonic to evaluate the characteristic material properties or cyclic to determine the fatigue life. Standard procedures available for testing common materials like metals and polymer matrix composites can be used, e.g. ASTM [[105]-[108]]. Bending tests can be conducted in either a three/four-point fixture depending on the stress distribution required at the loading location, [[109]-[112]]. The article by Munjal [[113]] is an excellent reference to determine the suitable static test methods for fibre reinforced composites. The output voltage of the integrated

piezoceramic can also be monitored during tensile or fatigue testing to detect for any loss in performance [[100],[114]]. For comparison, reference measurements can be obtained by isolating transducers from loading in both tensile and bending tests [[90],[115]]. Fractography and microscopy can be performed after loading tests to examine for cracks, debonding and defects surrounding the surface mounted or embedded piezoceramics [[115],[116]].

Paget and Levin [[117]] found that a 10 mm x 10 mm x 0.13 mm PZT transducer with an 18 μm copper interconnector and a 25.4 μm thick polyimide insulation film embedded centrally in a cross-ply carbon fibre-epoxy laminate did not reduce the strength of the composite by testing up to structural failure in either tension or compression loading. Optical microscopic examination of the sample's cross-section revealed only debonding at the conductive adhesive between the transducer element and the copper interconnector layer, which did not initiate the final failure of the composite. This observation proved that the vulnerability in the integrated transducer technology is not governed by the brittleness of the piezoceramic but rather the interface between the element and the host structure. Studies conducted by Mall and Coleman [[114]] also showed similar results where in tensile tests the average ultimate strength and Young's modulus of the $[0/\pm 45/90]_s$ quasi-isotropic laminate with a PZT located at a central cut-out region at the two middle 90° plies were within only 4% difference in the material properties of a blank reference specimen. In fatigue loading to failure, the rates of delamination growth were found to be comparable while the modulus reduction varied slightly higher within 5%-15%. In addition, it was found that below 1000 microstrains, the voltage output of the PZT remained constant for millions of fatigue cycles with the interface between PZT and surrounding plies showing no signs of damage up to 2000 microstrains. Increasing the fatigue loading above 1000 microstrains caused the voltage output to drop gradually and then sharply at 1500 microstrains, owing to a combination of depolarisation, sensor deformation and matrix cracking of the laminate. Mall [[118]] continued to support these results with further experiments of similar specimens with transducers either inserted in a cut-out hole or embedded directly in the middle plies of graphite-epoxy laminates where degradation in the embedded PZT was not found for quasistatic tensile loading up to 2000 microstrains and up to 1000 microstrains under tension-tension cyclic loading. Fatigue three-point bending tests of short composite specimens with embedded elements performed by Schaaf et al. [[45]] also showed that precision cut-out embedding can increase the fatigue life by nearly three folds compared to directly embedding the transducer in the laminate.

It is important to point out that loading in actual applications can be complex due to combinations of different loadings including environmental effects like temperature changes. It is thus necessary to conduct extensive testing with adequate coverage of expected loadings and influences prior to implementation. Bronowicki et al. [[99]] found no degradation from measured voltages in the actuating and sensing performance of a 0.19 mm thick piezoceramic wrapped in a thin epoxy insulating medium and embedded in graphite-epoxy composites under tensile strains of 1500 microstrains and then fatigue loading of 900 microstrains. 13% degradation in the transducer performance was detected only after fatigue loading at 1500 microstrains, which was equivalent to a further 13% degradation measured when $\pm 100^\circ\text{C}$ of cyclic thermal loading was introduced to the sample. Experiments performed by Mall and Hsu [[119]] considered both the mechanical and electrical cyclic loading on the embedded PZT in a laminate specimen. Mechanical fatigue was produced by loading up to 0.2% strain with stress ratio of 0.1 while simultaneously, the embedded PZTs were excited either by

triangular waveform from 10 V to 100 V out-of-phase to strain or from -10 V to -100V in-phase to strain. It was found that only the in-phase loading conditions caused a drop in the measured transducer output voltage and this was attributed to amplified electrical depolarisation due to the combined electromechanical effects.

Apart from the output voltage, capacitance and impedance are other methods of evaluating the performance of the integrated piezoceramic transducer. Capacitance can be measured before and after the transducer is embedded in composite manufacturing to ensure consistent functionality of the element [[42]]. However, Park et al. [[120]] reported that there can be capacitance losses up to 38% due to the difference in boundary conditions of the transducer before and after embedment. Experiments have also shown that fracture of a piezoceramic element does not necessarily lead to a loss in capacitance as long as all fragments still have considerable amount of contact among each other [[57]]. Therefore, the measurement of capacitance can only provide indicative information regarding the condition of the transducer and the contact of the electrodes. Impedance or admittance is a better transducer performance indicator, which gives valuable insight into the failure strain of the smart structure while assessing the quality of the adhesive layer and the strain transfer. The impedance method was pioneered by Liang et al. [[121]] and has been shown to be effective for ultrasonic frequencies, capturing local or near-field dynamics due to transducer damage and disbond. Novel ways to interpret the impedance spectra and identify structural damage have been developed for various applications [[122]-[124]]. Park et al. [[120]] explained that by referring to the admittance curve, bonding defects are related to a relative increase in its gradient while degradation or fracture of the transducer produces a relative decrease in its gradient. A similar method is to observe the slope of the load-strain and charge-strain performance curve in quasistatic loading before and after cyclic loading, as shown by Thielicke et al. [[125]]. Reduction in the slope of the charge-strain curve indicates electrical damage from depolarisation or disconnection from electrodes while a drop in the slope of the load-strain curve indicates physical damage of the transducer or disbond at the adhesive interface.

According to Paget et al. [[116]], monitoring the Lamb waves excited by the integrated transducer can be a method applied to examine the integrity of the smart structure. The wave response can be analysed by recognising changes in the response amplitude of the time domain or the power spectral density. The authors have used acoustic emission (AE) sensors but laser vibrometry is a non-contact system that can be an alternative technique for more accurate measurements [[126],[127]]. AE sensors were surface bonded to the specimens by silicone paste in quasistatic tests but in fatigue tests, the sensors were fixed to the specimen surface to avoid the influence of deteriorating coupling properties throughout the fatigue cycles. It was found that only matrix cracks around the embedded PZT element were detected from degradation of the frequency spectrum of the wave response in fatigue loading of ± 170 MPa or 2000 microstrains and ± 260 MPa or 3000 microstrains between 50,000 cycles and 100,000 cycles. However, onsets of disbonds between the transducer and the copper layer electrode found from fractographic examinations at below 50,000 fatigue cycles were not detected with this method. Other nondestructive testing techniques such as C-scan imaging by ultrasonic testing, radiography and thermography can also be used to provide further information regarding surface and internal damages surrounding the integrated transducers [[128],[129]].

Modelling Approaches

Crawley and Luis [[71]] produced one of the earliest known works in developing static and dynamic analytical models for a piezoelectric transducer pair that are located symmetrically about the structure neutral axis either bonded to an elastic beam or embedded in a laminated composite beam. The beam thickness in axial extension was assumed in uniform strain while the beam thickness in bending was assumed in linear strain distribution, following the one-dimensional Euler-Bernoulli beam model. A shear lag parameter, L , exists in the model due to imperfect strain transfer in a finite thickness of adhesive layer. It was found from the governing equations for piezoelectric strain that L is important to determine the behaviour and stiffness of the adhesive layer:

$$\begin{aligned}\frac{\varepsilon_p}{\varepsilon_i} &= \frac{\alpha E_p t_p}{E_b t_b + \alpha E_p t_p} \left(1 + \frac{E_b t_b \cosh L\bar{x}}{\alpha E_p t_p \cosh L} \right) \\ \frac{\varepsilon_b}{\varepsilon_i} &= \frac{\alpha E_p t_p}{E_b t_b + \alpha E_p t_p} \left(1 - \frac{\cosh L\bar{x}}{\cosh L} \right) \\ \frac{\gamma_a}{\varepsilon_i} &= -\frac{t_p}{t_a} \left(\frac{\sinh L\bar{x}}{L \cosh L} \right)\end{aligned}$$

where ε is the normal strain, γ is the shear strain, E is the Young's Modulus, t is the thickness, $\bar{x}$ is the non-dimensionalised coordinate along the length of the transducer, and α is 2 for extension and 6 for bending. Subscripts i , p and b refers to the induced strain, piezotransducer, and beam, respectively. Crawley and Anderson [[72]] examined the formulations of these models and found that the analytical solutions satisfactorily matched the results from experiments and a finite element model, which improved with increasing transducer-to-beam thickness ratio above unity. The Euler-Bernoulli model was extended by Hagood et al. [[130]] to include coupling between the electrical and mechanical components of the smart beam structure but only for surface bonded transducers. Coupled extension, bending and torsion loading conditions for single surface bonded transducers in beams were also considered in deriving the analytical models [[74],[75],[131]]. Molyet et al. [[76]] developed a two-dimensional model for the beam structure using the finite difference method and results obtained were used to identify the limitations of the one-dimensional analytical models.

Modelling for two-dimensional piezoceramic actuation in plates has also been studied extensively in literature. Analogous to the Euler-Bernoulli model for beams is the plate solution developed from the popular classical lamination theory [[132]-[134]]. Hong and Chopra [[136]] modified this formulation for a finite element model by including transverse shear effects with the analysis of coupled extension-twist and bending-twist loading. Another modelling approach was inspired by the laminate plate theory presented by Reddy [[137]] where the shear distribution was analysed separately in each lamina layer. This layer-wise theory is particularly suitable for analysing a piezoceramic sheet embedded in the composite, as shown by Mitchell and Reddy [[138]], where effects of coupling between mechanical deformation and electrostatic charge were included. References also show the consideration of coupling effects between thermal and mechanical loads [[139]-[141]] as well as dynamic

coupling between the transducer and the host structure [[142],[143]]. Higher order plate theories and three-dimensional models for more accurate simulation of the strain transfer have also been studied [[144]-[146]].

The finite element method is an important computational tool for generating simulations based on two- and three-dimensional models while validating analytical solutions and experimental results for integrated transducer systems. One of the earliest efforts to apply a finite element analysis in a smart structure was by Tzou and Tseng [[147]] where thin piezoelectric solid elements with individual internal degrees of freedom was applied for a vibration control application. Ha et al. [[148]] then presented a finite element model for a laminated plate integrated with piezoceramic transducers using a three-dimensional eight-node composite brick element. Shell, plate and beam elements were also used in modelling where a complete list of references using the different type of elements before the 21st century can be sourced from the survey conducted by Benjeddou [[149]]. The classical lamination theory was also explicitly applied in deriving finite element models in early works by Hwang and Park [[150]], and Lam et al. [[151]]. For the layer-wise shear deformation theory, the model was pioneered by Robbins and Reddy [[73]]. The efficiency of the model was later improved by incorporating variable kinematics and mesh superposition in the adhesive layer [[152]]. Reddy [[153]] then refined the formulations of the finite element models for both the classical laminate and layer-wise shear plate theories. Significant contributions in finite element modelling for embedded piezotransducers were presented by Chandrashekhara and Agarwal [[154]], and Saravanos and Heyliger [[155]]. Further developments in finite element modelling for integrated transducer technology can be referred from the bibliography compiled by Mackerle [[156],[157]].

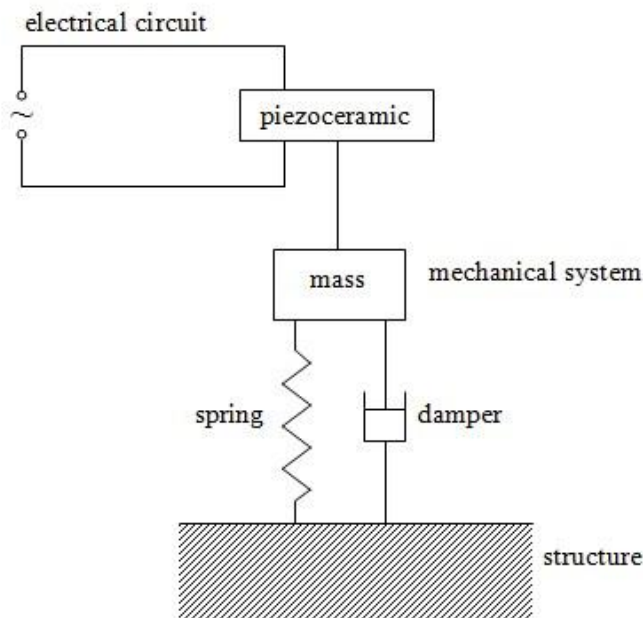


Figure 5. The electromechanical impedance model of a piezoceramic transducer.

Another important simulation method discussed in literature is the electromechanical impedance model, which describes the dynamic interaction between the piezoceramic transducer and the host structure based on electrical and mechanical impedance analysis for surface mounted transducers operating at high ultrasonic frequency regimes. The impedance model was first proposed by Liang et al. [[121]] as a one-dimensional simple mass-spring-damper system coupled to an electrical system to represent the bonded transducer on a substrate. The model is illustrated as in Figure , and according to Giurgiutiu et al. [[158]], the impedance, $Z(\omega)$ can be simply evaluated as follows:

$$Z(\omega) = \left[i\omega C \left(1 - \kappa_{31}^2 \frac{Z_s(\omega)}{Z_p(\omega) + Z_s(\omega)} \right) \right]^{-1}$$

where ω is the circular frequency, C is the zero-load capacitance, κ_{31} is the electromechanical cross coupling coefficient, and subscripts s and p denote the smart structure and the piezoelectric element, respectively. Zhou et al. [[159]] continued to develop the model by introducing a two-dimensional system. Later, Xu and Liu [[160]] included an additional degree of freedom into the model by taking into account the effect of the bonding layer. This was achieved by adding infinitesimal-length springs to represent the bond layer to model the joint between the transducer and the base structure. Thick adhesive layers can influence the measured impedance signal [[80]], hence Madhav and Soh [[95]] have taken this into account by formulating a three-dimensional interaction model that considered the mass of both the piezoceramic and the adhesive.

CONCLUSION

A comprehensive selection process needs to be undertaken to ensure for a feasible and effective design. The manufacturing concepts applied have to take into account the application and requirements of the host structure while producing a safe and reliable design. This includes the matching of CTE, temperature considerations, geometry, size and configuration of integrated transducers, electrical insulation, protection from damage, and wiring fabrication. For surface mounted piezoceramics, the influence of the adhesive layer needs to be carefully investigated to obtain the required strain transfer in actuating and sensing while reducing the likelihood for disbonds and transducer failure. In embedded piezoceramics, the embedding technique selected needs to ensure that the inclusion and connectors does not affect the strength and fatigue life of the structure past its design limit without compromising the integrity and durability of the load bearing transducer.

Testing of smart structures with integrated piezoceramics can be performed through a number of static and dynamic mechanical methods viz. tension and three/four-point bending tests depending on the expected loading conditions. Apart from mechanical loads, electrical and thermal effects can be included in the tests. Fractography and microscopy are techniques to examine the cross-section of the smart structure to identify the source of failure, consequently facilitating improvements in the design. Transducer performance can be gauged by monitoring the change in output voltage, impedance and ultrasonic signals from Lamb

waves and AE. Impedance measurements are also especially useful for assessing the bond quality of attached transducers. Test results shown in literature typically suggest that tested smart structure designs with integrated transducers did not report significant reduction in strength whilst the fatigue life was not affected at least up to 1000 microstrains of cyclic loading. When there was a decrease in the measured transducer performance, this was mainly attributed to the disbond at the transducer-substrate interface.

Modelling approaches in literature presented developments in either beam or plate theories with modifications to incorporate the integrated transducer. For beams, the uniform strain can be assumed for extensional loads while the Euler-Bernoulli theory can be used to model bending loads. In plates, models are commonly derived from either the classical laminate theory or the layer-wise shear deformation theory. There are continuing efforts to refine the models from more realistic and complex loadings by taking into account the coupling effects from mechanical, piezoelectric and thermal stresses. The electromechanical impedance method is an alternative analytical approach to specifically model for surface bonded transducers operating at high ultrasonic frequencies. Finite element methods are important computational tools in these modelling approaches for simulating complex two/three-dimensional systems, which can be applied to validate analytical solutions and experimental results.

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Chapter 5

PIEZO AND PYROELECTRIC COMPOSITE FILM FOR ACOUSTIC EMISSION AND X-RAY RADIATION INTENSITY DETECTION

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ABSTRACT

The continuous technological advances have required materials with properties that conventional material cannot display. Material property combinations are being the focus to the development of composite materials, which are considered a multiphase material that exhibits properties of the constituent phases. One interesting material to be studied as sensing material is the composite made of ferroelectric ceramic and polymeric matrix as a two phases composite material. In that case the combinations properties intended are the high piezo and pyroelectric activities of the dense ceramic with the impact resistant, flexibility, formability and low densities of the polymer. Considering the way that the ceramic is dispersed in the polymeric phase, according to Newnham [1] there are 10 possible connectivities, which indicate the way that each phase is self-connected in the composite. The composite studied here is made of a modified lead titanate (Pz34) ceramic and polyether-ether-ketone (PEEK) high performance polymer matrix. The composite film was obtained hot pressing both phases at 360 °C during 2 h, and 12 MPa of pressure, being essentially a 0-3 composite that means the dispersed ceramic grains are

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not self-connected. Using the piezoelectric property the composite film was used to detect acoustic emission (AE), which is a transient elastic wave generated by sudden deformation taken place in materials under stress. AE can be applied for evaluating the “health” of structures in a non-destructive way. The composite film was surface mounted on an aluminum panel and ball bearing drop and pencil lead break were used as simulating AE sources. On the pyroelectric side, the composite film was used as sensing element in a pyroelectric chamber to measure X-ray intensity in the orthovoltage range ($120 \text{ KV}_p - 300 \text{ KV}_p$). Experimental results indicate that the combination properties of ceramic and polymer phases provide an alternative sensing material to be used as piezo and pyroelectric sensor.

INTRODUCTION

Ferroelectric ceramics have been used for piezo and pyroelectric applications for many years. Using the piezo activity they have been used as sensing elements of hydrophones [2,3], mechanical power sources and shock sensors [4]. Also, the application can include the detection of acoustic emission (AE), which is a transient elastic wave generated in damaged mechanical structure. Due to the wide range of energy levels covered by AE signals, the sensor performance is evaluated using a plate-like specimen. The propagating waves can be analyzed with Lamb waves theory because the thickness is much smaller than the other two dimension in the plate-like geometry [5,6].

Some properties of ferroelectric ceramics, such as the high degree of stiffness which makes them fragile, brings limitations for these sensors and flexible piezo polymers were considered an alternative material [7,8], but its low melting point also becomes a problem. To overcome these problems, composite made of ferroelectric ceramic and polymer matrix with high melting point have been considered [9].

Due to the pyroelectric activity ferroelectric ceramics have application as infrared detector [10,11,12] and imaging device [13,14,15]. That way, ceramic/polymer composites also have been used for thermal detection and several papers, using different polymer matrix describing the pyroelectric activity of these composite can be found [16,17,18]. As thermal transducer the pyroelectric composite can also detect X-ray radiation [12].

In this work, a ferroelectric ceramic/polymer composite is presented as an alternative piezo and pyroelectric detector to sensing AE emission and measuring X-ray intensity in the orthovoltage range ($120 - 300 \text{ kV}_p$), respectively. The composite material combines good properties of each single material, offering some advantages in comparison with single crystalline materials or ferroelectric perovskite ceramics.

AE Sensors

An acoustic emission sensor senses the dynamic stress waves propagating through a structure. These waves were generated by the release of energy due to a sudden internal stress caused by some failure mechanism such as: crack, fatigue, delamination, etc. [19]. So, AE technique is a good tool for non-destructive evaluation (NDE) and the main difference between this technique and other NDE methods is that AE can monitor the activities inside

the materials while other NDE methods attempt to examine the internal structures of the materials.

For geometrically plate-like test specimen the plate theory can be used to analyze the sensor response. In this case the propagating waves are called plate waves and there are two modes of propagation, i.e., extensional and flexural. The extensional mode propagates with higher frequency and faster than the flexural mode [20]. Surface mounted sensor can detect these waves, which can be observed and recorded directly by a digital storage oscilloscope.

AE as non-destructive technique to monitor the condition of structures is very important not only on the economy point of view by reducing the costs of maintenance but also because this technology can detect structural defects before possible catastrophic failures. A comprehensive review on AE technique was recently reported by Jayakumar [21] showing the large range of application of this useful tool for monitoring various forming process.

X-Ray Intensity Detector

Cancer has been the worst health problem worldwide and radiation therapy is one of the most effective methods in cancer treatment. The dose radiation must be precisely and accurately delivered to avoid side effects as well as recurrence of the disease. According to Cameron [22] small exposure to radiation can reduce the risk of disease, but it is well known that high dose of radiation may damage healthy tissues, hence becoming very tragic. Determining the values of various dose quantities for use in research, in therapy and in a dose limiting system for protection is the final purpose of radiation dosimetry [23].

Among the commonly used dosimeters in radiation therapy, the ionization chamber is one of the most accurate and is considered as the gold standard for radiation dosimetry. However, there are some limitations related to the best spatial resolution and the signal-to-noise ratio [24]. Furthermore, ionization chambers are expensive, which is also a limitation.

Radiographic films and thermo luminescent materials have also been used for dosimetry of ionization radiation but these dosimeters do not provide a direct reading of the dose and the useful information about the dose only can be obtained after tedious set up procedures and further treatment after irradiation [25].

Semiconductor silicon diodes and solid-state scintillation detectors are widely used in dosimeters and in technological equipment for medical diagnosis. However the small area of the photosensitive surface of the photodiode is one of the major factors limiting the use of these detectors [26].

In 1938 Ta [27] proposed the idea of using the pyroelectric effect to detect radiation. A pyroelectric detector is usually a thin wafer of a ferroelectric material oriented with the electrode surface normal to its polarization vector. When subjected to radiation capable of transferring energy, its temperature increases, altering its polarization and causing a change in the surface charge across a parallel plate capacitor [28].

The pyroelectric detector differs from typical radiation detectors such as an ionization chamber, TLD, semiconductor, etc. in the following characteristics: 1) it is uniformly sensitive to a wide spectrum of radiation; b) it responds almost instantaneously with any change in radiation flux; c) it responds linearly with radiation intensity when the radiation is completely absorbed by the detector.

Material such as triglycine sulphate (TGS), lithium niobate, lithium tantalate, and lead zirconate titanate (PZT) have been used as pyroelectric elements in many devices. Radiant energy, both ionizing (X-ray, protons, neutrons) and non-ionizing (infrared, microwaves, UV) have been monitored by these detectors [29,30,31,32]

Recently, De Paula et. al. [33] proposed a microcontrolled pyroelectric instrument for measuring X-ray intensity as an alternative to the expensive ionization chamber. The authors have used lithium tantalate (LiTaO_3) and lithium niobate (LiNbO_3) ferroelectric crystals as detectors for monitoring X-ray intensity in the mammography X-ray range.

Experimental

Polymer matrix

Polyether-ether-ketone (PEEK) is an aromatic polyether. Because of characteristics such as mechanical strength, glass transition temperature around 143 °C, resistance to a large range of chemicals, PEEK is considered as high temperature engineering thermoplastic [34,35]. On the dielectric side, there is no relaxation in the frequency range of 1.0 Hz to 10^5 Hz and the relative permittivity is around 3.5 [36]. PEEK powder with mean particle size of 80 μm was purchased from Goodfellow.

Ferroelectric ceramic

Modified lead titanate (Pz34) powder, purchased from Ferroperm Piezoceramics A/S, exhibits large electromechanical anisotropy and low dielectric constant. Table I shows some properties of Pz34 [37]. The mean grain size of the ceramic is 2 μm .

Composite preparation

Pz34/PEEK composite films were obtained by pressing the mixture of both powders at 368 °C for 2 h. Samples in the thickness range of 150 – 300 μm were obtained after applying a pressure of 12 MPa. The ceramic volume fraction used to prepare the composite film was 40 vol.%, 50 vol.% and 60 vol.%. The amount of Pz34 and PEEK was calculated using the equation:

$$M_c = M_p \frac{\rho_c}{\rho_p} \left(\frac{\Phi_c}{1 - \Phi_c} \right) \quad (1)$$

where, M is the mass, ρ is the density and the subscripts p and c are related with polymer and ceramic, respectively. Φ_c denotes the volume fraction of ceramic.

Circular aluminum electrodes (1.0 cm diameter) were evaporated onto both sides of the samples. Poling of the films were performed by applying a suitable electric field across the thickness during 1.0 h at 130 °C. Due to the high ceramic content the composite films obtained could possess a mixed connectivity, i.e., 0-3 and 1-3 [38].

Surface mounted sensors

Composite films with three different compositions were surface mounted on 30 x 30 cm aluminum panel. Silver conductive adhesive fluid (Alfa Aesar) was used to bond the composite film on the panel. Two simulated AE sources were used to observe the responses of surface-mounted sensors. One source consists of the drop of a steel ball on the surface of the panel. With this experiment the behavior of the sensor with the impact energy and the distance of the source can be analyzed. Other source is the Hsu-Nielson source [39], which uses the break of a pencil lead above the panel surface.

Radiation detection system

Figure 1 shows the pyroelectric chamber constructed to measure X-ray intensity. The prototype instrument consists of a pyroelectric sensor (composite), a low-noise high sensitivity current-to-voltage converter, a microcontroller and a digital display.

The main component of the converter is an operational amplifier OPA 111 (Burr-Brown), which has very high input impedance ($10^{14} \Omega/3 \text{ pF}$), low input bias (0.8 pA) and offset currents (0.5 pF). In order to minimize noise from external sources, the current-to-voltage converter was placed inside the pyroelectric chamber, which is a grounded aluminum enclosure. Using guard-rings around the input pins of the operational amplifier minimized leakage currents across the fiberglass printed circuit board. The guard-ring acts also as shield against noise [40].

To protect the electronic circuit from radiation, it was placed inside a compartment in the pyroelectric chamber that was shielded with a 7 mm thick alloy (cerobend 70) composed of 50% of bismuth, 27% of lead, 13% of tin, and 10% of cadmium. The signals in the output of the converter are noise filtered and then fed into the microcontroller acquisition section. The microcontroller (Microchip PIC 16F777) sends the digital converted analogical signal to a display. Figure 2 shows the lateral view of the pyroelectric chamber and a block diagram of the radiation detector system.

The irradiations were carried out using an orthovoltage X-ray equipment (Siemens Stabilipan II).

Table II shows the filters used for each tube potential. Equivalent photon energy of the beam was determined from its measured half-value layer (HVL). The pyroelectric sensor was placed at 50 cm from the X-ray source, which is the commonly used distance in patients during treatment.

The energy fluence rate (Ψ) of the beam was calculated from the measured X-ray exposure rate ($\dot{X}$) and air energy absorption coefficient ($(\mu_{ab}/\rho)_{air}$) using the equation [28]:

$$\Psi = \dot{X} \frac{8.73 \times 10^{-3}}{(\mu_{ab} / \rho)_{air}} \text{ (W/m}^2\text{)} \quad (2)$$

where $\dot{X}$ was measured with a calibrated 0.6 cm³ cylindrical ionization chamber, connected to a FARMER model 2570 electrometer designed for clinical use in the orthovoltage range.

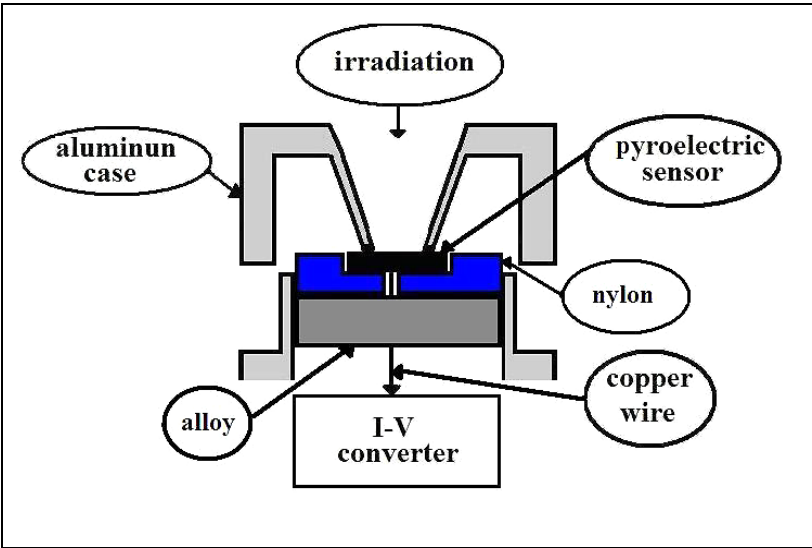


(a)

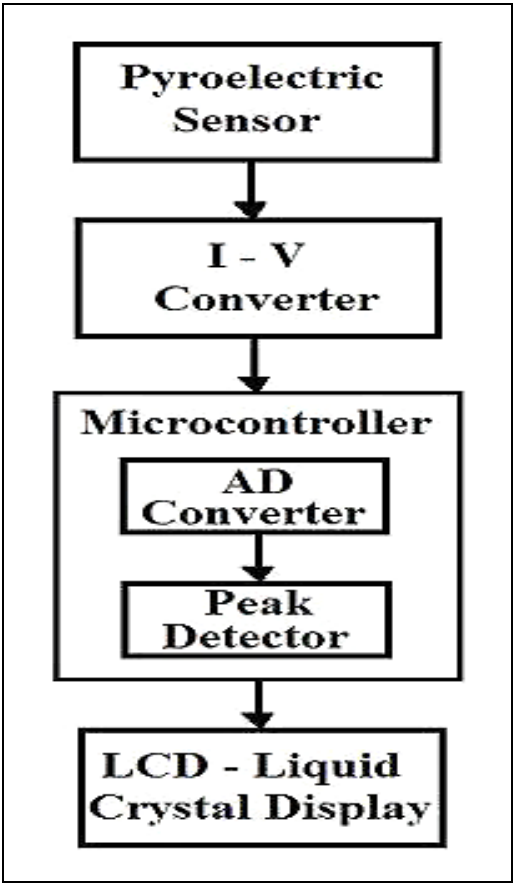


(b)

Figure 1. Photograph of the pyroelectric chamber (a) closed, (b) showing the composite sensor.



(a)



(b)

Figure 2. (a) Lateral view of pyroelectric chamber. (b) Block diagram of the final circuit.

Results and Discussion

Pz34/PEEK sensors in three different compositions of ceramic were surface-mounted on an aluminum panel. The panel was excited using two different simulated AE sources: ball bearing drop, which produces waves with large amplitude and low frequency and pencil lead break (Pentel 2H), the so-called Hsu-Nielson source, producing low amplitude and high frequency waves. The acoustic impedance of the sensor 40/60, 50/50 and 60/40 vol.% are 7.2, 9.0 and 9.8 (Pas/m³), respectively calculated using the longitudinal wave velocity of the composite films measured using the ultrasonic immersion technique [41]. The receiving sensitivity g_{33} of the sensors are 66.2, 110.2 and 133.8, respectively for the above composite compositions.

The sensor response with the impact energy and with the distance from the AE source was obtained by dropping a steel ball of 3.5 g and assuming the noise level equal to 2.0 mV_{pp}. Figures 3 and 4 show the results. For different energies the AE source distance was kept constant at 6 cm and for different distances the fixed impact energy was 1.75 mJ. The plotted values are the average values after 5 measurements. It can be seen from Figure 3 that there is no linearity in the sensor response with the increasing impact energy neither with the ceramic content. From Figure 4 it can be observed that even for lower Pz34 content the sensor can detect the stress wave 12 cm far from the AE source without any amplification of the signal response. In spite of the lower piezoelectric coefficient displayed by the composite in comparison with ferroelectric ceramic, its higher sensitivity parameter g_{33} due to lower dielectric constant gives to the composite film some advantages to use it as surface mounted sensor.

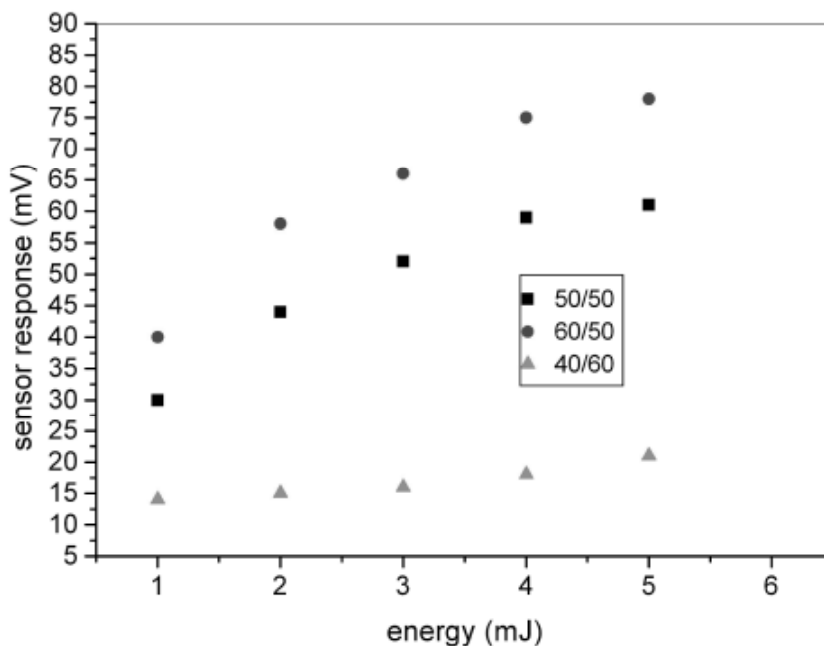


Figure 3. Sensor response to simulated AE source for different energies.

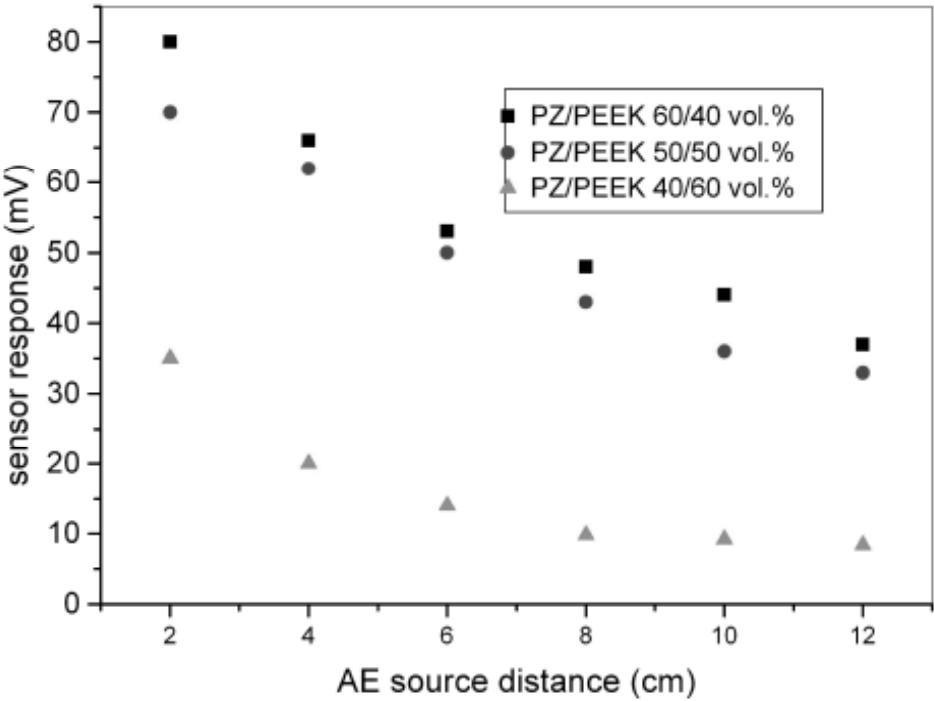


Figure 4. Response of the sensors at different distance from AE source.

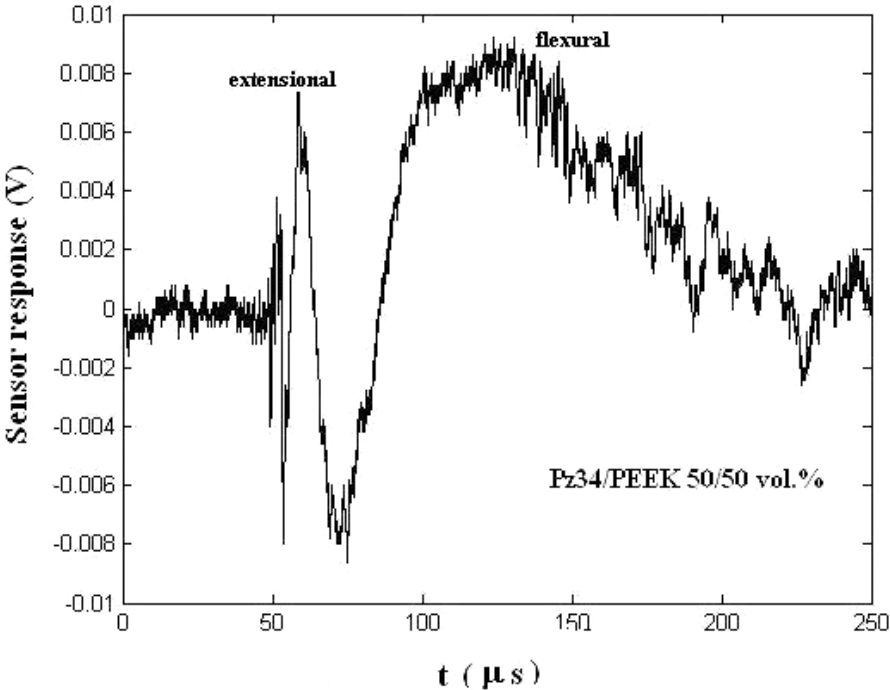


Figure 5. Time response of Pz34/PEEK 50/50 vol% sensor to a lead break test.

Figure 5 shows the time response of the Pz34/PEEK sensor to the lead break test. The pencil was broken on panel surface at about 6 cm from the sensor and the signals were recorded directly by a digital storage oscilloscope (Tektronix TDS 2022) without amplification. Clearly it can be seen both plate waves modes: extensional around 55 μs and the flexural at 120 μs. The sensor composition used for this test was 50 vol.% of ceramic because of the higher signal response.

Pz34/PEEK composite film with 50 vol. % of ceramic was poled at 130 °C for 1 h. By varying the applied electric field, the poling efficiency could be analyzed. To avoid breakdown, the polarization was carried out in a silicone oil bath. and a piezoelectric constant of 29 pC/N was obtained.

On the pyroelectric side, the 60 vol.% pre-poled sample with 15 MV/m was placed in a temperature-controlled chamber and the thermally stimulated discharge current (TSDC) was measured. The sample was heated at constant rate (1 °C/min) with its electrodes short-circuited and the current was monitored with an electrometer (Keithley model 610 C). The reversible pyroelectric current was established after removing all contributions to the irreversible TSDC current, which occurred after 3rd TSDC run.

The pyroelectric coefficient $p(T)$ as a function of temperature was calculated using the equation:

$$p(T) = \frac{1}{A} \frac{I_p}{dT/dt}$$

(3)

where A is the electroded area, I_p is the pyroelectric current and dT/dt the heating rate. At room temperature the value of the pyroelectric coefficient is 19 μC/m²K and the pyroelectric figure of merit (FOM) is 0.43 μC/m²K. No appreciable change in the piezo and pyroelectric activity was observed after the irradiation (0.2 C/kg ≈ 760 Roentgens), indicating that there is no degradation in the sensing element in the energy range used.

Table 1. Values of density, piezoelectric coefficient, dielectric constant and Curie temperature

Material	ρ (g/cm ³)	d ₃₃ (pC/N)	ε _r (1 kHz)	T _c (°C)
Pz34	7.55	50	210	400

Table 2. Filters used for each tube potential

Tube Potential (kV)	120	180	200	250	300
Filter	2 mm Al	4 mm Al	0.2 mm Cu	0.5 mm Cu	ThIII*

ThIII* = 1mm Sn + 0.25 mm Cu + 1 mm Al

To measure the absorbed energy by the sensor materials, a 7 mm thick lead foil with a 7 mm diameter hole was placed 50 cm from the X-ray tube. The exposure was measured with and without the sensors covering the hole. Table III shows the absorbed energy percentage of Pz34 ceramic, LiNbO₃ crystal and Pz34/PEEK composite for all tube potential ranges.

Table 3. Energy absorption by the sensor material

Material	Tube Potential (kV _p)	Exposure without sensor (R/min)	Exposure with sensor (R/min)	Absorption (%)
LiNbO ₃	120	7.58	1.22	83.91
Pz34	120	7.58	0.32	95.78
Pz34/PEEK	120	7.58	1.56	79.42
LiNbO ₃	180	13.44	4.62	65.63
Pz34	180	13.44	1.12	91.67
Pz34/PEEK	180	13.44	4.62	65.63
LiNbO ₃	200	10.06	5.00	50.30
Pz34	200	10.06	1.34	86.68
Pz34/PEEK	200	10.06	4.50	55.27
LiNbO ₃	250	10.64	7.14	32.89
Pz34	250	10.64	2.58	75.75
Pz34/PEEK	250	10.64	6.12	42.48
LiNbO ₃	300	4.84	4.02	16.94
Pz34	300	4.84	1.92	60.33
Pz34/PEEK	300	4.84	3.48	28.10

Table 4. Parameters for standard measure of radiation. Al and Cu are filters

Tube potential (kV _p)	HVL (mm)	Energy of Photon (keV)	μ_{ab}/ρ (m ² /kg)	Exposure rate (R/min)	Φ (W/m ²)
120	3.00 (Al)	34.69	0.0626	45.20	6.30
180	0.60 (Cu)	67.13	0.1030	82.50	6.99
200	1.10 (Cu)	83.67	0.0511	64.10	10.95
250	2.10 (Cu)	118.68	0.0223	66.80	26.10
300	4.20 (Cu)	178.19	0.0078	33.10	37.19

It can be seen from the data in Table III that the radiation was not completely absorbed by the sensor materials. The thickness of the Pz34 ceramic and the LiNbO₃ crystal were 1 mm.

Table IV shows the half-value layer (HVL), energy of the photons, absorption coefficient of mass energy (μ_{ab}/ρ), exposure rate and the energy fluence rate (Φ), which are parameters for standard measure of radiation. The half-value layer is related with the interaction probability (μ) of the photon in the thickness unit of the material. The rate μ_{ab}/ρ is usually called cross section.

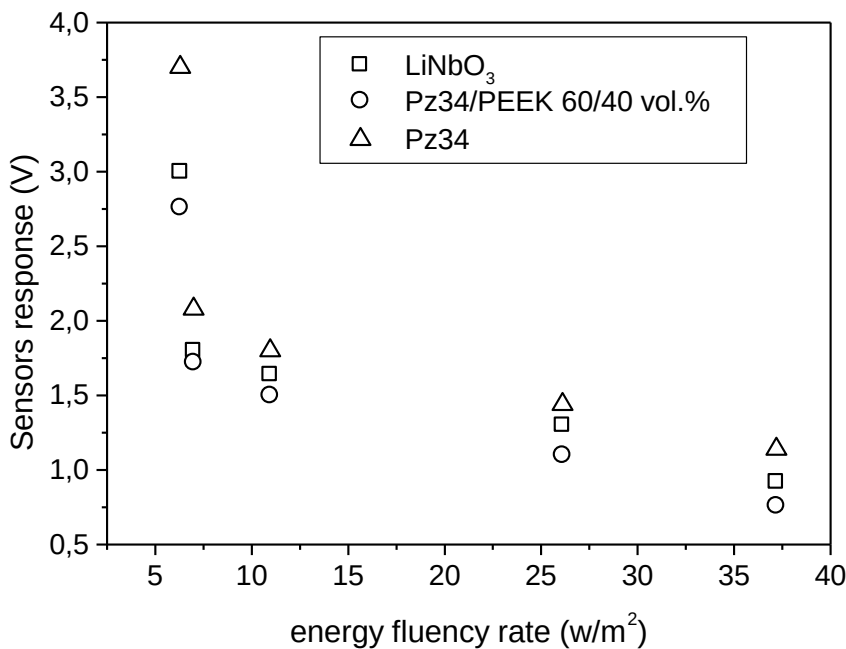


Figure 6. Sensors response to X-ray intensity in the orthovoltage range.

The behavior of the sensor response as a function of energy fluence rate or X-ray intensity is shown in Figure 6. For comparison the responses of Pz34 ferroelectric ceramic and LiNbO₃ crystal are also plotted.

The microcontroller is interfaced through a serial port to receive commands from and send data to a computer. The X-ray intensity was calculated by dividing energy fluence by X-ray exposure time. The exposure was measured with a Radcal system, model 9015 (Monrovia, CA, USA).

It can be seen from Figure 6 that for commonly energy used in radiation therapy (120 kV_p), the response of all sensing materials is higher. This behavior might be related with the energy absorption, which are 95.78 %, 83.91 % and 79.78 % for Pz34, LiNbO₃ and Pz34/PEEK, respectively. As the X-ray intensity increases, the absorption decreases and the sensor responses also decrease but the amplitude is sufficient to be accurately measured. The values presented in Figure 6 are the average values of 5 measurements and the repeatability was observed.

Using the composite material with 60 vol. % of ceramic we implemented a pyroelectric system to measure X-ray intensity in the orthovoltage range. In spite of the lower radiation absorption by the composite in comparison with Pz34 ceramic or LiNbO₃ crystal, the amplitude of its response is very similar.

Although the relationship between the response and the energy fluence rate is not linear, there is no double value of voltage for different energies, indicating that the composite Pz34/PEEK can be used as sensing material for radiation detection. Research is in progress in our lab to show that the composite film can be also used for lower energy range and for non-ionizing radiation as infrared.

CONCLUSIONS

Flexible composite films were fabricated as piezo and pyroelectric sensors. It was surface mounted on an aluminum panel and was able to detect stress waves provided by two simulated AE sources. Based on the ball bearing drop test, impacts with energy lower than 1.0 mJ can be detected even by sensor placed 12 cm far from the source and with lower ceramic content. In the Hsu-Nielson test both modes of waves propagation could be clearly observed without any amplification.

As pyroelectric sensor, a new instrument using a pyroelectric composite made of Pz34 ceramic and PEEK polymer matrix was described as an alternative system to an expensive ionization chamber for dosimetry. The instrument measures X-ray intensity in the orthovoltage range directly and can accurately monitor the application of radiation dose therapy. The response of the sensing element was not linear with the radiation intensity between 120 kV_p to 300kV_p but it does not become a problem because the instrument is microcontrolled. A linear relationship can be observed from 180 kV_p to 300 kV_p.

The experimental results show that no degradation was observed in the composite material after irradiation. The pyroelectric coefficient and the piezoelectric constant d_{33} and the pyroelectric coefficient remained practically the same after the sensor received a high exposure (about 0.2 C/kg = 760 Roentgens) of radiation in the orthovoltage range. In the energy fluence range of 6.30 to 37.19 W/m², the sensor response varies from 2.70 V to 0.80 V without double values of voltage for different energies, at least in the range used. Research is in progress to show that this composite material can also be used in the mammography x-ray range.

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Chapter 6

PIEZOELECTRIC FLOATING MASS TRANSDUCER FOR IMPLANTABLE MIDDLE EAR HEARING DEVICES

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ABSTRACT

A simple piezoelectric floating mass transducer (PFMT) for implantable middle ear hearing devices (IMEHDs) is studied. The transducer that is implanted in the middle ear consists of a PMN-PT multi-layered piezoelectric actuator and a metal case. The proposed PFMT can be attached to the long process of the incus using a metal clamp, and vibrates the ossicular chain according to the force originating from the expansion and contraction of the piezoelectric actuator. This chapter presents the design and an evaluation of the proposed PFMT as a new type of vibrating transducer. The design factors considered include the dimensions, generated force, weight, power consumption, and time-frequency characteristics. An *in vitro* experiment using a human temporal bone confirms that the PFMT excitation at 10 V_{pp} can generate vibration displacements equivalent to a sound pressure of about 94 dB SPL below 1 kHz and about 140 dB SPL at 8 kHz. The power consumption of the PFMT to produce a sound stimulation equivalent to 94 dB SPL is 65 μW at 1 kHz. Therefore, the proposed PFMT is expected to be useful as an IMEHD transducer with the advantages of simple surgical implantation, no interference from external electromagnetic fields, and improved productivity.

INTRODUCTION

Sensorineural hearing loss can result from a loss of function in the inner ear or auditory nerve and is the main cause of deafness in adults. As the hearing impaired population continues to increase due to our aging society and the influence of modern industrialization, convenient and comfortable hearing aids that can compensate for sensorineural hearing loss are now in demand all over the world. Several types of hearing aid are already available for treating sensorineural hearing loss, including conventional air conduction hearing aids, bone anchored hearing aids, and implantable middle ear hearing devices (IMEHDs). A cochlear implant is a surgically implanted electronic device that provides a sense of sound to a person who is profoundly deaf or severely hard of hearing and involves careful training after the implant surgery. In the case of IMEHDs, they directly stimulate the ossicular chain in the middle ear using a miniaturized vibration actuator attached to the ossicle and have been investigated since the 1970s as a new type of hearing aid to solve the problems of conventional air-conduction hearing aids [1-10].

IMEHDs can generally be divided into three parts: an implantable microphone, signal processor, and vibrating transducer, where the last is the most important component that directly stimulates the ossicular chain and can be divided into two types: piezoelectric and electromagnetic. Piezoelectric vibrators have already been studied by Yanagihara and Gyo [2-3] and Zenner and Leysieffer [4-5], resulting in better frequency characteristics, a lower power consumption, and higher sound fidelity than electromagnetic vibrators. However, the fixing plate used to implant a piezoelectric vibrator has to be installed at a specific position on the temporal bone using fixing screws. Plus, the vibrator tip must make precise contact with the ossicle surface with an appropriate pressure. Therefore, these systems require a surgical procedure, including a mastoidectomy and another form of manipulation of the ossicular chain, resulting in average surgical times of 2 to 5 hours and the requirement of experienced otologists [1].

A floating mass transducer (FMT), which is the only device currently available in the United States with FDA approval, is composed of one permanent magnet and two coils, and is simply attached with a titanium clamp to the incus through a facial recess approach [6]. However, despite the straightforwardness of its surgical procedure, an FMT has a potential safety problem due to interference from strong environmental electromagnetic fields, such as MRI [7]. Therefore, a differential floating mass transducer (DFMT) was proposed by Cho et al., which is minimally affected by environmental magnetic fields [8-9]. Yet, the fabrication process for this electromagnetic transducer is complex, consisting of about 10 components within a very small cylinder.

Accordingly, this chapter introduces a piezoelectric floating mass transducer (PFMT) with the advantages of conventional piezoelectric transducers, and demonstrates the feasibility of a PFMT with a multi-layered piezoelectric actuator. Even though the proposed PFMT is a floating mass type transducer, it does not suffer interference from electromagnetic fields. In addition, the PFMT can be attached to the long process of the incus using a similar surgical procedure to that used for a FMT. Finally, the PFMT is suitable for mass production, as it only consists of a multi-layered piezoelectric actuator and metal case. As the PFMT utilizes a multi-layered piezoelectric actuator that shows a flat frequency response for the

human audible frequency range and a voltage-displacement linearity [10], it displays good high-frequency vibration characteristics.

The important design considerations for a good transducer for IMEHDs include a suitable size for attachment to the ossicle, the force of the transducer, time and frequency characteristics, and power consumption. The vibration characteristics of the proposed PFMT are also measured and an *in vitro* experiment carried out using a human temporal bone in order to produce data under realistic mechanical loading conditions. The results are then compared with previously reported data for other hearing implants to determine the feasibility of the PFMT for use in implantable hearing aids.

DESIGN AND IMPLEMENTATION OF PFMT

Structure of PFMT

Several configurations of piezoelectric transducers for IMEHDs have already been proposed and investigated [1-5, 10]. However, while most have a simple construction and wide frequency characteristics, all require complex surgical procedures. Therefore, a piezoelectric floating mass type transducer is needed with a simple surgical procedure, similar to that used by an FMT. As such, the proposed PFMT is clamped to the long process of the incus, as illustrated in Figure 1, and positioned so that the center axis of the cylinder is aligned with the natural motion of the ossicular chain and vibrates the ossicular chain according to the force originating from the piezoelectric actuator. The proposed PFMT consists of a metal case and multi-layered piezoelectric actuator, as shown in Figure 1 (b), where one side of the piezoelectric actuator is fixed to the metal case, while the other side is connected to a rigid metal clamp that is used to attach the PFMT to the long process of the incus. The ossicle is then vibrated by the expansion and contraction of the piezoelectric actuator according to the applied voltage.

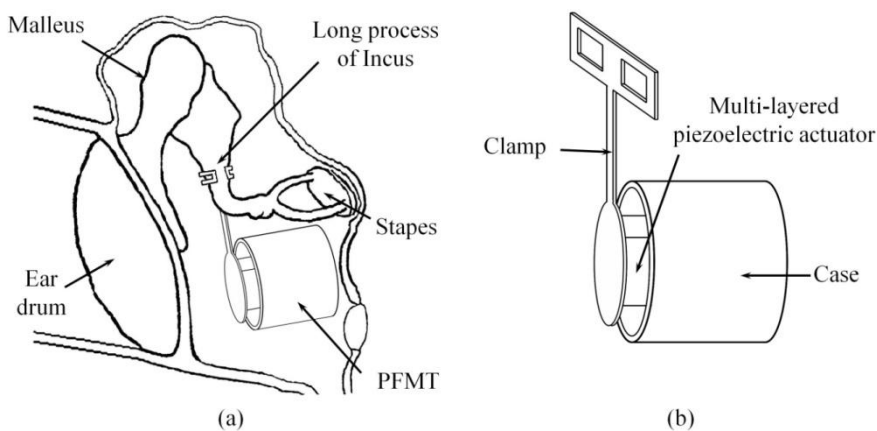


Figure 1. Structure of PFMT: (a) Schematic of PFMT for IMEHDs and (b) shape of PFMT.

When the PFMT is attached to the ossicle, the relationship between the metal case and the ossicular chain can be regarded as the movement of two objects ruled by the pair of action-reaction forces on both sides of the piezoelectric actuator. Thus, if the metal case is very heavy, all the displacement from the actuator is perfectly transferred to the incus side. However, since the ligaments of the incus are disregarded and the case mass is equal to the incus mass, the vibration displacement of the incus and the case is actually equal and the directions of the displacements are opposite.

Due to the limited space in the middle ear cavity, a multi-layered piezoelectric actuator is adopted to multiply the vibration force. In a multi-layered actuator, the displacement, ΔT , and driving force, F_p , produced from an actuator with n layers excited by the applied voltage, V are given by

$$\Delta T = nd_{33}V \text{ and } F_p = k_T\Delta T \tag{1}$$

where d_{33} is the strain coefficient [m/V] and k_T is the stiffness [N/m] of the piezoelectric material [10]. Piezoelectric actuators are traditionally characterized as high-voltage and high-impedance devices. However, large displacements can be generated at relatively low voltages in a multi-layered actuator, as the vibration displacement is the sum of the displacements of the individual layers, as expressed in equation (1).

This study used a PMN-PT multi-layered piezoelectric actuator that was designed and fabricated in a previous study [11]. Since a piezoelectric material with highly efficient vibration characteristics is required to achieve a large vibration displacement, the proposed actuator uses PMN-PT, which has much higher d_{33} and k_{33} values than conventional PZT materials, as shown in Table 1. Thus, based on eq. (1), a PMN-PT-based actuator can produce more vibration displacement than a PZT-based one. Furthermore, although a PZT-based ceramic has a superior Curie temperature characteristic than a PMN-PT-based ceramic, a PMN-PT-based ceramic can still be used in an actuator for IMEDDs, as the middle ear cavity maintains a relatively constant body temperature. The PMN-PT-based actuator used in this study had a cross-section area of $1\times1\text{ mm}^2$ and each layer was 0.2 mm thick. The PMN-PT actuators were fabricated by stacking nine layers of PMN-PT crystals, resulting in a thickness of 1.8 mm and total mass of 10 mg. Figure 2 shows the PMN-PT multi-layered actuator.

Table 1. Comparison of characteristics of piezoelectric ceramics

Type	Material	d_{33} (pC/N) (piezoelectric strain constant)	k_{33} (electromagnetic coupling factor)	Curie temperature (°C)
PZT-5A	PbTiO ₃	374	0.70	310
PZT-5H	PbTiO ₃ , PbZrO ₃	593	0.750	365
PMN-PT	Pb(Mg _{1/3} Nb _{2/3})O ₃ , PbTiO ₃	1,560	0.922	150

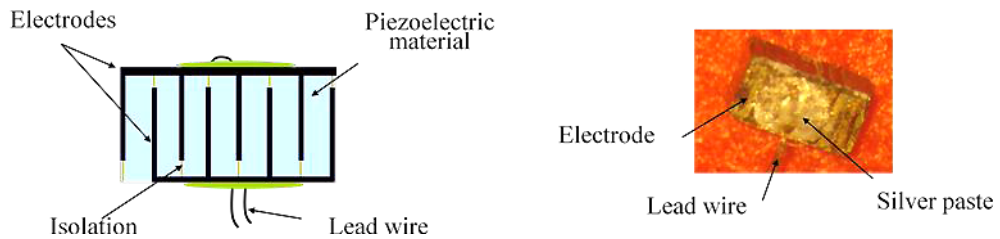


Figure 2. Profile and photograph of PMN-PT multi-layered actuator.

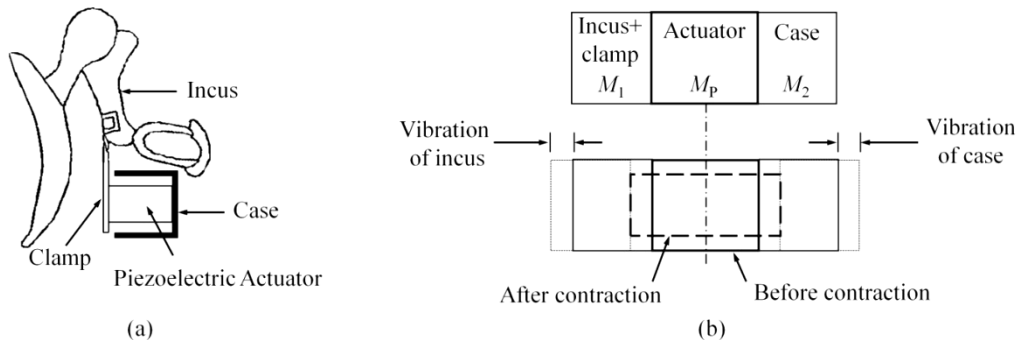


Figure 3. (a) Shape of PFMT attached to incus and (b) vibration due to expansion and contraction of piezoelectric actuator.

Figure 3 shows the vibration of the PFMT due to the expansion and contraction of the piezoelectric actuator. The PFMT is attached to the long process of the incus using an attachment clamp, as shown in Figure 3 (a). When the displacement is expanded by the actuator, the case moves to the right, while the incus moves to the left, and vice versa when the displacement is contracted. Therefore, the vibration due to the expansion and contraction of the actuator can be simply expressed as shown in figure 3 (b), where the ligaments of the incus are disregarded.

Figure 4 shows the displacement of the vibration driven by the force of the piezoelectric actuator, $2F_p$, when the PFMT is attached to the incus. When disregarding the damping and spring of the incus, M_1 , M_p , and M_2 move in the same straight line and the actuator expands the displacement, Δd , by the same amount on both sides. The displacement, Δd , can then be expressed by the following equation.

$$\Delta d = s_1 + s'_1 = s_2 + s'_2 \quad (2)$$

where,

- s_1 : displacement toward M_1 by action force of F_p ,
- s'_1 : displacement toward M_2 by reaction force of F_p ,
- s_2 : displacement toward M_1 by reaction force of F_p ,
- s'_2 : displacement toward M_2 by action force of F_p

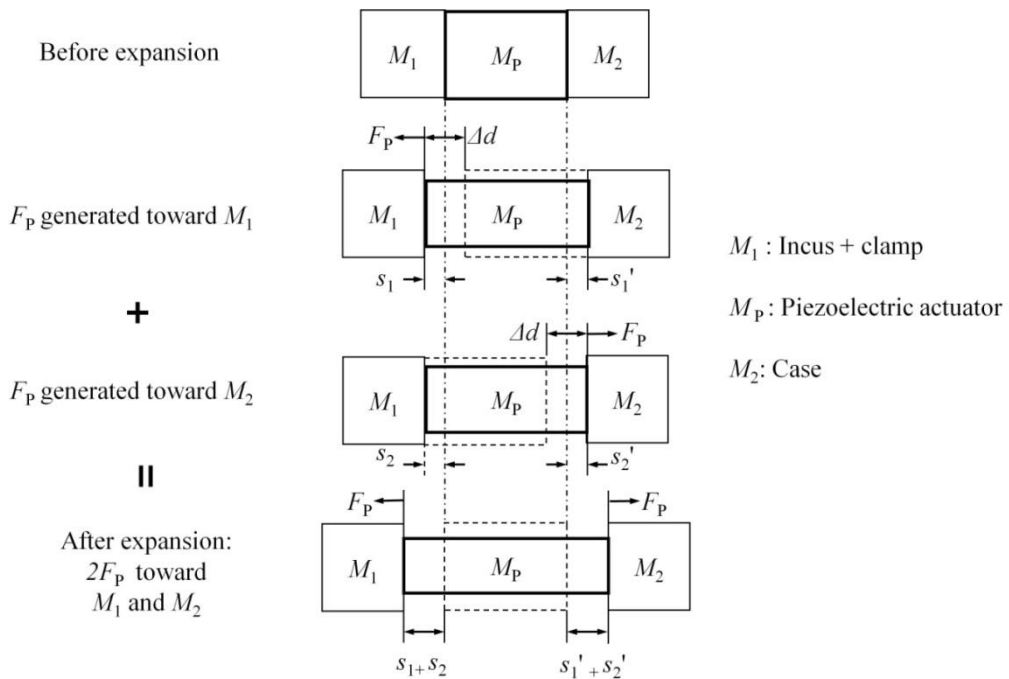


Figure 4. Schematic representation of vibration displacement originating from action and reaction to actuator's expansion.

When the actuator expands the displacement by Δd toward M_1 , the movement toward M_1 is only s_1 due to the action and reaction, and the movement toward the other side is s'_1 . Also, when the actuator expands the displacement by Δd toward M_2 , s_2 and s'_2 occur in the same way. Thus, the displacement toward M_1 is $s_1 + s_2$. The relationship between the weight and the displacement is given by Newton's 3rd law of motion

$$\frac{s_1}{s'_1} = \frac{s_1}{\Delta d - s_1} = \frac{M_p + M_2}{M_1} \quad (3)$$

$$\frac{s_2}{s'_2} = \frac{s_2}{\Delta d - s_2} = \frac{M_2}{M_p + M_1} \quad (4)$$

by combining equations (4) and (5), the displacement toward M_1 can be established as

$$s_1 + s_2 = \frac{M_p + 2M_2}{M_1 + M_p + M_2} \Delta d \quad (5)$$

From equation (5), the vibration displacement of the incus is proportional to the ratio of the masses and the displacement of the actuator. Thus, since it is evident that the vibration of the incus with the PFMT will change considerably when the weight of the case changes,

selecting the proper weight for the metal case and the vibration displacement of the actuator is a very important design consideration for the PFMT.

Figure 5 shows the vibration displacements of the incus using equation (5) when varying the case mass from 1 mg to 100 mg. The displacement originating from the transducer was presumed to be 200 nm, M_1 was 27 mg, and M_p was 10 mg. As shown by the results in Figure 5, the displacement of the incus increased when the weight of the metal case increased, indicating that a PFMT with a sufficient vibration displacement can be achieved by varying the mass of the metal case. A stapes displacement of 60 nm is adequate for a transducer in current IMEDs [4, 5, 10, 15]. As such, the vibration displacement of the incus was about 60 nm when the mass of the metal case was about 10 mg, as shown in Figure 5.

Design Considerations

The design parameters of a transducer for IMEDs are limited by number of factors, including the dimensions, frequency range, force of the actuator, weight, power consumption, and biocompatibility.

Dimensions of Transducer

The PFMT is surgically implanted and attached to the long process of the incus using a metal clamp. However, due to the confined space between the tympanic membrane and the incus, the dimensions and mass of the transducer are limited, thereby restricting the output. Thus, the size limitations for an actuator attached to the ossicle in the middle ear cavity are a diameter of 3-4 mm and height of 2.4 mm. In the case of an FMT, the maximum size is a height of 2.4 mm, while the minimum required sound pressure level is 100 dB SPL [6]. Therefore, to facilitate a similar surgical procedure to that used for an FMT, the height of the PFMT cylinder must be less than 2.4 mm and the diameter less than 3 mm, while the transducer must have a diameter smaller than 3 mm and height smaller than 2.2 mm to be implantable in most human middle ears.

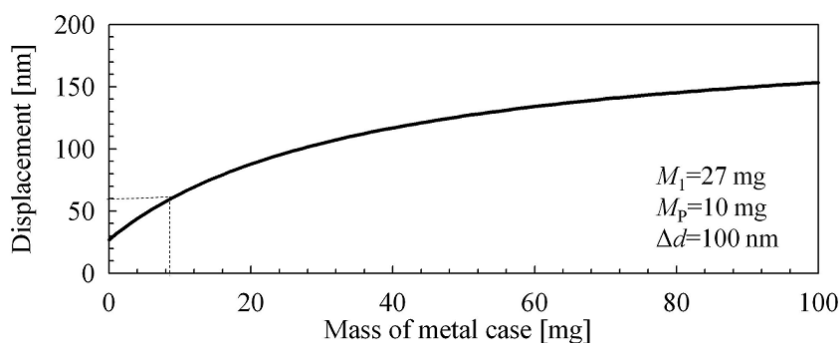


Figure 5. Vibration displacement of incus using equation (5) when varying case mass from 1 mg to 100 mg. Dashed line indicates vibration displacement of 60 nm.

When a floating mass type implantable hearing aid is turned off, the sound transmission is influenced by the mass of the transducer. It has already been reported that a loading of less than 110 mg is assumed to have a negligible effect on the behavior of the ossicular chain [12]. In another temporal bone study, adding mass to the ossicular chain was found to produce a frequency-dependent decrease in the stapes displacement, which was more pronounced at high frequencies [13]. It has also been demonstrated that the greater the loading mass, the less the displacement of the stapes footplate [14]. Therefore, a floating mass type transducer with a small mass is an important requirement for IMEHDs. Yet, since the mass of the PFMT is limited by the generation of vibration displacement, the PFMT should be designed to have the minimum mass that can generate a vibration displacement of 60 nm to 70 nm in order to keep the damping at an acceptably low level.

Force of Transducer

A practically useful implantable transducer should be able to generate forces corresponding to acoustic sound pressure levels of at least 100 dB SPL. According to the data presented by Maniglia [12], it is estimated that the force needed to drive the incus to the equivalent of 100 dB SPL is about 8.9×10^{-5} N. As expressed in equation (1), a relatively low voltage can generate a large displacement of a multi-layered piezoelectric actuator, as the total displacement ΔT is the sum of the displacements of the individual layers, unlike the case of traditional linear piezoelectric actuators. For the actuator used in this study, the displacement was 0.01 m per volt and the stiffness of the PMN-PT was 11.2×10^{10} N/m, resulting in a generated force of 1.12×10^{-3} N/V per 1 mm^2 , as shown in equation (1). Thus, if the driving voltage is 20 V_{pp}, the force originating from the PMN-PT actuator is 22.4×10^{-3} N, which is adequate for transducers in IMEHDs.

Time and Frequency Characteristics

Temporal bone studies have shown that a stapes vibration between 1 kHz and 1.5 kHz produces a vibration displacement between 60 nm and 100 nm and decay rate of about -6 dB/octave at higher frequencies. It has also been reported that a stapes vibration displacement of 100 nm below a frequency of 1 kHz can generate a vibration force equivalent to a sound pressure of about 100 dB SPL [12]. However, recent studies have shown that even a displacement of 60 nm can generate a vibration force equivalent to a sound pressure of 100 dB SPL [4, 5, 10, 15]. Therefore, this means that the PFMT could generate a vibration displacement of more than 100 nm, if half of the actuator's displacement can be transferred to the middle ear ossicle.

In the case of a floating mass type transducer, it has already been reported that the effect of mass loading on the ossicles is more pronounced at high frequencies [15], and most hearing-loss patients require more amplification of high frequencies than low frequencies. These patients experience a deficiency in speech intelligibility, as they suffer from hearing loss above 1 kHz and below 8 kHz. However, in practice, since a 100 Hz to 4 kHz bandwidth

is adequate for human communications, the transducers can provide a significant output within a range of 1 kHz to 4 kHz. Furthermore, as the vibration characteristic of the current piezoelectric actuator is flat above the audio frequency range [11], the actuator can transfer enough displacement to the ossicle without requiring a resonant system to magnify the vibration displacement. Therefore, even though the frequency response of the PFMT is affected by the frequency characteristics of the middle ear, the PFMT as a floating mass type transducer is able to provide adequate high-frequency compensation for any resonances within the audible frequency range and a sufficient driving force.

As transducers for IMEHDs must have a small ripple over a wide frequency range, the fundamental specifications for such transducers include a high spectral bandwidth of up to 10 kHz and small ripple for a frequency response of around ± 3 dB [5]. Thus, since multilayered piezoelectric actuators generate a very stable vibration displacement over the audible frequency range [11], the frequency response of the PFMT can be wide and stable, as the PFMT directly transfers the vibration forces from the actuator to the ossicle.

The total harmonic distortion (THD) is an indication of the sound quality, while the transient response of a transducer is an important design parameter for transferring the audio signal. Unlike the case of conventional air conduction hearing aids, only a few studies have investigated the transient response and THD for IMEHDs. Nonetheless, it is evident that a wide frequency response, short transient response, and low THD all contribute to the signal fidelity of a transducer [5]. Therefore, the THD characteristics and transient response of the PFMT must be evaluated.

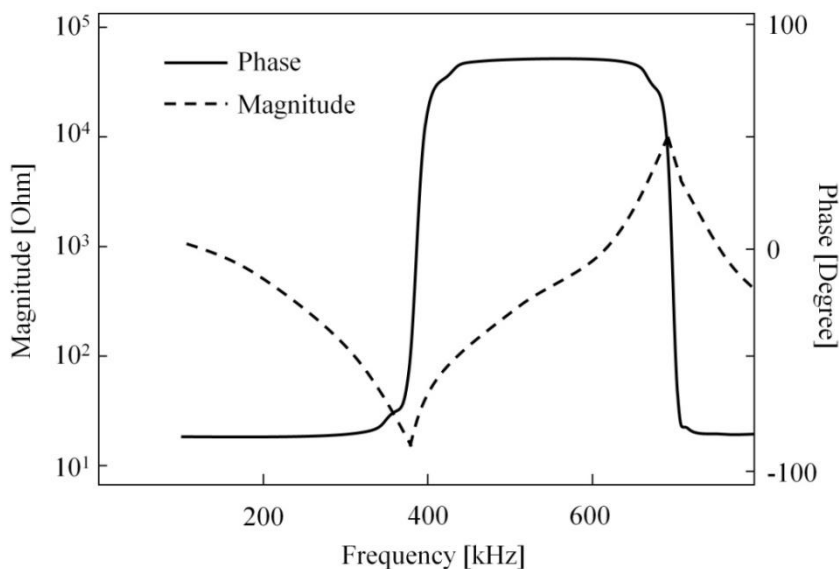


Figure 6. Impedance frequency responses of PMN-PT multi-layered actuator [11].

Power Consumption

The power consumption is a very important design parameter for implantable devices as regards generating a large vibration displacement. As such, while electromagnetic-type transducers require a power consumption of several tens of mW, piezoelectric-type transducers only use tens or hundreds of μW [5]. The behavior of a piezoelectric actuator can be approximated to that of a capacitor when the actuator operates at well below its resonant frequency. Thus, the capacitance of a multilayered actuator can be estimated by summing the capacitances of all the layers.

A multilayered actuator exhibits complicated impedance and phase characteristics, including a resonance over a wide frequency range. Figure 6 shows the impedance frequency response of the current PMN-PT multilayered actuator with the resonance and inverse resonance at 380 kHz and 700 kHz, respectively. This is a common plot for the impedance characteristic of a multilayered piezoelectric actuator [11]. As the PFMT operates over an audible frequency below 10 kHz, the impedance of the actuator continuously decreases as the frequency is increased, and the phase is about -90° .

In alternating current circuits, the power consumption can be calculated from the power factor, $\cos\theta$, when the current and voltage are both sinusoids with a known phase angle θ between them

$$P = V_{\text{RMS}} I_{\text{RMS}} \cos\theta \quad (6)$$

where,

V_{RMS} : effective value of voltage

I_{RMS} : effective value of current

The PFMT is designed to operate in a frequency range below 10 kHz, and the phase of the impedance is about -90° . Therefore, the power consumption of the PFMT is expected to be very small, as shown in equation (6).

Biocompatibility and Reliability

Biocompatibility is an essential condition for any implantable system. Thus, to achieve a similar implant reliability to that of an FMT and enhance the durability and biocompatibility of the PFMT, a 3 μm -thick parylene coating is uniformly deposited over the entire surface of the PFMT. Several studies have already been published on the benefits of an FMT after short and long-term use, and the results indicate a high level of reliability and satisfaction with an FMT as a treatment for sensorineural hearing impairment [16]. An FMT was the first device to be used routinely after its commercial release in Europe and USA. Therefore, floating mass type transducers have been one of the most reliable transducers over the past few decades.

Implementation of PFMT

In this study, the PFMT was designed and implemented to produce 70 nm of vibration displacement with a minimum mass. Multi-layered piezoelectric actuators are superior to other types of actuator due to their small size and large vibration displacement. As such, the piezoelectric actuator used in this study had a 1 mm×1mm cross-section and each layer was 0.2 mm thick, therefore, the final actuator with nine stacked layers was 1.8mm thick and weighed 10 mg. The vibration displacement of the actuator was about 200 nm with a driving voltage of 20 V_{pp}. Thus, given an incus mass of 27 mg, the case mass was designed to be 12mg with a vibration amplitude of about 70nm using equation (5). A cylinder-like PFMT was designed for use in the middle ear cavity, and implemented using a simple bonding process.

Figure 7 shows the implementation process and size of the PFMT. The case was 12 mg, the clamp was 2 mg, and the implemented PFMT was 28 mg, including the adhesive pastes. The lead wires were insulated copper with a 0.05mm diameter and stuck to the electrodes of the actuator using a silver paste. The titanium clamp was fabricated using precision processes and its thickness was 0.08 mm. The metal case was hand carved using micromachining, and had a diameter of 2.5 mm and height of 1.5mm. While one side of the actuator was attached to the metal case, the other side was stuck to the clamp. All the components were accurately assembled and a 3 µm-thick parylene coating uniformly deposited over the entire surface of the implemented PFMT. The resulting PFMT had a diameter of 2.5mm and height of 2.3 mm.

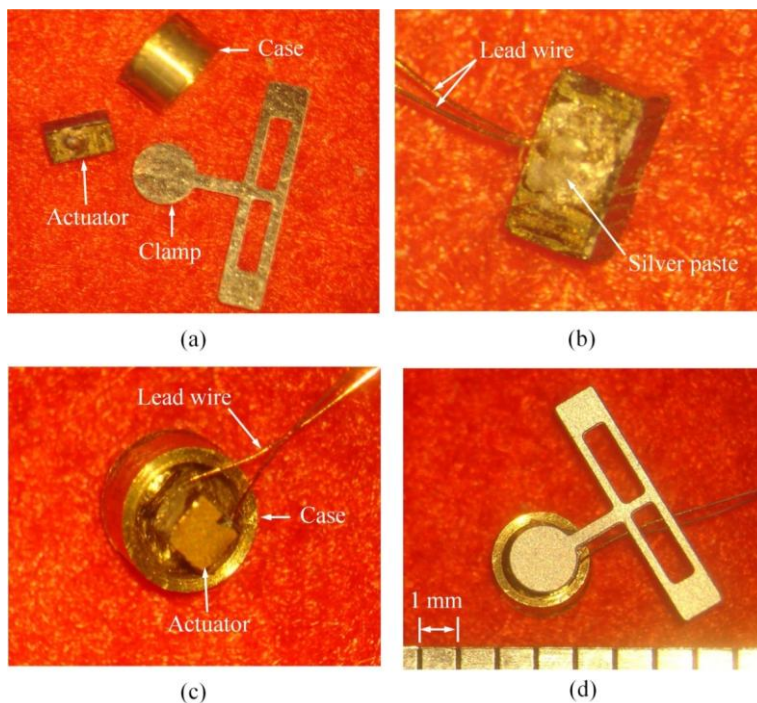


Figure 7. Photographs of PFMT fabrication process; (a) components of PFMT, (b) lead wire bonding, (c) case bonding, and (d) clamp bonding.

Experiments and Results

To certify the proposed PFMT for use in IMEHDs, the vibration displacement was analyzed by measuring the displacement of the actuator itself and an equivalent model, followed by an *in vitro* test using a human temporal bone to measure the stapes displacement. Laser Doppler vibrometry is frequently used in vibration studies of the middle ear, as it is a non-contact method. Thus, an OFV2200 (Polytec, Germany) as the laser vibrometer and OFV551 (Polytec, Germany) as the laser head were used to measure the vibration displacement. The output velocity signal from the vibrometer was digitized for a data acquisition system, System 3 (Tucker Davis Technologies, USA).

Measurement of Piezoelectric Actuator Vibration Displacement

As the vibration displacement of the PFMT is related to that of the piezoelectric actuator, the displacement of the piezoelectric actuator was measured by bonding one end of the actuator to an anti-vibration table and measuring the displacement of the other side when applying a sine wave voltage with a frequency range from 100 Hz to 10 kHz. As shown in Figure 8, the frequency characteristic of the actuator remained absolutely flat below 10 kHz, while the vibration displacement was about 108 nm at 10 V_{pp} and about 216 nm at 20 V_{pp}. The vibration displacement of the multi-layered piezoelectric actuator was proportional to the driving voltage, and the vibration displacement of the proposed actuator showed 10.8 nm per peak-to-peak volt. The vibration displacements were the same before and after the 3 μm-thick parylene coating.

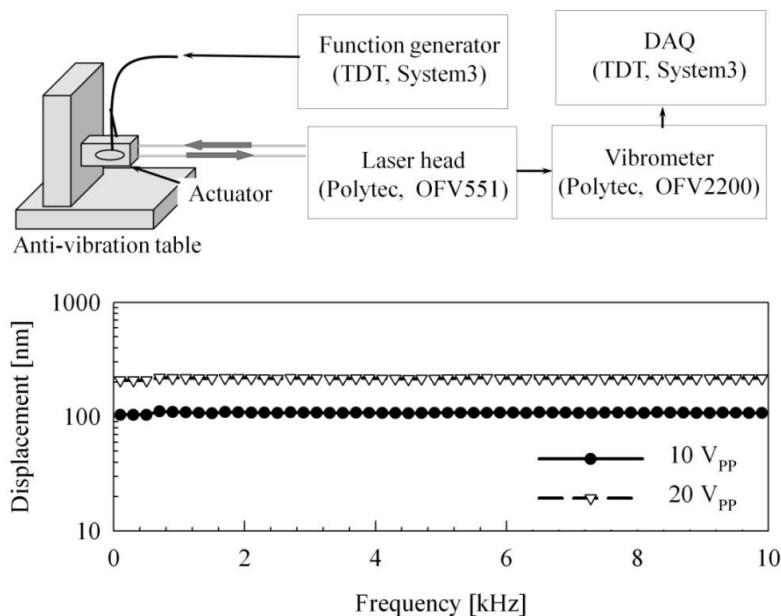


Figure 8. Vibration displacements of PMN-PT multi-layered piezoelectric actuator.

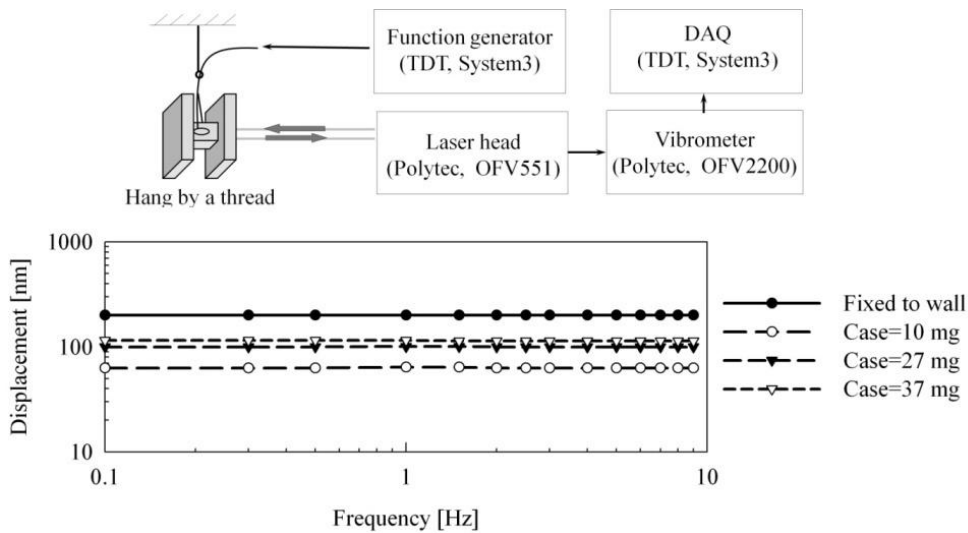


Figure 9. Vibration displacements of equivalent model of PFMT attached to incus with case weights of 10 mg, 27 mg, and 37 mg, respectively, when incus M_1 was 27 mg, actuator M_p was 10 mg, and sinusoidal input was 20 V_{pp}.

Experiment Using Simple Vibration Model

A simple model of the PFMT attached to the incus, as shown in Figure 2 (b), was realized to study the vibration characteristics. In equation (5), the PMFT vibration on the ossicle depends on the case mass M_2 and the mass of the other components. The incus mass used in this experiment was 27 mg and variable case weights were tested. The simple model was suspended in air using lead wires for the displacement measurement using the laser vibrometer. For the different case mass weights, the vibration displacement of the incus M_1 was measured within a frequency range from 100 Hz to 10 kHz.

The measured vibration displacements are shown in Figures 9 and 10 when the sinusoidal inputs were applied to the actuator. The vibration displacements of the incus M_1 varied from 40 to 120 nm according to the weight of the case M_2 . As shown by the results in Figures 9 and 10, when the weight of M_2 increased, the vibration displacement of M_1 also increased. The displacement was about half of the total displacement of the actuator itself when M_2 was equal to the weight of M_1 . Consequently, the experimental results demonstrated that when the PFMT is attached to an ossicular chain it can be vibrated by the action-and-reaction force originating from the actuator and its vibration characteristics controlled by the weight of the case M_2 .

Figure 10 shows a comparison of the calculated results using equation (5) and the experimental results using an equivalent model of the PFMT attached to the incus when varying the weights of the case M_2 from 0 to 60 mg and applying a sinusoidal input of 20 V_{pp} to the actuator. The experimental results were similar to the calculated results. When the metal case was not used in the simple model, the results showed a very low vibration displacement, as the force transferred to the incus M_1 by the action-and-reaction between the incus M_1 and the piezoelectric actuator was small, as in equation (5). The differences

between the calculated and experimental results were attributed, in part, to the stiffness of the lead wires.

Nonlinear Distortion and Transient Response

As frequency response measurements do not include specific information about sound fidelity, the total harmonic distortion (THD) and impulse response are also useful measures for the assessment of IMEDs [5]. Figure 11 shows the THD of the PFMT’s vibration displacement, which was measured as a function of the frequency between 100 Hz to 10 kHz using a 334A distortion meter (HP, USA). The THD values show that levels of less than 0.3% were achieved, and this improved at higher frequencies with a THD value of about 0.1% above 1 kHz.

The transient response of the PFMT plays as important a role as the wider frequency response. The impulse and step response, also called the transient response, provide insight into the characteristics of the system dynamics, including the peak response and settling time. Figure 12 shows the step response of the PMFT driven at 20 V_{pp}. Here, the peak rising time was about 2 μs, the settling time was about 55 μs, and the resonant frequency of the PFMT was 136 kHz.

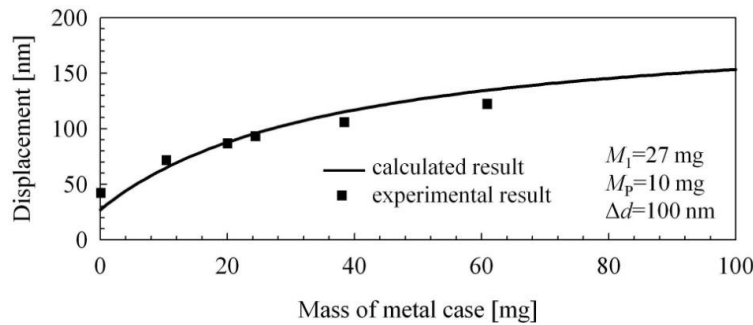


Figure 10. Comparison of vibration displacements: calculated results using equation (5) and experimental results using equivalent model of PFMT attached to incus.

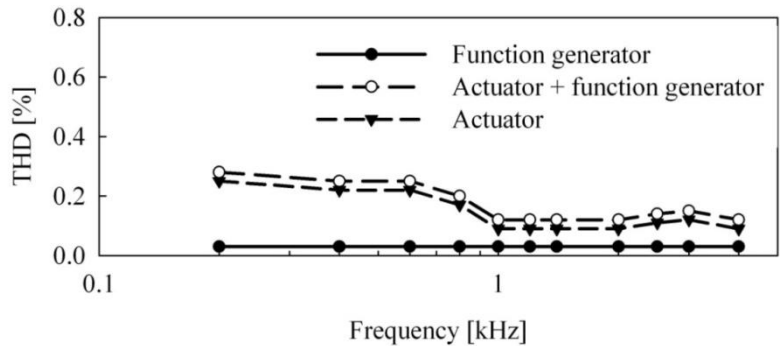


Figure 11. Total harmonic distortion of PMN-PT multi-layered actuator.

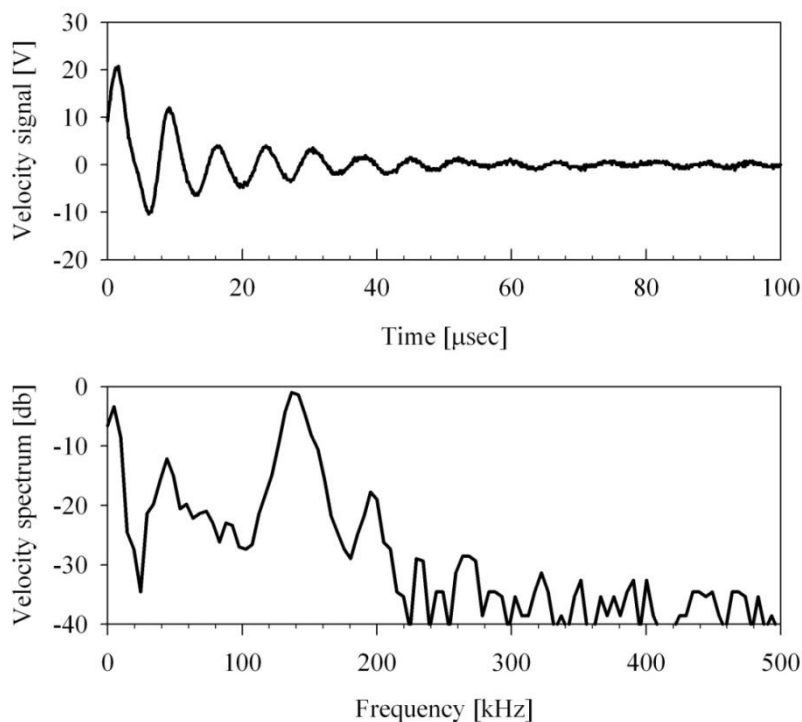


Figure 12. Step responses of PFMT.

In Vitro Experiment

To investigate the usefulness of the PFMT as a transducer for IMEHDs, an *in vitro* test was carried out using a human temporal bone. Generally, in temporal bone studies, the stapes vibration displacements from acoustic stimulation act as a reference so that the data from the actuator stimulation can be expressed in terms of the equivalent dB SPL at the eardrum to achieve the same stapes displacement. A comparison of the results can then be made with the reference stapes vibration displacement obtained from acoustic stimulation. Here, 94 dB SPL and 100 dB SPL acoustic stimulations were chosen as the references, as commonly used in other studies.

Using an ER-7C microphone (Etymotic Research,) and ER-2 loudspeaker (Etymotic Research,), sounds with an audible frequency of 94 dB SPL and 100 dB SPL were generated and applied to the eardrum of the temporal bone through a tube, and the vibration displacements of the stapes then measured, as shown in Figure 13. After measuring the ossicle vibration, the PFMT was attached to the long process of the incus. The PFMT implanting position was as close as possible to the incudostapedial joint to minimize any decrease in the displacement [15]. The PFMT was small enough to be attached to the long process of the incus using a clamp in the middle ear cavity. The vibration displacements of the stapes were then measured using a laser vibrometer when applying 10 V_{pp} and 20 V_{pp} sine wave signals from a TDT function generator (System3, USA) to the PFMT.

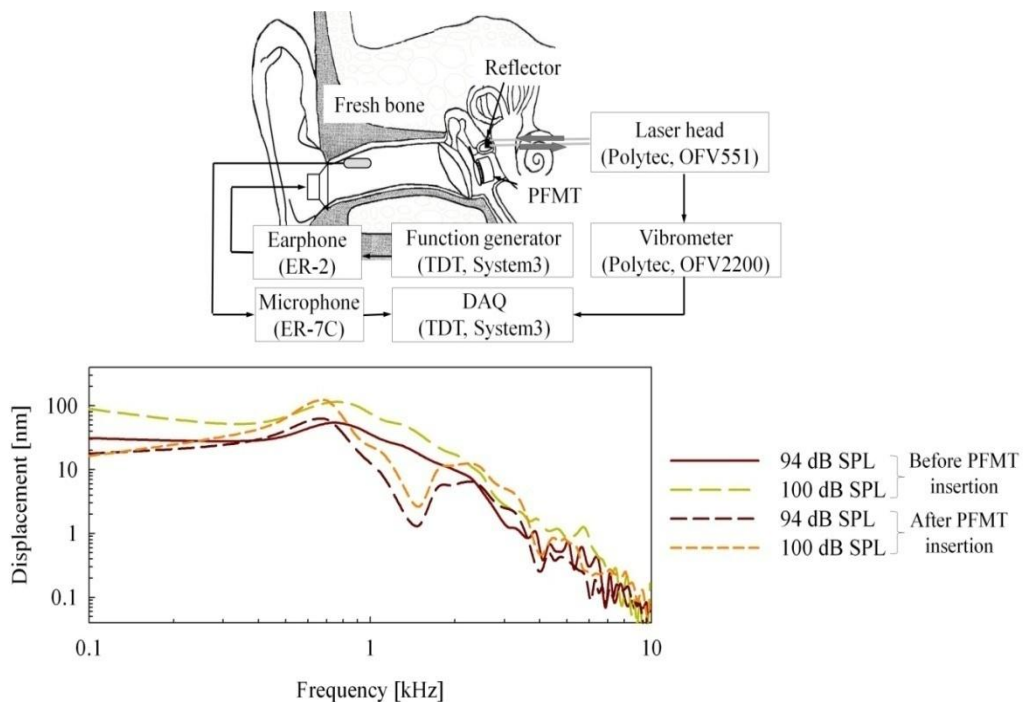


Figure 13. Loading effect of PFMT when applying acoustic stimulations to tympanic membrane.

The effect of mass loading the incus with the PFMT on the middle ear transfer function for the temporal bone is shown in Figure 13. Before the PFMT insertion, the peak-to-peak stapes displacement showed a similar response to that in previous studies. The bone showed a peak at around 0.7 kHz and 60 to 100 nm of stapes displacement with the acoustic input of 100 dB SPL. The stapes displacements were proportional to the acoustic inputs. A previous study showed that the effect of mass loading the incus was more prominent at high frequencies. Plus, the more the mass load increased, the more the stapes displacement decreased [15]. Here, the PFMT loading produced damps below 0.4 kHz and around 1.2 kHz.

Figure 14 shows the stapes displacements driven by the PFMT and acoustic stimulations at the tympanic membrane. The PFMT driven at 10 V_{pp} and 20 V_{pp} generated a vibration force equivalent to a sound pressure of about 94 dB SPL and 100 dB SPL, respectively. The primary resonant frequency appeared at approximately 0.8 kHz. From 2 kHz to 10 kHz, the stapes vibration from the PFMT was considerably greater than that from the acoustic stimulation. The stapes displacement was about 60 nm from 100 Hz to 500 Hz. The PFMT was designed to produce a vibration displacement of 70 nm when the actuator was driven at 20 V_{pp} and the incus mass was presumed to weigh 27 mg. Notwithstanding a 10 nm difference between the design and the experiment, the correctness of the proposed PFMT design was confirmed. Furthermore, the PFMT was shown to exhibit a voltage displacement linearity after inserting the PFMT into the long process of the incus. The stapes displacement per 1 V_{pp} of driving voltage was 4.3 nm at 1 kHz and 1.1 nm at 4 kHz.

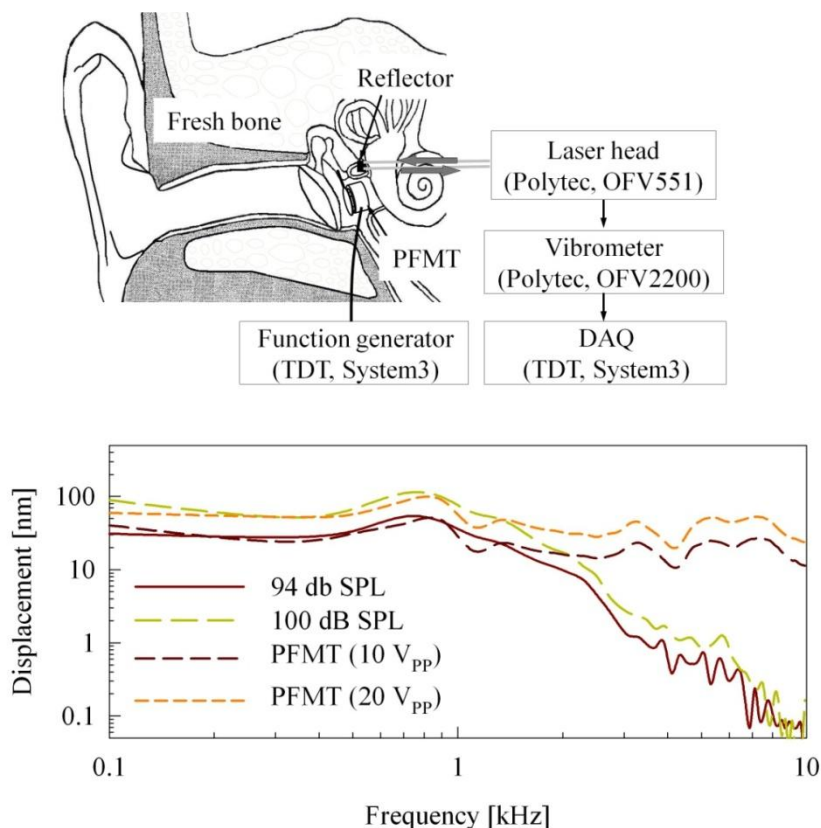


Figure 14. Comparison of stapes displacements driven by PFMT and acoustic stimulations at tympanic membrane.

Power Consumption

To determine the power factor of the PFMT, the impedance characteristics of the PFMT were measured using an impedance analyzer (HP, 4194A). Figure 15 shows the measured results of the impedance frequency response of the PFMT from 100 Hz to 10 kHz with a 25 Hz interval. The amplitude of the impedance decreased from 896 k Ω to 9 k Ω according to the frequency range between 100 Hz to 10 kHz, and the phase was about -89.3° . From equation (6), the power consumption of the PFMT is very small, due to the low power factor of 0.012. However, the low current consumption of the PFMT is mainly caused by the low total capacitance of the actuator. The PMN-PT material also shows a much higher value for d_{33} than the conventional PZT material. Thus, the PMN-PT-based actuator can produce the desired vibration displacement with fewer layers than a PZT-based actuator. In addition, when the actuator size is limited, the distance between adjacent layers in the stack becomes wider with a smaller number of layers. Thus, the lower capacitance of the PMN-PT-based actuator results in a low power consumption. Figure 16 shows the calculated power consumption of the PFMT to produce a sound pressure equivalent to 94 dB SPL and 100 dB SPL.

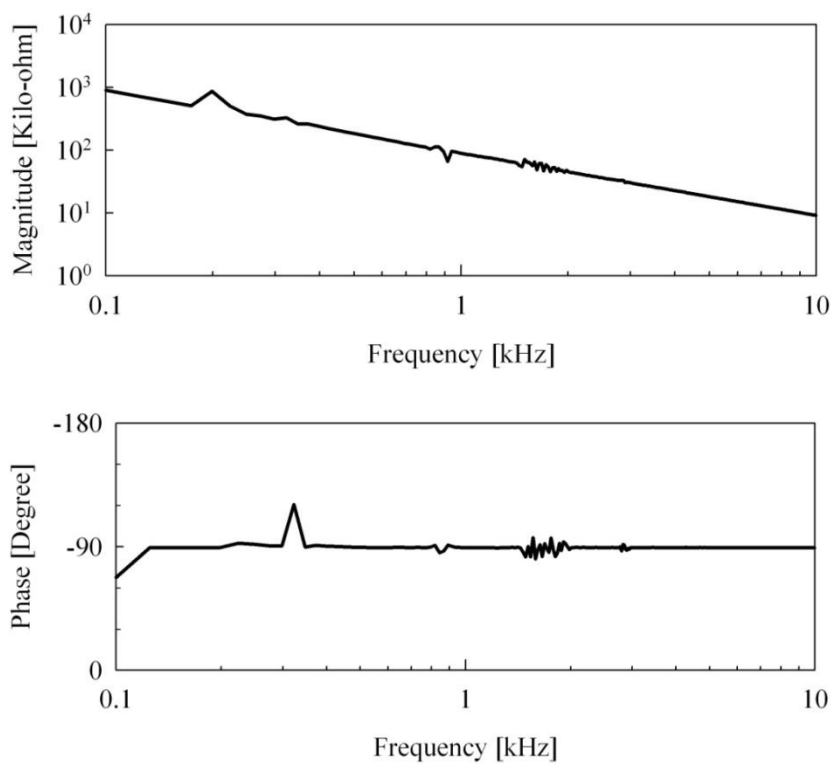


Figure 15. Impedance frequency response of magnitude and phase of actuator.

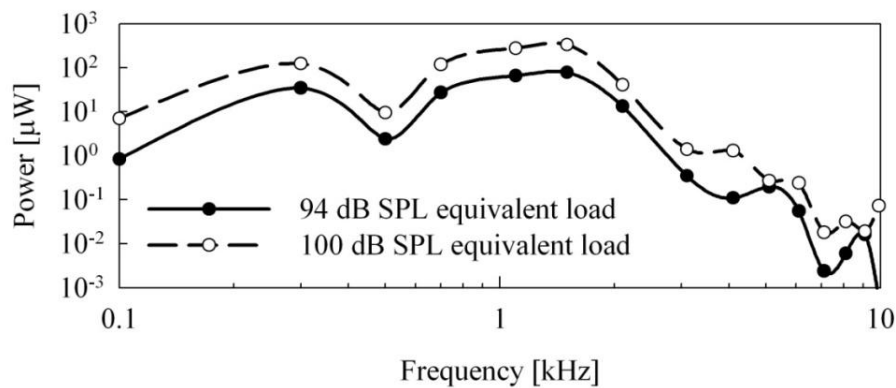


Figure 16. Power consumption of actuator attached to incus when driven by sound stimulations equivalent to 94 dB SPL and 100 dB SPL.

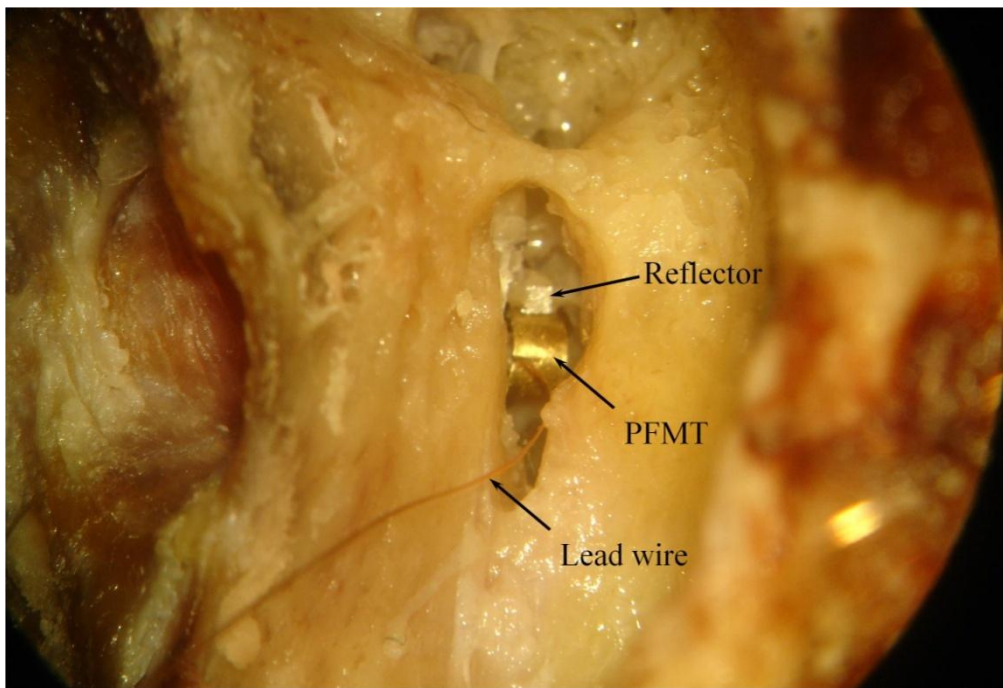


Figure 17. Photograph of implanted PFMT.

DISCUSSION

While the force generated from a piezoelectric actuator is enough to drive the ossicular chain, the transmission of the force from the actuator to the ossicle is inefficient when the actuator is floating rather than fixed in the middle ear. However, the proposed PFMT is able to deliver the force from the actuator to the ossicle using the metal case as a floating inertial mass. The proposed PFMT was implemented using a piezoelectric actuator and metal cap with an attachment clamp. The fabricated PFMT was 2.3 mm in length with a diameter of 2.5 mm and small enough to be attached to the long process of the incus in the middle ear cavity, as shown in Figure 17. Similar to an FMT, the transducer was also attached to the long process of the incus without any form of manipulation of the ossicular chain.

The vibration displacement by the PFMT attached to the ossicle was calculated and verified experimentally. The results of the calculation and experiment were in good agreement. As the stiffness of the ossicle ligaments is minimal, the vibration displacement in the case of practical implantation of the PMFT is mainly affected by the mass ratio of the incus and the case. The vibration displacement of the PFMT is proportional to the driving voltage of the piezoelectric actuator. In the experiments, the driving voltages were 10 V_{pp} and 20 V_{pp}. If more vibration displacement or a lighter transducer is needed, the driving voltage of the actuator can be increased. Practically, increasing the driving voltage is limited by the characteristics of the piezoelectric material. In this study, the piezoelectric actuator was able

to operate up to 40 V_{pp}, while the vibration displacement of the actuator itself could be magnified to about 430 nm.

To investigate the usefulness of a transducer that is implanted in the middle ear, a comparison of the vibration displacements should be made with stapes displacement data obtained from acoustic stimulation as a reference. Figure 18 shows the frequency response of the computed maximum equivalent sound pressure level driven by the PFMT described in Figure 14. As shown in Figure 18, the PFMT driven by 20 V_{pp} generated a sound pressure equivalent to about 100 dB SPL at the tympanic membrane below 1 kHz, 125 dB at 4 kHz, and about 140 dB SPL above 7 kHz. A totally integrated cochlea amplifier (TICA) excited by 100 dB SPL showed an equivalent sound pressure level of 125 dB SPL at 4 kHz and 150 dB SPL at 7 kHz, respectively [5]. Therefore, the proposed PFMT would seem to be suitable for individuals with sensorineural hearing loss, as it can compensate for displacement over a high frequency range, even though the PFMT is implanted as a floating mass type transducer.

Previously, it was shown that a PFMT attached to the long process of the incus using a piece of rubber and clamp can deliver the force from the actuator to the ossicle using the metal case as a floating inertial mass [17]. Plus, the vibration characteristics of the transducer can be controlled by adjusting the metal case and the elastic material of the transducer. As such, a transducer with sufficient vibration in a particular frequency range can be achieved by inserting the elastic material over the audible frequency range. The same rules also apply to the current PFMT, and the transducer can also be highly tuned with a resonance frequency over a high frequency range, like the TICA, which is tuned with a resonance frequency in the range of 7 to 10 kHz [5]. The reason for tuning a transducer's resonant frequency is to increase the vibration efficiency within the human audible frequency range. Thus, lowering the resonant frequency of the PFMT to an audible range can increase the vibration displacement of the transducer if the same input signal is supplied to the PFMT. As such, the TICA's tuned frequency is 7 kHz to 10 kHz in consideration of hearing impaired patients who have sensorineural hearing loss in a high frequency range.

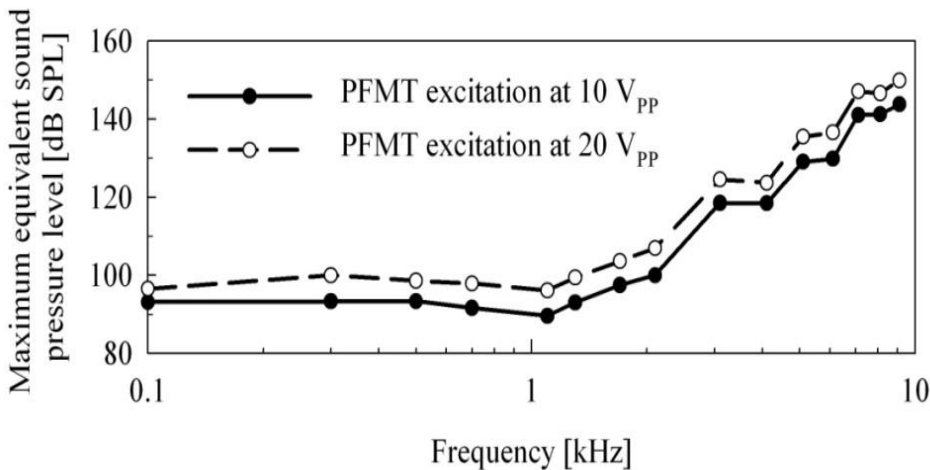


Figure 18. Frequency responses of computed maximum equivalent sound pressure level driven by PFMT described in Figure 14.

A multi-layered actuator with a wide frequency response, low non-linear distortion, and short transient response is used for the PFMT. Among the measured results for the implemented PFMT, the transient response is 50 μsec and the non-linear distortion is about 0.2 %. Due to the high fidelity nature of the multi-layered actuator, the ossicle is directly excited by the vibration displacements close to the original sound waves.

The power consumption of 60 μW is also significantly less than that of electromagnetic transducers when the PFMT produced a sound pressure equivalent to 94 dB SPL at 1 kHz. According to Zenner's reports [5], in order to produce a sound pressure equivalent to 94 dB SPL at 1 kHz, the power consumption of the electromagnetic transducers developed by Ball and Maniglia was over 10s mW and about 5mW, respectively, while the power consumption of the piezoelectric transducers developed by Wang [10] and Zener [5] was about 0.35 mW and 25 μW , respectively. Since a multi-layer device with n layers has n times the capacitance and vibration displacement and n times the power consumption, the PFMT implementation has a trade-off between the vibration displacement and the power consumption. Nonetheless, the low power consumption of the proposed PFMT could be suitable for realizing fully implantable hearing aids for the rehabilitation of moderate to severe sensorineural hearing loss.

CONCLUSION

This chapter presents a PFMT that combines the advantages of a piezoelectric vibration transducer and electromagnetic transducer for use in an IMEHD. As such, the proposed PFMT consists of a multi-layered piezoelectric actuator and metal case, and vibrates the ossicle. The nonlinear distortions of the transducer are small throughout the audio frequency range and the settling time of the transient response is extremely short. An *in vitro* experiment confirmed that the proposed PFMT can generate enough vibration displacements. The power consumption of the PFMT is also significantly lower than that of electromagnetic hearing implants. Therefore, the proposed PFMT can provide sufficient vibration displacements to the cochlea for utilization in an IMEHD for individuals with a mild-to-severe sensorineural hearing loss.

ACKNOWLEDGMENTS

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Chapter 7

THERMALLY INDUCED CRACKING OF PIEZOELECTRIC MATERIALS

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7.1 INTRODUCTION

Mechanical, electrical, and thermal fields are coupled in most of the physical problems. Thermal effects could be important when structures containing piezoelectric materials need to operate at high or low temperature environments. Such environments are quite possible in the applications of smart materials and structures. An area where piezoelectric materials may provide dramatic advantages is in the development of piezoelectric composite structures with the capability to sensor thermally induced distortions and to actively compensate for adverse thermo-mechanical conditions. Typical applications of such structures are envisioned in the thermal distortion management of propulsion components and/or space structures. The analysis of the thermal fracture process of piezoelectric materials could provide information to improve the design of piezoelectric devices operated in thermal environments. Thermoelastic behavior of piezoelectric solids without defects has been studied by many researchers [e.g. 1-9].

Thermo-piezoelectric materials with defects, such as holes and interface cracks, were studied [10, 11]. The problem of a two-dimensional piezoelectric material with an elliptic cavity under uniform heat flow was investigated [12]. Thermally induced fracture problem for a piezoelectric laminate having a crack subject to uniform electric and temperature fields was solved in [13]. Stress intensity factors for internal penny-shaped cracks in isotropic materials have been provided in Handbook [14], where the crack face temperature is maintained or the crack-face thermal flux is given. In the case of piezoelectric materials, crack tip singularities can include electric fields and electric displacements. The problem is important but is more complicate. Accordingly, there is a need to provide the thermal stress and electric displacement intensity factors for various thermal load conditions. The expressions should be in closed-form so that the material designers and scientists and directly use them.

This Chapter provides an exact solution for a thermal penny-shaped crack in a piezoelectric medium. An arbitrarily distributed temperature field is applied on the crack faces. Closed-form expressions for the stress and electric displacement intensity factors at the crack border are given. Both impermeable and permeable crack-face electric boundary conditions are considered. Further, crack tip field intensity factors for a given thermal flux on the crack faces are also obtained. The solutions are mainly provided for transversely isotropic materials, but the results for isotropic materials and dielectrics are also given. Also introduced in this chapter is the design of a smart functionally graded thermopiezoelectric composite structure.

7.2. PENNY-SHAPED CRACK SUBJECTED TO PRESCRIBED TEMPERATURE

A penny-shaped crack in piezoelectric media is considered by Wang [15, 16], for which the crack face temperature or thermal flux is given. Consider a cylindrical coordinate system r - θ - z . Imagine a penny-shaped crack of radius a located in the plane r - θ in the interior of an infinite medium, as shown in Figure 7.1. We first consider a temperature $T_0(r)$ maintained on the crack faces. A thermal flux q_0 applying on the crack faces will be considered in Section 7.3. The problem is axi-symmetric and symmetric with respect to the $z=0$ plane. Summarizing, on the boundary $z=0$ of the half-infinite medium $z \geq 0$, $\sigma_{rz}(r, z)=0$ for $0 \leq r < \infty$, and

(1) inside the (insulated) crack, for $0 \leq r < a$ and $z=0$,

$$T = T_0(r), \quad \sigma_{zz} = p_{01} = 0, \quad D_z = p_{02} = d_0(r), \quad (7.1)$$

where d_0 is the normal component of the electric displacement vector inside the crack. The crack face is free from mechanical tractions so that $p_{01}=0$. Because air or vacuum allows some penetration of electric flux through the crack interior, the electric displacement inside the crack may not be zero, which is denoted as $d_0(r)$.

(2) outside the crack, for $r > a$ and $z=0$,

$$w=0, \quad \phi=0, \quad \frac{\partial T}{\partial z} = 0. \quad (7.2)$$

where w and ϕ are z direction displacement and electric potential, respectively.

7.2.1. The Temperature Field

The temperature field T , in the absence of heat sources, is governed by

$$k_1 \left(\frac{\partial^2 T}{\partial r^2} + \frac{1}{r} \frac{\partial T}{\partial r} \right) + k_3 \frac{\partial^2 T}{\partial z^2} = 0, \quad (7.3)$$

which may be solved to give

$$T(r, z) = \int_0^\infty J_0(sr) \exp(-s\lambda_0 z) A_0 ds, \quad (7.4)$$

where $\lambda_0 = \sqrt{k_1/k_3}$, k_1 and k_3 are thermal conductivities, $A_0(s)$ is an unknown function of s . Hereafter, J_i ($i=0,1$) are the i -th order Bessel functions of the first kind. Note that equation (7.4) fulfills the condition that the local field vanishes at infinity. The derivation of (7.4) is:

$$\frac{\partial T(r, z)}{\partial z} = -\int_0^\infty s\lambda_0 J_0(sr) \exp(-s\lambda_0 z) A_0 ds. \quad (7.5)$$

We first consider crack face temperature boundary condition. From conditions (7.1) and (7.2), together with equations (7.4) and (7.5) we obtain

$$\int_0^\infty sJ_0(sr) A_0 ds = 0, \quad r > a, \quad (7.6a)$$

$$\int_0^\infty J_0(sr) A_0 ds = T_0(r), \quad 0 \leq r < a. \quad (7.6b)$$

The solution of the dual integral equations (7.6) is known to be (Referring to Appendix A of Chapter 4):

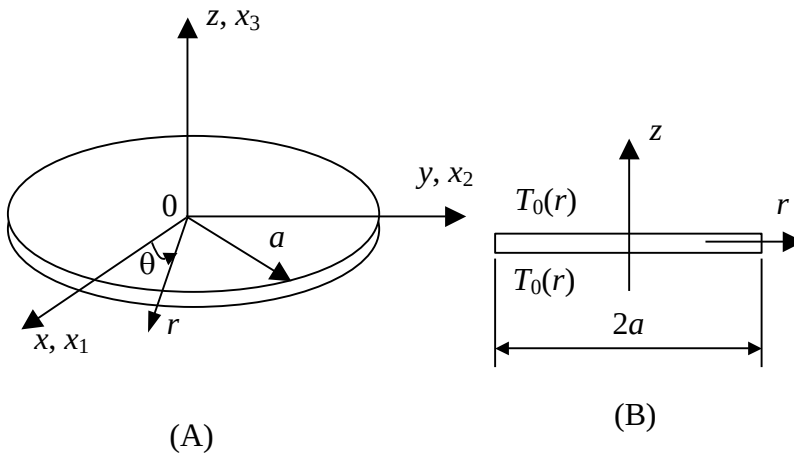


Figure 7.1. A penny-shaped crack in an infinite piezoelectric medium. (A) Crack geometry, (B) temperature on the crack faces is maintained.

$$A_0(s) = \int_0^a \overline{T}(x) \cos(sx) dx, \quad (7.7)$$

in which the function $\bar{T}(x)$ is:

$$\bar{T}(x) = \frac{2}{\pi} \frac{d}{dx} \int_0^x \frac{tT_0(t)}{\sqrt{x^2 - t^2}} dt. \quad (7.8)$$

Substituting Eq. (7.8) into Eq. (7.4), we arrive at the equation describing the temperature field generated by the crack

$$T(r, z) = \int_0^a \bar{T}(x) \left[\int_0^\infty J_0(sr) \cos(sx) \exp(-s\lambda_0 z) ds \right] dx. \quad (7.9)$$

For $z=0$ the foregoing integral represents a continuous function of r . Now

$$\int_0^\infty \cos(sx) J_0(sr) ds = \begin{cases} \frac{1}{\sqrt{r^2 - x^2}}, & r > x \\ 0, & r < x \end{cases}, \quad (7.10)$$

the temperature inside the crack is obtained as

$$T(r, 0) = \frac{2}{\pi} \int_0^r \left(\frac{d}{dx} \int_0^x \frac{tT_0(t)}{\sqrt{x^2 - t^2}} dt \right) \frac{1}{\sqrt{r^2 - x^2}} dx, \quad r < a. \quad (7.11)$$

This is an Abel type integral equation, which gives a solution $T(r, 0)=T_0(r)$ for $r < a$. Therefore, it follows from Eqs. (7.9) and (7.11) that

$$T(r, 0) = \begin{cases} T_0(r) & , r < a \\ \frac{2}{\pi} \int_0^a \left(\frac{d}{dx} \int_0^x \frac{tT_0(t)}{\sqrt{x^2 - t^2}} dt \right) \frac{1}{\sqrt{r^2 - x^2}} dx & , r \geq a \end{cases}. \quad (7.12)$$

If the applied temperature is a constant (this is, $T_0(r)=T_0$), then the temperature outside the crack obtained from Eq. (7.12) is $T(r, 0) = \frac{2}{\pi} \arcsin\left(\frac{a}{r}\right) T_0$, where $r \geq a$.

For $z=0$ the thermal flux is a discontinuous function of r . This can be put into the closed form from Eq. (7.9):

$$\begin{aligned} q_z(r, 0) &= \sqrt{k_1 k_3} \int_0^a \int_0^\infty \bar{T}(x) s J_0(sr) \cos(sx) ds dx \\ &= \sqrt{k_1 k_3} \int_0^a \int_0^\infty \bar{T}(x) J_0(sr) ds [\sin(sx)] \end{aligned}$$

$$= \sqrt{k_1 k_3} \bar{T}(a) \int_0^\infty J_0(sr) \sin(sa) ds - \sqrt{k_1 k_3} \int_0^a \int_0^\infty J_0(sr) \sin(sx) ds d\bar{T}, \quad (7.13)$$

which vanishes for $r > a$. Inside the crack, $r < a$, the thermal flux is

$$q_z(r, 0) = \sqrt{k_1 k_3} \bar{T}(a) \frac{1}{\sqrt{a^2 - r^2}} - \sqrt{k_1 k_3} \int_r^a \frac{d\bar{T}}{\sqrt{x^2 - r^2}}, \quad r < a. \quad (7.14)$$

Therefore, results (7.12)-(7.14) agree with the conditions imposed by equations (7.1) and (7.2).

7.2.2. Electro-Elasticity Field

Since the temperature field is axi-symmetric, the resulting electro-elasticity field is also axi-symmetric. Thermo-electro-elastic constitutive equations for piezoelectric materials polarized along the z direction can be written as

$$\boldsymbol{\sigma} = \mathbf{c}\boldsymbol{\varepsilon} - \mathbf{e}\mathbf{E} - \boldsymbol{\sigma}_T, \quad \mathbf{D} = \mathbf{e}^T \boldsymbol{\varepsilon} + \boldsymbol{\epsilon} \mathbf{E} - \mathbf{D}_T, \quad (7.15)$$

where $\boldsymbol{\sigma}$, $\boldsymbol{\varepsilon}$, $\mathbf{D}$, and $\mathbf{E}$ are stress, strain, electric displacement and electric field vectors, respectively, shown in Eqs. (4.91); $\boldsymbol{\sigma}_T$ and $\mathbf{D}_T$ are thermally induced stress and electric displacement vectors respectively:

$$\boldsymbol{\sigma}_T = (\lambda_{11}T, \lambda_{11}T, \lambda_{33}T, 0)^T, \quad \mathbf{D}_T = (0, \beta_3T)^T, \quad (7.16)$$

where λ_{ii} are the temperature-stress coefficients ($i=1,3$), and β_3 is the temperature-electric displacement coefficient. For a piezoelectric medium whose poling direction is coincident with the positive z -axis, material constant matrices $\mathbf{c}$, $\mathbf{e}$, and $\boldsymbol{\epsilon}$ have been given in Eqs.(4.92). In the absence of body forces and body charges, the equilibrium equations are Eqs. (4.93), whose solution, in present case, consists two parts: the homogeneous solution (which ignores the temperature effect), and a particular solution (which considers the temperature effect). It can be written in terms of the unknown functions $A_m(s)$ as follows:

$$\begin{Bmatrix} u(r, z) \\ w(r, z) \\ \phi(r, z) \end{Bmatrix} = \int_0^\infty \sum_{m=1}^3 \exp(-s\lambda_m z) \begin{Bmatrix} J_1(sr)a_{1m} \\ J_0(sr)a_{2m} \\ J_0(sr)a_{3m} \end{Bmatrix} A_m ds$$

$$+ \int_0^\infty \frac{1}{s} \exp(-s\lambda_0 z) J_0(sr) \begin{Bmatrix} f_1 \\ f_2 \\ f_3 \end{Bmatrix} A_0 ds. \quad (7.17)$$

On the RHS of Eqs. (7.17), the first term is the homogeneous solution, and the second term is the particular solution of Eqs. (4.93). By substituting Eqs. (7.17) into Eqs. (7.15) and (4.93), the constants (a_{1m} , a_{2m} , a_{3m}), the parameter λ , and the coefficients (f_1 , f_2 , f_3) are found to obey:

$$\begin{bmatrix} c_{11} - c_{44}\lambda_m^2 & -(c_{13} + c_{44})\lambda_m & -(e_{31} + e_{15})\lambda_m \\ -(c_{13} + c_{44})\lambda_m & c_{33}\lambda_m^2 - c_{44} & e_{33}\lambda_m^2 - e_{15} \\ -(e_{31} + e_{15})\lambda_m & e_{33}\lambda_m^2 - e_{15} & \epsilon_{11} - \epsilon_{33}\lambda_m^2 \end{bmatrix} \begin{Bmatrix} a_{1m} \\ a_{2m} \\ a_{3m} \end{Bmatrix} = 0, \quad (7.18)$$

$$\begin{Bmatrix} f_1 \\ f_2 \\ f_3 \end{Bmatrix} = \begin{bmatrix} c_{11} - c_{44}\lambda_0^2 & -(c_{13} + c_{44})\lambda_0 & -(e_{31} + e_{15})\lambda_0 \\ -(c_{13} + c_{44})\lambda_0 & c_{33}\lambda_0^2 - c_{44} & e_{33}\lambda_0^2 - e_{15} \\ -(e_{31} + e_{15})\lambda_0 & e_{33}\lambda_0^2 - e_{15} & \epsilon_{11} - \epsilon_{33}\lambda_0^2 \end{bmatrix}^{-1} \begin{Bmatrix} \lambda_{11} \\ -\lambda_{33}\lambda_0 \\ -\beta_3\lambda_0 \end{Bmatrix}. \quad (7.19)$$

Equation (7.18) is an eigen-value problem consisting of three equations. Non-trivial a_{jm} ($j=1, 2, 3$) exist if λ is a root of the determinant. It can be shown that if λ is a root of the determinant then $-\lambda$ is also a root of the determinant. We now arrange the order of roots λ_m ($m=1, \dots, 6$) such that $\text{Re}(\lambda_1) > 0$, $\text{Re}(\lambda_2) > 0$, $\text{Re}(\lambda_3) > 0$ and $\text{Re}(\lambda_4 = -\lambda_1) < 0$, $\text{Re}(\lambda_5 = -\lambda_2) < 0$, $\text{Re}(\lambda_6 = -\lambda_3) < 0$. Since all field quantities must be finite for any positive values of r and z , then only λ_1 , λ_2 and λ_3 should be used.

From constitutive relations (7.15), the stresses and electric displacement on the cracked plane can be obtained as:

$$\begin{Bmatrix} \sigma_{zz}(r, 0) \\ D_z(r, 0) \\ \sigma_{zr}(r, 0) \end{Bmatrix} = \sum_{m=1}^3 \int_0^\infty s \begin{Bmatrix} J_0(sr)C_{1m} \\ J_0(sr)C_{2m} \\ J_1(sr)C_{3m} \end{Bmatrix} A_m ds + \begin{Bmatrix} T(r, 0)D_1 \\ T(r, 0)D_2 \\ \int_0^\infty J_1(sr)D_3 A_0 ds \end{Bmatrix}. \quad (7.20)$$

The constants are:

$$C_{1m} = c_{13}a_{1m} - c_{33}\lambda_m a_{2m} - e_{33}\lambda_m a_{3m},$$

$$C_{2m} = e_{31}a_{1m} - e_{33}\lambda_m a_{2m} + \epsilon_{33}\lambda_m a_{3m},$$

$$C_{3m} = -c_{44}\lambda_m a_{1m} - c_{44}a_{2m} - e_{15}a_{3m},$$

$$D_1 = c_{13}f_1 - c_{33}\lambda_0 f_2 - e_{33}\lambda_0 f_3 - \lambda_{33},$$

$$D_2 = e_{31}f_1 - e_{33}\lambda_0 f_2 + e_{33}\lambda_0 f_3 - \beta_3,$$

$$D_3 = -c_{44}\lambda_0 f_1 - c_{44}f_2 - e_{15}f_3. \quad (7.21)$$

In obtaining Eq. (7.20), equation (7.4) for $z=0$ has been used.

Now, the only unknowns are the real constants A_m ($m=1, 2, 3$). They are determined from the mixed electro-mechanical boundary conditions on the $z=0$ plane.

7.2.3. Integral Equations and Crack Tip Field Intensity Factors

Define two auxiliary functions g_1 and g_2 such that $\sum_{m=1}^3 A_m a_{2m} + f_2 A_0 / s = g_1$ and $\sum_{m=1}^3 A_m a_{3m} + f_3 A_0 / s = g_2$. From the symmetry condition $\sigma_{rz} = 0$ on the $z=0$ plane, it follows that $\sum_{m=1}^3 C_{3m} A_m = -A_0 D_3 / s$. Thus, A_m ($m=1, 2, 3$) can be related to the unknown functions g_1 and g_2 through:

$$A_m(s) = \sum_{j=1}^2 \chi_m^j g_j(s) - \frac{1}{s} \delta_m A_0(s), \quad (m=1, 2, 3), \quad (7.22)$$

in which the material constants χ_m^j and δ_m ($m=1, 2, 3; j=1, 2$) are:

$$\begin{bmatrix} \chi_1^1 & \chi_1^2 & \delta_1 \\ \chi_2^1 & \chi_2^2 & \delta_2 \\ \chi_3^1 & \chi_3^2 & \delta_3 \end{bmatrix} = \begin{bmatrix} a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \\ C_{31} & C_{32} & C_{33} \end{bmatrix}^{-1} \begin{bmatrix} 1 & 0 & f_2 \\ 0 & 1 & f_3 \\ 0 & 0 & D_3 \end{bmatrix}. \quad (7.23)$$

From conditions (7.2) and Eqs. (7.17), it follows that $\int_0^\infty g_j(s) J_0(sr) ds = 0$ for $r \geq a$, where ($j=1, 2$). Hence, g_j ($j=1, 2$) have the solution of the form:

$$g_j(s) = \int_0^a \Phi_j(t) \sin(st) dt, \quad (j=1, 2), \quad (7.24)$$

provided that $\lim_{t \rightarrow 0} \Phi_j(t) = 0$. To obtain g_1 and g_2 , the crack face boundary conditions (7.1) must be considered. It follows from Eqs. (7.20), (7.22) and (7.24) that:

$$\int_0^\infty J_0(sr) \left[\int_0^a s \sin(st) \sum_{j=1}^2 \Lambda_{ij} \Phi_j(t) dt \right] ds + P_i = p_{0i}, \quad (7.25)$$

where $(i=1,2)$, $r < a$, Λ_{ij} are some material parameters: $\Lambda_{ij} = \sum_{m=1}^3 C_{im} \chi_m^j$, and P_1 and P_2 are, respectively, the thermally induced stress and electric displacement:

$$P_i(r) = \left(D_i - \sum_{m=1}^3 \delta_m C_{im} \right) T_0(r), \quad (i=1,2). \quad (7.26)$$

The solutions of the integral equations (7.25) are:

$$\sum_{j=1}^2 \Lambda_{ij} \Phi_j(l) = Q_i(l), \quad (i=1,2), \quad (7.27)$$

in which Q_i are general loads defined by:

$$Q_i(l) = \frac{2}{\pi} \int_0^l x \frac{(p_{0i}(x) - P_i(x))}{\sqrt{l^2 - x^2}} dx, \quad (i=1,2). \quad (7.28)$$

Equations (7.27) contain an unknown constant d_0 , which is the normal component of the electric displacement vector inside the crack (see Eqs. (7.1)). To obtain d_0 and Φ_j ($j=1,2$), an additional assumption is needed. This will be discussed in the following sub-sections (A), (B) and (C). Once d_0 and Φ_j ($j=1,2$) are known, the full-field solution is obtained. From the following expressions

$$K_1 = \lim_{r \rightarrow a} \sqrt{2\pi(r-a)} \sigma_{zz}(r,0), \quad K_D = \lim_{r \rightarrow a} \sqrt{2\pi(r-a)} D_z(r,0), \quad (7.29)$$

the stress and electric displacement intensity factors are obtained as:

$$K_1 = -\sqrt{\frac{\pi}{a}} \sum_{j=1}^2 \Lambda_{1j} \Phi_j(a), \quad K_D = -\sqrt{\frac{\pi}{a}} \sum_{j=1}^2 \Lambda_{2j} \Phi_j(a). \quad (7.30)$$

The normal displacement and the electric potential on the upper surface of the crack can be obtained by substituting $\sum_{m=1}^3 A_m a_{2m} + f_2 A_0 / s = g_1$, $\sum_{m=1}^3 A_m a_{3m} + f_3 A_0 / s = g_2$, and Eqs. (7.24) into Eqs. (7.17). The results are

$$\begin{Bmatrix} w(r,0) \\ \phi(r,0) \end{Bmatrix} = \int_r^a \frac{1}{\sqrt{l^2 - r^2}} \begin{Bmatrix} \Phi_1(l) \\ \Phi_2(l) \end{Bmatrix} dl. \quad (7.31)$$

(A) Impermeable crack solution

The simplest way to solve the problem is to consider the crack as electrically impermeable and the electric displacement inside the crack is zero. Therefore, in equations (7.28) $d_0=0$. Thus, Φ_j ($j=1,2$) can be obtained directly from Eqs. (7.27). The field intensity factors obtained from Eqs. (7.30) are as follows:

$$K_1 = \frac{2}{\sqrt{\pi a}} \left(D_1 - \sum_{m=1}^3 \delta_m C_{1m} \right) \int_0^a \frac{x T_0(x)}{\sqrt{a^2 - x^2}} dx, \quad (7.32a)$$

$$K_D = \frac{2}{\sqrt{\pi a}} \left(D_2 - \sum_{m=1}^3 \delta_m C_{2m} \right) \int_0^a \frac{x T_0(x)}{\sqrt{a^2 - x^2}} dx. \quad (7.32b)$$

Therefore, once the temperature distribution on the crack faces, $T_0(r)$, is given, the stress and electric field intensity factors are determined.

(B) Permeable crack solution

Sub-section A gives the solution for the electrically impermeable crack face boundary condition. In that case the crack face is free of electric charges. In piezoelectric fracture problems, the electrically permeable crack-face boundary condition is also used. In such case, the electric displacement on the crack faces is unknown but the electric potential jump between the upper and lower faces of the crack is zero (i.e., $\Phi_2=0$). The only unknown for this problem is Φ_1 which can be determined from the first equation of Eqs. (7.27):

$$\Phi_1(l) = -\frac{2}{\pi} \frac{1}{\Lambda_{11}} \left(D_1 - \sum_{m=1}^3 \delta_m C_{1m} \right) \int_0^l \frac{x T_0(x)}{\sqrt{l^2 - x^2}} dx. \quad (7.33)$$

It can be shown that the thermally induced electrical load P_2 has no effect on the solution for the permeable crack problem. The stress intensity factor at the permeable crack tip remains the same as in the impermeable case, Eq. (7.32a). The electric displacement intensity factor obtained from Eqs. (7.30) is:

$$K_D = \frac{\Lambda_{21}}{\Lambda_{11}} K_1. \quad (7.34)$$

For a permeable crack, the electric potential jump across the crack is zero and the electric charges will be accumulated on the crack surfaces. From Eqs. (7.27), it follows that:

$$\frac{\Lambda_{21}\Phi_1(l) - \frac{2}{\pi} \int_0^l \frac{x d_0(x)}{\sqrt{l^2 - x^2}} dx}{\Lambda_{11}\Phi_1(l)} = \frac{D_2 - \sum_{m=1}^3 \delta_m C_{2m}}{D_1 - \sum_{m=1}^3 \delta_m C_{1m}}. \quad (7.35)$$

The condition $p_{01}=0$, $p_{02}=d_0$ (see Eqs. (7.1)) and Eqs. (7.28) have been used. Substituting Eq. (7.33) into Eq. (7.35), we obtain the normal component of the thermally induced electric displacement vector inside the crack

$$d_0(r) = \bar{\beta} T_0(r), \quad r < a, \quad (7.36)$$

in which the equivalent material parameter $\bar{\beta}$ is

$$\bar{\beta} = D_2 - \frac{\Lambda_{21}}{\Lambda_{11}} D_1 - \sum_{m=1}^3 \delta_m \left(C_{2m} - \frac{\Lambda_{21}}{\Lambda_{11}} C_{1m} \right). \quad (7.37)$$

Clearly, if the temperature on the crack faces is a constant, then the thermally induced electric displacement inside the crack is also a constant.

(C) A notch of finite thickness

The preceding sub-sections (A) and (B) considered both impermeable and permeable crack face electrical boundary conditions. Of these extreme cases, the permeable crack assumption simply treats the crack as electrically conductive, while the impermeable assumption considers the crack as electrically insulated. In engineering practices, however, flaws in a medium are not like cleavage cracks of zero gap thickness, but rather like notches with a finite thickness. Strictly, even if the permittivity of air or vacuum inside the crack is quite small, the flux of an electric field through the crack should not be zero. Hence, it is more reasonable to consider the electric field inside the notch and the electric jump across the notch simultaneously.

Suppose the notch thickness δ is sufficiently small. Along the z -direction the electric displacement d_0 inside the notch can be obtained from

$$d_0(r) = -\epsilon_0 \frac{2\phi(r,0)}{\delta(r)}, \quad (7.38)$$

where ϵ_0 is the dielectric permittivity of the medium inside the notch, which is usually air or vacuum. If the temperature inside the notch T_0 is a constant, and the notch profile is elliptic such that

$$\delta(r) = \sqrt{a^2 - r^2} (\delta_0 / a), \quad (7.39)$$

then the electric displacement inside the notch d_0 is also a constant. It follows from Eqs. (7.26), (7.28), (7.31), (7.38) and (7.39) that

$$d_0 = \frac{D_2 - \frac{\Lambda_{21}}{\Lambda_{11}} D_1 - \sum_{m=1}^3 \delta_m \left(C_{2m} - \frac{\Lambda_{21}}{\Lambda_{11}} C_{1m} \right)}{1 + \frac{\pi}{4} \frac{1}{\epsilon_0} \frac{\delta_0}{\Lambda_{22}^{-1} a}} T_0. \quad (7.40)$$

The stress intensity factor is obtained from Eqs. (7.30), and is found to be the same as the impermeable crack and permeable crack solutions:

$$K_1 = 2\sqrt{\frac{a}{\pi}} \left(D_1 - \sum_{m=1}^3 \delta_m C_{1m} \right) T_0. \quad (7.41)$$

The electric displacement intensity factor at the notch tip is obtained from Eqs. (7.32) as

$$K_D = 2\sqrt{\frac{a}{\pi}} \left[\left(D_2 - \sum_{m=1}^3 \delta_m C_{2m} \right) - d_0 \right] T_0. \quad (7.42)$$

Clearly, the permittivity of the medium inside the notch enters the electric displacement intensity factor, but does not enter the mechanical stress intensity factor.

It can be shown that if the permittivity of the notch is ignored so that ϵ_0 equals zero, Eq. (7.42) is reduced to the impermeable crack solution, Eq. (7.32b) (for a constant T_0). On the other hand, if the notch interior is filled by conductive medium, such that ϵ_0 equals infinity, Eq. (7.42) is reduced to the permeable crack solution, Eq. (7.34). The actual values of K_D reside between the impermeable crack solution and the permeable crack solution.

7.2.4. Applying to Isotropic Materials and Dielectrics

Although the solutions above are provided for transversely isotropic materials, the results for isotropic materials and dielectrics (orthotropic or isotropic) can also be obtained from the proposed analytical model. We discuss this below:

(A) Isotropic materials

The material constants of the isotropic materials are calculated from: $c_{11} = c_{33} = E(1-\nu)/[(1+\nu)(1-2\nu)]$, $c_{12} = c_{13} = \nu E/[(1+\nu)(1-2\nu)]$, $c_{44} = E/[2(1+\nu)]$, and $\lambda_{11} = \lambda_{33} = E\alpha/(1-2\nu)$, where E is the Young's modulus, ν the Poisson's ratio, and α the coefficient of thermal expansion. In order to use the above analytical model, the value of c_{44} in the calculation can be considered to be a little smaller

than the actual value (e.g., 0.001% smaller than the actual value). It is found that such a change in elastic modulus has little influence on the results. As a result, the quantity:

$$D_1 - \sum_{m=1}^3 \delta_m C_{1m} = -\frac{E\alpha}{2(1-\nu)}, \quad (7.43)$$

is obtained. Then, the stress intensity factor for an arbitrarily distributed temperature field $T_0(r)$ on the crack faces in an isotropic material is obtained by substituting Eq. (7.43) into Eq. (7.32a). The result is:

$$K_1 = -\frac{E\alpha}{\sqrt{\pi a}(1-\nu)} \int_0^a \frac{xT_0(x)}{\sqrt{a^2-x^2}} dx. \quad (7.44)$$

If a uniform temperature T_0 is applied on a concentric area $0 < r < b$ of a penny-shaped crack in an isotropic material, the stress intensity factor is:

$$K_1 = -\frac{E\alpha T_0 \sqrt{a}}{\sqrt{\pi}(1-\nu)} \left[1 - \sqrt{1 - \left(\frac{b}{a}\right)^2} \right]. \quad (7.45)$$

Equation (7.45) is same as that in the Handbook [14]. It can be shown that the stress intensity factor is positive if the crack is cooled.

(B) Dielectrics

There is no piezoelectric effect in dielectrics. However, because of the pyroelectric effect and thermal expansion, as in the case of piezoelectric materials, the stress and electric displacement intensity factors may still be developed simultaneously in the dielectrics under thermal loads. The values of the stress and electric displacement intensity factors in dielectrics can be easily obtained from the corresponding equation in Subsections 7.2.2 and 7.2.3 by setting the material constants e_{31} , e_{33} and e_{15} to zero. Further, if the dielectrics are isotropic, the stress intensity factor can also be determined from Eq. (7.44).

Since e_{31} , e_{33} and e_{15} are zero, it is possible to obtain the electric displacement intensity factor in the dielectrics in closed form. From the above analysis, it can be shown that

$$D_2 - \sum_{m=1}^3 \delta_m C_{2m} = -\frac{\beta_3}{1 + \sqrt{\frac{\epsilon_{33} k_1}{\epsilon_{11} k_3}}}. \quad (7.46)$$

The electric displacement intensity factor K_D vanishes for electrically permeable cracks in dielectrics. For an arbitrarily distributed temperature field $T_0(r)$ on the surfaces of an electrically impermeable crack in dielectrics, K_D is obtained by substituting Eq. (7.46) into Eq. (7.32b). The result is:

$$K_D = -\frac{2}{\sqrt{\pi a}} \frac{\beta_3}{1 + \sqrt{\frac{\epsilon_{33} k_1}{\epsilon_{11} k_3}}} \int_0^a \frac{x T_0(x)}{\sqrt{a^2 - x^2}} dx. \quad (7.47)$$

For isotropic dielectrics, $\epsilon_{11} = \epsilon_{33}$ and $k_1 = k_3$ should be used in Eq. (7.47).

7.3 PENNY-SHAPED CRACK SUBJECTED TO PRESCRIBED THERMAL FLUX

Section 7.2 established the solutions for a known temperature on the crack faces. In the cases of a positive thermal flux q on the upper crack face and a negative $-q$ on the lower crack faces (Figure 7.2(A)), and a constant thermal flux on the upper and lower crack faces (Figure 7.2(B)), the analytic and closed-form solutions can also be obtained. This is considered in this section.

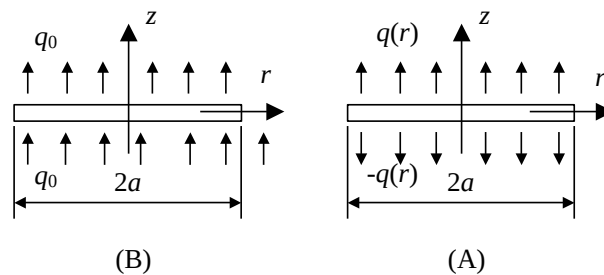


Figure 7.2. A penny-shaped crack in an infinite piezoelectric medium. (A): thermal fluxes on the crack faces are maintained; (B): a constant thermal flux applied on the crack faces^(*).

7.3.1. Crack Surfaces Subjected to Known Thermal Fluxes (Figure 7.2(A))

By direct analogy, it can be seen that the problem of known fluxes on the crack faces shown in Figure 7.2(A) is similar to that of a known temperature on the crack faces given in Section 7.2. The only different is that in the present problem the temperature on the crack faces T_0 is unknown and it must be solved from the prescribed thermal flux boundary

^(*) In most cases, the crack is thermally insulated, and there is a uniform thermal flow, q_0 , in the medium in the direction of the negative z -axis, acting at infinity (original problem). The entire solution is the sum of two solutions: a uniform thermal flow q_0 in the negative z -direction through out the medium including the crack region, and a uniform thermal flow q_0 in the positive direction applied on the crack faces (this is, Fig. 6.2(B)). It is well known that a uniform thermal flow in a fully free homogeneous material does not produce any thermal stresses and electric displacements. Therefore, the near crack tip solution for a thermally insulated crack in piezoelectric materials subjected to a uniform thermal flow at infinity (original problem) is same as the solution of Fig. 6.2(B).

conditions on the crack faces. Once T_0 becomes known, the whole solution can be directly obtained from Section 7.2.

Because the problem is symmetric about $z=0$, it is sufficient to consider the upper half ($z>0$) of the medium. The thermal boundary conditions on the cracked plane are expressed as:

$$\frac{\partial T}{\partial z} = 0, \quad r > a, \quad (7.48a)$$

$$-k_3 \frac{\partial T(r,0)}{\partial z} = q(r), \quad r < a. \quad (7.48b)$$

Eq. (7.48a) indicates that the normal component of the thermal flux vector outside of the crack is zero. Once again, the temperature is given by Eq. (7.4), where the constants $A_0(s)$ is determined from Eqs. (7.48) via the inversion of Hankel transformation. Giving:

$$A_0(s) = \frac{1}{\sqrt{k_1 k_3}} \int_0^a r J_0(sr) q(r) dr. \quad (7.49)$$

The temperature on the crack faces T_0 is obtained by substituting Eq. (7.49) into Eq. (7.4). This yields

$$T_0(x) = \frac{1}{\sqrt{k_1 k_3}} \int_0^a r q(r) \left(\int_0^\infty J_0(sx) J_0(sr) ds \right) dr, \quad x < a. \quad (7.50)$$

Using Eq. (7.50) and the results [17]

$$\int_0^a \frac{x J_0(sx)}{\sqrt{a^2 - x^2}} dx = \frac{\sin(sa)}{s}, \quad (7.51)$$

$$\int_0^\infty \frac{\sin(sa)}{s} J_0(sr) ds = \begin{cases} \pi/2, & a > r \\ \arcsin(a/r), & a < r \end{cases}, \quad (7.52)$$

we obtain

$$\int_0^a \frac{x T_0(x)}{\sqrt{a^2 - x^2}} dx = \frac{1}{\sqrt{k_1 k_3}} \frac{\pi}{2} \int_0^a r q(r) dr. \quad (7.53)$$

Substituting Eq. (7.53) into Eqs. (7.32) and (7.34), the stress and electric displacement intensity factors can be obtained. Summarizing, for a crack subjected to given thermal fluxes $q_0(r)$ on its upper face, and $-q_0(r)$ on its lower face, the thermal stress and electric displacement intensity factors are as follows:

(A) Impermeable crack solution

$$K_1 = \sqrt{\frac{\pi}{a}} \frac{1}{\sqrt{k_1 k_3}} \left(D_1 - \sum_{m=1}^3 \delta_m C_{1m} \right) \int_0^a r q(r) dr, \quad (7.54a)$$

$$K_D = \sqrt{\frac{\pi}{a}} \frac{1}{\sqrt{k_1 k_3}} \left(D_2 - \sum_{m=1}^3 \delta_m C_{2m} \right) \int_0^a r q(r) dr. \quad (7.54b)$$

(B) Permeable crack solution

$$K_1 = \sqrt{\frac{\pi}{a}} \frac{1}{\sqrt{k_1 k_3}} \left(D_1 - \sum_{m=1}^3 \delta_m C_{1m} \right) \int_0^a r q(r) dr, \quad (7.55a)$$

$$K_D = \frac{\Lambda_{21}}{\Lambda_{11}} K_1. \quad (7.55b)$$

Equations (7.54) and (7.55) finalize the analysis for piezoelectric materials contain an embedded penny-shaped crack subjected to thermal fluxes $q(r)$ on its upper face, and $-q$ on its lower face. For isotropic materials and dielectrics, the closed form solutions are obtained as follows:

(A) Isotropic materials

For the special case of isotropic materials, the thermal stress intensity factor is obtained from Eqs. (7.54a) and (7.43) as:

$$K_1 = -\sqrt{\frac{\pi}{a}} \frac{1}{k} \frac{E\alpha}{2(1-\nu)} \int_0^a r q(r) dr, \quad (7.56)$$

where k is the thermal conductivity of the medium. Equation (7.56) is the same as that in the Handbook [14]. The stress intensity factor is positive if the crack is cooled.

(B) Dielectrics

There is no piezoelectric effect in dielectrics. The values of the stress and electric displacement intensity factors in dielectrics can be easily obtained from the corresponding equations above by setting the material constants e_{31} , e_{33} and e_{15} to zero. Further, if the dielectrics are isotropic, the stress intensity factor can also be determined from Eq. (7.56) On the other hand, for an electrically impermeable crack in dielectrics, the electric displacement intensity factor is obtained from Eqs. (7.54b) and (7.46) as

$$K_D = -\sqrt{\frac{\pi}{a}} \frac{1}{\sqrt{k_1 k_3}} \frac{\beta_3}{1 + \sqrt{\frac{\epsilon_{33}}{\epsilon_{11}} \frac{k_1}{k_3}}} \int_0^a r q(r) dr. \quad (7.57)$$

The electric displacement intensity factor for an electrically permeable crack in dielectrics is zero. For isotropic dielectrics $\epsilon_{11}=\epsilon_{33}$ and $k_1=k_3$ should be used in Eq. (7.57).

7.3.2. A Constant Thermal Flux on the Crack Faces ((Figure 7.2(B))

Referring again to the crack geometry shown in Figure 7.1(A). Here we consider a constant thermal flux q_0 maintained on the crack faces, as shown in Figure 7.2(B)^(*). The problem is axi-symmetric and anti-symmetric with respect to the $z=0$ plane. Summarizing, on the boundary $z=0$ of the half-infinite medium $z \geq 0$, $\sigma_{zz}(r, z)=D_z(r, z)=0$ for $0 \leq r < \infty$, and

(1) inside the (insulated) crack, for $0 \leq r < a$ and $z=0$,

$$q = q_0, \quad \sigma_{rz} = 0, \quad (7.58a)$$

where d_0 is the normal component of the electric displacement vector inside the crack. The crack face is free from mechanical tractions so that $p_{01}=0$.

(2) outside the crack, for $r > a$ and $z=0$,

$$u=0, \quad \frac{\partial T}{\partial z} = 0. \quad (7.58b)$$

where u is the radial component of the displacement vector.

(1) The temperature field

The temperature field is still governed by Eq. (7.4). After using thermal boundary conditions (7.58), the dual integral equations for $A_0(s)$ are obtained as follows:

$$\int_0^\infty J_0(sr) A_0 ds = 0, \quad r > a, \quad (7.59a)$$

$$\int_0^\infty s J_0(sr) A_0 ds = \frac{1}{\sqrt{k_1 k_3}} q_0, \quad 0 \leq r < a. \quad (7.59b)$$

* As pointed above, a constant positive thermal flux on the crack faces represents the crack tip solution for a thermally insulated crack in an infinite medium subjected to a far-field uniform thermal flux in the negative z -direction.

From Eq. (7.59a), it follows that A_0 has the solution of the form:

$$A_0(s) = \int_0^{a-} \bar{T}(t) \sin(st) dt, \quad (7.60)$$

provided that $\lim_{x \rightarrow 0} \bar{T}(x) = 0$. Substituting Eq. (7.60) into Eq. (7.59b), we obtain:

$$\int_0^\infty \left(s \int_0^{a-} \bar{T}(t) \sin(st) dt \right) J_0(sr) ds = \frac{1}{\sqrt{k_1 k_3}} q_0, \quad 0 \leq r < a. \quad (7.61)$$

From above equation, the function $\bar{T}(t)$ is found to be:

$$\bar{T}(t) = \frac{2}{\pi} \frac{1}{\sqrt{k_1 k_3}} t q_0, \quad (7.62)$$

Therefore, once the thermal flux on the crack faces is given, the function $\bar{T}(t)$ can be calculated from Eq. (7.62). $A_0(s)$ is given by Eq. (7.60). Then the temperature field is obtained from Eq. (7.4).

(2). *Electro-elasticity field*

In present case, the electro-elastic field is same as Eqs. (7.17). The stresses and electric displacement are in Eqs. (7.20). To use the mixed mode mechanical boundary conditions

(7.58), an auxiliary functions g is defined such that $\sum_{m=1}^3 A_m a_{1m} + f_1 A_0 / s = g$. From the

symmetry conditions $\sigma_{zz} = D_z = 0$ on the $z=0$ plane, it follows that

$\sum_{m=1}^3 C_{jm} A_m = -A_0 D_j / s$, where $(j=1,2)$. Thus, A_m can be related to the unknown

function g through:

$$A_m(s) = \chi_m g(s) - \frac{1}{s} \delta_m A_0(s), \quad (m=1,2,3), \quad (7.63)$$

in which the material constants χ_m^j and δ_m ($m=1,2,3; j=1,2$) are:

$$\begin{bmatrix} \chi_1 & \delta_1 \\ \chi_2 & \delta_2 \\ \chi_3 & \delta_3 \end{bmatrix} = \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ C_{11} & C_{12} & C_{13} \\ C_{21} & C_{22} & C_{23} \end{bmatrix}^{-1} \begin{bmatrix} 1 & f_1 \\ 0 & D_1 \\ 0 & D_2 \end{bmatrix}. \quad (7.64)$$

From conditions (7.58b) and Eqs. (7.17), it follows that $\int_0^\infty g(s)J_1(sr)ds = 0$ for $r \geq a$. Hence, g has the solution of the form:

$$g(s) = s^{1/2} \int_0^a \Phi(t)J_{3/2}(st)dt, \quad (j=1,2), \quad (7.65)$$

provided that $\lim_{t \rightarrow 0} \sqrt{t}\Phi_j(t) = 0$. To solve g , the second of the crack face boundary conditions (7.58a) must be considered. Substituting Eqs. (7.63) and (7.65) into Eqs. (7.20), from the boundary conditions (7.58a) we know that:

$$\Lambda \int_0^\infty J_1(sr) \left[\int_0^a s^{3/2} J_{3/2}(st) \Phi(t) dt \right] ds = -P_0(r), \quad (7.66)$$

where ($i=1,2$), $r < a$, Λ is a material parameter: $\Lambda = \sum_{m=1}^3 C_{3m} \chi_m$, and P_0 is the thermally induced stress:

$$P_0(r) = \left(D_3 - \sum_{m=1}^3 \delta_m C_{3m} \right) \int_0^\infty J_1(sr) A_0(s) ds. \quad (7.67)$$

Substituting of Eqs. (7.60) and (7.62) into Eq. (7.67) gives:

$$P_0(r) = \left(D_3 - \sum_{m=1}^3 \delta_m C_{3m} \right) \frac{1}{2} \frac{1}{\sqrt{k_1 k_3}} q_0 r. \quad (7.68)$$

In obtaining Eq. (7.68), the following integral equality [17] has been used:

$$\int_0^\infty J_1(sr) \sin(st) ds = \begin{cases} \frac{t}{r} \frac{1}{\sqrt{r^2 - t^2}}, & t < r \\ 0, & t > r \end{cases}, \quad (7.69)$$

The solution of Eq. (7.66) is (see Appendix A):

$$\Phi(t) = -\frac{1}{3} \sqrt{\frac{2}{\pi}} \frac{1}{\Lambda} \left(D_3 - \sum_{m=1}^3 \delta_m C_{3m} \right) \frac{1}{\sqrt{k_1 k_3}} q_0 t^{5/2}. \quad (7.70)$$

(3) The stress intensity factor

Once Φ is known, the full-field solution is obtained. Because the applied thermal load is anti-symmetric about $z=0$, only the mode II (shear) stress intensity factor is developed by the thermal flux. From the following expressions

$$K_2 = \lim_{r \rightarrow a} \sqrt{2\pi(r-a)} \sigma_{rz}(r,0), \quad (7.71)$$

the stress intensity factor is obtained as (See Appendix B):

$$K_2 = -\sqrt{2}\Lambda \frac{1}{a} \Phi(a). \quad (7.72)$$

Thus, for a constant thermal flux q_0 applied on the crack faces, the thermal stress intensity factor is obtained from Eqs. (7.70) and (7.72) as follows:

$$K_2 = \frac{2}{3\sqrt{\pi}} \left(D_3 - \sum_{m=1}^3 \delta_m C_{3m} \right) \frac{1}{\sqrt{k_1 k_3}} q_0 a^{3/2}. \quad (7.73)$$

The radial displacement on the upper surface of the crack can be obtained by substituting $\sum_{m=1}^3 A_m a_{1m} + f_1 A_0 / s = g$ and Eq. (7.65) into Eqs. (7.17). The result is

$$u(r,0) = \sqrt{\frac{2}{\pi}} r \int_r^a \frac{1}{\sqrt{l^2 - r^2}} \Phi(l) dl. \quad (7.74)$$

Equation (A2) in Appendix A has been used when deriving Eq. (7.74). For a constant thermal flux, the function Φ is given by Eq. (7.70).

This finalizes the analysis for piezoelectric materials contain an embedded penny-shaped crack subjected to a constant thermal flux on its faces. For isotropic materials, the quantity

$D_3 - \sum_{m=1}^3 \delta_m C_{3m}$ is found to be $E\alpha/[2(1-\nu)]$. The thermal stress intensity factor is obtained from Eqs. (7.73) as:

$$K_2 = \frac{1}{3\sqrt{\pi}} \frac{1}{k} \frac{E\alpha}{1-\nu} q_0 a^{3/2}, \quad (7.75)$$

which is the same as that in the Handbook [14]. It can be shown that the stress intensity factor is positive if q_0 is positive (i.e., a thermally insulated crack in a piezoelectric medium subjected to a thermal flux in the negative z direction).

7.4. DESIGN OF A SMART FUNCTIONALLY GRADED THERMOPIEZOELECTRIC COMPOSITE STRUCTURE

Functionally graded materials feature gradual transitions in microstructure and composition that are engineered so as to meet functional performance requirements that vary with location within a single component and to optimize the overall performance of the component. FGMs may also be deemed to represent piezoelectric materials. In this section, analysis and design of a smart functionally graded piezoelectric structure [18] is introduced. The smart structure consists of three layers: one layer of metal, one layer of piezoelectric material (PZT) used as an actuator, and a graded metal/PZT layer between the metal layer and the PZT layer. A finite element code is developed in the programming environment MATLAB, with each finite element having varied material properties through space coordinates. The results reveal that the both stress discontinuity and thermal deformation of the structure can be controlled. By introducing a functionally graded layer between the PZT actuator layer and the metal beam layer, both stress discontinuity and the edge local stresses can be essentially reduced. The proposed method is expected to be useful for functionally graded thermopiezoelectric composite structures under operating environments where the thermal and piezoelectric effects are important.

7.4.1. Finite Element Formulation of Thermal Equation

The fundamental equations for piezoelectric materials have been given in Chapter 1 of this book. For convenience, the symbol θ will be used to represent the temperature change in the medium.

From the transient thermal equilibrium and constitutive equations, in conjunction with the divergence theorem, the finite element approximation of the heat equation can be obtained as (after assemblage)

$$[C_{\theta\theta}]\{\dot{\theta}\} + [K_{Tu}]\{\dot{u}\} + [K_{T\phi}]\{\dot{\phi}\} + [K_{\theta\theta}]\{\theta\} = \{\bar{p}_\theta\}. \quad (7.76)$$

The element matrices and external heat vector are given by

$$\begin{aligned} [K_{Tu}] &= \int_{\Omega} \theta_0 [N_\theta]^T [\lambda]^T [B_u] d\Omega, \\ [C_{\theta\theta}] &= \int_{\Omega} C_\varepsilon [N_\theta]^T [N_\theta] d\Omega, \\ [K_{T\phi}] &= - \int_{\Omega} \theta_0 [N_\theta]^T \{\beta\}^T [B_\phi] d\Omega, \end{aligned} \quad (7.77)$$

$$[K_{\theta\theta}] = \int_{\Omega} [B_{\theta}]^T [K] [B_{\theta}] d\Omega,$$

$$\{\bar{p}_{\theta}\} = - \int_{s_h} [N_{\theta}]^T \bar{h} ds + \int_{\Omega} \gamma [N_{\theta}]^T d\Omega,$$

where $C_{\theta} = \theta_0 \alpha$ is the thermal capacitance, θ_0 is the reference temperature, Ω is the volume of the element considered, s is the element area; N is the shape function matrix with the subscripts u , ϕ , and θ represent mechanical field, electric field, and thermal field, respectively; $[B_u] = [L_u][N_u]$ with $[L_u]$ denoting a differential operator matrix, $\bar{h}$ is heat flux normal to boundary surface s_h . For an isentropic process for which $\eta=0$, Eq. (7.76) reduces to

$$[K_{\theta\theta}]\{\theta\} = \{\bar{p}_{\theta}\}. \quad (7.78)$$

In this study, all the material property matrices are assumed to be functions of space coordinate.

7.4.2. Finite Element Formulation of Piezoelectric Equation

Define

$$H_{\pi} = \int_{\Omega} (G + \eta\theta) d\Omega - \int_{\Omega} f_i u_i d\Omega - \int_{s_p} \bar{p}_i u_i ds + \int_{s_{\omega}} \omega \phi ds, \quad (7.79)$$

where $\bar{p}_i$ is the vector of surface forces applied to surface s_p , ω the surface charge on the surface s_{ω} , G the thermodynamic potential defined by

$$G = \frac{1}{2} \sigma_{ij} \varepsilon_{ij} - \frac{1}{2} D_i E_i - \eta \theta. \quad (7.80)$$

Through using of the divergence theorem and neglecting body charges, the equilibrium equations can be expressed in an equivalent variational form (Hamilton's principle) as

$$\delta \int_{t_0}^t (\xi - H_{\pi}) dt = 0, \quad (7.81)$$

where the kinetic energy ξ is given by

$$\xi = \frac{1}{2} \int_{t_0}^{t_1} \rho \dot{u}_j \dot{u}_j d\Omega, \quad (7.82)$$

which yields

$$\int_{t_0}^{t_1} \delta \xi dt = \int_{t_0}^{t_1} \int_{\Omega} \rho \delta \dot{u}_j \delta \dot{u}_j dt d\Omega = - \int_{t_0}^{t_1} \int_{\Omega} \rho \delta u_j \delta \ddot{u}_j dt d\Omega. \quad (7.83)$$

Using Eq. (7.81), and with substituting of σ_{ij} and D_i from the governing equations, the following piezoelectric finite element equations can be obtained (after assemblage)

$$[M_u] \{\ddot{u}\} + [K_{uu}] \{u\} + [K_{u\phi}] \{\phi\} + [K_{u\theta}] \{\theta\} = \{\bar{P}_u\}, \quad (7.84)$$

$$[K_{\phi u}] \{u\} + [K_{\phi\phi}] \{\phi\} + [K_{\phi\theta}] \{\theta\} = \{\bar{p}_\phi\}. \quad (7.85)$$

The element matrices and vectors are defined by

$$\begin{aligned} [M_u] &= \int_{\Omega} \rho [N_u]^T [N_u] d\Omega, \quad [K_{uu}] = \int_{\Omega} [B_u]^T [C] [B_u] d\Omega, \\ [K_{u\phi}] &= \int_{\Omega} [B_u]^T [e] [B_\phi] d\Omega, \\ [K_{\phi u}] &= \int_{\Omega} [B_\phi]^T [e]^T [B_u] d\Omega, \quad [K_{u\theta}] = - \int_{\Omega} [B_u]^T [\lambda] [N_\theta] d\Omega, \\ [K_{\phi\phi}] &= \int_{\Omega} [B_\phi]^T [-\epsilon] [B_\phi] d\Omega, \quad [K_{\phi\theta}] = \int_{\Omega} [B_\phi]^T \{\beta\} [N_\theta] d\Omega, \\ \{\bar{p}_\phi\} &= - \int_{s_\omega} [N_\phi]^T \bar{\omega} ds, \quad \{\bar{p}_u\} = \int_{s_p} [N_u]^T \{\bar{p}\} ds + \int_{\Omega} [N_u]^T \{\bar{f}\} d\Omega. \end{aligned} \quad (7.86)$$

The third term in left-hand of Eq. (7.85) represents the pyroelectric effect.

7.4.3. Modal and Material Properties

Figure 7.3(a) shows a piezoelectric actuator pasted on a metal beam. The polarized direction of the PZT material is along the y -axis. The structure subjected to a uniform temperature of T_1 on its low surface and the top surface is kept at zero. For the electric boundary conditions, an electric potential, ϕ_0 was imposed on the upper surface of the PZT layer, while the beam was grounded. The stress distribution induced by temperature gradient and piezoelectric effect is very important to ensure that the structure will not crack and no

plastic deformation will appear within the materials. Due to the discontinuity of the material parameters, large stress discontinuity, which is the direct cause of delamination, is induced on the interface of the two layers. In order to reduce the stress discontinuity as well as the high local stress, a functionally graded piezoelectric layer is introduced between the PZT and the beam, as shown in model 2 of Figure 7.3(b). The material properties of the new smart functionally graded material vary continuously in the thickness direction.

The non-zero material properties of the beam and the PZT are shown in table 7.1. The FGM layer is composed of two components: metal, which is the same as the adjacent beam layer, and PZT, which is the same as the adjacent PZT actuator layer. It is assumed that the material properties of the FGM layer vary linearly from the metal on the lower surface to the PZT on the upper surface.

In Figure 7.3(c) we present an illustration of the finite element model used in this study. Considering symmetry, one half of the structure was analyzed. A two-dimensional 8-node element is utilized for the analysis. The elements near the interface were of size $h_g/10$ along the thickness direction. Mesh refinement to $h_g/20$ with 80 elements along x directions were conducted without appreciable changes in the primary variables.

7.4.4. Active Thermal Deformation Control for Model 1

The resulting bending deformation with volts applied on the surface of the piezoceramic layer is shown in Figure 7.4. Though application of increasing active voltages in the PZT layer, the thermally induced deformation of the plate can be gradually eliminated. Figure 7.4 also shows the reduction in deflection achieved by applying active voltage of 100V, 300V, and 400V, respectively. The active voltage is applied on the upper surface of PZT layer, while the beam layer is kept grounded.

Table 7.1. material properties

	Beam	PZT
Stiffness coefficient c_{11} (10^{10} Pa)	22.6	13.9
Stiffness coefficient c_{12} (10^{10} Pa)	6.78	7.43
Stiffness coefficient c_{33} (10^{10} Pa)	22.6	11.3
Stiffness coefficient c_{44} (10^{10} Pa)	8.7	2.56
Thermal expansion factor α (10^{-6})	12	7
Heat conductivity k_{11} (W/mk)	248	0.52
Heat conductivity k_{33} (W/mk)	248	0.12
Piezoelectric constant e_{31} (C/m ²)		-6.98
Piezoelectric constant e_{33} (C/m ²)		13.84
Piezoelectric constant e_{15} (C/m ²)		13.44
Dielectric permittivity ϵ_{11} (10^{10} C/Vm)		60
Dielectric permittivity ϵ_{33} (10^{10} C/Vm)		54.7

Stress distribution for model 1

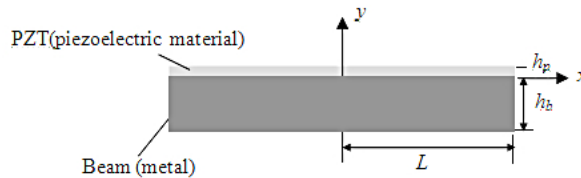
Figures 7.5 show the normal stress (σ_{xx}), normal stress (σ_{yy}), and out-of-plane shear stress (σ_{xy}) along the x direction under different applied voltages. All stresses increase as higher active voltages are applied, reflecting the increase in piezoelectric strains. The results indicate that one potential tradeoff for minimizing thermal deformations would be increased stresses.

Stress distribution for smart FGM

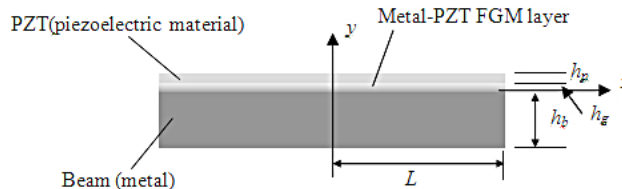
Figures 7.6 illustrate stress distribution of the smart FGM structure (model 2). The maximum stresses σ_{xx} are almost the same as those of model 1, however the stress discontinuity is reduced, as shown in Figure 7.6 (a).

Figures 7.6 (b) and 7.6(c) show the stress distributions near the free edge of the smart FGM structure. It is clear that the maximum stresses are much smaller than those of the model 1.

The results in this section show that it is possible to control the deformation of the structure by choosing specific values of the applied voltage on the PZT actuator layer. By introducing a functionally graded layer between the PZT actuator layer and the metal beam layer, both stress discontinuity and the edge local stresses can be essentially reduced.



(a) Model 1 ($L=100\text{mm}$, $h_b=12\text{mm}$, $h_p=3\text{mm}$)



(b) Model 2 ($L=100\text{mm}$, $h_b=12\text{mm}$, $h_p=2.5\text{mm}$, $h_g=0.5\text{mm}$)



(c) Finite element discretization mesh of the smart structure

Figure 7.3. A piezoelectric actuator pasted on a metal beam.

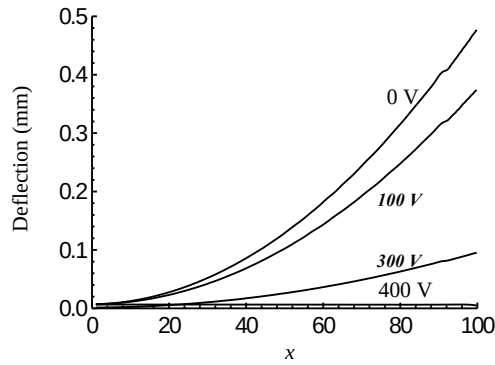


Figure 7.4. Bending deformation of the piezoelectric layer for various active voltages ($y=12\text{mm}$).

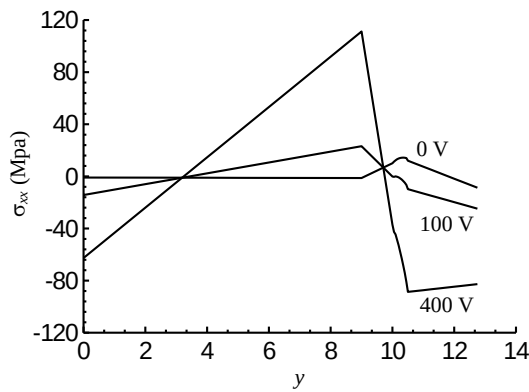


Figure 7.5(a). Normal stress σ_{xx} distribution for active voltages of 0V, 100V, and 400V ($x=0$).

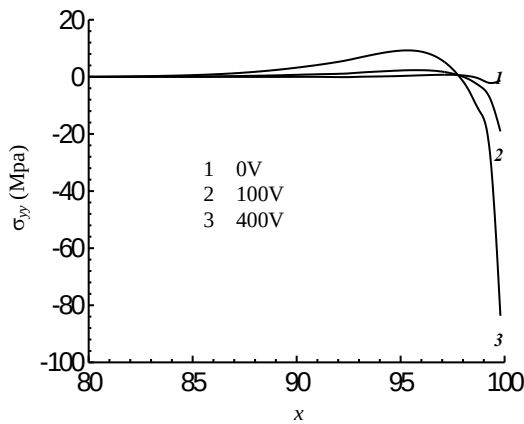


Figure 7.5(b). Normal stress σ_{yy} distribution for active voltages of 0V, 100V, and 400V ($y=12\text{mm}$).

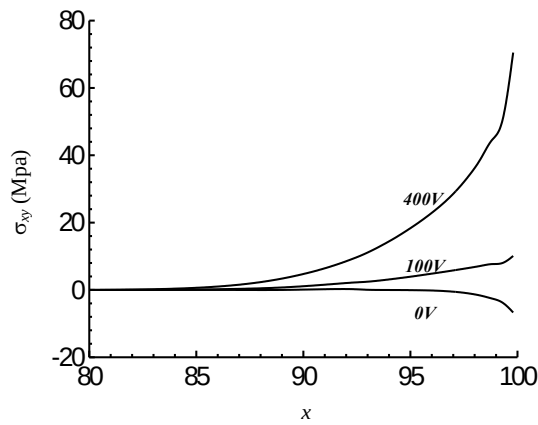


Figure 7.5(c). Shear stress σ_{xy} distribution for active voltages of 0V, 100V, and 400V ($y=12\text{mm}$).

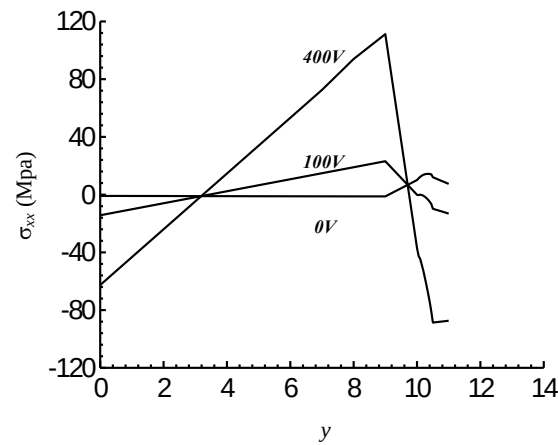


Figure 7.6(a). Normal stress σ_{xx} distribution for active voltages of 0V, 100V, and 400V ($x=0$).

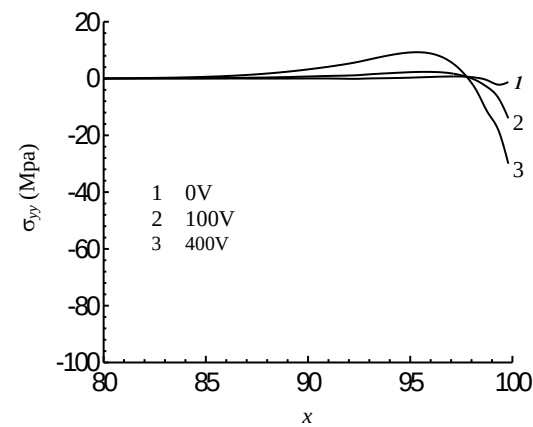


Figure 7.6(b). Normal stress σ_{yy} distribution for active voltages of 0V, 100V, and 400V ($y=12\text{mm}$).

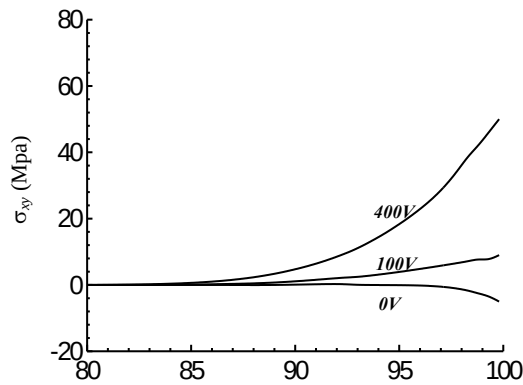


Figure 7.6(c). Shear stress σ_{xy} distribution for active voltages of 0V, 100V, and 400V ($y=12\text{mm}$).

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APPENDIX A

Integrating the inner integral on the LHS of Eqs. (7.66) by parts, we obtain

$$\begin{aligned} \int_0^a s J_{3/2}(st) \Phi(t) dt = \\ - J_{1/2}(sa) \Phi(a) + \int_0^a t^{-1/2} \frac{d[\Phi(t)t^{1/2}]}{dt} J_{1/2}(st) dt. \end{aligned} \quad (A1)$$

Substituting Eq. (A1) into Eqs. (7.66) and using [17]:

$$\int_0^\infty J_\lambda(sr) J_\nu(st) s^{1+\nu-\lambda} ds = \begin{cases} 0 & , \quad 0 < r < t \\ \frac{t^\nu (r^2 - t^2)^{\lambda-\nu-1}}{2^{\lambda-\nu-1} r^\lambda \Gamma(\lambda-\nu)} & , \quad 0 < t < r \end{cases} \quad (A2)$$

where $\lambda > \nu > -1$, we obtain

$$\Lambda \sqrt{\frac{2}{\pi}} \frac{1}{r} \int_0^r \left[\frac{1}{\sqrt{r^2 - t^2}} \frac{d[\Phi(t)t^{1/2}]}{dt} \right] dt = -P_0(r). \quad (A3)$$

This is an Abel integral equation whose solution gives:

$$\Phi(t) = -\sqrt{\frac{2}{\pi}} \frac{t^{-1/2}}{\Lambda} \int_0^t \frac{x^2 P_0(x)}{\sqrt{t^2 - x^2}} dx. \quad (A4)$$

Substituting of Eq. (7.68) into Eq. (A4) gives Eq (7.70).

APPENDIX B

The shear stress on the cracked plane can be obtained by substituting Eqs (7.63) and (7.65) into Eq. (7.20). This gives

$$\sigma_{rz}(r,0) = \Lambda \int_0^\infty J_1(sr) \left[\int_0^a s^{3/2} J_{3/2}(st) \Phi(t) dt \right] ds + P_0(r), \quad (\text{B1})$$

To obtain the singularity at the crack front, Eqs. (A1) and (A2) are considered. The integral in Eq. (B1) is evaluated by neglecting the term $P_0(r)$ in Eq. (B1) and the integrand involving the second term on RHS of Eq. (A1). This leads to the near tip stress solution:

$$\sigma_{rz}(r,0) = - \frac{\sqrt{2a}\Lambda}{\sqrt{\pi r} \sqrt{r^2 - a^2}} \Phi(a), \quad (\text{B2})$$

From the definition of the stress intensity factor, Eq. (7.71), we obtain Eq. (7.72).

Chapter 8

TRANSIENT THERMAL FRACTURE OF PIEZOELECTRIC MATERIALS STRUCTURES

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8.1 INTRODUCTION

It is generally well understood that thermal shock conditions, which arise during sudden heating or cooling of a solid, can result in very high stresses. If the thermal transient is severe enough, sudden fracture may occur. The degree of damage and strength degradation of ceramics subjected to fluctuating thermal environments is a major limiting factor in relation to service requirements and lifetime performance. Early studies to quantify the severity of thermal shock in a plate of finite thickness attempted to draw a correlation between the magnitude of maximum tensile stress on the plate surface and the likelihood of failure. This approach assumes that there are no pre-existing surface flaws. A more sophisticated fracture analysis incorporates an edge crack in the transient stress analysis and thus the degree of severity of any given thermal shock is characterized in terms of the stress intensity factor which is a function of crack size and relevant heat transfer and thermo-elastic coefficients [1-3].

The strength of a material under a transient thermal stress state depends on a number of material properties including thermal expansion coefficient, thermal conductivity, thermal diffusivity, fracture toughness and tensile strength, elastic properties, etc. For piezoelectric ceramics, more parameters can affect transient thermal fracture. Generally, the application of piezothermoelasticity to the fracture of piezoelectric materials under thermal environment involves the solution of two distinct boundary-value problems. The first is a standard potential problem in the classical theory of thermal conduction, and the second is the determination of the mechanical and electrical fields caused by the uneven heating of the various points within the solid. This chapter is mainly concerned with the stress and electric fields in piezoelectric materials subjected to steady state or transient thermal loads. Section 8.2 gives a case study for piezoelectric material strip suddenly exposed to an environment of

different temperature. The poling direction of the strip is parallel to its surfaces. In section 8.3, a piezoelectric material strip whose poling direction is perpendicular to its surfaces is investigated. Section 8.4 investigated an electrically conductive surface crack in a piezoelectric half space under a transient heating (or cooling) on its surface. Crack propagation behavior is discussed and the thermal shock strength of the piezoelectric medium is established. In section 8.5, an internal crack is studied for a piezoelectric cylinder under a transient heating on its outer surface. The transient thermal stress and electric displacement fields, and the associated thermal fracture problem are obtained. The maximum thermal stress and electric displacement, and the maximum thermal stress and electric displacement intensity factors are given. Also obtained are the transient crack opening displacement and the transient electric displacement inside the crack. The strength and fracture of the cylinder are evaluated. Finally, some conclusions are drawn in Section 8.6.

8.2 STRENGTH EVALUATION OF PIEZOELECTRIC CERAMICS UNDER TRANSIENT THERMAL ENVIRONMENTS

This section considers a piezoelectric material strip under a sudden heating or cooling (thermal shock) on its surfaces [4], as shown in Figure 8.1. Both the impermeable and permeable crack assumptions are considered. The temperature distribution is obtained firstly and the thermal stresses and electric displacements of the un-cracked problem are evaluated. The superposition technique is used to solve the governing equations. This is formulated by using the thermal stresses and electric displacements from the un-cracked problem with an opposite sign as the crack surface traction and electric charge loads. Lower bound solutions are obtained for the maximum thermal shock that the material can sustain without catastrophic failure according to a stress-based criterion and a fracture mechanics-based criterion, respectively. The parameters that control the transient thermal stress and electric displacement are also identified.

8.2.1 Temperature Distribution

The problem under consideration is described in Figure 8.1. The coordinates x , y and z are the principal axes of the material symmetry. The x -axis is oriented in the poling direction of the piezoelectric medium (poling direction perpendicular to the x -axis will be considered in Section 8.3). The medium contains a crack normal to the surfaces of the plate. The external thermal loads are assumed to be such that the plane of the crack is a plane of symmetry. The crack problem is solved by superposition. That is, the problem of thermal stresses and electrical displacements in the absence of any cracks is solved first, and then the isothermal crack problem are studied by superposing equal and opposite thermal stresses and electrical displacements to the crack surface tractions and electric charges, respectively. In the problem considered, the heat conduction is one-dimensional, and a straight crack in the z -direction does not obstruct the heat flow in this arrangement, the temperature field T is then governed by

$$k_2 \frac{\partial^2 T(z, t)}{\partial z^2} = \rho c_v \frac{\partial T(z, t)}{\partial t}, \quad (8.1)$$

where k_2 is the coefficient of thermal conductivity of the medium in the z -direction, ρ is the density, and c_v is the specific heat. The heat flow is

$$q_z = -k_2 \frac{\partial T}{\partial z}. \quad (8.2)$$

Suppose the piezoelectric medium is initially at a uniform temperature zero. Then

$$T(z, 0) = 0. \quad (8.3)$$

The following two kinds of boundary conditions are considered:

Boundary Condition (a):

$$T(0, t) = T_0 H(t), \quad (8.4)$$

$$T(h, t) = 0. \quad (8.5)$$

Boundary Condition (b):

$$T(0, t) = T_0 H(t), \quad (8.6)$$

$$\partial T(h, t) / \partial z = 0. \quad (8.7)$$

Equations (8.4) and (8.6) define the thermal shock conditions on the top surface of the medium where $H(t)$ is the Heaviside function.

Equation (8.1) is solved with heat transfer boundary conditions (a) and (b) by a standard separation-of-variables technique, giving

Boundary Condition (a):

$$T(z) = -\sum_{m=1}^{\infty} \frac{2T_0}{m\pi} \sin\left(\frac{m\pi}{h} z\right) \exp\left(-m^2 \pi^2 \frac{t}{t_0}\right) + \frac{T_0}{h} (h - z), \quad (8.8)$$

Boundary Condition (b):

$$T(z) = T_0 \left(1 - 4 \sum_{m=1,3,5}^{\infty} \frac{\sin(m\pi/2)}{m\pi} \exp\left(-\left(\frac{m\pi}{2}\right)^2 \frac{t}{t_0}\right) \cos \frac{m\pi}{2h} (h - z) \right), \quad (8.9)$$

where $t_0 = \rho c_v h^2 / k_2$. It can be seen immediately that the thermal diffusivity $k_2 / \rho c_v$ dictates the time scale for the transient temperature distribution, but does not affect the level of the temperature.

8.2.2 Thermal Stresses and Electric Displacements of the Un-Cracked Medium

Once $T(z)$ is known, the stresses and the electric displacements can be found from the constitutive relations of the linear thermopiezoelectric medium poled along the x -axis:

$$\sigma_{zz} = c_{11}\varepsilon_{zz} + c_{13}\varepsilon_{xx} - e_{31}E_x - \lambda_{11}T, \quad (8.10a)$$

$$\sigma_{xx} = c_{13}\varepsilon_{zz} + c_{33}\varepsilon_{xx} - e_{33}E_x - \lambda_{33}T, \quad (8.10b)$$

$$D_x = e_{31}\varepsilon_{zz} + e_{33}\varepsilon_{xx} - \epsilon_{33}E_x - \beta_3T, \quad (8.10c)$$

where σ_{ij} and D_i ($i, j=x, z$) are stresses and electrical displacements, ε_{ij} and E_i ($i, j=x, z$) are strains and electrical fields, and λ_{ij} , β_i , c_{ij} , e_{ij} , and ϵ_{ii} represent stress-temperature coefficients, pyroelectric constants, elastic constants, piezoelectric constants, and dielectric permittivities, respectively.

If we now assume that the plate is infinite along x - and y -axis, (i.e., $-\infty < (x, y) < \infty$), free of surface tractions and electric charges at $z=0$ and $z=h$, it may be shown that

$$\sigma_{zz}=0, D_z=0, \quad (8.11)$$

and all non-vanishing field quantities are independent of x and y . By solving ε_{zz} from (8.10a), in terms of ε_{xx} , E_x and T , and then substituting it into (8.10b,c), it is found that

$$\sigma_{xx} = \bar{c}_{33}\varepsilon_{xx} - \bar{e}_{33}E_x - \bar{\lambda}_{33}T, \quad (8.12)$$

$$D_x = \bar{e}_{33}\varepsilon_{xx} + \bar{\epsilon}_{33}E_x - \bar{\beta}_3T, \quad (8.13)$$

where

$$\bar{c}_{33} = c_{33} - c_{13}^2 / c_{11}, \quad (8.14)$$

$$\bar{e}_{33} = e_{33} - c_{13}e_{31} / c_{11}, \quad (8.15)$$

$$\bar{\lambda}_{33} = \lambda_{33} - c_{13}\lambda_{11} / c_{11}, \quad (8.16)$$

$$\bar{\epsilon}_{33} = \epsilon_{33} + e_{31}^2 / c_{11}, \quad (8.17)$$

$$\bar{\beta}_3 = \beta_3 - e_{31} \lambda_{11} / c_{11}. \quad (8.18)$$

The compatibility conditions that need to be satisfied are:

$$d^2 \epsilon_{xx} / dz^2 = 0, \quad (8.19)$$

$$\partial E_x / \partial z = \partial E_z / \partial x = 0, \quad (8.20)$$

giving

$$\epsilon_{xx} = Az + B, \quad E_x = \text{constant}. \quad (8.21)$$

Therefore

$$\sigma_{xx} = \bar{c}_{33}(Az + B) - \bar{e}_{33}E_x - \bar{\lambda}_{33}T, \quad (8.22)$$

$$D_x = \bar{e}_{33}(Az + B) + \bar{\epsilon}_{33}E_x - \bar{\beta}_3T, \quad (8.23)$$

where A , B and E_x are unknown constants to be obtained from electromechanical equilibrium conditions at the left and the right ends of the plate. The mechanical equilibrium conditions can be any one of the following:

(1) If the plate is mechanically unconstrained, then

$$\int_0^h \sigma_{xx}(z) dz = 0, \quad \int_0^h \sigma_{xx}(z) z dz = 0, \quad (8.24a)$$

(2) If the plate stretches uniformly but does not bend, then $A=0$, B is determined from the first of Eqs. (8.24a). That is,

$$A=0, \quad \int_0^h \sigma_{xx}(z) dz = 0, \quad (8.24b)$$

(3) For a fully constrained plate, $\epsilon_{xx}=0$, giving

$$A=0, \quad B=0, \quad (8.24c)$$

Likewise, the electrical equilibrium conditions can be one of the following two conditions:

(i) If the plate is electrically unconstrained, then

$$\int_0^h D_x(z) dz = 0, \quad (8.25)$$

(ii) For an electrically constrained plate $E_x=0$.

Therefore, there are $3 \times 2 = 6$ groups of possible electromechanical equilibrium conditions. Either one of these six groups can be used to determine the unknown constants A , B and E_x . From the crack problem under consideration, equal and opposite stresses and electric displacements given by Eqs. (8.22) and (8.23) will be superposed, respectively, to the tractions and electric charges, on the crack surfaces, which constitute the applied loads on the crack faces. The medium is assumed to be under plane strain condition.

8.2.3. The Crack Problem

Assume that the crack surface is free from any mechanical traction. Denote D_0 as the normal component of the electric displacement on the crack faces. Consider also the crack problem as isothermal and quasi-static, and equal and opposite thermal stresses and electric displacements obtained from Eqs. (8.22) and (8.23) are superposed the existing crack surface tractions and electric displacements, respectively. Therefore, the following boundary conditions on the crack faces become:

$$\sigma_{xx}(0, z) = -\sigma_{\infty}(z), \quad z \in [b, c], \quad (8.26)$$

$$D_x(0, z) = D_0(z) - D_{\infty}(z), \quad z \in [b, c], \quad (8.27)$$

where $\sigma_{\infty}(z)$ and $D_{\infty}(z)$ are the corresponding values of stress and electric displacement given by Eqs. (8.22) and (8.23). Due to symmetry of the problem with respect to the $x=0$ plane, we have

$$u(0, z) = 0, \quad \phi(0, z) = 0, \quad z \notin [b, c], \quad (8.28)$$

$$\sigma_{xz}(0, z) = 0, \quad z \in [0, h]. \quad (8.29)$$

On the top surface $z=h$, the traction and electric charge boundary conditions give:

$$\sigma_{xz}(x, 0) = 0, \quad \sigma_{zz}(x, 0) = 0, \quad D_z(x, 0) = 0. \quad (8.30)$$

At the bottom surface $z=h$, two kinds of boundary conditions are considered: the first is the traction and electric charge free conditions:

$$\sigma_{xz}(x, h) = 0, \quad \sigma_{zz}(x, h) = 0, \quad D_z(x, h) = 0, \quad (8.31a)$$

and the second is that the plate stretches freely but without bending:

$$\sigma_{xz}(x, h) = 0, \partial w(x, h) / \partial x = 0, D_z(x, h) = 0. \quad (8.31b)$$

In the absence of temperature change, the equilibrium equations for the plane problem of a piezoelectric medium poled along x -direction are:

$$\left. \begin{aligned} c_{11}w_{,zz} + c_{44}w_{,xx} + (c_{13} + c_{44})u_{,xz} + (e_{31} + e_{15})\phi_{,xz} &= 0 \\ (c_{13} + c_{44})w_{,xz} + c_{44}u_{,zz} + c_{33}u_{,xx} + e_{15}\phi_{,zz} + e_{33}\phi_{,xx} &= 0 \\ (e_{31} + e_{15})w_{,xz} + e_{15}u_{,zz} + e_{33}u_{,xx} - \epsilon_{11}\phi_{,zz} - \epsilon_{33}\phi_{,xx} &= 0 \end{aligned} \right\}. \quad (8.32)$$

Since the problem is symmetric with respect to the $x=0$ plane, it is sufficient to consider one-half ($x>0$) of the medium only. The solutions of the governing equations (8.32) can be expressed in terms of unknown coefficients $A_m(s)$ and $B_m(\xi)$ in the following forms:

$$\left\{ \begin{aligned} u(x, z) \\ w(x, z) \\ \phi(x, z) \end{aligned} \right\} = \frac{2}{\pi} \sum_{m=1}^6 \int_0^\infty \left\{ \begin{aligned} a_{1m} \sin(sx) \\ a_{2m} \cos(sx) \\ a_{3m} \sin(sx) \end{aligned} \right\} A_m \exp(s\lambda_m z) ds +$$

$$\frac{1}{2\pi} \sum_{m=1}^3 \int_{-\infty}^\infty \left\{ \begin{aligned} a_{1m} \\ i \operatorname{sgn}(\xi) a_{2m} \\ a_{3m} \end{aligned} \right\} B_m \exp\left(\frac{|\xi|}{\lambda_m} x\right) \exp(-i\xi z) d\xi, \quad (8.33)$$

where $\operatorname{sgn}(\xi)$ equals 1 for positive values of ξ , and -1 for negative values of ξ ; A_m and B_n ($m=1, \dots, 6$; $n=1, 2, 3$) are unknown functions of variables s and ξ , respectively; λ_m and the corresponding eigenvectors (a_{1m} , a_{2m} , a_{3m}) are determined from the following characteristic equations:

$$\begin{bmatrix} c_{44}\lambda_m^2 - c_{33} & -(c_{13} + c_{44})\lambda_m & e_{15}\lambda_m^2 - e_{33} \\ -(c_{13} + c_{44})\lambda_m & c_{44} - c_{11}\lambda_m^2 & -(e_{31} + e_{15})\lambda_m \\ e_{15}\lambda_m^2 - e_{33} & -(e_{31} + e_{15})\lambda_m & \epsilon_{33} - \epsilon_{11}\lambda_m^2 \end{bmatrix} \begin{Bmatrix} A_{1m} \\ A_{2m} \\ A_{3m} \end{Bmatrix} = 0, \quad (8.34)$$

In Eq. (8.34), there are six roots for λ , forming three conjugate pairs. The order of roots λ_m are arranged such that $\operatorname{Re}(\lambda_1)<0$, $\operatorname{Re}(\lambda_2)<0$, $\operatorname{Re}(\lambda_3)<0$ and $\operatorname{Re}(\lambda_4)>0$, $\operatorname{Re}(\lambda_5)>0$, $\operatorname{Re}(\lambda_6)>0$. It can be shown that Eqs. (8.33) satisfy the conditions that the displacements and electric potential vanish as x approaches positive infinity.

In the absence of temperature variation, the stresses and electric displacements associated with Eqs. (8.33) are:

$$\begin{aligned}
\begin{Bmatrix} \sigma_{xx}(x, z) \\ D_x(x, z) \\ \sigma_{zz}(x, z) \\ \sigma_{xz}(x, z) \\ D_x(x, z) \end{Bmatrix} &= \frac{2}{\pi} \sum_{m=1}^6 \int_0^\infty s \begin{Bmatrix} C_{1m} \cos(sx) \\ C_{2m} \cos(sx) \\ C_{3m} \cos(sx) \\ C_{4m} \sin(sx) \\ C_{5m} \sin(sx) \end{Bmatrix} \exp(s\lambda_m z) A_m ds + \\
\frac{1}{2\pi} \sum_{m=1}^3 \int_{-\infty}^\infty \frac{1}{\lambda_m} &\begin{Bmatrix} |\xi| C_{1m} \\ |\xi| C_{2m} \\ |\xi| C_{3m} \\ -i\xi C_{4m} \\ -i\xi C_{5m} \end{Bmatrix} \exp\left(\frac{|\xi|x}{\lambda_m}\right) \exp(-i\xi z) B_m d\xi, \quad (8.35)
\end{aligned}$$

where the coefficients C_{jm} ($j=1, \dots, 5$) are:

$$\begin{aligned}
C_{1m} &= c_{33}a_{1m} + c_{13}\lambda_m a_{2m} + e_{33}a_{3m}, \\
C_{2m} &= e_{33}a_{1m} + e_{31}\lambda_m a_{2m} - \epsilon_{33} a_{3m}, \\
C_{3m} &= c_{13}a_{1m} + c_{11}\lambda_m a_{2m} + e_{31}a_{3m}, \\
C_{4m} &= c_{44}\lambda_m a_{1m} - c_{44}a_{2m} + e_{15}\lambda_m a_{3m}, \\
C_{5m} &= e_{15}\lambda_m a_{1m} - e_{15}a_{2m} - \epsilon_{11} \lambda_m a_{3m}. \quad (8.36)
\end{aligned}$$

(1) The singular integral equation

We now introduce two new unknown functions

$$\begin{Bmatrix} g_1(z) \\ g_2(z) \end{Bmatrix} = \frac{\partial}{\partial z} \begin{Bmatrix} u(0, z) \\ \phi(0, z) \end{Bmatrix}. \quad (8.37)$$

From the continuity condition (8.28) it follows that

$$g_i(z) = 0, \text{ for } z \notin [b, c], \quad (i = 1, 2). \quad (8.38)$$

In the case of an embedded crack ($b > 0$, $c < h$), $g_i(z)$ should satisfy the following single-value conditions

$$\int_b^c g_i(z) dz = 0, (i = 1, 2). \quad (8.39)$$

In the case of an edge crack ($b=0, c<h$), the single-value conditions are no longer satisfied.

Substituting Eqs. (8.33) into Eqs. (8.37), using conditions (8.38), and inverting the corresponding Fourier integrals, we obtain

$$\sum_{m=1}^3 a_{1m} B_m = i\xi^{-1} \int_b^c g_1(r) e^{i\xi r} dr, \quad (8.40)$$

$$\sum_{m=1}^3 a_{3m} B_m = i\xi^{-1} \int_b^c g_2(r) e^{i\xi r} dr. \quad (8.41)$$

From the symmetry condition (8.29), we know that

$$\sum_{m=1}^3 \lambda_m^{-1} C_{4m} B_m = 0. \quad (8.42)$$

It follows from Eqs. (8.40)–(8.42) that:

$$B_m(\xi) = i\xi^{-1} \int_b^c [b_{m1} g_1(r) + b_{m2} g_2(r)] \exp(i\xi r) dr, \quad (8.43)$$

where the material constants b_{m1} and b_{m2} ($m=1, 2, 3$) are given by:

$$\begin{bmatrix} b_{11} & b_{12} \\ b_{21} & b_{22} \\ b_{31} & b_{32} \end{bmatrix} = \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{31} & a_{32} & a_{33} \\ C_{41} & C_{42} & C_{43} \end{bmatrix}^{-1} \begin{bmatrix} 1 & 0 \\ 0 & 1 \\ 0 & 0 \end{bmatrix}. \quad (8.44)$$

By applying Eqs. (8.35) to the homogeneous boundary conditions (8.30) and (8.31), and using Eq. (8.44), we obtain:

$$[G]\{A\} = \frac{1}{2s} \int_b^c [\{f_1(s, r)\} g_1(r) + \{f_2(s, r)\} g_2(r)] dr, \quad (8.45)$$

where $\{A\}=\{A_m\}$, the 6×6 matrix $[G]$ and the vectors $\{f_1\}$ and $\{f_2\}$ are given in Appendix A.

The linear algebraic equations (8.45) can be used to determine the unknowns A_m ($m=1, \dots, 6$), in terms of g_1 and g_2 , the results are

$$A_m = \frac{1}{2s} \int_b^c [\chi_{m1}(s, r) g_1(r) + \chi_{m2}(s, r) g_2(r)] dr, (m=1, \dots, 6), \quad (8.46)$$

in which the coefficients $\chi_{mj}(s, r)$ ($j=1,2$) are the elements of the vectors $\{\chi_j\}$:

$$\{\chi_j\} = [G]^{-1} \{f_j\}. \quad (8.47)$$

By substituting Eqs. (8.43) and (8.46) into Eqs. (8.35), the stress σ_{xx} and electric displacement D_x may be expressed in terms of $g_1(z)$ and $g_2(z)$. At the cracked plane ($x=0$), we have

$$\frac{[\Lambda]}{\pi} \int_b^c \frac{1}{r-z} \begin{Bmatrix} g_1(r) \\ g_2(r) \end{Bmatrix} dr + \int_b^c [H(z, r)] \begin{Bmatrix} g_1(r) \\ g_2(r) \end{Bmatrix} dr = \begin{Bmatrix} \sigma_{xx}(0, z) \\ D_x(0, z) \end{Bmatrix}, \quad (8.48)$$

where the kernels $[H]$ and $[\Lambda]$ are two by two matrices, whose elements are:

$$H_{ij}(z, r) = \frac{1}{\pi} \int_0^\infty \sum_{m=1}^6 C_{im} \chi_{mj}(s, r) \exp(s \lambda_m z) ds, \quad (8.49)$$

and

$$\Lambda_{ij} = - \sum_{m=1}^3 \lambda_m^{-1} C_{im} b_{mj}. \quad (8.50)$$

The singular integral equations for the as-yet-unknown functions g_1 and g_2 are obtained by applying Eqs. (8.48) to the crack face boundary conditions (8.26) and (8.27), the results are

$$\frac{1}{\pi} [\Lambda] \int_b^c \frac{1}{r-z} \begin{Bmatrix} g_1(r) \\ g_2(r) \end{Bmatrix} dr + \int_b^c [H(z, r)] \begin{Bmatrix} g_1(r) \\ g_2(r) \end{Bmatrix} dr = \begin{Bmatrix} -\sigma_\infty(z) \\ D_0(z) - D_\infty(z) \end{Bmatrix}, \quad z \in [b, c]. \quad (8.51)$$

As time t only enters the problem via $\sigma_\infty(z)$ and $D_\infty(z)$ in Eq. (8.51), the singular integral equation is solved separately for individual values of t . By normalizing the integral interval $[b, c]$ in the following way:

$$z = \frac{c-b}{2} \bar{z} + \frac{c+b}{2}, \quad (8.52)$$

$$r = \frac{c-b}{2} \bar{r} + \frac{c+b}{2}, \quad (8.53)$$

the integral equation (8.51) would then reduce to a standard form, which can be used to solve the auxiliary functions g_1 and g_2 .

(2) Stress and electric displacement intensity factors

The singular Integral equation (8.51) contains Cauchy-type kernels. In the case of edge crack problem, the solutions may be expressed as [5]:

$$\begin{Bmatrix} g_1(r) \\ g_2(r) \end{Bmatrix} = \frac{1}{\sqrt{1-r}} \sum_{n=0}^{\infty} \begin{Bmatrix} a_{n1} \\ a_{n2} \end{Bmatrix} T_n(\bar{r}), \quad (8.54)$$

where T_n is the Chebyshev polynomial of the first kind, a_{n1} and a_{n2} are unknown coefficients. Applying Eq. (8.54) to Eq. (8.51) and following the procedure outlined in [5], we obtain

$$\begin{aligned} & [\Lambda] \frac{1}{\pi} \frac{\log|B(\bar{z})|}{\sqrt{1-\bar{z}}} \sum_{n=0}^{\infty} \begin{Bmatrix} a_{n1} \\ a_{n1} \end{Bmatrix} T_n(\bar{z}) + [\Lambda_0] \frac{1}{\pi} \sum_{n=0}^{\infty} \begin{Bmatrix} a_{n1} \\ a_{n2} \end{Bmatrix} \int_{-1}^1 \frac{T_n(\bar{r}) - T_n(\bar{z})}{\sqrt{1-\bar{r}}(\bar{r}-\bar{z})} d\bar{r} \\ & + \frac{c}{2} \sum_{n=0}^{\infty} \left(\int_{-1}^1 [H(z, r)] \frac{T_n(\bar{r})}{\sqrt{1-\bar{r}}} d\bar{r} \right) \begin{Bmatrix} a_{n1} \\ a_{n2} \end{Bmatrix} = \begin{Bmatrix} -\sigma_{\infty}(z) \\ D_0(z) - D_{\infty}(z) \end{Bmatrix}, \end{aligned} \quad (8.55)$$

where

$$B(\bar{z}) = \frac{1 + \sqrt{(1-\bar{z})/2}}{1 - \sqrt{(1-\bar{z})/2}}. \quad (8.56)$$

For the embedded crack problem the solution of Eq. (8.51) has the following form

$$\begin{Bmatrix} g_1(r) \\ g_2(r) \end{Bmatrix} = \frac{1}{\sqrt{1-r^2}} \sum_{n=1}^{\infty} \begin{Bmatrix} a_{n1} \\ a_{n2} \end{Bmatrix} T_n(\bar{r}). \quad (8.57)$$

Application of Eq. (8.57) to Eq.(8.51) yields

$$[\Lambda] \sum_{n=1}^{\infty} \begin{Bmatrix} a_{n1} \\ a_{n2} \end{Bmatrix} U_{n-1}(\bar{z}) + a \sum_{n=1}^{\infty} \left(\int_{-1}^1 [H(z, r)] \frac{T_n(\bar{r})}{\sqrt{1-\bar{r}^2}} d\bar{r} \right) \begin{Bmatrix} a_{n1} \\ a_{n2} \end{Bmatrix} = \begin{Bmatrix} -\sigma_{\infty}(z) \\ D_0(z) - D_{\infty}(z) \end{Bmatrix}, \quad (8.58)$$

where U_n is the Chebyshev polynomial of the second kind.

Equations (8.55) and (8.58) can be solved by applying the proper electrical boundary conditions on the crack faces. As mentioned previously, two kinds of electrical boundary conditions on the crack surfaces are usually considered. One commonly used electrical boundary condition is that the crack is assumed to be impermeable to electric fields. That is, the crack face is charge free and the electric displacement $D_0(z)$ vanishes everywhere inside the crack. The other electrical boundary condition assumes the crack is electrically permeable, i.e., no electric potential jump across the crack. Therefore, $a_{n2}=0$ and the normal component

$D_0(z)$ of the electric displacement inside of the crack is unknown. In both cases, equations (8.55) and (8.58) can be solved by truncating the series and using a collocation technique [5]. After determining the coefficients a_{n1} and a_{n2} , the stress intensity factor K_I and the electric displacement intensity factor K_D ahead of the crack tips at $z=b$ and $z=c$ can be defined and calculated as

$$\begin{Bmatrix} K_I(c) \\ K_D(c) \end{Bmatrix} = \left(\sqrt{2\pi[z-c]} \right)_{z \rightarrow c+0} \begin{Bmatrix} \sigma_{xx}(z) \\ D_x(z) \end{Bmatrix} = -\sqrt{\pi c} [\Lambda] \sum_{n=0}^{\infty} \begin{Bmatrix} a_{n1} \\ a_{n2} \end{Bmatrix}. \quad (8.59)$$

for the edge crack, and

$$\begin{Bmatrix} K_I(b) \\ K_D(b) \end{Bmatrix} = \left(\sqrt{2\pi[b-z]} \right)_{z \rightarrow b-0} \begin{Bmatrix} \sigma_{xx}(z) \\ D_x(z) \end{Bmatrix} = \sqrt{\pi a} [\Lambda] \sum_{n=1}^{\infty} (-1)^n \begin{Bmatrix} a_{n1} \\ a_{n2} \end{Bmatrix}, \quad (8.60)$$

$$\begin{Bmatrix} K_I(c) \\ K_D(c) \end{Bmatrix} = \left(\sqrt{2\pi[z-c]} \right)_{z \rightarrow c+0} \begin{Bmatrix} \sigma_{xx}(z) \\ D_x(z) \end{Bmatrix} = -\sqrt{\pi a} [\Lambda] \sum_{n=1}^{\infty} \begin{Bmatrix} a_{n1} \\ a_{n2} \end{Bmatrix}, \quad (8.61)$$

for the embedded crack.

8.2.4. Cracking due to Thermal Shock

Consider a piezoelectric plate of width h . The plate is initially at a zero temperature environment and the temperature at its surfaces $z=0$ and h are suddenly changed to T_0 . This thermal shock problem may be used to study the thermal stresses in a quenched specimen. The temperature field in the plate is symmetric with respect to the $z=h/2$ plane where zero thermal flow condition holds. Therefore, the temperature is determined by Eq. (8.9) with h replaced by $h/2$:

$$T(z) = T_0 \left(1 - 4 \sum_{m=1,3,5}^{\infty} \frac{1}{m\pi} \exp\left(- (m\pi)^2 \frac{t}{t_0}\right) \sin \frac{m\pi}{h} z \right), \quad (8.62)$$

where $t_0 = \rho c_v h^2 / k_2$.

Suppose the plate is mechanically and electrically unconstrained at the far ends. The thermal stress $\sigma(z, t) = \sigma_{xx}(z, t)$ and the electric displacement $D(z, t) = D_x(z, t)$ are obtained directly from equations (8.22), (8.23), (8.24a) and (8.25), and are written in non-dimensional form as

$$\bar{\sigma} = \frac{\sigma_{xx}(z, t)}{\lambda_{33} T_0} = \frac{4}{\pi} \sum_{m=1,3,5}^{\infty} \frac{1}{m} \exp\left(- (m\pi)^2 \frac{t}{t_0}\right) \left[\sin\left(\frac{m\pi}{h} z\right) - \frac{2}{m\pi} \right], \quad (8.63)$$

$$\bar{D} = \frac{D_x(z, t)}{\beta_3 T_0} = \frac{4}{\pi} \sum_{m=1,3,5}^{\infty} \frac{1}{m} \exp\left(- (m\pi)^2 \frac{t}{t_0}\right) \left[\sin\left(\frac{m\pi}{h} z\right) - \frac{2}{m\pi} \right]. \quad (8.64)$$

Thus, among many thermo-electro-mechanical parameters in piezoelectric materials, the parameters that govern the transient thermal stress and electric displacement levels in the plate are identified as $\bar{\lambda}_{33}$ and $\bar{\beta}_3$, respectively. The time scale for the transient thermal stress and electric displacement state is governed by the thermal diffusivity $\rho c_v / k_2$, which does not affect the levels of thermal stresses and electric displacements.

The dimensionless thermal stresses $\bar{\sigma}(z, t)$ and electric displacements $\bar{D}(z, t)$ are plotted against dimensionless Fourier number $F_0 = t/t_0 = tk_2 / \rho c_v h^2$ in Figures 8.2 and 8.3 for different position in the plate. It is clear that if $T_0 < 0$ (cold shock), the region near the surface of the plate experiences a tensile stress while a compressive zone is developed at the center of the plate. The maximum tensile stress is attained at the surface and the compressive stress is largest at the center of the plate. The maximum values of

$$\sigma = -\bar{\lambda}_{33} T_0, \quad D = -\bar{\beta}_3 T_0, \quad (8.65)$$

are achieved at the plate surface at $t=0$. The maximum compressive stress and electric displacement

$$\sigma = 0.308 \bar{\lambda}_{33} T_0, \quad D = 0.308 \bar{\beta}_3 T_0, \quad (8.66)$$

are attained at the plate center at a time $t/t_0 = 0.0290$. These values are valid for all piezoelectric materials.

Under a hot shock ($T_0 > 0$), at any location of the plate, the stress and electric displacement will change their signs from the cold shock. Hence the surface of the plate will become compressive and the center will be tensile. The magnitudes of the thermal stresses and electric displacements subjected to a hot shock will be same as those in a cold shock.

As described above, the surface of the plate undergoes a transient tensile stress under a cold shock, while the center of the plate has the largest tensile stress under a hot shock. It is then expected that a crack may initiate at the surface under cold shocking or at the center under hot shocking. The crack grows unstably until it enters the compressive region. The problem can be synthesized from the general solution for an embedded crack in a finite plate, with crack tips located at $z=b$ and $z=c$, as shown in Figure 8.1. Since the crack plane is normal to the top and bottom surfaces of the plate, it does not perturb the transient temperature distribution in this arrangement, determination of the temperature distribution and resulting thermal stresses and electric displacements would be quite straightforward. The stresses and electric displacements obtained from Eqs. (8.63) and (8.64), with opposite signs, become the crack surface tractions and electric charges in the quasi-static problem. The thermal stress and electrical displacement intensity factors are solved by the method outlined in the last section.

(1) A double-edge cracked plate under cold shocking

The problem for a cold shock can be idealized to the case of a double-edge-cracked plate, as shown in Figure 8.4. For this crack geometry the stress intensity factor K_I and electric displacement intensity factor K_D are calculated during a cold shock event. A PZT-5H piezoelectric ceramic is considered. The properties are given in Table 4.1. The stress intensity factors are found to be independent of the crack surface electrical boundary condition assumption (whether impermeable or permeable). Hence, the material property $\bar{\beta}_3$ does not influence the thermal stress intensity factors. The normalized stress intensity factor $\bar{K}_I = K_I / K_{I0}$, where $K_{I0} = -\bar{\lambda}_{33}\sqrt{\pi h}T_0$, is plotted against dimensionless time t/t_0 in Figure 8.5, for selected normalized crack depth c/h . The results are normalized such that they do not depend on crack length c . At any given crack depth c , the stress intensity factor increases from an initial zero with time, displays a peak value and then decreases to zero as time approaches infinity. This means that the thermal stress intensity factor occurs only at transient state and that its steady state value is zero. It is further noted that the magnitude of K_I depends on crack length, and achieves a peak values at a crack of length $c \approx 0.093h$.

Figure 8.6 shows the variation of stress intensity factor K_I with crack depth c , at different time. If we consider a crack of given initial length, we see that the temporal behavior is alike for all crack depths. The envelope of this family of curves is the locus of the highest stress intensity factors reached for every crack depths at any time. It is seen that K_I attains an overall maximum value

$$K_{I_{\max}} = 0.160K_{I0} = -0.160\bar{\lambda}_{33}\sqrt{\pi h}T_0, \quad (8.67)$$

at a particular time $t^* \approx 0.01t_0$ and a particular crack depth $c^* = 0.093h$. Note that these expressions are only valid for material parameters used in the numerical analysis.

The electric displacement intensity factor K_D for the impermeable crack assumption is shown in Figures 8.7 and 8.8, for selected values of crack depth. Since K_D can be caused by thermally induced stresses or electric displacements, they are plotted respectively in these two figures. The total electrical displacement intensity factors should be the combination of those in Figures 8.7 and 8.8. At any given crack depth c , the electric displacement intensity factor increases from the initial zero value, displays a peak value after a finite time and decreases to zero as time approaches infinity.

As for the permeable crack assumption, the electrical displacement intensity factor K_D does not depend on the electrical displacement load $\bar{\beta}_3 T_0$. Values of K_D for several crack depths are plotted in Figure 8.8.

(2) A center-cracked plate under hot shock

Consider the piezoelectric ceramic plate shown in Figure 8.1 subjected to a hot shock: initially the temperature of the plate is zero. At time $t=0$ the top and bottom surfaces of the plate are exposed suddenly to temperature T_0 (>0). The material properties for this problem are same as those for the cold shock. As discussed before, hot shock induces transient tensile stresses at the center of the plate, with peak value specified by $\bar{\sigma} = 0.308$ at time $t/t_0 = 0.0290$. This stress is most likely to develop a mode I crack in the center of the plate. Now assume the

plate contains a pre-existing center crack of length $2a$, symmetrical with respect to the central line of the plate, as shown in Figure 8.10. The stress intensity factor and electrical displacement intensity factor associated with the transient thermal stress and electric displacement loads (8.63) and (8.64) are determined from Eqs. (8.60) and (8.61) by solving the singular integral equation (8.58).

The non-dimensional stress intensity factors K_I / K_{I0} , where $K_{I0} = \bar{\lambda}_{33} \sqrt{\pi h T_0}$, are plotted against time t/t_0 for selected crack length in Figure 8.11. At any crack length, the stress intensity factor increases from zero to a maximum value, and then decays to zero. Figure 8.12 illustrates the values of stress intensity factors K_I with crack length, for selected times. If we consider a crack of given initial length, we can see that the temporal behavior is alike for all depths. The stress intensity factors increases with crack length to a maximum and then slowly decreases. Those curves form an envelope that represents the relationship between crack length and the largest stress intensity factor. It is noted that K_I attains an overall maximum value

$$K_{I_{max}} = 0.131 K_{I0} = 0.131 \bar{\lambda}_{44} \sqrt{\pi h T_0}, \quad (8.68)$$

at a particular time $t^* \approx 0.017 t_0$ and a particular crack length $2a^* = 0.56h$. These expressions are again only valid for material parameters used in the numerical analysis.

For completeness of results, the electric displacement intensity factors K_D are also calculated and plotted in Figures 8.13 and 8.14 for the impermeable crack assumption, and in Figure 8.15 for the permeable crack assumption. Again K_D values do not depend on the electrical displacement load $\bar{\beta}_3 T_0$ for the permeable crack.

8.2.5. Thermal Shock resistance

Thermal shock resistance is a major issue in the design of engineering ceramics for applications at high temperatures. A central problem is the identification of appropriate material selection criterion in order to select the most shock-resistant material for a given application. Material performance indices are summarized for both stress-controlled and fracture mechanics-controlled failure criteria.

A stress-based failure criterion for heat shock is that the maximum thermal stress appears on the surface (for cold shock) or the center (for hot shock) of the plate equals or exceeds the strength value σ_b . The maximum temperature jump sustainable by the material ΔT_c follows from equations (8.65) and (8.66) as

$$\Delta T_c = \Delta T_c^{\text{stress}} = \sigma_b / \bar{\lambda}_{33}, \text{ for cold shock}, \quad (8.69a)$$

$$\Delta T_c = \Delta T_c^{\text{stress}} = 3.25 \sigma_b / \bar{\lambda}_{33}, \text{ for hot shock}. \quad (8.69b)$$

Equations (8.69) are valid for all materials.

A similar strategy can be employed to rank the materials on the basis of failure from a dominant crack by thermal shock. The fracture mechanics-based criterion for hot and cold shock is taken to be that the maximum thermal stress intensity factor $K_{\max}(a^*, t^*)$ equals or exceeds the toughness of the solid K_{IC} . Obviously, the maximum temperature jump sustainable by the solid ΔT_c depends on the depth of the pre-existing crack, since the maximum stress intensity factor depends on the crack length. As the envelope in Figures 8.6 and 8.12 represent the maximum stress intensity factor $K_{\max}$ for different crack lengths, it follows from $K_{\max}=K_{IC}$ that

$$\Delta T_c = \Delta T_c^{\text{fracture}} = \frac{K_{IC}}{\overline{K\lambda_{33}}\sqrt{\pi h}}. \quad (8.70)$$

It is instructive to plot the sustainable temperature jump ΔT_c as a function of crack length. The results of ΔT_c as a function of crack size are plotted in Figures 8.16 and 8.17 for cold shock and hot shock, respectively. Clearly, if the stress-based criterion is used, ΔT_c does not depend on pre-existing crack lengths. However, ΔT_c predicted by the fracture mechanics-based criterion shows a strong dependence on pre-existing crack sizes. The permissible temperature jump with increasing crack length goes through a minimum with a region of crack instability bounded by two values of crack length. Since the thermal stresses are in self-equilibrium, the stress intensity factors are zero for both an un-cracked plate ($a=0$, or $c=0$) and a fully cracked plate ($a=h/2$, or $c=h/2$). Accordingly, if we use the fracture mechanics-controlled failure as a design criterion, the admissible temperature jump ΔT_c will be infinity both at $c=0$ and $c=h/2$ for cold shock (or $a=0$ and $a=h/2$ for hot shock). ΔT_c decreases with crack length from infinity to a minimum and then increases again to infinity. The crack length at which ΔT_c is minimum is denoted by $c_{\min}$ for cold shock, and $a_{\min}$ for hot shock. The minimum temperature jump sustainable by the material $\Delta T_c^{\min}$ is obtained from Eqs. (8.67) and (8.68):

$$\Delta T_c^{\min} = 6.25K_{IC} / \overline{\lambda_{33}}\sqrt{\pi h}, \text{ for cold shock,} \quad (8.71a)$$

$$\Delta T_c^{\min} = 7.63K_{IC} / \overline{\lambda_{33}}\sqrt{\pi h}, \text{ for hot shock.} \quad (8.71b)$$

Since thermal stress intensity factors are zero for a fully cracked plate, the failure of the plate will be governed by the stress-based criterion for small and large crack lengths. A transient crack length c_t for cold shock (or $2a_t$ for hot shock) exists for which ΔT_c is equal for toughness-controlled failure and for stress-controlled failure. This length can be found by equaling (8.69) and (8.70):

$$\frac{K_{IC}}{\overline{K(c_t)}\lambda_{33}\sqrt{\pi h}} = \frac{\sigma_b}{\lambda_{33}}, \text{ for cold shock,} \quad (8.72a)$$

$$\frac{K_{IC}}{\bar{K}(a_t)\lambda_{33}\sqrt{\pi h}} = 3.25 \frac{\sigma_b}{\lambda_{33}}, \text{ for hot shock.} \quad (8.72b)$$

Since $\bar{K}(c)$ and $\bar{K}(a)$ are complex functions of crack length, c_t and a_t can only be found numerically. It can be shown from Figures. 8.16 and 8.17 that, except at $c_{\min}$ and $a_{\min}$, there are two roots in each of Eqs. (8.72), namely, (c_1, c_2) for c_t and (a_1, a_2) for a_t .

Consideration of Eqs. (8.69) and (8.71) reveals that the admissible temperature jumps for both cold and hot shocks are smaller for fracture mechanics-based than stress-based failure criterion, at a sufficiently large plate thickness. A transient plate thickness h_t exists for which $\Delta T_c^{\min}$ is equal for both failure criteria. Hence, by equating $\Delta T_c^{\min}$ values for the stress-based criterion (8.69) and for the fracture mechanics-based criterion (8.71), it is found that

$$h_t = 12.4 \left(\frac{K_{IC}}{\sigma_b} \right)^2, \text{ for cold shock,} \quad (8.73a)$$

$$h_t = 1.75 \left(\frac{K_{IC}}{\sigma_b} \right)^2, \text{ for hot shock.} \quad (8.73b)$$

Therefore, the thermal shock resistance of a piezoelectric plate with thickness smaller than h_t will be controlled by the stress-based criterion; a plate with thickness larger than h_t will be controlled by the fracture-based criterion. Thus, a piezoelectric material whose thickness is h_t has the same thermal shock resistance according to the stress criterion and the fracture criterion. The fracture-based criterion is conservative for a plate whose thickness is above h_t , the stress-based criterion is conservative for a plate whose thickness is below h_t .

8.3. POLING PERPENDICULAR TO THE STRIP SURFACES

Section 8.2 studied the transient fracture of a piezoelectric strip whose poling direction is parallel to its surfaces. In the case of poling direction perpendicular to its surfaces, the analysis procedure is analogously same as that shown in above, provided some basic equations are different. We discuss this below [6].

8.3.1. Temperature Distribution

Referring to Figure 8.18, a transversely isotropic piezoelectric ceramic plate with the isotropic plane parallel to the z -axis is given. The plate thickness is denoted as H in this section, since h will be used as another parameter. Initially, the plate is at a uniform temperature T_i , and at time $t=0$ its top and bottom surfaces (at $z=0, h$) are suddenly exposed to a convective medium of temperature T_∞ . Suppose the thermal conductivity along the z -axis is

k_3 , and heat transfer coefficient between the plate and environmental medium is h . Thus, the heat flow is one-dimensional and its values on the plate surfaces take the following form

$$k_3 \frac{\partial T}{\partial z} = \pm h(T_\infty - T), \text{ at } z=0, H, \quad (8.74)$$

where $T(z, t)$ is the temperature of the piezoelectric plate.

The temperature field T is governed by a same Fourier heat conductivity equation as Eq. (8.1), provided the coefficient k_2 is replaced by k_3 . The time scale in present case is $t_0 = \rho c_v H^2 / k_3$. The solution satisfying the heat transfer boundary condition (8.74) is:

$$\frac{T(z, t) - T_i}{T_i - T_\infty} = -1 + 2 \sum_{m=1,3,5}^{\infty} \exp\left(-4\beta_m^2 \frac{t}{t_0}\right) \frac{\sin \beta_m \cos(\beta_m - 2\beta_m z / H)}{\beta_m + \sin \beta_m \cos \beta_m}, \quad (8.75)$$

where β_m are the positive roots of $\beta \tan \beta = \text{Bi}$, and the non-dimensional heat transfer coefficient Bi is known as the Biot number: $\text{Bi} = h(H/2)/(k_3/\rho c_v)$. It can be shown that the thermal diffusivity $\rho c_v / k_3$ dictates the time scale for the transient temperature distribution, but does not affect the level of the temperature. Conversely, the Biot number affects only the temperature level but does not affect the time scale for the temperature distribution. In the limit of perfect thermal insulation, $\text{Bi}=0$, $\beta_m = m\pi$ and $T(z, t) = T_i$. At the other limit of perfect heat transfer, $\text{Bi}=\infty$, $\beta_m = (m+1/2)\pi$, and equation (8.75) reduces to

$$\frac{T(z, t) - T_i}{T_i - T_\infty} = -1 + \frac{4}{\pi} \sum_{m=1,3,5}^{\infty} \frac{1}{m} \exp\left(-(m\pi)^2 \frac{t}{t_0}\right) \sin\left(m\pi \frac{z}{H}\right), \quad (8.76)$$

for finite values of Biot number, the coefficient β_m is determined numerically and is bounded by the two limiting values

$$(m+1/2)\pi > \beta_m > m\pi. \quad (8.77)$$

8.3.2. TRANSIENT THERMAL STRESS IN THE PIEZOELECTRIC CERAMIC PLATE

Consider plane strain deformation in the x - z plane. Once $T(z, t)$ is known, the thermal stresses can be obtained from the constitutive equations of the piezoelectric ceramics:

$$\begin{Bmatrix} \sigma_{xx} \\ \sigma_{zz} \\ \sigma_{xz} \\ D_x \\ D_z \end{Bmatrix} = \begin{bmatrix} c_{11} & c_{13} & 0 & 0 & -e_{31} \\ c_{13} & c_{33} & 0 & 0 & -e_{33} \\ 0 & 0 & c_{44} & -e_{15} & 0 \\ 0 & 0 & e_{15} & \epsilon_{11} & 0 \\ e_{31} & e_{33} & 0 & 0 & \epsilon_{33} \end{bmatrix} \begin{Bmatrix} \epsilon_{xx} \\ \epsilon_{zz} \\ 2\epsilon_{xz} \\ E_x \\ E_z \end{Bmatrix} - \begin{Bmatrix} \lambda_{11} \\ \lambda_{33} \\ 0 \\ 0 \\ \beta_3 \end{Bmatrix} (T - T_i), \quad (8.78)$$

Suppose the plate is infinite along x - and y -axis, (i.e., $-\infty < (x, y) < \infty$), free of surface traction and electric charge at $z=0$ and $z=h$, it may be shown that

$$\sigma_{zz}=0, \sigma_{xz}=0, D_x=0, D_z=0, \quad (8.79)$$

and all non-vanishing field quantities are independent of x and y . By solving ϵ_{zz} and E_z from the second and the fifth of Eqs. (8.78), in terms of ϵ_{xx} and T , and then substituting into the first of Eqs. (8.78), it is found that:

$$\sigma_{xx} = \bar{c}_{11} \epsilon_{xx} - \bar{\lambda}_{11} (T - T_i), \quad (8.80)$$

where

$$\bar{c}_{11} = c_{11} - \frac{(c_{13} \epsilon_{33} + e_{31} e_{33}) c_{13} - (e_{31} c_{33} - c_{13} e_{33}) e_{31}}{c_{33} \epsilon_{33} + e_{33}^2}, \quad (8.81a)$$

$$\bar{\lambda}_{11} = \lambda_{11} - \frac{(c_{13} \epsilon_{33} + e_{31} e_{33}) \lambda_{33} - (e_{31} c_{33} - c_{13} e_{33}) \beta_3}{c_{33} \epsilon_{33} + e_{33}^2}. \quad (8.81b)$$

The compatibility condition that needs to be satisfied is:

$$d^2 \epsilon_{xx} / dz^2 = 0, \quad (8.82)$$

giving

$$\epsilon_{xx} = Az + B, \quad (8.83)$$

$$\sigma_{xx} = \bar{c}_{11} (Az + B) - \bar{\lambda}_{11} (T - T_i), \quad (8.84)$$

where A and B are unknown constants to be obtained from the boundary conditions at the left and right ends of the plate. Assume the plate is unconstrained along its far ends, we obtain

$$\int_0^h \sigma_{xx}(z, t) dz = 0, \quad \int_0^h \sigma_{xx}(z, t) z dz = 0. \quad (8.85)$$

The thermal stress $\sigma(z, t) = \sigma_{xx}(z, t)$ is obtained directly from equations (8.75), (8.84) and (8.85), and is written in non-dimensional form as:

$$\bar{\sigma} = \frac{\sigma_{xx}(z, t)}{\lambda_{11}(T_i - T_\infty)} = 2 \sum_{m=1,3,5}^{\infty} \exp\left(-4\beta_m^2 \frac{t}{t_0}\right) \frac{\sin \beta_m}{\beta_m + \sin \beta_m \cos \beta_m} \left(\cos(\beta_m - 2\beta_m \frac{z}{H}) - \frac{\sin \beta_m}{\beta_m} \right). \quad (8.86)$$

Thus, among many thermo-electro-mechanical parameters in the piezoelectric materials, the parameter that governs the transient thermal stress level in the plate is identified as $\bar{\lambda}_{11}$. The time scale for the transient thermal stress state is governed by the thermal diffusivity $\rho C_V/k_3$ that does not affect the level of thermal stresses.

In general, the higher the Biot number the more severe the resultant transient thermal stress, with $Bi = \infty$ representing an instantaneous temperature change T_∞ on the surfaces of the plate [1]. Thus $Bi = \infty$ gives the maximum thermal stress in the plate. In what follows, we consider only this extreme case. The thermal stress in Eq. (8.86), after replacing β_m with $(m+1/2)\pi$ for the infinite Biot number, takes the form:

$$\bar{\sigma} = \frac{\sigma_{xx}(z, t)}{\lambda_{11}(T_i - T_\infty)} = \frac{4}{\pi} \sum_{m=1,3,5}^{\infty} \frac{1}{m} \exp\left(- (m\pi)^2 \frac{t}{t_0}\right) \left[\frac{2}{m\pi} - \sin\left(m\pi \frac{z}{H}\right) \right]. \quad (8.87)$$

The values of the dimensionless stress, $\bar{\sigma}(z, t) = \sigma(z, t) / \sigma_0$, where $\sigma_0 = \bar{\lambda}_{11}(T_i - T_\infty)$, are same as those shown in Figures. 8.2 and 8.3. The only differences are the non-dimensional quantities σ_0 and t_0 . Once again, if $T_\infty < T_i$ (cold shock), the region near the surface of the plate is in tension while a compressive zone is developed at the plate center. The maximum tensile stress is attained at the surface while the compressive stress is largest at the center of the plate. A maximum value of

$$\bar{\sigma} = 1, \quad \sigma = \bar{\lambda}_{11}(T_i - T_\infty), \quad (8.88)$$

is achieved at the surface of the plate at $t=0$. The maximum compressive stress

$$\bar{\sigma} = 0.308, \quad \sigma = 0.308 \bar{\lambda}_{11}(T_i - T_\infty), \quad (8.89)$$

is achieved at the plate center at a time $t/t_0 = 0.0290$.

For a hot shock ($T_\infty > T_i$), at any location of the plate, the stress will change its sign from the cold shock. Hence the surface of the plate will become compressive and the center will be tensile. The magnitudes of the thermal stresses subjected to a hot shock will be same as those in a cold shock.

8.3.3. Cracking due to Thermal Shock

As described above, the surface of the plate undergoes a transient tensile stress under cold shock, while the center of the plate has the largest tensile stress under hot shock. It is then expected that a crack may initiate at the surface under cold shocking or at the center under hot shocking. The crack grows unstably until it enters the compressive region. The problem can be synthesized from the general solution for an embedded crack in a finite plate, with crack tips located at $z=b$ and $z=c$, as shown in Figure 8.18. Since the crack plane is normal to the surface of the plate, it does not perturb the transient temperature distribution in this arrangement, determination of the temperature distribution and the resulting thermal stresses would be quite straightforward and the related crack problem would be one of mode I. The stress given by Eq. (8.87), with opposite sign, becomes the crack surface tractions in the quasi-static, mixed boundary-value problem. The homogeneous boundary conditions used in formulating the mixed problem are given by:

$$\sigma_{xz}(x, 0)=0, \sigma_{zz}(x, 0)=0, D_z(x, 0)=0, \quad (8.90)$$

$$\sigma_{xz}(x, h)=0, \sigma_{zz}(x, h)=0, D_z(x, h)=0. \quad (8.91)$$

Assuming $x=0$ is a plane of symmetry, the additional mixed boundary conditions along the $x=0$ plane are

$$u(0, z)=0, \quad z \notin (b, c), \quad (8.92)$$

$$\sigma_{xx}(0, z)=\sigma_0(z, t), \quad z \in (b, c), \quad (8.93)$$

where u is the x component of the displacement vector, $\sigma_0(z, t)$ is the equal and opposite value of $\sigma_{xx}(z, t)$ given in eq. (8.86).

The mixed boundary-value problem of Eqs. (8.90) to (8.93) is solved by introducing a dislocation density function along $x=0$

$$g(z) = \partial u(0, z) / \partial z. \quad (8.94)$$

The solution of the equations of piezoelectricity subjected to the conditions specified by Eqs. (8.90)-(8.94) results in an integral equation for a crack subjected to relatively general loading conditions. Algebraic details concerning derivation of the integral equation for the crack problem are similar to Subsection 8.2.3 and can also be found in [7]. The resulting singular integral equation is of the form

$$\frac{\Lambda_0}{\pi} \int_b^c \frac{1}{r-z} g(r) dr + \int_b^c h_1(z, r) g(r) dr = \sigma_0(z, t), \quad c \geq z \geq b, \quad (8.95)$$

where Λ_0 is a material constant which depends on material properties but not on the geometry of the structure, h_1 is a Fredholm kernel. Since time t only enters the problem through $\sigma_0(z, t)$

in Eq. (8.95), the singular integral equation is solved separately for individual values of t . Note that the solution of the singular integral Eq. (8.97) is:

$$g(r) = \frac{1}{\sqrt{1-r^2}} \sum_{n=1}^{\infty} \alpha_n T_n(\bar{r}), \quad (8.96)$$

for an embedded crack, and

$$g(r) = \frac{1}{\sqrt{1-r^2}} \sum_{n=0}^{\infty} \alpha_n T_n(\bar{r}), \quad (8.97)$$

for an edge crack, where $z = \bar{a}z + (c+b)/2$, $r = \bar{a}r + (c+b)/2$, T_n is the Chebyshev polynomial of the first kind and $\alpha_0, \alpha_1, \dots$ are unknown constants. Equation (8.95) can be solved by using a collocation technique [5]. For the embedded crack problem, the stress intensity factor K ahead of the crack tips at $z=b$ and $z=c$ can be defined and calculated by:

$$K(b) = \left(\sqrt{2\pi(b-z)} \right)_{z \rightarrow b^-} \sigma_{xx}(z) = \Lambda_0 \sqrt{\pi a} \sum_{n=1}^{\infty} (-1)^n \alpha_n, \quad (8.98a)$$

$$K(c) = \left(\sqrt{2\pi(z-c)} \right)_{z \rightarrow c^+} \sigma_{xx}(z) = -\Lambda_0 \sqrt{\pi a} \sum_{n=1}^{\infty} \alpha_n. \quad (8.98b)$$

For the edge crack problem the stress intensity factor is obtained as:

$$K(c) = \left(\sqrt{2\pi(z-c)} \right)_{z \rightarrow c^+} \sigma_{xx}(z) = -\Lambda_0 \sqrt{\pi c} \sum_{n=0}^{\infty} \alpha_n. \quad (8.99)$$

The numerical results of the stress intensity factor K have been obtained for an edge-cracked plate under cold shock, and a center-cracked plate under hot shock. To verify the correctness and accuracies of the derived equations, we have calculated the transient thermal stress intensity factors by finite element method as shown in Appendix B.

(1) A single-edge cracked plate under cold shock

To gain insights on crack growth in piezoelectric ceramics, a PZT-5H piezoelectric ceramic is again considered. For the infinite plate of Figure 8.18 subjected to a cold shock. The problem is idealized to the case of a single-edge cracked plate, as shown in Figure 8.19, and the stress intensity factor K is calculated for a cold shock event. Let the surface heat transfer condition be infinity. The normalized stress intensity factor $\bar{K} = K/K_0$, where

$K_0 = \bar{\lambda}_{11} \sqrt{\pi H} (T_i - T_\infty)$, is plotted against dimensionless time t/t_0 in Figure 8.20, for

selected normalized crack depth c/H , where c is the crack depth. At any given c/H , the stress intensity increases from an initial zero with time, displays a peak value and then decreases to zero as time approaches infinity. This means that the thermal stress intensity factor occurs only at transient state and that its steady state value is zero. It is further noted that the magnitude of K depends on crack length, and achieves a peak value at a crack length of $c \approx 0.122H$. Thus, K achieves a global maximum value at a time t_* and a crack depth c_* .

Figure 8.21 shows the variation of stress intensity factor K with crack depth c , at different time. If we consider a crack of given initial length, we see that the temporal behavior is alike for all crack depths. The envelope of this family of curves is the locus of the highest stress intensity factors reached for every crack depths at any time. It is seen that K attains an overall maximum value

$$K_{\max} = 0.177K_0 = 0.177\bar{\lambda}_{11}\sqrt{\pi H}(T_i - T_\infty), \quad (8.100)$$

at a particular time $t_* \approx 0.015t_0$ and a particular crack depth $c_* = 0.122H$. Note that these expressions are only valid for material parameters used in the numerical analysis.

(2) A center crack under hot shock

As discussed above, a hot shock induces transient tensile stresses at the center of the plate, with a peak value specified by $\bar{\sigma} = 0.308$ at time $t/t_0 = 0.0290$. This stress is most likely to develop a mode I crack in the center of the plate. Now assume the plate contains a pre-existing center crack of length $2a$, symmetrical with respect to the central line of the plate, as shown in figure 8.22. The piezoelectric ceramic plate is subjected to a hot shock: initially the plate is at a uniform temperature T_i . At time $t=0$ the top and bottom surface of the plate are exposed to an environment at temperature $T_\infty (>T_i)$, and the surface heat transfer condition is assumed to be infinity. The stress intensity factor associated with the transient thermal stress Eq. (8.87) is determined from Eqs. (8.98) by solving the integral equation (8.95).

The non-dimensional stress intensity factor $\bar{K} = K/K_0$, where $K_0 = \bar{\lambda}_{11}\sqrt{\pi H}(T_\infty - T_i)$, is evaluated against time t/t_0 for selected crack lengths in Figure 8.23. At any crack length, the stress intensity factor increases from zero to a maximum value, and then decays to zero. Figure 8.24 illustrates values of K versus crack length for selected times. The stress intensity factor increases with crack length to a maximum and then decreases. Those curves form an envelope that represents the relationship between crack length and the largest stress intensity factor. It is noted that K attains an overall maximum value: $K_{\max} = 0.123K_0 = 0.123\bar{\lambda}_{11}\sqrt{\pi H}(T_\infty - T_i)$ at a particular time $t_* \approx 0.020t_0$ and a particular crack length $2a_* = 0.48H$. These expressions are again only valid for material parameters used in the numerical analysis.

8.3.4 Thermal Shock Resistance

Thermal shock resistance is a major issue in the design of engineering ceramics for thermal applications. A central problem in designing against thermal shock is the

identification of an appropriate material selection criterion in order to select the most shock-resistant material for a given application.

A stress-based failure criterion for cold shock is that the maximum thermal stress that appears on the plate surface $\sigma_{\max}(0, t)$ or $\sigma_{\max}(H, t)$ attains the value σ_b . The maximum temperature jump sustainable by ΔT_c follows from Eqs. (8.88) and (8.89) as

$$\Delta T_c = \Delta T_c^{\text{stress}} = \sigma_b / \bar{\lambda}_{11}, \text{ for cold shock,} \quad (8.101a)$$

$$\Delta T_c = \Delta T_c^{\text{stress}} = 3.25\sigma_b / \bar{\lambda}_{11}, \text{ for hot shock.} \quad (8.101b)$$

Similarly, for the fracture-based failure criterion, the maximum thermal stress intensity factor $K_{\max}(a^*, t^*)$ must equal the fracture toughness of the ceramic K_{IC} . It is clear that the maximum temperature jump sustainable by the ceramic ΔT_c depends on the depth of the pre-existing crack, since the maximum stress intensity factor depends on the crack length. Since the envelopes in Figures. 8.21 and 8.24 represent the maximum stress intensity factor $K_{\max}$ for different crack length, it follows from $K_{\max}=K_{IC}$ that

$$\Delta T_c = \Delta T_c^{\text{fracture}} = \frac{K_{IC}}{K\bar{\lambda}_{11}\sqrt{\pi H}}. \quad (8.102)$$

It is instructive to plot the sustainable temperature jump ΔT_c as a function of crack length. The results of ΔT_c as a function of half crack length are given in Figures. 8.25 and 8.26, for cold shock and hot shock, respectively. It is clear that if the stress-based criterion is used, ΔT_c does not depend on the pre-existing crack length. On the other hand, ΔT_c predicted by the fracture-based criterion exhibits a strong dependence on pre-existing crack length. The permissible temperature jump with increasing crack length goes through a minimum with a region of crack instability bounded by two values of crack length. Since the thermal stresses are in self-equilibrium, the stress intensity factors are zero for both an un-cracked plate ($a=0$, or $c=0$) and a fully cracked plate ($c=H$, or $2a=H$). Accordingly, if we use the fracture-controlled failure as a design criterion, the admissible temperature jump ΔT_c will be infinity both at $c=0$ and $c=H$ (or $2a=0$ and $2a=H$ for hot shock). ΔT_c decreases with crack length from infinity to a minimum and then increases again to infinity. The crack length at which ΔT_c is minimized is denoted by $c_{\min}$ for cold shock, and $a_{\min}$ for hot shock. The minimum temperature jump sustainable by the ceramic is:

$$\Delta T_c^{\min} = 5.65K_{IC} / \bar{\lambda}_{11}\sqrt{\pi H}, \text{ at } c=c_{\min}, \text{ for cold shock,} \quad (8.103a)$$

$$\Delta T_c^{\min} = 8.13K_{IC} / \bar{\lambda}_{11}\sqrt{\pi H}, \text{ at } a=a_{\min}, \text{ for hot shock.} \quad (8.103b)$$

Because thermal stress intensity factors are zero at $c=0$ and $c=H$ (or $2a=0$ and $2a=H$ for hot shock), the failure of the plate should be predicted by the stress-based criterion for small crack length and large crack length. A transient crack length c_t (or a_t) exists for which ΔT_c is

equal for fracture-controlled failure and for stress-controlled failure. By equating Eqs. (8.101) and (8.102), this length can be determined from the relation

$$\frac{K_{IC}}{\overline{K}(c_t)\lambda_{11}\sqrt{\pi H}} = \frac{\sigma_b}{\lambda_{11}}, \text{ for cold shock,} \quad (8.104a)$$

$$\frac{K_{IC}}{\overline{K}(a_t)\lambda_{11}\sqrt{\pi H}} = 3.25 \frac{\sigma_b}{\lambda_{11}}, \text{ for hot shock,} \quad (8.104b)$$

As $\overline{K}(c)$ and $\overline{K}(a)$ are complex functions of crack length, c_t and a_t can only be found numerically. It can be shown from Figures. 8.25 and 8.26 that, except at $c_{\min}$ and $a_{\min}$, there are two roots in each of Eqs. (8.104), namely, (c_1, c_2) for c_t and (a_1, a_2) for a_t .

Consideration of Eqs. (8.101) and (8.103) reveals that the admissible temperature jumps for both cold shock and hot shock are less for fracture-controlled failure than for stress-controlled failure, at a sufficiently large plate thickness. A transient plate thickness value H_t exists for which $\Delta T_c^{\min}$ is equal for fracture-controlled failure and stress-controlled failure. Then, upon equating $\Delta T_c^{\min}$ values for the stress-based criterion (8.101) and for the fracture-based criterion (8.103), it is found that

$$H_t = 10.2(K_{IC} / \sigma_b)^2, \text{ for cold shock,} \quad (8.105)$$

$$H_t = 2.00(K_{IC} / \sigma_b)^2, \text{ for hot shock.} \quad (8.106)$$

Therefore, a piezoelectric ceramic plate with thickness smaller than H_t should be predicted by the stress-based criterion; a plate with thickness larger than H_t should be predicted by the fracture-based criterion. Thus, a piezoelectric material whose thickness is H_t has the same thermal shock resistance according to the stress criterion and the fracture criterion. The fracture-based criterion is conservative for a plate whose thickness is above H_t , the stress-based criterion is conservative for a plate whose thickness is below H_t .

8.4. TRANSIENT THERMAL FRACTURE OF A SEMI-INFINITE PIEZOELECTRIC MEDIUM WITH A CONDUCTIVE SURFACE CRACK

Section 8.2 investigated the electrically insulated cracks in piezoelectric materials. Also interested in piezoelectric fracture is the electrically conductive crack problems. In such case, the electric field inside the crack is zero. This section considers the mechanical problem of a semi-infinite piezoelectric medium with a surface crack under sudden thermal load [8]. The medium contains an electrically conducting crack perpendicular to its surface. The transient stresses and electric fields in an un-cracked medium are calculated. These stresses and electric fields are used as the crack surface traction and electric field loads with opposite signs to

formulate the mixed boundary value problem. Numerically results for the stress and electric field intensity factors are calculated as a function of normalized time and crack size. The parameters that control the transient thermal stress and electric fields are also identified. Crack propagation behavior is discussed and the thermal shock strength of the piezoelectric medium is established.

8.4.1. Temperature Distribution

The problem under consideration is a semi-infinite piezoelectric medium shown in Figure 8.28. The coordinates x and y (which are coincident with x_1 and x_2 , respectively) are the principal axes of the material symmetry. The poling direction of the piezoelectric medium is oriented in the x -axis. The medium contains a crack normal to the surface of the medium. The external thermal load is assumed to be such that the plane of the crack is a plane of symmetry. In the problem considered, the heat conduction is one-dimensional, and a straight crack in the x -direction does not obstruct the heat flow in this arrangement, the temperature field T is then governed by:

$$k_3 \frac{\partial^2 T(x, t)}{\partial x^2} = \rho c_v \frac{\partial T(x, t)}{\partial t}, \quad (8.107)$$

The heat flow is $q_x = -k_3 \partial T / \partial x$. Suppose the piezoelectric medium is initially at a uniform temperature zero and there is a sudden temperature change on its surface. Then the initial and boundary conditions are:

$$T(x, 0) = 0, \quad T(0, t) = T_0 H(t), \quad (8.108)$$

where $H(t)$ is the Heaviside function.

Equation (8.107) is solved with heat transfer boundary conditions (8.108) by a standard separation-of-variables technique, giving

$$T(x) = T_0 \operatorname{erfc} \left(\frac{x}{2\sqrt{(k_3 / \rho c_v) t}} \right) = \frac{2}{\pi} T_0 \int_0^\infty \left(1 - \exp \left(-\frac{k_3 s^2 t}{\rho c_v} \right) \right) \sin(sx) \frac{ds}{s}, \quad (8.109)$$

where erfc is the complementary error function. Once again, the thermal diffusivity $k_3 / \rho c_v$ dictates the time scale for the transient temperature distribution, but does not affect the level of the temperature.

8.4.2. Thermal Stresses and Electric Fields in the un-Cracked Medium

Once $T(x)$ is known, the stresses and the electric fields can be found from the constitutive relations of the linear thermopiezoelectric medium poled along the x -axis:

$$\sigma_{xx} = c_{33}\varepsilon_{xx} + c_{13}\varepsilon_{yy} - e_{33}E_x - \lambda_{33}T, \quad (8.110a)$$

$$\sigma_{yy} = c_{13}\varepsilon_{xx} + c_{11}\varepsilon_{yy} - e_{31}E_x - \lambda_{11}T, \quad (8.110b)$$

$$D_x = e_{33}\varepsilon_{xx} + e_{31}\varepsilon_{yy} + \epsilon_{33}E_x - \beta_3T, \quad (8.110c)$$

Since the plate is infinite along y -axis, free of tractions and electric charges at surface $x=0$, it may be shown that

$$\varepsilon_{yy}=0, \sigma_{xx}=0, D_x=0, \quad (8.111)$$

and all non-vanishing field quantities are independent of y . By solving ε_{xx} and E_x from Eqs. (8.110a) and (8.110c), in terms of T , and then substituting them into Eq. (8.110b), we obtain

$$\sigma_{yy} = -\tilde{\lambda}_{11}T, \quad E_x = \tilde{\beta}_3T, \quad (8.112)$$

where

$$\tilde{\lambda}_{11} = \lambda_{11} - \frac{(c_{13}\epsilon_{33} + e_{31}e_{33})\lambda_{33} - (e_{31}c_{33} - c_{13}e_{33})\beta_3}{c_{33}\epsilon_{33} + e_{33}^2}, \quad (8.113a)$$

$$\tilde{\beta}_3 = \frac{c_{33}\beta_3 - e_{33}\lambda_{33}}{c_{33}\epsilon_{33} + e_{33}^2}. \quad (8.113b)$$

For the crack problem under consideration, the equal and opposite stresses and electric fields given by Eqs. (8.112) will be superposed, respectively, to the tractions and electric fields, on the crack surfaces. This constitutes the applied loads on the crack surfaces in the isothermal problem. The medium is assumed to be under a plane strain condition.

8.4.3. The Crack Problem

The crack problem is solved by treating it as isothermal and quasi-static, and by using the equal and opposite values of thermal stress and electric field obtained from Eqs. (8.112) as crack surface traction and electric field loads. Therefore, we have the following boundary conditions on the crack faces:

$$\sigma_{yy}(x,0) = -\sigma_0(x), \quad E_x(x,0) = -E_0(x), \quad x \in [0, c]. \quad (8.114)$$

where $\sigma_0(x)$ and $E_0(x)$ are the corresponding values of stress and electric field given by Eqs. (8.112).

For a two-dimensional piezoelectric medium whose poling direction is parallel to the x -axis, its constitutive equations are same as Eqs. (1.22), provided that the subscripts x and y get exchanged. Since the electric displacement D is divergence-free in the absence of space charge, there exists a potential function $\Phi(x_1, x_2)$ which gives the electric displacement according to Eqs. (4.64). The governing equations for u_1 and u_2 (which are displacements in x_1 and x_2 directions) and Φ are Eqs. (4.67).

Denote u and v as displacements in x - and y -directions, respectively. Due to symmetry of the problem with respect to the $y=0$ plane, we have:

$$\sigma_{xy}(x, 0)=0, x \in [0, \infty], \quad (8.115)$$

$$v(x, 0)=0, \Phi(x, 0)=0, x \in [c, \infty], \quad (8.116)$$

The surface (the $x=0$ plane) of the medium is assumed to be traction free and has a constant electric potential. Hence,

$$\sigma_{xy}(0, y)=0, \sigma_{xx}(0, y)=0, E_y(0, y)=0, \quad (8.117)$$

Since the problem is symmetric about the $y=0$ plane, it suffices to consider one-half ($y>0$) of the medium. The solutions of the governing equations (4.67) can be expressed in terms of unknown coefficients $A_m(s)$ and $B_m(\xi)$ in the following form:

$$\begin{aligned} \begin{Bmatrix} u(x, y) \\ v(x, y) \\ \Phi(x, y) \end{Bmatrix} &= \frac{2}{\pi} \int_0^\infty \exp(s\lambda_m x) \begin{Bmatrix} \cos(sy)a_{1m} \\ \sin(sy)a_{2m} \\ \sin(sy)a_{3m} \end{Bmatrix} A_m ds \\ &+ \frac{1}{2\pi} \int_{-\infty}^\infty \exp\left(\frac{|\xi|}{\lambda_m} y\right) \exp(-i\xi x) \begin{Bmatrix} a_{1m} \\ \text{sgn}(\xi)a_{2m} \\ a_{3m} \end{Bmatrix} B_m d\xi, \end{aligned} \quad (8.118)$$

where ($u=u_1, v=u_2$), sgn is the signum function ($\text{sgn}(\xi)$ equals 1 for positive values of ξ , and -1 for negative values of ξ). By substituting Eqs. (8.118) into Eqs.(4.67), we obtain the following characteristic equations to determine the eigen-values λ_m , and the corresponding eigen-vectors (a_{1m}, a_{2m}, a_{3m}):

$$\begin{bmatrix} \bar{c}_{33}\lambda_m^2 - \bar{c}_{44} & (\bar{c}_{13} + \bar{c}_{44})\lambda_m & (\bar{e}_{33} - \bar{e}_{15})\lambda_m \\ (\bar{c}_{13} + \bar{c}_{44})\lambda_m & \bar{c}_{11} - \bar{c}_{44}\lambda_m^2 & \bar{e}_{31} + \bar{e}_{15}\lambda_m^2 \\ (\bar{e}_{33} - \bar{e}_{15})\lambda_m & \bar{e}_{31} + \bar{e}_{15}\lambda_m^2 & \bar{\epsilon}_{33} - \bar{\epsilon}_{11}\lambda_m^2 \end{bmatrix} \begin{Bmatrix} a_{1m} \\ a_{2m} \\ a_{3m} \end{Bmatrix} = 0. \quad (8.119)$$

In Eqs. (8.119), there are six roots for λ . We now arrange the order of roots λ_m such that $\text{Re}(\lambda_1) < 0$, $\text{Re}(\lambda_2) < 0$, $\text{Re}(\lambda_3) < 0$ and $\text{Re}(\lambda_4) > 0$, $\text{Re}(\lambda_5) > 0$, $\text{Re}(\lambda_6) > 0$. Since the mechanical displacements and electric potential must vanish as x or y approaches infinity, then $A_m = B_m = 0$ for $m=4, 5$ and 6 . The stress and electrical field components can be obtained by differentiating Eqs. (8.118) to obtain the strains and electric displacements, and substituting them into the electro-elastic constitutive equations. This gives:

$$\begin{Bmatrix} \sigma_{xy}(x, y) \\ \sigma_{xx}(x, y) \\ E_y(x, y) \\ \sigma_{yy}(x, y) \\ E_x(x, y) \end{Bmatrix} = \frac{2}{\pi} \sum_{m=1}^3 \int_0^\infty s \begin{Bmatrix} \sin(sy)C_{1m} \\ \cos(sy)C_{2m} \\ \sin(sy)C_{3m} \\ \cos(sy)D_{1m} \\ \cos(sy)D_{2m} \end{Bmatrix} \exp(s\lambda_m x) A_m ds$$

$$+ \frac{1}{2\pi} \sum_{m=1}^3 \int_{-\infty}^\infty \frac{1}{\lambda_m} \exp(|\xi| \frac{y}{\lambda_m}) \begin{Bmatrix} -i\xi C_{1m} \\ |\xi| C_{2m} \\ -i\xi C_{3m} \\ |\xi| D_{1m} \\ |\xi| D_{2m} \end{Bmatrix} \exp(-i\xi x) B_m d\xi, \quad (8.120)$$

where the coefficients C_{jm} and D_{jm} are:

$$C_{1m} = -\bar{c}_{44}a_{1m} + \bar{c}_{44}\lambda_m a_{2m} - \bar{e}_{15}\lambda_m a_{3m}, \quad (8.121a)$$

$$C_{2m} = \bar{c}_{33}\lambda_m a_{1m} + \bar{c}_{13}a_{2m} + \bar{e}_{33}a_{3m}, \quad (8.121b)$$

$$C_{3m} = -\bar{e}_{15}a_{1m} + \bar{e}_{15}\lambda_m a_{2m} + \bar{\epsilon}_{11}\lambda_m a_{3m}, \quad (8.121c)$$

$$D_{1m} = \bar{c}_{13}\lambda_m a_{1m} + \bar{c}_{11}a_{2m} + \bar{e}_{31}a_{3m}, \quad (8.121d)$$

$$D_{2m} = \bar{e}_{33}\lambda_m a_{1m} + \bar{e}_{31}a_{2m} + \bar{\epsilon}_{33}a_{3m}. \quad (8.121e)$$

(A) The singular integral equation

To satisfy conditions (8.116), we introduce two auxiliary functions g_1 and g_2 defined along the crack plane (the $y=0$ plane):

$$g_1(x) = \frac{\partial v(x, 0)}{\partial x}, \quad g_2(x) = \frac{\partial \Phi(x, 0)}{\partial x}. \quad (8.122)$$

From the symmetry of the problem, it can be shown that g_1 and g_2 vanish for $x \notin [0, c]$. By applying Eqs. (8.118) to Eqs. (8.122) and Eqs. (8.120) to the symmetry condition (8.115), one obtains a system of linear algebraic equations. Using conditions (8.116) and inverting the Fourier integrals to these linear algebraic equations, the constants B_m can be expressed in terms of g_1 and g_2 as:

$$B_m(\xi) = i\xi^{-1} \int_0^c \sum_{N=1}^2 b_m^N g_N(r) \exp(i\xi r) dr, \quad (m=1,2,3), \quad (8.123)$$

where the material constants b_m^N ($N=1, 2; m=1,2,3$) are

$$\begin{bmatrix} b_1^1 & b_1^2 \\ b_2^1 & b_2^2 \\ b_3^1 & b_3^2 \end{bmatrix} = \begin{bmatrix} a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \\ C_{11}/\lambda_1 & C_{12}/\lambda_2 & C_{13}/\lambda_3 \end{bmatrix}^{-1} \begin{bmatrix} 1 & 0 \\ 0 & 1 \\ 0 & 0 \end{bmatrix}. \quad (8.124)$$

By applying Eqs. (8.120) to the homogeneous boundary conditions (8.117), and using Eq. (8.123) we obtain

$$[C]\{A\} = \frac{1}{2s} \int_0^c \sum_{N=1}^2 \{f^N(s, r)\} g_N(r) dr, \quad (8.125)$$

in which $\{A\} = \{A_m\}$ is a three-element column, $[C] = [C_{ij}]$ is a 3×3 matrix, where C_{ij} are given in Eqs. (8.121), the elements of the columns $\{f^1\}$ and $\{f^2\}$ are:

$$f_1^N(s, r) = -\frac{1}{\pi} \sum_{m=1}^3 \int_{-\infty}^{\infty} \frac{s\lambda_m}{\xi^2 + s^2\lambda_m^2} C_{1m} b_m^N e^{i\xi r} d\xi = \sum_{m=1}^3 \text{Re}(C_{1m} b_m^N e^{sr\lambda_m}), \quad (8.126a)$$

$$f_2^N(s, r) = \frac{i}{\pi} \sum_{m=1}^3 \int_{-\infty}^{\infty} \frac{\xi}{\xi^2 + s^2\lambda_m^2} C_{2m} b_m^N e^{i\xi r} d\xi = -\sum_{m=1}^3 \text{Re}(C_{2m} b_m^N e^{sr\lambda_m}), \quad (8.126b)$$

$$f_3^N(s, r) = -\frac{1}{\pi} \sum_{m=1}^3 \int_{-\infty}^{\infty} \frac{s\lambda_m}{\xi^2 + s^2\lambda_m^2} C_{3m} b_m^N e^{i\xi r} d\xi = \sum_{m=1}^3 \text{Re}(C_{3m} b_m^N e^{sr\lambda_m}). \quad (8.126c)$$

The linear algebraic equations (8.125) can be used to determine the unknown constants A_m ($m=1,2,3$), in terms of g_1 and g_2 , the results are

$$A_m = \frac{1}{2s} \int_0^c \sum_{N=1}^2 \chi_m^N(s, r) g_N(r) dr, \quad m=1,2,3, \quad (8.127)$$

where the coefficients $\chi_m^N(s, r)$ are the elements of vectors $\{\chi^N(s, r)\}$, which are obtained as:

$$\{\chi^N\} = [C]^{-1} \{f^N\} = \sum_{j=1}^3 \sum_{l=1}^3 C_{ml}^{-1} \operatorname{Re} [\Pi_{lj} b_j^N \exp(sr\lambda_j)], \quad (N=1,2), \quad (8.128)$$

where C_{ml}^{-1} and Π_{lj} are, respectively, the elements of the following matrices:

$$[C^{-1}] = \begin{bmatrix} C_{11} & C_{12} & C_{13} \\ C_{21} & C_{22} & C_{23} \\ C_{31} & C_{32} & C_{33} \end{bmatrix}^{-1}, \quad [\Pi] = \begin{bmatrix} C_{11} & C_{12} & C_{13} \\ -C_{21} & -C_{22} & -C_{23} \\ C_{31} & C_{32} & C_{33} \end{bmatrix}. \quad (8.129)$$

By substituting Eqs. (8.123) and (8.127) into Eqs. (8.120), the stress σ_{yy} and the electric field E_x can be expressed in terms of g_1 and g_2 . At the cracked plane (the $y=0$ plane), we have:

$$\int_0^c [H(x, r)] \begin{Bmatrix} g_1(r) \\ g_2(r) \end{Bmatrix} dr + \int_0^c [\Xi(x, r)] \begin{Bmatrix} g_1(r) \\ g_2(r) \end{Bmatrix} dr = \begin{Bmatrix} \sigma_{yy}(x, 0) \\ E_x(x, 0) \end{Bmatrix}, \quad (8.130)$$

where the kernels H and Ξ are 2×2 matrices whose elements are ($i, k=1, 2$):

$$\begin{aligned} H_{ik}(x, r) &= \frac{1}{\pi} \int_0^\infty \sum_{m=1}^3 D_{im} \chi_m^k(s, r) \exp(s\lambda_m x) ds \\ &= -\frac{1}{2\pi} \sum_{m=1}^3 \sum_{j=1}^3 \sum_{l=1}^3 D_{im} b_j^k \frac{C_{ml}^{-1} \Pi_{lj}}{\lambda_m x + \lambda_j r} - \frac{1}{2\pi} \sum_{m=1}^3 \sum_{j=1}^3 \sum_{l=1}^3 D_{im} \bar{b}_j^k \frac{C_{ml}^{-1} \bar{\Pi}_{lj}}{\lambda_m x + \lambda_j r}, \end{aligned} \quad (8.131a)$$

and

$$\Xi_{ik}(x, r) = \frac{1}{\pi} \Lambda_{ik} \frac{1}{r - x}, \quad (8.131b)$$

in which an over-bar represents the complex conjugate, and

$$\Lambda_{ik} = -\sum_{m=1}^3 \frac{1}{\lambda_m} D_{im} b_m^k, \quad (8.132)$$

is a material constant matrix which can be shown to be symmetric.

The singular integral equations for the unknown functions g_1 and g_2 are obtained by applying Eq. (8.130) to the crack face boundary conditions (8.114). This leads to:

$$\frac{1}{\pi}[\Lambda] \int_0^c \frac{1}{r-x} \begin{Bmatrix} g_1(r) \\ g_2(r) \end{Bmatrix} dr + \int_0^c [H(x, r)] \begin{Bmatrix} g_1(r) \\ g_2(r) \end{Bmatrix} dr = \begin{Bmatrix} -\sigma_0(x) \\ -E_0(x) \end{Bmatrix}, \quad x \in [0, c]. \quad (8.133)$$

These integral equations will be used to determine the auxiliary functions g_1 and g_2 .

(B) Crack tip field intensity factors

The solution of the singular integral equation (8.133) may be expressed as:

$$\begin{Bmatrix} g_1(r) \\ g_2(r) \end{Bmatrix} = \frac{1}{\sqrt{1-r}} \sum_{n=0}^{\infty} \begin{Bmatrix} a_{1n} \\ a_{2n} \end{Bmatrix} T_n(\bar{r}), \quad (8.134)$$

where $\bar{x} = 2x/c - 1$, $\bar{r} = 2r/c - 1$, T_n is the Chebyshev polynomial of the first kind, a_{1n} and a_{2n} are unknown coefficients. Equation (8.133) can be solved by truncating the series in Eq. (8.134) and using a collocation technique. Because of the symmetry of the problem, only mode I stress intensity factor K_I and electric field intensity K_E exist. They are defined and obtained as:

$$\begin{Bmatrix} K_I(c) \\ K_E(c) \end{Bmatrix} = \left(\sqrt{2\pi(x-c)} \right)_{x \rightarrow c+0} \begin{Bmatrix} \sigma_{22}(x) \\ E_1(x) \end{Bmatrix} = -\sqrt{\pi c} [\Lambda] \sum_{n=0}^{\infty} \begin{Bmatrix} a_{1n} \\ a_{2n} \end{Bmatrix}. \quad (8.135)$$

The energy release rate can be obtained from the virtual crack closure integral:

$$G = \frac{1}{2} \lim_{\delta \rightarrow 0} \frac{1}{\delta} \int_0^{\delta} [\sigma_{yy}(x+a)\Delta v(x+a-\delta) + E_x(x+a)\Delta q(x+a-\delta)] dx, \quad (8.136)$$

where Δv and Δq are, respectively, the relatively opening of, and the accumulated charges on, the crack surfaces, which can be obtained by integrating Eq. (8.134). It follows from Eq. (8.136) that:

$$G = \frac{1}{2} (\Lambda_{11}^{-1} K_I^2 + (\Lambda_{12}^{-1} + \Lambda_{21}^{-1}) K_I K_E + \Lambda_{22}^{-1} K_E^2), \quad (8.137)$$

in which Λ_{ij}^{-1} ($i, j=1, 2$) are elements of the matrix $[\Lambda^{-1}]$.

8.4.4. Numerical Results

Suppose the piezoelectric medium undergoes a sudden temperature decrease on the top surface, the numerical results for the temperature $T(x, t)$, the thermal stress $\sigma_{xx}(x, t)$, and electric field $E_x(x, t)$, the stress intensity factor K_I , and the electric displacement intensity factor K_{IV} , have been obtained for various crack lengths. The following material properties are used [7]: $c_{11}=7.41 \times 10^{10} \text{ N/m}^2$, $c_{33}=8.36 \times 10^{10} \text{ N/m}^2$, $c_{12}=4.52 \times 10^{10} \text{ N/m}^2$, $c_{13}=3.93 \times 10^{10} \text{ N/m}^2$, $c_{44}=1.32 \times 10^{10} \text{ N/m}^2$, $e_{31}=-0.160 \text{ C/m}^2$, $e_{33}=0.347 \text{ C/m}^2$, $e_{15}=-0.138 \text{ C/m}^2$, $\epsilon_{11}=0.826 \times 10^{-10} \text{ C/Vm}$, $\epsilon_{33}=0.903 \times 10^{-10} \text{ C/Vm}$.

(A) Thermal stresses and electric fields

The calculated distributions of the stress and electric displacement for the un-cracked medium are shown in Figures. 8.28. Time dependence is formally replaced by the penetration depth of cooling, $\delta(t) = 2\sqrt{tk_3 / \rho C_v}$. Figure 8.28 clearly shows that the stresses and electric fields increase with time and their peak values appear at steady state. It is clear that if $T_0 < 0$ (cold shock), the medium experiences a tensile stress. The maximum tensile stress is attained at the surface of the medium. The maximum values of the stress and electric field:

$$\sigma = -\tilde{\lambda}_{11}T_0, \quad E = \tilde{\beta}_3T_0 \quad (8.138)$$

are achieved at the surface of the medium at $t=0$. As time goes on, the interior medium begins to experience the stress and electric fields whose values increase with time and achieve their maximum values at steady state.

(B) Thermal stress and electric field intensity factors

Since the medium undergoes a tensile stress under a cold shock, it is then expected that a crack may initiate at the surface of the medium. The problem can be synthesized from the general solution for a surface crack problem, with crack tip located at $x=c$, as shown in Figure 8.27. Since the crack plane is normal to surface of the medium, it does not perturb the transient temperature distribution in this arrangement, determination of the temperature distribution and resulting thermal stresses and electric field would be quite straightforward. The stresses and electric fields obtained from Eqs. (8.112), with opposite signs, become the crack surface traction and electric field loads in the quasi-static problem. The thermal stress and electrical field intensity factors are solved by the method outlined in the last section.

Some numerical results for the thermal stress and electric field intensity factors are shown in Figures. 8.29-8.31. In all cases, the stress and electric field intensity factors increase with time. In Figure 8.31, K_I and K_E are normalized such that they do not depend on the crack length c . As time increases, the normalized stress intensity factor increases and reaches the value for a surface crack under a uniform tensile stress $\sigma = -\tilde{\lambda}_{11}T_0$:

$$\frac{K_I}{-\tilde{\lambda}_{11}T_0\sqrt{\pi c}} = 1.101. \quad (8.139)$$

In fact, as cooling time approaches infinity, the crack surfaces will undergo a uniform tensile stress $\sigma = -\tilde{\lambda}_{11}T_0$ and a uniform electric field $E = \tilde{\beta}_3T$. It is well known that for an isotropic homogeneous half-plane, the normalized stress intensity factor is 1.122. It can be show that the influence of piezoelectric effect on the stress field intensity factor is less then 2%.

(C) Thermal crack growth analysis

Figure 8.32 plots the variations of thermal stress intensity factors with crack lengths. In each curve the peak value of the thermal stress intensity factor increases with crack length to a maximum value and then decreases. Therefore, a pre-existing crack of length c_0 will start propagating when the thermal stress intensity $K(c_0)$ reaches the fracture toughness K_{IC} of the medium, at a certain time. The crack will grow instantaneously at least to a new length c_1 , where the thermal stress intensity factor $K(c_1)$ again reaches the fracture toughness K_{IC} . After $c=c_1$, since the stress intensity factor keeps increasing with increasing time t (or δ), the crack will continue to grow in a stable manner. Such a fact can also be found in Figure 8.33 for crack growth trajectory $\delta(t)$ vs. crack length c , for an initial crack length $c_0=0.1\text{mm}$. For each given fracture toughness of the medium, there is a time t_0 at which the crack begins to grow, suddenly from c_0 to a new value c_1 , then stable propagate as time goes on. Apparently, the crack will be longer for a medium with smaller fracture toughness. The values of c_0 and c_1 depend on the initial crack length, the fracture toughness of the medium, and the applied temperature load.

(D) Thermal shock resistance curve

Thermal shock resistance is a major issue in the design of engineering ceramics for the thermal application. A central problem in designing against thermal shock is the identification of appropriate material selection criterion in order to select the most shock resistant material for a given application. Material performance indices are summarized for both stress-controlled failure and toughness-controlled failure.

A stress based failure criterion for thermal shock is that the maximum thermal stress appears on the surface of the medium attains the strength value σ_b . The maximum temperature jump sustainable by the material ΔT_c follows from the first of equations (8.138) as:

$$\Delta T_c = \Delta T_c^{\text{stress}} = \frac{\sigma_b}{\tilde{\lambda}_{11}}, \quad (8.140)$$

which is valid for any kind of material.

A similar strategy can be employed to rank the materials on the basis of failure from a dominant crack by thermal shock. The toughness-based fracture criterion is taken to be that the maximum thermal stress intensity factor $K_{\max}(c^*, t^*)$ attains the fracture toughness of the solid K_{IC} . Obviously, the maximum temperature jump sustainable by the solid ΔT_c depends on the length of the pre-existing crack, since the maximum stress intensity factor depends on the crack length. As Eq. (8.139) gives the maximum stress intensity factor $K_{\max}$ for different crack lengths, it follows from $K_{\max}=K_{IC}$ that

$$\Delta T_c = \Delta T_c^{\text{fracture}} = \frac{K_{IC}}{1.101 \tilde{\lambda}_{11} \sqrt{\pi c}}. \quad (8.141)$$

It is instructive to plot the sustainable temperature jump ΔT_c as a function of crack length. The result of ΔT_c as a function of crack size is depicted in Figure 8.34. It is clear that if the stress-based criterion is used, ΔT_c does not depend on the pre-existing crack length. On the other hand, ΔT_c predicted by the fracture-based criterion exhibits a strong dependence on pre-existing crack length. Since the stress intensity factor is zero for an un-cracked medium ($c=0$), the admissible temperature jump ΔT_c based on fracture-controlled failure will be infinity at $c=0$. $\Delta T_c^{\text{fracture}}$ decreases with crack length from infinity to zero. Hence, the failure of the medium will be governed by the stress-based criterion for small crack length. A transient crack length c_t exists for which ΔT_c is equal for toughness-controlled failure and for stress-controlled failure. This length can be found by equaling Eqs. (8.140) and (8.141):

$$c_t = 0.263 \left(\frac{K_{IC}}{\sigma_b} \right)^2. \quad (8.142)$$

Therefore, the thermal shock strength of a piezoelectric medium with a crack smaller than c_t will be controlled by the stress-based criterion; a medium with a crack larger than c_t will be controlled by the fracture-based criterion. A piezoelectric medium with a crack of length c_t has the same thermal shock resistance according to the stress criterion and the fracture criterion. The fracture-based criterion is conservative for a medium whose initial crack length is above c_t , the stress-based criterion is conservative for a medium whose crack length is below c_t .

In summary, the maximum temperature that the material can sustain without catastrophic failure is analyzed according to the maximum local tensile stress criterion and the maximum stress intensity factor criterion. A transient crack length c_t was obtained for which the admissible temperature jump is equal for fracture-controlled failure and stress-controlled failure. Thermal shock strength of a piezoelectric material with crack smaller than c_t will be controlled by the stress-based criterion, and larger than c_t will be controlled by the fracture-based criterion.

The results show that the influence of the piezoelectric effects on the stress intensity factors is relatively insignificant. This suggests that the thermal fracture of piezoelectric materials can be analysis in a similar way as ordinary brittle solids.

8.5. FRACTURE OF A PIEZOELECTRIC CYLINDER UNDER A TRANSIENT TEMPERATURE FIELD

The preceding sections mainly studied the transient thermal fracture for two-dimensional plates or strips. Due to mathematical difficulties, three-dimensional transient thermal fracture of piezoelectric ceramics has not received much attention. For the case of internal embedded

cracks, the penny-shaped crack geometry may be considered. This section provides a strength evaluation method for piezoelectric ceramics. A cylindrical specimen is used for the case study. Two distinct boundary-value problems are involved. The first is a classical theory of thermal conduction, and the second is the determination of the stress and electric field intensity factors at the crack tip in the cylinder caused by the uneven heating. Once the temperature distribution is obtained, the thermal stresses and electric fields in the un-cracked medium are evaluated. The thermal stresses and electric displacements from the un-cracked problem with opposite signs are used as crack surface traction and electric field loads. The crack problem is solved by means of integral equation technique. The results of the stress and electric displacement intensity factors are obtained analytically for an infinitesimal crack, and numerically for a finite-sized crack. Thermal strength is also evaluated.

8.5.1. The Temperature Field

The problem under consideration is a piezoelectric cylinder subjected to a sudden temperature T_0 on its outer surface ($r=b$), as described in Figure 8.35. The cylindrical coordinates r , θ and z are coincident with the principal axes of the material symmetry. The z -axis is oriented in the poling direction of the piezoelectric medium. The cylinder has a radius b and is infinite along its axial direction. Investigated is an axi-symmetric problem such that all the field variables are functions of the radial coordinate r and axial coordinate z only.

In the problem under consideration, the external temperature is only a function of the radial coordinate r and the time variable t . Therefore, the heat conduction is one-dimensional and the temperature field T is governed by

$$k_2 \frac{\partial^2 T(r,t)}{\partial r^2} = \rho c_v \frac{\partial T(r,t)}{\partial t}, \quad (8.143)$$

where k_2 is the coefficient of thermal conductivity of the medium in the radial direction, ρ the density, and c_v the specific heat. Suppose the piezoelectric medium is initially at a uniform temperature zero, and its outer surface undergoes a sudden temperature variation T_0 . That is

$$T(r, 0)=0. \quad (8.144)$$

$$T(b, t)=T_0 H(t). \quad (8.145)$$

Equations (8.144) and (8.145) define a thermal shock condition on the outer surface of the medium where $H(t)$ is the Heaviside function.

The thermal equation (8.143) is solved with the heat transfer boundary and initial conditions (8.144) and (8.145) by a standard separation-of-variables technique, giving

$$T(r,t) = T_0 - T_0 \sum_{n=1}^{\infty} \frac{2}{\gamma_n J_1(\gamma_n)} J_0(\gamma_n \frac{r}{b}) \exp(-\gamma_n^2 \frac{t}{t_0}), \quad (8.146)$$

where $t_0 = \rho c_v b^2 / k_2$, J_i ($i=0, 1$) are the i -th Bessel functions of the first kind, and γ_n are the roots of the equation $J_0(\gamma)=0$. It can be seen immediately that the thermal diffusivity $k_2/\rho c_v$ dictates the time scale for transient temperature distribution, but does not affect the level of the temperature.

8.5.2. Electro-Elasticity Fields in the un-Cracked Piezoelectric Medium

The thermo-electro-elastic constitutive relations for piezoelectric materials in a cylindrical coordinate system have been given in Eqs. (8.15). Due to symmetry of the problem, the shear stress σ_{rz} and the electric displacement D_r are zero. It follows from the equilibrium equations, Eqs. (4.93), that

$$\frac{\partial(r\sigma_{rr})}{\partial r} - \sigma_{\theta\theta} = 0, \quad \frac{\partial\sigma_{zz}}{\partial z} = 0, \quad \frac{\partial D_z}{\partial z} = 0. \quad (8.147)$$

With substituting of constitutive relations, Eq. (8.147a) gives

$$c_{11} \left(\frac{\partial^2 u}{\partial r^2} + \frac{1}{r} \frac{\partial u}{\partial r} - \frac{u}{r^2} \right) = \lambda_{11} \frac{\partial T}{\partial r}, \quad (8.148)$$

which gives the solution

$$u = \frac{\lambda_{11}}{c_{11}} \frac{1}{r} \int_0^r T r dr + C_1 r + \frac{C_2}{r}. \quad (8.149)$$

Since the displacement is finite at $r=0$, the coefficient C_2 must be zero. The radial stress σ_{rr} associated with Eq. (8.149) can be obtained from thermoelasticity constitutive relations, Eqs. (8.15), as:

$$\sigma_{rr} = -(c_{11} - c_{12}) \frac{\lambda_{11}}{c_{11}} \frac{1}{r^2} \int_0^r T r dr + (c_{11} + c_{12}) C_1 + c_{13} \frac{\partial w}{\partial z} + e_{31} \frac{\partial \phi}{\partial z}, \quad (8.150)$$

Since the outer surface of the medium is stress free, the radial stress σ_{rr} must vanish at $r=b$. Using this condition, the constant C_1 can be solved from Eq. (8.150). From constitutive relations Eqs. (8.15), the radial displacement Eq. (8.149), and with substituting of C_1 , the axial stress σ_{zz} and the axial electric displacement D_z are obtained as follows:

$$\sigma_{zz} = \bar{\lambda}_{11} \frac{2}{b^2} \int_0^b T r dr + \bar{c}_{33} \frac{\partial w}{\partial z} + \bar{e}_{33} \frac{\partial \phi}{\partial z} - \bar{\lambda}_{33} T, \quad (8.151a)$$

$$D_z = \bar{\beta}_1 \frac{2}{b^2} \int_0^b T r dr + \bar{e}_{33} \frac{\partial w}{\partial z} - \bar{\epsilon}_{33} \frac{\partial \phi}{\partial z} - \bar{\beta}_3 T, \quad (8.151b)$$

where $\bar{\lambda}_{11}$, $\bar{\beta}_1$, $\bar{c}_{33}$, $\bar{e}_{33}$ and $\bar{\epsilon}_{33}$ are some equivalent material constants, and

$$\bar{\lambda}_{33} = \lambda_{33} - \frac{c_{13}}{c_{11}} \lambda_{11}, \quad \bar{\beta}_3 = \beta_3 - \frac{e_{31}}{c_{11}} \lambda_{11}. \quad (8.152)$$

From the overall equilibrium of the cylinder, the total force and electric charge along the axial direction are zero. Hence:

$$\int_0^b \sigma_{zz} r dr = 0, \quad \int_0^b D_z r dr = 0, \quad (8.153)$$

It follows from Eqs. (8.151) and (8.153) that

$$\bar{c}_{33} \frac{\partial w}{\partial z} + \bar{e}_{33} \frac{\partial \phi}{\partial z} = \frac{2}{b^2} (\bar{\lambda}_{33} - \bar{\lambda}_{11}) \int_0^b T r dr, \quad (8.154a)$$

$$\bar{e}_{33} \frac{\partial w}{\partial z} - \bar{\epsilon}_{33} \frac{\partial \phi}{\partial z} = \frac{2}{b^2} (\bar{\beta}_3 - \bar{\beta}_1) \int_0^b T r dr. \quad (8.154b)$$

Finally, the axial thermal stress σ_0 and the axial electric displacement D_0 can be obtained by substituting Eqs. (8.154) into Eqs. (8.151). This leads to:

$$\sigma_0(r, t) = \sigma_{zz}(r, 0, t) = \bar{\lambda}_{33} \frac{2}{b^2} \int_0^b T r dr - \bar{\lambda}_{33} T, \quad (8.155a)$$

$$D_0(r, t) = D_z(r, 0, t) = \bar{\beta}_3 \frac{2}{b^2} \int_0^b T r dr - \bar{\beta}_3 T. \quad (8.155b)$$

The transient thermal stress and electric displacement are obtained by substituting the temperature field, Eq. (8.146), into Eqs. (8.155). The result is

$$\sigma_0(r, t) = -\bar{\lambda}_{33} T_0 \sum_{n=1}^{\infty} \left[\frac{4}{\gamma_n^2} - \frac{2}{\gamma_n J_1(\gamma_n)} J_0(\gamma_n \frac{r}{b}) \right] \exp(-\gamma_n^2 \frac{t}{t_0}), \quad (8.156a)$$

$$D_0(r, t) = -\bar{\beta}_3 T_0 \sum_{n=1}^{\infty} \left[\frac{4}{\gamma_n^2} - \frac{2}{\gamma_n J_1(\gamma_n)} J_0(\gamma_n \frac{r}{b}) \right] \exp(-\gamma_n^2 \frac{t}{t_0}). \quad (8.156b)$$

Thus, among many thermo-electro-mechanical parameters in the piezoelectric materials, the parameters that govern the transient thermal stress and electric displacement levels in the plate are identified as $\bar{\lambda}_{33}$ and $\bar{\beta}_3$. The time scale for the transient thermal stress and electric displacement state is governed by the thermal diffusivity $\rho c_v/k_2$, which does not affect the levels of thermal stresses and electric displacements.

The evaluation of the dimensionless thermal stresses and electric displacements are plotted against the dimensionless Fourier number $F_0=t/t_0$, where $t_0=\rho c_v b^2/k_2$, in Figures. 8.36 and 8.37 for different positions in the cylinder. It is clear that if $T_0>0$, the region near the surface of the cylinder experience a compressive stress while a tensile zone is developed at the center of the cylinder. The maximum compressive stress is attained at the surface and the tensile stress is largest at the center of the cylinder. The maximum values of

$$\sigma_0 = -\bar{\lambda}_{33}T_0, \quad D_0 = -\bar{\beta}_3T_0, \quad (8.157)$$

are achieved at the surface of the cylinder at $t=0$. The maximum compressive stress and the electric displacement

$$\sigma_0 = 0.471\bar{\lambda}_{33}T_0, \quad D_0 = 0.471\bar{\beta}_3T_0, \quad (8.158)$$

are attained at the center of the cylinder at a time $t/t_0=0.0761$. These expressions are valid for any piezoelectric materials.

Under a cooling condition ($T_0<0$), at any location of the plate, the stress and electric displacement will change their signs from the heating condition. Hence the surface of the cylinder will become tensile and the center will be compressive. The magnitudes of the thermal stresses and electric displacements subjected to a cooling will be same as those in a heating condition.

8.5.3. Cracking of the Piezoelectric Cylinder Under Transient Heating on its Outer Surface

The above analysis shows that under the heating condition, the cylinder has the largest axial tensile stress at its center, while the axial stress at its outer surface is compressive. It is then expected that a crack may initiate at the center of the cylinder under the heating condition. The crack grows unstably until it enters the compressive region. The problem can be synthesized from the general solution for an embedded crack in a cylinder, with crack front located at $r=a$, as shown in Figure 8.38. Since the crack plane is normal to the outer surfaces of the cylinder, it does not perturb the transient temperature distribution in this arrangement, determination of the temperature distribution and the resulting thermal stresses and electric displacements would be quite straightforward. The stress and electric displacement given by Eqs. (8.156), with opposite signs, are superposed to the crack surface mechanical and electrical loads in the quasi-static problem. Generally, the crack surfaces are traction free. However, because air or vacuum allows some penetration of electric flux through the crack

interior, the electric displacement inside the crack may not be zero and its normal component (along the z direction) is denoted as d_0 . Then the mixed boundary conditions on the $z=0$ plane can be stated as follows

$$\sigma_{zz}(r,0) = p_{01}(r) = -\sigma_0(r), \quad D_z(r,0) = p_{02}(r) = d_0(r) - D_0(r), \quad r < a, \quad (8.159a)$$

$$w(r,0)=0, \phi(r,0)=0, \quad r \geq a. \quad (8.159b)$$

where $\sigma_0(r)$ and $D_0(r)$ are the values obtained from the solution without any cracks. They are caused by the thermal loads given by Eqs. (8.156).

Algebraic details concerning solution for a crack in a cylinder under coupled electro-mechanical loads can be found from Ref. [10]. The resulting integral equations are obtained as:

$$\sum_{j=1}^2 \left[\Lambda_{ij} \Phi_j(t) + \int_0^a K_{ij}(t,r) \Phi_j(r) dr \right] = Q_i(t), \quad (8.160)$$

in which the general loads are:

$$Q_1(l) = -\frac{2}{\pi} \int_0^l \frac{r \sigma_0(r,t)}{\sqrt{l^2 - r^2}} dr = \frac{4}{\pi} \bar{\lambda}_{33} T_0 b \sum_{n=1}^{\infty} \frac{1}{\gamma_n^2} \exp(-\gamma_n^2 \frac{t}{t_0}) \left[2 \frac{l}{b} - \frac{1}{J_1(\gamma_n)} \sin(\gamma_n \frac{l}{b}) \right], \quad (8.161a)$$

$$\begin{aligned} Q_2(l) &= \frac{2}{\pi} \int_0^l \frac{r d_0(r,t)}{\sqrt{l^2 - r^2}} dr - \frac{2}{\pi} \int_0^l \frac{r D_0(r,t)}{\sqrt{l^2 - r^2}} dr \\ &= \frac{2}{\pi} \int_0^l \frac{r d_0(r,t)}{\sqrt{l^2 - r^2}} dr + \frac{4}{\pi} \bar{\beta}_3 T_0 b \sum_{n=1}^{\infty} \frac{1}{\gamma_n^2} \exp(-\gamma_n^2 \frac{t}{t_0}) \left[2 \frac{l}{b} - \frac{1}{J_1(\gamma_n)} \sin(\gamma_n \frac{l}{b}) \right]. \end{aligned} \quad (8.161b)$$

Note that in the problem under consideration time t enters into the analysis through $Q_i(l)$ only, and Eqs. (8.160) must be solved for each value of t separately. Once integral equations (8.160) are solved numerically, the stress and electric field intensity factors can be obtained for electrically impermeable cracks and for electrically permeable cracks. In particular, if the crack is infinitesimal, the closed-form expressions can be obtained. This is discussed below.

8.5.4. A Crack Embedded in an Infinite Piezoelectric Medium

In the case of an infinitesimal crack, the integral kernels K_{ij} in Eqs (8.160) vanish. As a result, the solution can be obtained analytically for impermeable and permeable crack face electrical boundary conditions.

Impermeable crack

The electric displacement inside the crack d_0 is zero for an impermeable crack. The expressions for Φ_j are obtained by substituting Eqs. (8.161) into Eqs. (8.160). Since l/b is small, the function $\sin(\gamma_n l/b)$ equals $\gamma_n l/b$. It follows that:

$$\Phi_j(l) = M_j f(t) T_0 l, \quad (j = 1, 2), \quad (8.162)$$

where

$$M_j = \bar{\lambda}_{33} \Lambda_{j1}^{-1} + \bar{\beta}_3 \Lambda_{j2}^{-1}, \quad (j = 1, 2), \quad (8.163)$$

are material constants, and

$$f(t) = \frac{4}{\pi} \sum_{n=1}^{\infty} \frac{1}{\gamma_n^2} \exp(-\gamma_n^2 \frac{t}{t_0}) \left[2 - \frac{1}{J_1(\gamma_n)} \gamma_n \right], \quad (8.164)$$

is a function of time t and is independent of material properties.

The stress and electric field intensity factors are obtained as follows:

$$k_1 = -\sqrt{\pi a} \bar{\lambda}_{33} T_0 f(t), \quad k_D = -\sqrt{\pi a} \bar{\beta}_3 T_0 f(t). \quad (8.165)$$

Permeable crack

The electric potential jump across an electrically permeable crack is zero. Therefore, the function Φ_2 vanishes for a permeable crack. The solution of Φ_1 is obtained from the first of Eqs. (8.160):

$$\Phi_1(l) = \frac{\bar{\lambda}_{33}}{\Lambda_{11}} T_0 l f(t). \quad (8.166)$$

The stress intensity factor is the same as Eq. (8.165a) for an impermeable crack. The electric displacement intensity factor is related to k_1 through:

$$k_D = \frac{\Lambda_{21}}{\Lambda_{11}} k_1. \quad (8.167)$$

For a permeable crack, the electric potential jump across the crack is zero and there will be electric charges accumulated on the crack surfaces. Note that the kernel K_{21} vanishes since the crack is infinitesimal. Further, the function $D_0(x)$ obtained from Eq. (8.156b) is uniform:

$$D_0 = -\frac{\pi}{2} \bar{\beta}_3 T_0 f(t). \quad (8.168)$$

where the approximation $J_0(\gamma_n x/b) \approx 1$ for small values of x/b has been used. Finally, the electric displacement inside the crack is obtained as follows:

$$d_0 = \frac{\pi}{2} \left(\frac{\Lambda_{21} \bar{\lambda}_{33}}{\Lambda_{11}} - \bar{\beta}_3 \right) T_0 f(t), \quad (8.169)$$

which is uniform (it does not depend on the radial coordinate r).

Time-dependent solutions

From the solution established above, it is seen that the time-dependency of the full field solution is solely governed by the function $f(t)$, which is depicted in Figure 8.39. Clearly, the thermally induced electro-mechanical field is zero initially, increases to a peak as time goes on, and then decreases to zero as time approaches infinity. The function $f(t)$ is negative for any t . Its maximum absolute value is:

$$f_{\max} = 0.3, \quad (8.170)$$

which is attained at a time $t/t_0=0.0761$.

8.5.5. A Numerical Example

Section 8.5.4 gives the analytical and closed-form solution for an infinitesimal crack in a piezoelectric cylinder under a transient thermal load. These results are useful for the strength evaluation of piezoelectric materials under transient thermal environments.

For the problem of crack growth under transient temperature field, the crack may not necessary to be infinitesimal. As a result, the solutions can only be obtained numerically for given materials. Three kinds of particular materials, PZT-4, PZT-5H and P-7, are considered. The material properties have been summarized in Table 4.1.

After solving the problem for the poling direction parallel to the z -axis, the material constant matrix $[\Lambda]$ are obtained and are tabulated in Table 8.1.

(A) Stress intensity factors

From the numerical results, it is found that the stress intensity factor k_1 caused by a thermally induced electric displacement load D_0 is negligible. Further, the values of k_1 are same for an electrically impermeable crack and for an electrically permeable crack.

Table 8.1. The elements of matrix $[\Lambda]$

Materials	PZT-4	PZT-5H	P-7
Λ_{11}	-4.328×10^{10}	-5.107×10^{10}	-3.860×10^{10}
$\Lambda_{21}=\Lambda_{12}$	-10.96	-14.37	-11.91
Λ_{22}	86.61×10^{-10}	179.7×10^{-10}	216.3×10^{-10}

In Figure 8.40, the normalized stress intensity factors for PZT-4 piezoelectric materials $\bar{k} = k_1 / k_0$, where $k_0 = \bar{\lambda}_{33} \sqrt{\pi b T_0}$, are plotted against dimensionless time t/t_0 , for selected normalized crack radius a/b , where a is the crack radius. At any given a/b , the stress intensity factor increases from an initial zero with time, displays a peak value and then decreases to zero as time approaches infinity. This means that the thermal stress intensity factor occurs only at transient state and that its steady value is zero.

Referring to Figure 8.40 for a PZT-4 material, if we consider a crack of given initial radius, we see that the temporal behavior is alike for all crack radiuses. The envelope of this family of curves is the locus of the highest thermal stress intensity factors reached for every crack radiuses at any time. It is seen that k attains an overall maximum value:

$$k_{\max} \approx 0.165 \bar{\lambda}_{33} \sqrt{\pi b T_0}, \quad (8.171)$$

at a particular time $t^* \approx 0.05 t_0$ and a particular crack depth $a^* \approx 0.5 b$. The peak values of the stress intensity factors, and the corresponding crack radii and times for different materials are summarized in Table 8.2.

(B) Electric displacement intensity factors for an electrically impermeable crack

Figures 8.41 and 8.42 plot the electric displacement intensity factors produced, respectively, by thermally induced stress loads and electric displacement loads. The results in Figure 8.41 show that if crack size a is small, the electric displacement intensity factor K_D caused by the thermal stress load σ_0 is negligible. K_D becomes significant only when crack size a is larger than $0.4b$.

(C) Electric displacement intensity factors for an electrically permeable crack

For an electrically permeable crack, it does not response to thermally induced electrical load D_0 . Once the stress intensity factors (which are same for impermeable cracks and permeable crack) are obtained, the electric displacement intensity factors can be directly found from Eq. (8.167). Using material constants Λ_{ij} in Table 8.1, we obtained:

$$k_D = 2.534 \times 10^{-10} k_1, \text{ for PZT-4.} \quad (8.172a)$$

$$k_D = 2.814 \times 10^{-10} k_1, \text{ for PZT-5H.} \quad (8.172b)$$

$$k_D = 3.086 \times 10^{-10} k_1, \text{ for P-7.} \quad (8.172c)$$

Table 8.2. The maximum stress intensity factors, and the corresponding crack radii and times

Materials	PZT-4	PZT-5H	P-7
$k_{\max}$	$0.165 \bar{\lambda}_{33} \sqrt{\pi b T_0}$	$0.166 \bar{\lambda}_{33} \sqrt{\pi b T_0}$	$0.169 \bar{\lambda}_{33} \sqrt{\pi b T_0}$
t^*	$0.05 t_0$	$0.048 t_0$	$0.04 t_0$
a^*	$0.5 b$	$0.52 b$	$0.58 b$

The value of the electric displacement intensity factor in Eq. (8.172a) is the same as that obtained for a Griffith crack in a PZT-4 piezoelectric ceramic [9].

(D) Thermal shock resistance of the piezoelectric cylinder

In the following, solutions are obtained for the maximum thermal shock that the medium can sustain without catastrophic failure according to the two conditions that: (i) maximum local tensile stress equals the tensile strength of the ceramic, and (ii) maximum stress intensity factor equals the fracture toughness of the ceramic.

A stress-based failure criterion for thermal shock is that the maximum thermal stress that appears on the center of the cylinder $\sigma_{\max}(0, t)$ attains the value σ_b , which is the strength limit of the piezoelectric medium. The maximum temperature jump sustainable by ΔT_c follows from Eq. (8.158a) as

$$\Delta T_c = \Delta T_c^{\text{stress}} = 2.12\sigma_b / \bar{\lambda}_{33}. \quad (8.173)$$

Equation (8.173) represents the minimum temperature jump sustainable by the medium based on the stress-failure criterion. ΔT_c does not depend on the cylinder radius b , and are valid for all materials.

As pointed above, the stress intensity factors for the electrically impermeable cracks and the electrically permeable cracks are same. Therefore, if the stress intensity factor is used as a failure criterion for piezoelectric ceramics, there is no need to consider the electrical boundary conditions on the crack faces (the effect of the applied electric field on the fracture can be taken into consideration by considering the dependency of the fracture toughness on the electric field). For the fracture-based failure criterion, the maximum thermal stress intensity factor $k_{\max}$ must equal the fracture toughness of the ceramic k_{1c} . From the results in Table 8.2, we know that

$$\Delta T_c = \Delta T_c^{\text{fracture}} = 6.06 \frac{k_{1c}}{\lambda_{33} \sqrt{\pi b}}, \text{ for PZT-4,} \quad (8.174a)$$

$$\Delta T_c = \Delta T_c^{\text{fracture}} = 6.02 \frac{k_{1c}}{\lambda_{33} \sqrt{\pi b}}, \text{ for PZT-5H,} \quad (8.174b)$$

$$\Delta T_c = \Delta T_c^{\text{fracture}} = 5.92 \frac{k_{1c}}{\lambda_{33} \sqrt{\pi b}}, \text{ for P-7,} \quad (8.174c)$$

Each of equations (8.174) represents the minimum temperature jump sustainable by the medium based on the fracture-failure criterion. Clearly, ΔT_c has a strong dependency on the cylinder radius b . ΔT_c decreases as b increases.

Consideration of Eqs. (8.173) and (8.174) reveals that the admissible temperature jump is less for fracture-controlled failure than for stress-controlled failure, at a sufficiently large cylinder radius. A transient cylinder radius value b_t exists for which ΔT_c is equal for fracture-

based failure and stress-based failure. Then, upon equating ΔT_c values for the stress-based criterion (8.173) and for the fracture-based criterion (8.174), it is found that

$$b_t = 2.60 \left(\frac{K_{1c}}{\sigma_b} \right)^2, \text{ for PZT-4,} \quad (8.175a)$$

$$b_t = 2.57 \left(\frac{K_{1c}}{\sigma_b} \right)^2, \text{ for PZT-5H,} \quad (8.175b)$$

$$b_t = 2.48 \left(\frac{K_{1c}}{\sigma_b} \right)^2, \text{ for P-7,} \quad (8.175c)$$

Therefore, the thermal strength of a piezoelectric ceramic cylinder with radius smaller than b_t should be evaluated by the stress-based criterion; a cylinder with thickness larger than b_t should be evaluated by the fracture-based criterion. Thus, a piezoelectric material whose radius is b_t has the same thermal strength according to the stress criterion and the fracture criterion. The fracture-based criterion is conservative for a cylinder whose thickness is above b_t , the stress-based criterion is conservative for a cylinder whose thickness is below b_t .

In conclusion, this Chapter mainly concerns the fracture of piezoelectric materials under transient thermal environments. Various kinds of fracture mechanics specimen have been investigated.

Thermal shock fracture behavior of piezoelectric materials has been investigated theoretically for a piezoelectric plate containing an embedded crack or an edge crack.

The surfaces of the plate undergo sudden heating (for the embedded crack problem) or cooling (for the edge crack problem). The temperature and stress histories in the plate are given. For the case of a cold shock, the stress field is tensile near the surface of the plate and gives rise to a mode I stress intensity factor for a pre-existing crack. Vice versa, for the case of a hot shock, the stress field is tensile at the center of the plate and gives rise to a mode I stress intensity factor there. For the geometry and boundary conditions considered, the thermally induced stress and electric displacement intensity factors occur only at transient state. They steady values are zero.

Among many material parameters, thermal stresses and thermal electric displacements in an un-cracked piezoelectric plate are only controlled by some equivalent parameters. For parallel poling, those parameter are $\bar{\lambda}_{33}$ and $\bar{\beta}_3$, which are defined in Eqs. (8.16) and (8.18), respectively. The maximum stresses are $\sigma = -\bar{\lambda}_{33}T_0$ at the surface of the plate, and $\sigma = 0.308\bar{\lambda}_{33}T_0$ at the center of the plate. For perpendicular poling, the parameter is $\bar{\lambda}_{11}$, which is defined in Eq. (8.81). The maximum stresses are: $\sigma = \bar{\lambda}_{11}(T_i - T_\infty)$ which is tensile at the surface of the plate, and $\sigma = 0.308\bar{\lambda}_{11}(T_i - T_\infty)$ which is compressive at the center of the plate, respectively.

Thermal shock strength of the piezoelectric materials is evaluated. The maximum temperature that the material can sustain without catastrophic failure is analyzed according to the maximum local tensile stress criterion and the maximum stress intensity factor criterion. A transient plate thickness value h_t (or H_t) was obtained for which the admissible temperature jump is equal for fracture-based criterion and stress-based criterion. The thermal strength of the piezoelectric ceramic plates with thickness smaller than h_t (or H_t) should be predicted by the stress-based criterion, the plates with thickness larger than h_t (or H_t) should be predicted by the fracture-based criterion.

Comparing the results in Section 8.2 and 8.3, it is evident that the poling direction of piezoelectric media has a significant influence on their thermal shock strength.

The thermal shock fracture of a semi-infinite piezoelectric medium was studied.

The medium contains an electrically conductive surface crack. The medium surface undergoes a sudden cooling. The stress and electric field histories in the medium are given. For the geometry and boundary conditions considered, the maximum thermally induced stress and electric field intensity factors occur only at steady state.

Among many material parameters, the thermal stresses and the thermal electric fields in an un-cracked medium are only controlled, respectively, by equivalent parameters $\tilde{\lambda}_{11}$ and $\tilde{\beta}_3$, which are defined in Eqs. (8.113). The maximum thermal stress and the electric field are $\sigma = -\tilde{\lambda}_{11}T_0$ and $E = \tilde{\beta}_3T_0$, respectively.

Thermally induced stress and fracture behavior of piezoelectric ceramics have been investigated theoretically for a piezoelectric cylinder

The poling axis of the piezoelectric cylinder is parallel to its axial direction. The outer surface of the cylinder undergoes a sudden heating. Exact and closed-form solutions for the thermally induced crack tip field are obtained for an infinitesimally cracking. For the geometry and boundary conditions considered, the thermal stress intensity factors occur only at transient state. The steady thermal stress intensity factor is always zero. It is found that:

- (1) Among many material parameters, the thermal stresses and the thermal electric displacements in an un-cracked piezoelectric plate are only controlled, respectively, by equivalent parameters $\bar{\lambda}_{33}$ and $\bar{\beta}_3$, which are defined in Eqs. (8.152).
- (2) The maximum stresses and electric displacements are, respectively, $\sigma_0 = -\bar{\lambda}_{33}T_0$ and $D_0 = -\bar{\beta}_3T_0$, at the outer surface of the cylinder, and $\sigma_0 = 0.471\bar{\lambda}_{33}T_0$ and $D_0 = 0.471\bar{\beta}_3T_0$ at the center of the cylinder.
- (3) Analytical and closed-form solution for an infinitesimal crack is obtained. The maximum stress intensity factor is $k_{1\max} = 0.3\bar{\lambda}_{33}T_0\sqrt{\pi a}$, and the maximum electric displacement intensity factors are $k_{D\max} = 0.3\bar{\beta}_3T_0\sqrt{\pi a}$ for the electrically

impermeable cracks and $k_{\text{Dmax}} = 0.3(\Lambda_{21}/\Lambda_{11})\bar{\lambda}_{33}T_0\sqrt{\pi a}$ for the electrically permeable cracks.

- (4) Numerical solution technique for a finite-sized crack is given. The maximum temperature that the material can sustain without catastrophic failure is investigated according to the maximum local tensile stress criterion and the maximum stress intensity factor criterion.
- (5) A transient cylinder radius value b_t is obtained for which the admissible temperature jump is equal for fracture-based criterion and stress-based criterion. The admissible temperature of a piezoelectric ceramic cylinder with thickness smaller than b_t should be evaluate by the stress-based criterion, a cylinder with radius larger than b_t should be evaluated by the fracture-based criterion.

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APPENDIX A

If boundary conditions (7.31a) are used, then the j -th column in the 6×6 matrix $[G]$ is

$$(C_{4i}e^{s\lambda_i h}, C_{4i}, C_{3i}e^{s\lambda_i h}, C_{3i}, C_{5i}e^{s\lambda_i h}, C_{5i})^T, (i=1, \dots, 6), \quad (\text{A1})$$

and the elements in vector $\{f_u\}$ are

$$f_{1j}(s, r) = -\frac{1}{\pi} \sum_{m=1}^3 \int_{-\infty}^{\infty} \frac{s\lambda_m}{\xi^2 + s^2\lambda_m^2} C_{4m}b_{mj} \exp[i\xi(r-h)]d\xi$$

$$= \text{Re} \left(\sum_{m=1}^3 C_{4m}b_{mj} \exp[s(h-r)\lambda_m] \right),$$

$$f_{2j}(s, r) = -\frac{1}{\pi} \sum_{m=1}^3 \int_{-\infty}^{\infty} \frac{s\lambda_m}{\xi^2 + s^2\lambda_m^2} C_{4m}b_{mj} \exp(i\xi r)d\xi$$

$$= \text{Re} \left(\sum_{m=1}^3 C_{4m}b_{mj} \exp[sr\lambda_m] \right),$$

$$f_{3j}(s, r) = \frac{i}{\pi} \sum_{m=1}^3 \int_{-\infty}^{\infty} \frac{\xi}{\xi^2 + s^2\lambda_m^2} C_{3m}b_{mj} \exp[i\xi(r-h)]d\xi$$

$$= \text{Re} \left[\sum_{m=1}^3 C_{3m}b_{mj} \exp[s(h-r)\lambda_m] \right],$$

$$f_{4j}(s, r) = \frac{i}{\pi} \sum_{m=1}^3 \int_{-\infty}^{\infty} \frac{\xi}{\xi^2 + s^2\lambda_m^2} C_{3m}b_{mj} \exp[i\xi r]d\xi$$

$$= -\text{Re} \left[\sum_{m=1}^3 C_{3m}b_{mj} \exp[sr\lambda_m] \right],$$

$$f_{5j}(s, r) = -\frac{1}{\pi} \sum_{m=1}^3 \int_{-\infty}^{\infty} \frac{s\lambda_m}{\xi^2 + s^2\lambda_m^2} C_{5m}b_{mj} \exp[i\xi(r-h)]d\xi$$

$$\begin{aligned}
 &= \operatorname{Re} \left(\sum_{m=1}^3 C_{5m} b_{mj} \exp[s(h-r)\lambda_m] \right), \\
 f_{6j}(s, r) &= -\frac{1}{\pi} \sum_{m=1}^3 \int_{-\infty}^{\infty} \frac{s\lambda_m}{\xi^2 + s^2\lambda_m^2} C_{5m} b_{mj} \exp(i\xi r) d\xi \\
 &= \operatorname{Re} \left(\sum_{m=1}^3 C_{5m} b_{mj} \exp[sr\lambda_m] \right). \tag{A2}
 \end{aligned}$$

If boundary conditions (7.31b) are used, then C_{3i} in the third row of matrix $[G]$ should be replaced by a_{2j} , and f_{3j} should be replaced by

$$\begin{aligned}
 f_{3j}(s, r) &= -\frac{1}{\pi} \sum_{m=1}^3 \int_{-\infty}^{\infty} \frac{s\lambda_m}{\xi^2 + s^2\lambda_m^2} a_{2m} b_{mj} \exp[i\xi(r-h)] d\xi \\
 &= \operatorname{Re} \left(\sum_{m=1}^3 a_{2m} b_{mj} \exp[s(h-r)\lambda_m] \right). \tag{A3}
 \end{aligned}$$

Obviously, f_{ij} are pure real numbers.

APPENDIX B: MODEL VERIFICATION

The transient thermal stress expressions (7.86) and (7.87) are obtained in a quite straightforward manner. The main issue that needs to be verified is the proposed analytical model for the stress intensity factors. It is noted that there are no published results in the literature for comparison. However, the transient thermal stress intensity factors can be readily obtained by appropriate finite element method. Thus, we calculate the transient thermal stress intensity factors using the proposed model, and then compare the results with those obtained from the finite element method, both considering and ignoring the piezoelectric effect.

Consider the crack geometry shown in Figure 7.18. According to fracture mechanics theory, the stress near the crack tip is:

$$\sigma_{xx}(0, z) = K / \sqrt{2\pi r}, \text{ for } r = z - c \rightarrow +0. \tag{B1}$$

Therefore, the stress intensity can be calculate from

$$K = \lim_{z \rightarrow +c} \sigma_{xx}(0, z) \sqrt{2\pi r}. \quad (\text{B2})$$

We will now use the finite element code ABAQUS to calculate the stress σ_{xx} at the crack front. Due to symmetry with respect to $x=0$ plane, only one-half of the medium needs to be considered ($d > x > 0$ part, where d is chosen to be 10 times greater than strip thickness H such that the strip is nearly infinite length). Plane four-node element is utilized for the analysis. The largest size of the element is $H/10$. The size of the elements near the crack tip is refined and the smallest element size is $c/200$. Care should be taken to the selection of r in Eq. (B2). If r is too small the stress will show singularity, hence the FEM results will be not accurate. On the other hand, if r is large the definition of Eq. (B2) will be invalid. In this calculation, r is typical between $0.01c$ and $0.05c$, where c is the total length of the crack.

Three sets of results are obtained. (1) Results from proposed analytical model considering piezoelectric effect. (2) Results from finite element method considering piezoelectric effect. (3) Results obtained from finite element method without considering piezoelectric effect. Figures B1 and B2 show normalized stress intensity factors plotted against dimensionless time t/t_0 at selected crack lengths for an edge crack under heating, and a central crack under cooling. It is found that:

- (i) The proposed analytical model gives quite reasonable results. This means that the derived equations are correct and accurate.
- (ii) The piezoelectric effect can increase or decrease the stress intensity factors, depending on the time. But the influence is not significant if the crack is shorter than the size considered (e.g., $c < 0.1H$ or $a < 0.1H$).
- (iii) The finite element method is very time-consuming since it needs to repeat the whole calculation at each time.

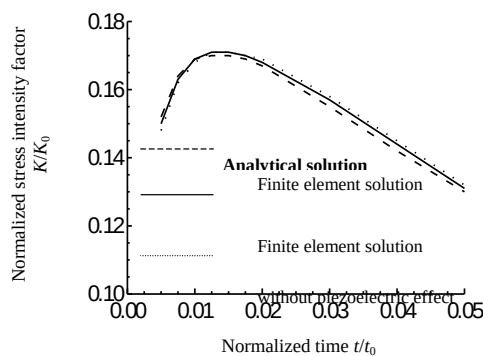


Figure B1. Model verification: Normalized stress intensity factor $\bar{K} = K/K_0$ obtained from analytical solution and finite element solution; edge crack under cold shock, $c/H=0.1$,

$$K_0 = \bar{\lambda}_{11} \sqrt{\pi H (T_i - T_\infty)}.$$

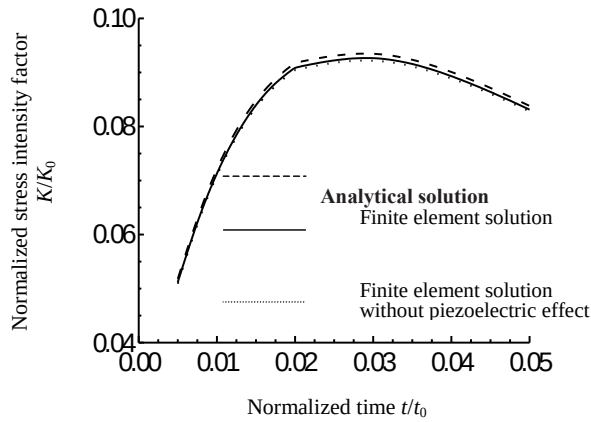


Figure B2. Model verification: Normalized stress intensity factor $\bar{K} = K/K_0$ obtained from analytical solution and finite element solution; central crack under hot shock, $a/H=0.1$,
 $K_0 = \bar{\lambda}_{11} \sqrt{\pi H (T_\infty - T_i)}$.

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